

Topological Energy Condensate Theory (TECT): 34

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I. REFERENCE-PDE FINITE EXTRACTION ON THE EXPLICIT SEED PROFILE

II. GOAL

We keep the reference reduced PDE fixed as

$$\boxed{\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = \beta = -\frac{\beta_X \gamma_{\perp}}{M_X^2} I_c,} \quad (1)$$

and perform the next nonredundant step: insert the explicit seed profile

$$\boxed{\theta(\xi) = \theta_0 \operatorname{sech}(\kappa\xi), \quad f_+(\xi) = \sqrt{\frac{\kappa}{2}} \operatorname{sech}(\kappa\xi), \quad f_-(\xi) = \sqrt{\frac{3\kappa}{2}} \tanh(\kappa\xi) \operatorname{sech}(\kappa\xi),} \quad (2)$$

into the overlap integrals

$$\boxed{I_3 = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3, \quad I_c = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) \cos \theta(\xi).} \quad (3)$$

III. EXPLICIT BACKGROUND DATA

Write

$$s(\xi) := \operatorname{sech}(\kappa\xi), \quad t(\xi) := \tanh(\kappa\xi). \quad (4)$$

Then

$$\theta(\xi) = \theta_0 s(\xi), \quad \theta'(\xi) = -\theta_0 \kappa s(\xi) t(\xi), \quad (5)$$

and

$$f_+(\xi) f_-(\xi) = \sqrt{\frac{\kappa}{2}} \sqrt{\frac{3\kappa}{2}} t(\xi) s(\xi)^2 = \frac{\sqrt{3}\kappa}{2} t(\xi) s(\xi)^2. \quad (6)$$

IV. EXACT EVALUATION OF I_3

Substituting the explicit profiles gives

$$I_3 = \int d\xi \frac{\sqrt{3}\kappa}{2} t s^2 (-\theta_0 \kappa s t)^3 \quad (7)$$

$$= -\frac{\sqrt{3}}{2} \theta_0^3 \kappa^4 \int d\xi t^4 s^5. \quad (8)$$

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Now set

$$u = \tanh(\kappa\xi), \quad du = \kappa s^2 d\xi, \quad s = \sqrt{1 - u^2}. \quad (9)$$

Then

$$I_3 = -\frac{\sqrt{3}}{2} \theta_0^3 \kappa^3 \int_{-1}^1 u^4 (1 - u^2)^{3/2} du. \quad (10)$$

The remaining integral is elementary:

$$\int_{-1}^1 u^4 (1 - u^2)^{3/2} du = \frac{3\pi}{128}. \quad (11)$$

Therefore

$$I_3 = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3. \quad (12)$$

Hence the longitudinal reduced coefficient becomes

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3 = -\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3. \quad (13)$$

V. IMMEDIATE EXACT LONGITUDINAL MATCHING

The canonical mixed odd-active target sheet requires

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3. \quad (14)$$

Comparing (13) with (14) yields

$$\lambda_c = \alpha_X L_*. \quad (15)$$

So on the explicit seed profile the longitudinal matching is exact.

VI. EXACT FINITE REPRESENTATION OF I_c

Now evaluate

$$I_c = \int d\xi \frac{\sqrt{3}\kappa}{2} t s^2 (-\theta_0 \kappa s t) \cos(\theta_0 s) \quad (16)$$

$$= -\frac{\sqrt{3}\theta_0 \kappa^2}{2} \int d\xi t^2 s^3 \cos(\theta_0 s). \quad (17)$$

Using again $u = \tanh(\kappa\xi)$, one obtains

$$I_c = -\frac{\sqrt{3}\theta_0 \kappa}{2} \int_{-1}^1 u^2 \sqrt{1 - u^2} \cos(\theta_0 \sqrt{1 - u^2}) du. \quad (18)$$

This is an exact finite one-dimensional representation of the transverse overlap.

VII. SMALL- θ_0 EXPANSION OF I_c

Using

$$\cos \theta = 1 - \frac{\theta^2}{2} + O(\theta^4), \quad (19)$$

we get

$$I_c = I_\theta - \frac{1}{2} I_{\theta^3} + O(\theta_0^5 \kappa), \quad (20)$$

where

$$I_\theta := \int d\xi f_+ f_- \theta', \quad I_{\theta^3} := \int d\xi f_+ f_- \theta' \theta^2. \quad (21)$$

The already known explicit value is

$$I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa. \quad (22)$$

Now compute I_{θ^3} :

$$I_{\theta^3} = \int d\xi \frac{\sqrt{3}\kappa}{2} t s^2 (-\theta_0 \kappa s t) (\theta_0^2 s^2) \quad (23)$$

$$= -\frac{\sqrt{3}\theta_0^3 \kappa^2}{2} \int d\xi t^2 s^5 \quad (24)$$

$$= -\frac{\sqrt{3}\theta_0^3 \kappa}{2} \int_{-1}^1 u^2 (1-u^2)^{3/2} du. \quad (25)$$

Since

$$\int_{-1}^1 u^2 (1-u^2)^{3/2} du = \frac{\pi}{16}, \quad (26)$$

we obtain

$$I_{\theta^3} = -\frac{\sqrt{3}\pi}{32} \theta_0^3 \kappa. \quad (27)$$

Therefore

$$I_c = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa + \frac{\sqrt{3}\pi}{64} \theta_0^3 \kappa + O(\theta_0^5 \kappa) \quad (28)$$

$$= -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left(1 - \frac{\theta_0^2}{4} + O(\theta_0^4) \right). \quad (29)$$

VIII. TRANSVERSE MATCHING

The target mixed odd-active transverse sheet is

$$\alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa. \quad (30)$$

From the reference route,

$$\alpha = \beta = -\frac{\beta_X \gamma_\perp}{M_X^2} I_c. \quad (31)$$

Hence the exact matching condition is

$$\boxed{\frac{\gamma_{\perp}}{M_X^2} = \frac{\frac{\sqrt{3}\pi}{16} \theta_0 \kappa}{I_c}}. \quad (32)$$

Using (29),

$$\boxed{\frac{\gamma_{\perp}}{M_X^2} = -\frac{1}{1 - \theta_0^2/4 + O(\theta_0^4)} = -\left(1 + \frac{\theta_0^2}{4} + O(\theta_0^4)\right)}. \quad (33)$$

So at leading order,

$$\boxed{\gamma_{\perp} \approx -M_X^2}. \quad (34)$$

IX. RESULTING EXPLICIT REFERENCE COEFFICIENT SHEET

The reference branch therefore becomes

$$\boxed{\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3}, \quad (35)$$

$$\boxed{\alpha = \beta = -\frac{\beta_X \gamma_{\perp}}{M_X^2} I_c, \quad I_c = -\frac{\sqrt{3} \theta_0 \kappa}{2} \int_{-1}^1 u^2 \sqrt{1-u^2} \cos(\theta_0 \sqrt{1-u^2}) du}. \quad (36)$$

In the small-defect regime this simplifies to

$$\boxed{\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3, \quad \alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa + O(\theta_0^3 \kappa)}. \quad (37)$$

X. HONEST STATUS

At the level of the explicit seed profile:

1. the longitudinal overlap is fully closed and yields the exact match

$$\boxed{\lambda_c = \alpha_X L_*};$$

2. the transverse overlap is reduced to the exact finite integral (18);
3. in the small-defect regime, the transverse matching gives

$$\boxed{\gamma_{\perp}/M_X^2 = -1 + O(\theta_0^2)};$$

4. therefore the reference route is now analytically bridged to the canonical mixed odd-active target, up to the remaining finite audits on the target branch itself.

XI. FINAL SUMMARY

On the explicit seed profile

$$\theta(\xi) = \theta_0 \operatorname{sech}(\kappa \xi), \quad f_+(\xi) = \sqrt{\frac{\kappa}{2}} \operatorname{sech}(\kappa \xi), \quad f_-(\xi) = \sqrt{\frac{3\kappa}{2}} \tanh(\kappa \xi) \operatorname{sech}(\kappa \xi),$$

the overlap integrals satisfy

$$I_3 = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3,$$

and

$$I_c = -\frac{\sqrt{3}\theta_0\kappa}{2} \int_{-1}^1 u^2 \sqrt{1-u^2} \cos(\theta_0 \sqrt{1-u^2}) du.$$

Hence

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3, \quad \alpha = \beta = -\frac{\beta_X \gamma_{\perp}}{M_X^2} I_c.$$

Matching to the canonical mixed odd-active target gives the exact longitudinal identity

$$\lambda_c = \alpha_X L_*.$$

For small θ_0 ,

$$I_c = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left(1 - \frac{\theta_0^2}{4} + O(\theta_0^4) \right),$$

so

$$\frac{\gamma_{\perp}}{M_X^2} = -\left(1 + \frac{\theta_0^2}{4} + O(\theta_0^4) \right),$$

and in particular

$$\gamma_{\perp} \approx -M_X^2$$

at leading order.

Therefore the reference PDE is now analytically matched to the target mixed branch as far as the explicit seed profile allows; the genuinely remaining open items are the two target-branch finite audits (Audit A and Audit B) and the actual microscopic insertion of the remaining same-shell/mixed-block constants.

XII. EXACT MEANING OF “AUDIT-FREE” AND IMMEDIATE CLOSURE OF AUDIT A

XIII. GOAL

The phrase “audit-free” should be understood sharply and operationally.

At the present stage, the canonical mixed odd-active branch is already structurally fixed by

$$T_E = T_O = 1, \quad R_{\perp} = \sqrt{2}, \quad \Delta_{00} = \frac{1}{\sqrt{2}},$$

and its reduced branch-level coefficient sheet is already

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad \alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa.$$

What is still missing is not a structural theorem, but the insertion of the two remaining finite microscopic audits. Therefore the next nonredundant step is to convert Audit A into a completely explicit linear package.

XIV. 1. EXACT OPERATIONAL MEANING OF AUDIT-FREE

A branch is *audit-free* if there remains no unresolved finite PASS/FAIL slot of the following kind:

1. an uninserted support inequality;
2. an uncomputed finite overlap package;
3. an unresolved cancellation/comparison correction;
4. a coefficient sheet that still depends on placeholders such as

$$\mathcal{S}_{\text{diag}}^{(\pi/4)}, \quad \varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)}, \quad \Delta e_{\text{mix}}, \quad \Delta o_{\text{mix}}.$$

Equivalently, the branch is audit-free iff the final microscopic coefficient point

$$(\lambda_{\parallel}, \alpha, \beta)$$

is fully populated with no remaining conditional insertion.

Thus the present mixed odd-active branch is *not* yet audit-free, because exactly two finite microscopic audits remain:

Audit A: mixed same-shell support insertion, Audit B: mixed-block transverse cancellation audit.

XV. 2. EXACT STATEMENT OF AUDIT A

The branch-consistent sufficient support condition is

$$\boxed{|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > \frac{4\beta_X^2}{M_X^2} \mathcal{S}_{\text{diag}}^{(\pi/4)} + \varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)}} \quad (38)$$

The previously isolated unresolved same-shell package consists of seven constants:

$$\boxed{\mathcal{S}_{\text{diag}}^{TT}, \quad L_+^{TT}, L_-^{TT}, L_0^{TT}, \quad C_+^{TT}, C_-^{TT}, C_m^{TT}} \quad (39)$$

The point of the present step is that these are not seven independent new mysteries. On the mixed branch they are exact linear combinations of the already available *EE*, *OO*, and *EO* Taylor packages.

XVI. 3. EXACT MIXED SAME-SHELL OVERLAP FAMILY

The exact branch-consistent mixed same-shell overlap family is

$$\boxed{\mathcal{S}_{ab}^{TT}(p; \varphi) = \cos^2 \varphi \mathcal{S}_{ab}^{EE}(p) + \sin^2 \varphi \mathcal{S}_{ab}^{OO}(p) + 2 \sin \varphi \cos \varphi \mathcal{S}_{ab}^{EO}(p)}, \quad (40)$$

for $a, b \in \{+, -\}$.

On the canonical mixed branch,

$$\varphi = \frac{\pi}{4}, \quad \cos^2 \varphi = \sin^2 \varphi = \frac{1}{2}, \quad 2 \sin \varphi \cos \varphi = 1.$$

Therefore

$$\boxed{\mathcal{S}_{ab}^{TT}(p; \pi/4) = \frac{1}{2} \mathcal{S}_{ab}^{EE}(p) + \frac{1}{2} \mathcal{S}_{ab}^{OO}(p) + \mathcal{S}_{ab}^{EO}(p)}. \quad (41)$$

This is the key exact reduction formula.

XVII. 4. TAYLOR EXPANSION AND THE SEVEN-CONSTANT PACKAGE

Expand each block near $p = 0$ in the already used same-shell Taylor form:

$$\mathcal{S}_{ab}^{XY}(p) = \mathcal{S}_{ab}^{XY}(0) + L_{ab}^{XY} \cdot p + Q_{ab}^{XY}(p), \quad Q_{ab}^{XY}(p) = O(|p|^2), \quad (42)$$

where $XY \in \{EE, OO, EO\}$.

Then by linearity of (41), every Taylor coefficient of TT is the same linear combination of the corresponding EE , OO , and EO coefficients.

In particular, the diagonal zero-momentum term is

$$\mathcal{S}_{\text{diag}}^{(\pi/4)} = \frac{1}{2} \mathcal{S}_{\text{diag}}^{EE} + \frac{1}{2} \mathcal{S}_{\text{diag}}^{OO} + \mathcal{S}_{\text{diag}}^{EO}. \quad (43)$$

Likewise the linear Taylor data become

$$L_+^{TT} = \frac{1}{2} L_+^{EE} + \frac{1}{2} L_+^{OO} + L_+^{EO}, \quad (44)$$

$$L_-^{TT} = \frac{1}{2} L_-^{EE} + \frac{1}{2} L_-^{OO} + L_-^{EO}, \quad (45)$$

$$L_0^{TT} = \frac{1}{2} L_0^{EE} + \frac{1}{2} L_0^{OO} + L_0^{EO}. \quad (46)$$

And the quadratic drift constants become

$$C_+^{TT} = \frac{1}{2} C_+^{EE} + \frac{1}{2} C_+^{OO} + C_+^{EO}, \quad (47)$$

$$C_-^{TT} = \frac{1}{2} C_-^{EE} + \frac{1}{2} C_-^{OO} + C_-^{EO}, \quad (48)$$

$$C_m^{TT} = \frac{1}{2} C_m^{EE} + \frac{1}{2} C_m^{OO} + C_m^{EO}. \quad (49)$$

Therefore the seven unresolved Audit-A constants are exactly reducible to the $EE/OO/EO$ Taylor package.

XVIII. 5. EXPLICIT DRIFT BOUND IN MIXED FORM

The small- p drift term entering Audit A has the same form as in the pure branch, but now with the mixed TT constants:

$$\varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)} \leq \left[(\rho A_+^{TT} + C_+^{TT} \rho^2)^2 + (\rho A_-^{TT} + C_-^{TT} \rho^2)^2 + 2(\rho B^{TT} + C_m^{TT} \rho^2)^2 \right]^{1/2}, \quad (50)$$

where the linear budgets $A_{\pm}^{TT}, B^{TT}$ are extracted from $L_{\pm}^{TT}, L_0^{TT}$ in the same way as in the already established same-shell drift estimate.

So after (44)–(49), the only remaining work in Audit A is literal insertion.

XIX. 6. AUDIT A IS NOW REDUCED TO A TABLE-FILL PROBLEM

Substituting (43)–(49) into (38) gives the exact mixed-branch same-shell support condition:

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > \frac{4\beta_X^2}{M_X^2} \left(\frac{1}{2} \mathcal{S}_{\text{diag}}^{EE} + \frac{1}{2} \mathcal{S}_{\text{diag}}^{OO} + \mathcal{S}_{\text{diag}}^{EO} \right) + \varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)}, \quad (51)$$

with

$$\varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)}$$

itself determined by the linear combinations (44)–(49).

Hence Audit A is no longer conceptually open. It is reduced exactly to the insertion of the already defined EE , OO , and EO Taylor tables.

XX. 7. SHARP EXECUTION ORDER FOR AUDIT B

Once Audit A is reduced as above, the next step is not to reopen any structural discussion. It is to execute Audit B in the minimal order:

$$\text{(i) compute } e_{0;K}^{(1)}, o_{0;K}^{(1)}, \quad \text{(ii) compute } \Delta e_{\text{mix}}, \Delta o_{\text{mix}}, \quad (52)$$

from the mixed blocks

$$TD, DT, OD, DO, DD.$$

The actual transverse pair is

$$\Lambda_E = e_{0;K}^{(1)} + \Delta e_{\text{mix}}, \quad \Lambda_O = o_{0;K}^{(1)} + \Delta o_{\text{mix}}. \quad (53)$$

For mere survival, a strong sufficient condition is

$$|\Delta e_{\text{mix}}| < |e_{0;K}^{(1)}|, \quad |\Delta o_{\text{mix}}| < |o_{0;K}^{(1)}|. \quad (54)$$

For the canonical symmetric mixed target, one needs in addition

$$\Lambda_E = \Lambda_O, \quad (55)$$

that is,

$$\Delta o_{\text{mix}} - \Delta e_{\text{mix}} = e_{0;K}^{(1)} - o_{0;K}^{(1)}. \quad (56)$$

So Audit B naturally splits into two independent checks:

1. **non-cancellation check:** $(\Lambda_E, \Lambda_O) \neq (0, 0)$;
2. **symmetric-target check:** $\Lambda_E = \Lambda_O$.

XXI. 8. HONEST STATUS AFTER THE PRESENT STEP

After the exact reduction (41)–(49), Audit A is no longer a conceptual audit. It is a finite table-fill audit.

Therefore the only genuinely uncomputed object left after Audit A is Audit B: the mixed-block transverse correction pair

$$(\Delta e_{\text{mix}}, \Delta o_{\text{mix}}).$$

So the sharp current statement is:

$$\text{not audit-free yet, but Audit A has now been algebraically closed.} \quad (57)$$

XXII. FINAL SUMMARY

Audit-free means:

$$\boxed{\text{no remaining unresolved finite PASS/FAIL insertion slot.}}$$

Equivalently, the final microscopic coefficient point

$$(\lambda_{\parallel}, \alpha, \beta)$$

must be fully populated with no placeholder constants or pending cancellation checks.

At the present stage, the mixed odd-active branch is still *not* audit-free because exactly two finite audits remain:

Audit A: mixed same-shell support insertion, Audit B: mixed-block transverse cancellation audit.

However, Audit A is now algebraically closed.

On the canonical mixed branch $\varphi = \pi/4$,

$$\boxed{\mathcal{S}_{ab}^{TT}(p) = \frac{1}{2}\mathcal{S}_{ab}^{EE}(p) + \frac{1}{2}\mathcal{S}_{ab}^{OO}(p) + \mathcal{S}_{ab}^{EO}(p).}$$

Therefore the full seven-constant same-shell package is exactly

$$\boxed{\mathcal{S}_{\text{diag}}^{(\pi/4)} = \frac{1}{2}\mathcal{S}_{\text{diag}}^{EE} + \frac{1}{2}\mathcal{S}_{\text{diag}}^{OO} + \mathcal{S}_{\text{diag}}^{EO},}$$

$$\boxed{L_{\pm}^{TT} = \frac{1}{2}L_{\pm}^{EE} + \frac{1}{2}L_{\pm}^{OO} + L_{\pm}^{EO}, \quad L_0^{TT} = \frac{1}{2}L_0^{EE} + \frac{1}{2}L_0^{OO} + L_0^{EO},}$$

$$\boxed{C_{\pm}^{TT} = \frac{1}{2}C_{\pm}^{EE} + \frac{1}{2}C_{\pm}^{OO} + C_{\pm}^{EO}, \quad C_m^{TT} = \frac{1}{2}C_m^{EE} + \frac{1}{2}C_m^{OO} + C_m^{EO}.}$$

Thus Audit A is no longer structurally open. It is reduced to literal insertion of the $EE/OO/EO$ Taylor table into the support inequality.

The only genuinely remaining unresolved object after that is Audit B:

$$\boxed{\Lambda_E = e_{0;K}^{(1)} + \Delta e_{\text{mix}}, \quad \Lambda_O = o_{0;K}^{(1)} + \Delta o_{\text{mix}}.}$$

For the canonical symmetric target one must check both

$$(\Lambda_E, \Lambda_O) \neq (0, 0)$$

and

$$\Lambda_E = \Lambda_O.$$

So the project is not audit-free yet, but one of the two final audits has now been reduced to a direct table-fill computation.

XXIII. AUDIT B: EXACT REDUCTION TO PRINCIPAL PAIR PLUS RESIDUAL CORRECTION

XXIV. GOAL

The remaining mixed-block transverse audit is to determine whether the actual transverse coefficient pair

$$(\Lambda_E, \Lambda_O)$$

survives after the mixed completion channels are inserted, and whether it matches the canonical symmetric mixed target

$$\Lambda_E = \Lambda_O \neq 0.$$

The exact conservative branch-level form is

$$\boxed{\Lambda_E = e_{0;K}^{(1)} + \Delta e_{\text{mix}}, \quad \Lambda_O = o_{0;K}^{(1)} + \Delta o_{\text{mix}}.} \tag{58}$$

XXV. 1. EXACT BLOCKWISE DECOMPOSITION

For each candidate mixed block

$$AB \in \{TD, DT, OD, DO, DD\},$$

define the leading even/odd transverse coefficients by

$$e_{0;AB}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{1}{4} E^\mu{}_a \text{Tr} [\Sigma_a \mathcal{M}_{AB, \mu}^{\text{inv}}], \quad (59)$$

$$o_{0;AB}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{1}{4} J^\mu{}_a \text{Tr} [\Sigma_a \mathcal{M}_{AB, \mu}^{\text{inv}}]. \quad (60)$$

Then the full leading transverse coefficient point is

$$\Lambda_E = \sum_{AB \in \{TD, DT, OD, DO, DD\}} e_{0;AB}^{(1)}, \quad \Lambda_O = \sum_{AB \in \{TD, DT, OD, DO, DD\}} o_{0;AB}^{(1)}. \quad (61)$$

XXVI. 2. PRINCIPAL PAIR AND RESIDUAL CORRECTION VECTOR

Define the principal pair

$$\Lambda_E^{\text{pr}} := e_{0;TD}^{(1)} + e_{0;DT}^{(1)}, \quad \Lambda_O^{\text{pr}} := o_{0;TD}^{(1)} + o_{0;DT}^{(1)}. \quad (62)$$

Define the residual correction vector

$$r_{\text{rem}} := (r_E, r_O), \quad r_E := e_{0;OD}^{(1)} + e_{0;DO}^{(1)} + e_{0;DD}^{(1)} + \dots, \quad r_O := o_{0;OD}^{(1)} + o_{0;DO}^{(1)} + o_{0;DD}^{(1)} + \dots. \quad (63)$$

Then

$$\Lambda_E = \Lambda_E^{\text{pr}} + r_E, \quad \Lambda_O = \Lambda_O^{\text{pr}} + r_O. \quad (64)$$

At the current conservative bookkeeping level one may identify

$$e_{0;K}^{(1)} \equiv \Lambda_E^{\text{pr}}, \quad o_{0;K}^{(1)} \equiv \Lambda_O^{\text{pr}}, \quad \Delta e_{\text{mix}} \equiv r_E, \quad \Delta o_{\text{mix}} \equiv r_O. \quad (65)$$

XXVII. 3. LEADING MINIMAL TRUNCATION

At the minimal mixed-block truncation,

$$OD = DO = DD = 0,$$

so

$$\Lambda_E = \Lambda_E^{\text{pr}}, \quad \Lambda_O = \Lambda_O^{\text{pr}}. \quad (66)$$

If the branch-summary invariant form is written as

$$V_{TD}^\mu = \theta'(\xi) [A_E E^\mu{}_a \Sigma_a + A_O J^\mu{}_a \Sigma_a], \quad V_{DT}^\mu = \theta'(\xi) [B_E E^\mu{}_a \Sigma_a + B_O J^\mu{}_a \Sigma_a], \quad (67)$$

then

$$\Lambda_E^{\text{pr}} = \frac{1}{2}(A_E + B_E), \quad \Lambda_O^{\text{pr}} = \frac{1}{2}(A_O + B_O). \quad (68)$$

Hence in the minimal truncation the reduced coefficients are

$$\alpha = -\frac{\sqrt{3}\pi}{32} \theta_0 \kappa (A_E + B_E), \quad \beta = -\frac{\sqrt{3}\pi}{32} \theta_0 \kappa (A_O + B_O). \quad (69)$$

XXVIII. 4. SURVIVAL CONDITION

The exact no-cancellation survival condition is

$$\boxed{\Delta e_{\text{mix}} \neq -e_{0;K}^{(1)}, \quad \Delta o_{\text{mix}} \neq -o_{0;K}^{(1)}} \quad (70)$$

A stronger sufficient condition is

$$\boxed{|\Delta e_{\text{mix}}| < |e_{0;K}^{(1)}|, \quad |\Delta o_{\text{mix}}| < |o_{0;K}^{(1)}|} \quad (71)$$

Under (71), the signs of Λ_E, Λ_O are inherited from the principal K -sector pair.

XXIX. 5. SYMMETRIC-TARGET REDUCTION

For the canonical mixed odd-active target, one needs

$$\boxed{\Lambda_E = \Lambda_O} \quad (72)$$

Using (58), this is equivalent to

$$\boxed{\Delta o_{\text{mix}} - \Delta e_{\text{mix}} = e_{0;K}^{(1)} - o_{0;K}^{(1)}} \quad (73)$$

This is the sharp exact equality test.

Thus the symmetric-target audit is not a two-dimensional comparison problem anymore. It is a single scalar difference equation.

XXX. 6. SUM/DIFFERENCE NORMAL FORM

Define

$$s_K := \frac{1}{2}(e_{0;K}^{(1)} + o_{0;K}^{(1)}), \quad d_K := \frac{1}{2}(e_{0;K}^{(1)} - o_{0;K}^{(1)}), \quad (74)$$

$$s_{\text{mix}} := \frac{1}{2}(\Delta e_{\text{mix}} + \Delta o_{\text{mix}}), \quad d_{\text{mix}} := \frac{1}{2}(\Delta e_{\text{mix}} - \Delta o_{\text{mix}}). \quad (75)$$

Then

$$\boxed{\Lambda_E = (s_K + s_{\text{mix}}) + (d_K + d_{\text{mix}}), \quad \Lambda_O = (s_K + s_{\text{mix}}) - (d_K + d_{\text{mix}})} \quad (76)$$

Therefore:

1. symmetric target

$$\Lambda_E = \Lambda_O \iff d_K + d_{\text{mix}} = 0;$$

2. nonzero survival on the symmetric target

$$\Lambda_E = \Lambda_O \neq 0 \iff d_K + d_{\text{mix}} = 0 \quad \text{and} \quad s_K + s_{\text{mix}} \neq 0.$$

Equivalently,

$$\boxed{d_{\text{mix}} = -d_K, \quad s_{\text{mix}} \neq -s_K} \quad (77)$$

This is the cleanest normal form of Audit B.

XXXI. 7. FINAL POPULATED COEFFICIENT SHEET AFTER AUDIT B

Once Audit B is inserted, the mixed odd-active branch coefficient point is

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{256}\theta_0^3\kappa^3\ell_{0;OO}^{(3)}, \quad (78)$$

$$\alpha = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\left(e_{0;K}^{(1)} + \Delta e_{\text{mix}}\right), \quad (79)$$

$$\beta = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\left(o_{0;K}^{(1)} + \Delta o_{\text{mix}}\right). \quad (80)$$

For the canonical symmetric mixed target one further imposes

$$e_{0;K}^{(1)} + \Delta e_{\text{mix}} = o_{0;K}^{(1)} + \Delta o_{\text{mix}} \neq 0. \quad (81)$$

XXXII. 8. HONEST STATUS

Audit B is now reduced to two finite microscopic objects:

1. the principal pair

$$(\Lambda_E^{\text{pr}}, \Lambda_O^{\text{pr}}) = (e_{0;TD}^{(1)} + e_{0;DT}^{(1)}, o_{0;TD}^{(1)} + o_{0;DT}^{(1)});$$

2. the residual correction vector

$$r_{\text{rem}} = (r_E, r_O) = \left(\sum_{AB \in \{OD, DO, DD\}} e_{0;AB}^{(1)}, \sum_{AB \in \{OD, DO, DD\}} o_{0;AB}^{(1)} \right).$$

So the actual remaining work is no longer conceptual. It is the literal evaluation of

$$r_{\text{rem}}.$$

Once r_{rem} is known, both survival and the symmetric-target condition are decided immediately by (70) and (73).

XXXIII. FINAL SUMMARY

The mixed-block transverse audit is exactly reducible to

$$(\Lambda_E, \Lambda_O) = (\Lambda_E^{\text{pr}}, \Lambda_O^{\text{pr}}) + r_{\text{rem}},$$

where

$$\Lambda_E^{\text{pr}} = e_{0;TD}^{(1)} + e_{0;DT}^{(1)}, \quad \Lambda_O^{\text{pr}} = o_{0;TD}^{(1)} + o_{0;DT}^{(1)},$$

and

$$r_{\text{rem}} = \left(\sum_{AB \in \{OD, DO, DD\}} e_{0;AB}^{(1)}, \sum_{AB \in \{OD, DO, DD\}} o_{0;AB}^{(1)} \right).$$

The exact survival condition is

$$\Delta e_{\text{mix}} \neq -e_{0;K}^{(1)}, \quad \Delta o_{\text{mix}} \neq -o_{0;K}^{(1)},$$

and a strong sufficient condition is

$$|\Delta e_{\text{mix}}| < |e_{0;K}^{(1)}|, \quad |\Delta o_{\text{mix}}| < |o_{0;K}^{(1)}|.$$

For the canonical mixed odd-active target, the equality condition is reduced to the single scalar test

$$\Delta o_{\text{mix}} - \Delta e_{\text{mix}} = e_{0;K}^{(1)} - o_{0;K}^{(1)}.$$

In sum/difference variables

$$s_K = \frac{e_{0;K}^{(1)} + o_{0;K}^{(1)}}{2}, \quad d_K = \frac{e_{0;K}^{(1)} - o_{0;K}^{(1)}}{2}, \quad s_{\text{mix}} = \frac{\Delta e_{\text{mix}} + \Delta o_{\text{mix}}}{2}, \quad d_{\text{mix}} = \frac{\Delta e_{\text{mix}} - \Delta o_{\text{mix}}}{2},$$

the canonical target condition becomes

$$d_{\text{mix}} = -d_K, \quad s_{\text{mix}} \neq -s_K.$$

Therefore Audit B is no longer conceptually open. It is reduced completely to the finite evaluation of the residual correction vector

$$r_{\text{rem}} = \left(\sum_{AB \in \{OD, DO, DD\}} e_{0;AB}^{(1)}, \quad \sum_{AB \in \{OD, DO, DD\}} o_{0;AB}^{(1)} \right).$$

XXXIV. AUDIT B REDUCED TO TWO EXPLICIT RESIDUAL PROJECTOR COMPUTATIONS

XXXV. GOAL

The exact residual mixed-block audit has already been reduced to the blocks

$$OD, \quad DO, \quad DD.$$

The next nonredundant step is to reduce this further by the two available structural simplifications:

1. reflection parity under $\xi \mapsto -\xi$;
2. $O \leftrightarrow D$ exchange symmetry.

This yields the sharpest executable form of Audit B.

XXXVI. 1. EXACT RESIDUAL VECTOR

For each residual block $AB \in \{OD, DO, DD\}$, define

$$e_{0;AB}^{(1)} = \lim \mathfrak{P}_E[\mathcal{M}_{AB}^{\text{inv}}], \quad o_{0;AB}^{(1)} = \lim \mathfrak{P}_O[\mathcal{M}_{AB}^{\text{inv}}]. \quad (82)$$

Then the exact residual correction vector is

$$r_{\text{rem}} = (\Delta e_{\text{rem}}, \Delta o_{\text{rem}}) = \sum_{AB \in \{OD, DO, DD\}} (e_{0;AB}^{(1)}, o_{0;AB}^{(1)}). \quad (83)$$

The full transverse coefficient point is therefore

$$(\Lambda_E, \Lambda_O) = r_{\text{pr}} + r_{\text{rem}}, \quad (84)$$

with r_{pr} the already isolated principal $TD + DT$ source.

XXXVII. 2. PARITY ELIMINATION RULE

For a given residual block AB , the exact next test is the reflection parity of

$$\mathcal{M}_{AB,\text{inv}}^\mu(\xi)$$

under

$$\xi \mapsto -\xi.$$

The sharp rule is:

$$\boxed{\mathcal{M}_{AB,\text{inv}}^\mu(-\xi) = +\mathcal{M}_{AB,\text{inv}}^\mu(\xi) \implies (e_{0;AB}^{(1)}, o_{0;AB}^{(1)}) = (0, 0),} \quad (85)$$

while

$$\boxed{\mathcal{M}_{AB,\text{inv}}^\mu(-\xi) = -\mathcal{M}_{AB,\text{inv}}^\mu(\xi) \implies \text{the block is parity-allowed and must be computed.}} \quad (86)$$

Hence the true residual vector is

$$\boxed{r_{\text{rem}} = \sum_{\substack{AB \in \{OD, DO, DD\} \\ \mathcal{M}_{AB} \text{ odd}}} (e_{0;AB}^{(1)}, o_{0;AB}^{(1)})}. \quad (87)$$

XXXVIII. 3. EXCHANGE REDUCTION

If the projected mixed sector preserves the $O \leftrightarrow D$ exchange symmetry, then DO is not independent of OD , and one may reduce at the symmetry-reduced level to

$$\boxed{e_{0;DO}^{(1)} = e_{0;OD}^{(1)}, \quad o_{0;DO}^{(1)} = o_{0;OD}^{(1)}}. \quad (88)$$

Therefore the residual vector becomes

$$\boxed{r_{\text{rem}} = 2(e_{0;OD}^{(1)}, o_{0;OD}^{(1)}) + (e_{0;DD}^{(1)}, o_{0;DD}^{(1)})}, \quad (89)$$

again with the understanding that any even-parity block drops out exactly.

Thus the practical audit is reduced from three raw projector computations to:

$$\boxed{\text{compute only the parity-allowed part of } OD \text{ and } DD,} \quad (90)$$

provided exchange symmetry holds.

XXXIX. 4. CANONICAL SYMMETRIC MIXED-TARGET CONDITION

The conservative coefficient sheet is

$$\boxed{\alpha = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa(e_{0;K}^{(1)} + \Delta e_{\text{rem}}), \quad \beta = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa(o_{0;K}^{(1)} + \Delta o_{\text{rem}}).} \quad (91)$$

For the canonical mixed odd-active target one needs

$$\boxed{\Lambda_E = \Lambda_O \neq 0.} \quad (92)$$

Using the exchange-reduced form

$$\Delta e_{\text{rem}} = 2e_{0;OD}^{(1)} + e_{0;DD}^{(1)}, \quad \Delta o_{\text{rem}} = 2o_{0;OD}^{(1)} + o_{0;DD}^{(1)},$$

the equality condition becomes

$$\boxed{2(o_{0;OD}^{(1)} - e_{0;OD}^{(1)}) + (o_{0;DD}^{(1)} - e_{0;DD}^{(1)}) = e_{0;K}^{(1)} - o_{0;K}^{(1)}}. \quad (93)$$

This is the sharp final scalar condition for the symmetric target.

XL. 5. SURVIVAL INEQUALITIES

The strong componentwise sufficient survival condition is

$$\boxed{|\Delta e_{\text{rem}}| < |\Lambda_E^{\text{pr}}|, \quad |\Delta o_{\text{rem}}| < |\Lambda_O^{\text{pr}}|.} \quad (94)$$

A weaker Euclidean sufficient condition is

$$\boxed{\|r_{\text{rem}}\|_2 < \|r_{\text{pr}}\|_2.} \quad (95)$$

Under exchange reduction, these become

$$\boxed{\left| 2e_{0;OD}^{(1)} + e_{0;DD}^{(1)} \right| < |\Lambda_E^{\text{pr}}|, \quad \left| 2o_{0;OD}^{(1)} + o_{0;DD}^{(1)} \right| < |\Lambda_O^{\text{pr}}|,} \quad (96)$$

or, more coarsely,

$$\boxed{\left\| 2(e_{0;OD}^{(1)}, o_{0;OD}^{(1)}) + (e_{0;DD}^{(1)}, o_{0;DD}^{(1)}) \right\|_2 < \|r_{\text{pr}}\|_2.} \quad (97)$$

XLI. 6. FINAL EXECUTABLE DECISION TREE

Thus the residual transverse audit is now exactly:

1. test the parity of $\mathcal{M}_{OD,\text{inv}}^\mu$ and $\mathcal{M}_{DD,\text{inv}}^\mu$;
2. discard whichever block is reflection-even;
3. compute the projector traces only for the parity-allowed blocks;
4. insert the result into (96) and (93).

If both OD and DD are even, then

$$\boxed{r_{\text{rem}} = 0,} \quad (98)$$

and the canonical target reduces immediately to

$$\boxed{e_{0;K}^{(1)} = o_{0;K}^{(1)} \neq 0.} \quad (99)$$

If exactly one of them is odd, only that one must be computed.

If both are odd, then both must be computed and inserted through (93).

XLII. FINAL SUMMARY

The residual transverse correction is no longer a three-block open problem in practice.

Its exact starting form is

$$r_{\text{rem}} = \sum_{AB \in \{OD, DO, DD\}} (e_{0;AB}^{(1)}, o_{0;AB}^{(1)}).$$

Using reflection parity, only the $\xi \mapsto -\xi$ odd kernels survive:

$$r_{\text{rem}} = \sum_{\substack{AB \in \{OD, DO, DD\} \\ \mathcal{M}_{AB} \text{ odd}}} (e_{0;AB}^{(1)}, o_{0;AB}^{(1)}).$$

Using $O \leftrightarrow D$ exchange symmetry, one gets

$$e_{0;DO}^{(1)} = e_{0;OD}^{(1)}, \quad o_{0;DO}^{(1)} = o_{0;OD}^{(1)},$$

so the residual vector reduces to

$$r_{\text{rem}} = 2(e_{0;OD}^{(1)}, o_{0;OD}^{(1)}) + (e_{0;DD}^{(1)}, o_{0;DD}^{(1)}),$$

with any even-parity block removed exactly.

Therefore the practical Audit B is reduced to two literal residual projector computations:

$$OD, \quad DD.$$

The canonical symmetric mixed-target condition

$$\Lambda_E = \Lambda_O$$

becomes the single scalar equation

$$2(o_{0;OD}^{(1)} - e_{0;OD}^{(1)}) + (o_{0;DD}^{(1)} - e_{0;DD}^{(1)}) = e_{0;K}^{(1)} - o_{0;K}^{(1)}.$$

The strong sufficient survival inequalities become

$$\left| 2e_{0;OD}^{(1)} + e_{0;DD}^{(1)} \right| < |\Lambda_E^{\text{pr}}|, \quad \left| 2o_{0;OD}^{(1)} + o_{0;DD}^{(1)} \right| < |\Lambda_O^{\text{pr}}|.$$

So the project is now at the sharpest possible finite stage:

$$\boxed{\text{compute the parity-allowed part of } OD \text{ and } DD, \text{ then insert them once.}}$$

XLIII. RESIDUAL AUDIT B REDUCED TO THE ODD PARTS OF OD AND DD

XLIV. GOAL

The next sharp step is not to re-open the block list

$$OD, \quad DO, \quad DD,$$

but to split each residual block into its reflection-even and reflection-odd pieces, and keep only the odd pieces that can survive the graded projection.

XLV. 1. GENERAL PARITY DECOMPOSITION

For any residual block $AB \in \{OD, DO, DD\}$, write

$$\boxed{\mathcal{M}_{AB,\text{inv}}^\mu(\xi) = \mathcal{M}_{AB,\text{even}}^\mu(\xi) + \mathcal{M}_{AB,\text{odd}}^\mu(\xi),} \tag{100}$$

with

$$\mathcal{M}_{AB,\text{even}}^\mu(-\xi) = +\mathcal{M}_{AB,\text{even}}^\mu(\xi), \quad \mathcal{M}_{AB,\text{odd}}^\mu(-\xi) = -\mathcal{M}_{AB,\text{odd}}^\mu(\xi). \tag{101}$$

For the reflection-symmetric defect background, the graded low modes satisfy

$$u_a^+(-\xi) = +u_a^+(\xi), \quad u_b^-(-\xi) = -u_b^-(\xi).$$

Hence the projected mixed matrix element

$$(M_{AB}^\mu)_{ab} = \int d\xi (u_a^+)^\dagger \mathcal{M}_{AB,\text{inv}}^\mu u_b^-$$

receives no contribution from the even part:

$$\boxed{\int d\xi (u_a^+)^\dagger \mathcal{M}_{AB,\text{even}}^\mu u_b^- = 0.} \quad (102)$$

Therefore

$$\boxed{e_{0;AB}^{(1)} = \lim \mathfrak{P}_E[\mathcal{M}_{AB,\text{odd}}], \quad o_{0;AB}^{(1)} = \lim \mathfrak{P}_O[\mathcal{M}_{AB,\text{odd}}].} \quad (103)$$

XLVI. 2. SHARP *OD*-CLASSIFICATION

The *OD*-block is not a direct $J^{(1)}/K^{(1)}$ odd transport source. Instead it is generated from the mixed quartic interaction

$$\lambda_{OD} \rho_O \rho_D$$

on the background

$$\bar{\Phi}_{\min} = (0, \bar{n}_B(\xi), \bar{\chi}_B(\xi)).$$

Its background-generated cubic vertices are

$$\boxed{\Gamma_{\nu\nu\zeta}^{(3)}[\bar{\Phi}_{\min}] \sim \lambda_{OD} \bar{\chi}_B(\xi), \quad \Gamma_{\nu\zeta\zeta}^{(3)}[\bar{\Phi}_{\min}] \sim \lambda_{OD} \bar{n}_B(\xi).} \quad (104)$$

Therefore the *OD*-kernel is not theorem-level odd by itself. Its honest decomposition is

$$\boxed{\mathcal{M}_{OD,\text{inv}}^\mu = \mathcal{M}_{OD,\text{pot/even}}^\mu + \mathcal{M}_{OD,\text{trans/odd}}^\mu.} \quad (105)$$

Here:

1. $\mathcal{M}_{OD,\text{pot/even}}^\mu$ denotes the part built only from background amplitudes and non-gradient projector contractions; it is reflection-even and vanishes after graded projection;
2. $\mathcal{M}_{OD,\text{trans/odd}}^\mu$ denotes the part that acquires one transverse background-gradient insertion through the projected kernel extraction, derivative-anisotropy response, or background-to-worldvolume transport conversion; this is the parity-allowed piece.

Thus the exact projector-visible *OD*-coefficients are

$$\boxed{e_{0;OD}^{(1)} = \lim \mathfrak{P}_E[\mathcal{M}_{OD,\text{trans/odd}}^\mu], \quad o_{0;OD}^{(1)} = \lim \mathfrak{P}_O[\mathcal{M}_{OD,\text{trans/odd}}^\mu].} \quad (106)$$

XLVII. 3. SHARP *DD*-CLASSIFICATION

The *DD*-sector contains

$$\boxed{\mathcal{H}_{DD} = -Z_{D,k} \partial^2 + \mathcal{M}_{DD}^{\text{pot}}(\xi) + Y_k \mathcal{K}_{DD}^{\nabla \text{ani}}(\xi, \partial) + \mathcal{T}_{DD}(\xi, \partial_y).} \quad (107)$$

Hence its first-order projected kernel also splits naturally:

$$\boxed{\mathcal{M}_{DD,\text{inv}}^\mu = \mathcal{M}_{DD,\text{pot/even}}^\mu + \mathcal{M}_{DD,\nabla \text{ani/odd}}^\mu + \mathcal{M}_{DD,\text{trans/odd}}^\mu.} \quad (108)$$

The honest status is therefore:

1. the defect-background-dependent potential tensor contributes an even piece, killed by the graded projection;
2. derivative-anisotropy terms may generate an odd piece once one transverse background-gradient insertion is present;
3. the explicit transport block $\mathcal{T}_{DD}$ is the most natural source of a parity-allowed odd contribution.

Therefore

$$\boxed{e_{0;DD}^{(1)} = \lim \mathfrak{P}_E[\mathcal{M}_{DD,\text{odd}}^\mu], \quad o_{0;DD}^{(1)} = \lim \mathfrak{P}_O[\mathcal{M}_{DD,\text{odd}}^\mu]}, \quad (109)$$

where

$$\mathcal{M}_{DD,\text{odd}}^\mu := \mathcal{M}_{DD,\nabla\text{ani/odd}}^\mu + \mathcal{M}_{DD,\text{trans/odd}}^\mu.$$

XLVIII. 4. UNIVERSAL MONOMIAL NORMAL FORM OF EVERY SURVIVING ODD RESIDUAL PIECE

The full principal transport kernel is already classified by the monomial families

$$\theta' X_A n^\mu \sigma_3, \quad \theta'^3 X_A n^\mu \sigma_3, \quad \theta' X_A E^\mu{}_a \sigma_a, \quad \theta' X_A J^\mu{}_a \sigma_a,$$

where X_A are parity-even local defect invariants.

Therefore any residual contribution that survives the parity filter and contributes to the transverse pair must necessarily reduce to

$$\boxed{\mathcal{M}_{AB,\text{odd}}^\mu = \theta'(\xi) \left[X_E^{(AB)}(\xi) E^\mu{}_a \Sigma_a + X_O^{(AB)}(\xi) J^\mu{}_a \Sigma_a \right], \quad AB \in \{OD, DD\},} \quad (110)$$

with

$$X_E^{(AB)}(-\xi) = X_E^{(AB)}(\xi), \quad X_O^{(AB)}(-\xi) = X_O^{(AB)}(\xi).$$

Thus the parity-allowed residual sector is not arbitrary: every surviving transverse correction must contain exactly one odd defect-gradient factor $\theta'(\xi)$, multiplied by parity-even local invariants.

XLIX. 5. EXCHANGE REDUCTION

If the background and projected regulator preserve the $O \leftrightarrow D$ exchange symmetry to leading order, then

$$\boxed{e_{0;DO}^{(1)} = e_{0;OD}^{(1)}, \quad o_{0;DO}^{(1)} = o_{0;OD}^{(1)}} \quad (111)$$

So the residual correction vector becomes

$$\boxed{\Delta e_{\text{rem}} = 2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)}, \quad \Delta o_{\text{rem}} = 2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)}} \quad (112)$$

L. 6. FINAL SCALAR CONDITION FOR THE CANONICAL SYMMETRIC TARGET

The canonical mixed odd-active target requires

$$\Lambda_E = \Lambda_O \neq 0.$$

Using

$$\Lambda_E = e_{0;K}^{(1)} + \Delta e_{\text{rem}}, \quad \Lambda_O = o_{0;K}^{(1)} + \Delta o_{\text{rem}},$$

together with (112), the exact equality condition becomes

$$\boxed{2\left(o_{0;OD,\text{odd}}^{(1)} - e_{0;OD,\text{odd}}^{(1)}\right) + \left(o_{0;DD,\text{odd}}^{(1)} - e_{0;DD,\text{odd}}^{(1)}\right) = e_{0;K}^{(1)} - o_{0;K}^{(1)}} \quad (113)$$

The strong sufficient survival conditions become

$$\boxed{\left|2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)}\right| < |e_{0;K}^{(1)}|,} \quad (114)$$

$$\boxed{\left|2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)}\right| < |o_{0;K}^{(1)}|.} \quad (115)$$

LI. 7. HONEST CLOSURE STATUS

At the present stage the sharpest closed statement is:

$$\boxed{OD \text{ and } DD \text{ are not theorem-level zero,}} \quad (116)$$

but

$$\boxed{\text{their even parts are killed exactly, and only their odd } \theta' X_A\text{-type parts remain load-bearing.}} \quad (117)$$

Therefore the final residual work is no longer “compute the full OD and DD blocks”. It is only:

$$\boxed{\text{extract the coefficients of the odd normal forms } \theta' X_A E^\mu{}_a \Sigma_a, \quad \theta' X_A J^\mu{}_a \Sigma_a \text{ in } OD \text{ and } DD.} \quad (118)$$

LII. FINAL SUMMARY

The residual blocks OD and DD do not vanish automatically.

However, each one splits as

$$\mathcal{M}_{AB,\text{inv}}^\mu = \mathcal{M}_{AB,\text{even}}^\mu + \mathcal{M}_{AB,\text{odd}}^\mu, \quad AB \in \{OD, DD\},$$

and the reflection-even part is killed exactly by the graded projection:

$$e_{0;AB}^{(1)} = o_{0;AB}^{(1)} = 0 \quad \text{for } \mathcal{M}_{AB,\text{even}}^\mu.$$

The OD -block is generated from the mixed quartic interaction

$$\lambda_{OD} \rho_O \rho_D$$

through the cubic vertices

$$\Gamma_{\nu\nu\zeta}^{(3)} \sim \lambda_{OD} \bar{\chi}_B, \quad \Gamma_{\nu\zeta\zeta}^{(3)} \sim \lambda_{OD} \bar{n}_B.$$

Therefore it is not theorem-level odd by itself; only its transport-producing odd part survives.

The DD -block contains

$$\mathcal{M}_{DD}^{\text{pot}}, \quad \mathcal{K}_{DD}^{\nabla \text{ani}}, \quad \mathcal{T}_{DD},$$

so its potential part is even, while its derivative-anisotropy and explicit transport parts can carry an odd contribution.

Every surviving residual transverse contribution must reduce to the universal odd normal form

$$\boxed{\theta'(\xi) \left[X_E(\xi) E^\mu{}_a \Sigma_a + X_O(\xi) J^\mu{}_a \Sigma_a \right],}$$

with X_E, X_O parity-even local defect invariants.

Under $O \leftrightarrow D$ exchange symmetry,

$$\Delta e_{\text{rem}} = 2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)}, \quad \Delta o_{\text{rem}} = 2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)}.$$

Therefore the canonical symmetric mixed-target condition is reduced to the single scalar equation

$$2\left(o_{0;OD,\text{odd}}^{(1)} - e_{0;OD,\text{odd}}^{(1)}\right) + \left(o_{0;DD,\text{odd}}^{(1)} - e_{0;DD,\text{odd}}^{(1)}\right) = e_{0;K}^{(1)} - o_{0;K}^{(1)}.$$

So the remaining microscopic work is now maximally sharp:

$$\text{extract only the odd } \theta' X_A\text{-type coefficients in } OD \text{ and } DD.$$

LIII. EXACT EXTRACTION FORMULAS FOR THE RESIDUAL ODD COEFFICIENTS

LIV. GOAL

The transverse coefficient point is already split as

$$(\Lambda_E, \Lambda_O) = v_{\text{pr}} + r_{\text{rem}}, \quad (119)$$

where

$$v_{\text{pr}} = \left(e_{0;TD}^{(1)} + e_{0;DT}^{(1)}, o_{0;TD}^{(1)} + o_{0;DT}^{(1)}\right),$$

and the residual correction is carried by

$$OD, \quad DO, \quad DD.$$

The next nonredundant step is to write the exact executable formulas for the parity-allowed residual coefficients

$$e_{0;OD,\text{odd}}^{(1)}, \quad o_{0;OD,\text{odd}}^{(1)}, \quad e_{0;DD,\text{odd}}^{(1)}, \quad o_{0;DD,\text{odd}}^{(1)}.$$

LV. 1. UNIVERSAL PROJECTOR EXTRACTION RULE

For any block $AB \in \{OD, DO, DD\}$, define the tangent-projected first-order kernel

$$\mathcal{M}_{AB}^\mu(\xi) := \Pi_Q \mathcal{V}_{AB}^\mu(\xi) \Pi_Q. \quad (120)$$

Its parity-odd part is

$$\mathcal{M}_{AB,\text{odd}}^\mu(\xi) := \frac{1}{2} \left(\mathcal{M}_{AB}^\mu(\xi) - \mathcal{M}_{AB}^\mu(-\xi) \right). \quad (121)$$

Then the surviving transverse coefficients are

$$e_{0;AB,\text{odd}}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{1}{4} E^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_{AB,\mu,\text{odd}}), \quad (122)$$

$$o_{0;AB,\text{odd}}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{1}{4} J^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_{AB,\mu,\text{odd}}). \quad (123)$$

LVI. 2. EXACT OD -ODD EXTRACTION

On the minimal admissible background

$$\bar{\Phi}_{\min} = (0, \bar{n}_B(\xi), \bar{\chi}_B(\xi)),$$

the mixed quartic interaction

$$\lambda_{OD} \rho_O \rho_D$$

generates the cubic vertices

$$\Gamma_{\nu\nu\zeta}^{(3)}[\bar{\Phi}_{\min}] \sim \lambda_{OD} \bar{\chi}_B(\xi), \quad \Gamma_{\nu\zeta\zeta}^{(3)}[\bar{\Phi}_{\min}] \sim \lambda_{OD} \bar{n}_B(\xi). \quad (124)$$

Let

$$\mathcal{H}_{OO} u_m = E_m^O u_m, \quad \mathcal{H}_{DD} v_n = E_n^D v_n,$$

with projected regulated propagators

$$G_O^{(m)}(q), \quad G_D^{(n)}(q).$$

The projected OD -self-energy kernel on the O -sector is

$$\Sigma_O^{(OD)}(p; \bar{\Phi}_{\min}) = - \int \frac{d^d q}{(2\pi)^d} \sum_{m,n} \mathcal{V}_{O;mn}^{(OD)}(p, q) G_D^{(n)}(q) \mathcal{V}_{O;nm}^{(OD)}(-p, q) G_O^{(m)}(p+q) \partial_t R_k(q), \quad (125)$$

where

$$\mathcal{V}_{O;mn}^{(OD)}(p, q) = \langle u_m | \Gamma_{\nu\nu\zeta}^{(3)}(p, q; \bar{\Phi}_{\min}) | v_n \rangle. \quad (126)$$

The parity-allowed first-order transport kernel extracted from this loop is

$$\mathcal{M}_{OD,\text{odd}}^\mu(\xi) := \frac{1}{2} \left[\Pi_Q \left(\frac{\partial}{\partial p_\mu} \Sigma_O^{(OD)}(p; \xi) \right)_{p=0} \Pi_Q - \Pi_Q \left(\frac{\partial}{\partial p_\mu} \Sigma_O^{(OD)}(p; -\xi) \right)_{p=0} \Pi_Q \right]. \quad (127)$$

Therefore the exact executable formulas are

$$e_{0;OD,\text{odd}}^{(1)} = \frac{1}{4} E_a^\mu \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_{OD,\mu,\text{odd}}), \quad (128)$$

$$o_{0;OD,\text{odd}}^{(1)} = \frac{1}{4} J_a^\mu \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_{OD,\mu,\text{odd}}). \quad (129)$$

Equivalently, inserting (125),

$$e_{0;OD,\text{odd}}^{(1)} = -\frac{1}{4} E_a^\mu \int \frac{d^d q}{(2\pi)^d} \sum_{m,n} \text{Tr}_{Q_B} \left[\Sigma_a \partial_{p_\mu} \left(\mathcal{V}_{O;mn}^{(OD)}(p, q) G_D^{(n)}(q) \mathcal{V}_{O;nm}^{(OD)}(-p, q) G_O^{(m)}(p+q) \right)_{p=0}^{\text{odd}} \right] \partial_t R_k(q), \quad (130)$$

$$o_{0;OD,\text{odd}}^{(1)} = -\frac{1}{4} J_a^\mu \int \frac{d^d q}{(2\pi)^d} \sum_{m,n} \text{Tr}_{Q_B} \left[\Sigma_a \partial_{p_\mu} \left(\mathcal{V}_{O;mn}^{(OD)}(p, q) G_D^{(n)}(q) \mathcal{V}_{O;nm}^{(OD)}(-p, q) G_O^{(m)}(p+q) \right)_{p=0}^{\text{odd}} \right] \partial_t R_k(q). \quad (131)$$

Thus the only actual work in the OD -sector is the p -derivative at $p=0$ followed by odd parity projection.

LVII. 3. EXACT DD -ODD EXTRACTION

The projected DD -kernel already has the invariant decomposition

$$\mathcal{M}_{DD,\text{inv}}^\mu = a_{||;DD}(\xi) n^\mu \Sigma_3 + a_{E;DD}(\xi) E^\mu{}_a \Sigma_a + a_{O;DD}(\xi) J^\mu{}_a \Sigma_a. \quad (132)$$

Define the odd parts

$$a_{E;DD}^{\text{odd}}(\xi) := \frac{1}{2} \left(a_{E;DD}(\xi) - a_{E;DD}(-\xi) \right), \quad a_{O;DD}^{\text{odd}}(\xi) := \frac{1}{2} \left(a_{O;DD}(\xi) - a_{O;DD}(-\xi) \right). \quad (133)$$

Then the transverse residual coefficients from DD are directly

$$e_{0;DD,\text{odd}}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} a_{E;DD}^{\text{odd}}, \quad o_{0;DD,\text{odd}}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} a_{O;DD}^{\text{odd}}. \quad (134)$$

Equivalently,

$$e_{0;DD,\text{odd}}^{(1)} = \frac{1}{4} E^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_{DD,\mu,\text{odd}}), \quad o_{0;DD,\text{odd}}^{(1)} = \frac{1}{4} J^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_{DD,\mu,\text{odd}}), \quad (135)$$

with

$$\mathcal{M}_{DD,\text{odd}}^\mu = a_{E;DD}^{\text{odd}} E^\mu{}_a \Sigma_a + a_{O;DD}^{\text{odd}} J^\mu{}_a \Sigma_a.$$

LVIII. 4. EXCHANGE-REDUCED RESIDUAL PAIR

If the background and projected regulator preserve $O \leftrightarrow D$ exchange symmetry to leading order, then

$$e_{0;DO,\text{odd}}^{(1)} = e_{0;OD,\text{odd}}^{(1)}, \quad o_{0;DO,\text{odd}}^{(1)} = o_{0;OD,\text{odd}}^{(1)}.$$

Hence

$$\Delta e_{\text{rem}} = 2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)}, \quad \Delta o_{\text{rem}} = 2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)}. \quad (136)$$

LIX. 5. FINAL TRANSVERSE COEFFICIENTS

The final reduced coefficients are therefore

$$\alpha = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left(\Lambda_E^{\text{pr}} + 2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)} \right), \quad (137)$$

$$\beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left(\Lambda_O^{\text{pr}} + 2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)} \right). \quad (138)$$

LX. 6. CANONICAL SYMMETRIC MIXED-TARGET CONDITION

For the canonical mixed odd-active target one needs

$$\Lambda_E = \Lambda_O \neq 0.$$

Thus the exact scalar equality condition is

$$2 \left(o_{0;OD,\text{odd}}^{(1)} - e_{0;OD,\text{odd}}^{(1)} \right) + \left(o_{0;DD,\text{odd}}^{(1)} - e_{0;DD,\text{odd}}^{(1)} \right) = \Lambda_E^{\text{pr}} - \Lambda_O^{\text{pr}}. \quad (139)$$

A strong sufficient survival condition is

$$\left| 2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)} \right| < |\Lambda_E^{\text{pr}}|, \quad (140)$$

$$\left| 2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)} \right| < |\Lambda_O^{\text{pr}}|. \quad (141)$$

LXI. 7. HONEST STATUS

At this point nothing conceptual remains in the residual transverse audit. The actual remaining work is exactly:

1. evaluate the p -derivative of the projected OD -self-energy kernel at $p = 0$;
2. take its odd part in ξ ;
3. read off $e_{0;OD,\text{odd}}^{(1)}$ and $o_{0;OD,\text{odd}}^{(1)}$ using (128)–(129);
4. split $a_{E;DD}$, $a_{O;DD}$ into odd/even parts and keep only (133);
5. insert the result into (137), (138), and (139).

LXII. FINAL SUMMARY

The residual transverse audit is now fully reduced to four explicit numbers:

$$e_{0;OD,\text{odd}}^{(1)}, \quad o_{0;OD,\text{odd}}^{(1)}, \quad e_{0;DD,\text{odd}}^{(1)}, \quad o_{0;DD,\text{odd}}^{(1)}.$$

They are extracted by the universal projector rule

$$e_{0;AB,\text{odd}}^{(1)} = \frac{1}{4} E^\mu{}_a \text{Tr}_{Q_B} (\Sigma_a \mathcal{M}_{AB,\mu,\text{odd}}), \quad o_{0;AB,\text{odd}}^{(1)} = \frac{1}{4} J^\mu{}_a \text{Tr}_{Q_B} (\Sigma_a \mathcal{M}_{AB,\mu,\text{odd}}),$$

with $AB \in \{OD, DD\}$.

For OD , the odd kernel is obtained from the projected self-energy by

$$\mathcal{M}_{OD,\text{odd}}^\mu = \frac{1}{2} \left[\Pi_Q \left(\partial_{p_\mu} \Sigma_O^{(OD)} \right)_{p=0,\xi} \Pi_Q - \Pi_Q \left(\partial_{p_\mu} \Sigma_O^{(OD)} \right)_{p=0,-\xi} \Pi_Q \right].$$

For DD , the invariant decomposition is already

$$\mathcal{M}_{DD,\text{inv}}^\mu = a_{\parallel;DD} n^\mu \Sigma_3 + a_{E;DD} E^\mu{}_a \Sigma_a + a_{O;DD} J^\mu{}_a \Sigma_a,$$

so only

$$a_{E;DD}^{\text{odd}} = \frac{1}{2} (a_{E;DD}(\xi) - a_{E;DD}(-\xi)), \quad a_{O;DD}^{\text{odd}} = \frac{1}{2} (a_{O;DD}(\xi) - a_{O;DD}(-\xi))$$

matter.

Under leading $O \leftrightarrow D$ exchange symmetry,

$$\Delta e_{\text{rem}} = 2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)}, \quad \Delta o_{\text{rem}} = 2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)}.$$

Therefore the final reduced coefficients are

$$\alpha = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left(\Lambda_E^{\text{pr}} + 2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)} \right),$$

$$\beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left(\Lambda_O^{\text{pr}} + 2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)} \right).$$

The canonical symmetric mixed-target condition is reduced to the single scalar equation

$$2 \left(o_{0;OD,\text{odd}}^{(1)} - e_{0;OD,\text{odd}}^{(1)} \right) + \left(o_{0;DD,\text{odd}}^{(1)} - e_{0;DD,\text{odd}}^{(1)} \right) = \Lambda_E^{\text{pr}} - \Lambda_O^{\text{pr}}.$$

So the remaining microscopic task is no longer structural: it is the literal evaluation of those four odd residual coefficients.

LXIII. COMBINED RESIDUAL INSERTION FROM $OD + DO + DD$

LXIV. GOAL

The principal transverse pair has already been isolated from the mixed shell–defect blocks:

$$\boxed{\Lambda_E^{\text{pr}} = \mathfrak{P}_E[\mathcal{M}_{TD+DT}^{\text{inv}}], \quad \Lambda_O^{\text{pr}} = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}].} \quad (142)$$

Hence the full residual completion problem is now exactly:

$$\boxed{(\Lambda_E, \Lambda_O) = (\Lambda_E^{\text{pr}}, \Lambda_O^{\text{pr}}) + r_{OD/DO} + r_{DD}.} \quad (143)$$

LXV. 1. UNIVERSAL RESIDUAL PROJECTOR RULE

For any candidate residual block $AB \in \{OD, DO, DD\}$, define the tangent-projected first-order kernel

$$\boxed{\mathcal{M}_{AB}^\mu(\xi) := \Pi_Q \mathcal{V}_{AB}^\mu(\xi) \Pi_Q.} \quad (144)$$

Its reflection-odd part is

$$\boxed{\mathcal{M}_{AB,\text{odd}}^\mu(\xi) = \frac{1}{2} \left(\mathcal{M}_{AB}^\mu(\xi) - \mathcal{M}_{AB}^\mu(-\xi) \right).} \quad (145)$$

The projector-visible transverse coefficients are then

$$\boxed{e_{0;AB,\text{odd}}^{(1)} = \frac{1}{4} E^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_{AB,\mu,\text{odd}}),} \quad (146)$$

$$\boxed{o_{0;AB,\text{odd}}^{(1)} = \frac{1}{4} J^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_{AB,\mu,\text{odd}}).} \quad (147)$$

LXVI. 2. EXACT OD -SECTOR INSERTION

On the minimal admissible background

$$\bar{\Phi}_{\text{min}} = (0, \bar{n}_B(\xi), \bar{\chi}_B(\xi)),$$

the mixed quartic interaction

$$\lambda_{OD} \rho_O \rho_D$$

generates the background-induced cubic vertices

$$\Gamma_{\nu\nu\zeta}^{(3)}[\bar{\Phi}_{\text{min}}] \sim \lambda_{OD} \bar{\chi}_B(\xi), \quad \Gamma_{\nu\zeta\zeta}^{(3)}[\bar{\Phi}_{\text{min}}] \sim \lambda_{OD} \bar{n}_B(\xi). \quad (148)$$

Choose orthonormal projected mode bases

$$\mathcal{H}_{OO} u_m = E_m^O u_m, \quad \mathcal{H}_{DD} v_n = E_n^D v_n,$$

and define the regulated projected propagators

$$G_O(q) = \left(P_O [\mathcal{H}_{OO} + R_{O,k}] P_O \right)^{-1}, \quad G_D(q) = \left(P_D [\mathcal{H}_{DD} + R_{D,k}] P_D \right)^{-1}. \quad (149)$$

The explicit projected mixed self-energies are

$$\Sigma_O^{(OD)}(p) = -\lambda_{OD}^2 \int_q \sum_{m,n} \mathcal{B}_{mn}^\chi \mathcal{P}_O(p, q) G_D^{(n)}(q) \mathcal{B}_{nm}^\chi \mathcal{P}_O(-p, q) G_O^{(m)}(p+q) \partial_t R_k(q), \quad (150)$$

$$\Sigma_D^{(OD)}(p) = -\lambda_{OD}^2 \int_q \sum_{m,n} \mathcal{B}_{nm}^n \mathcal{P}_D(p, q) G_O^{(m)}(q) \mathcal{B}_{mn}^n \mathcal{P}_D(-p, q) G_D^{(n)}(p+q) \partial_t R_k(q). \quad (151)$$

The correct residual transport kernel is the first momentum derivative of the *projected* mixed self-energy:

$$\mathcal{M}_{OD/DO, \text{odd}}^\mu(\xi) := \frac{1}{2} \Pi_Q \left[\partial_{p_\mu} \left(\Sigma_O^{(OD)}(p; \xi) + \Sigma_D^{(OD)}(p; \xi) \right)_{p=0} - \partial_{p_\mu} \left(\Sigma_O^{(OD)}(p; -\xi) + \Sigma_D^{(OD)}(p; -\xi) \right)_{p=0} \right] \Pi_Q. \quad (152)$$

Hence the exchange-type residual coefficients are

$$e_{0;OD/DO, \text{odd}}^{(1)} = \frac{1}{4} E^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_{OD/DO, \mu, \text{odd}}), \quad (153)$$

$$o_{0;OD/DO, \text{odd}}^{(1)} = \frac{1}{4} J^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_{OD/DO, \mu, \text{odd}}). \quad (154)$$

If $O \leftrightarrow D$ exchange symmetry is valid at the projected level, then

$$e_{0;DO, \text{odd}}^{(1)} = e_{0;OD, \text{odd}}^{(1)}, \quad o_{0;DO, \text{odd}}^{(1)} = o_{0;OD, \text{odd}}^{(1)}, \quad (155)$$

so

$$r_{OD/DO} = \left(2e_{0;OD, \text{odd}}^{(1)}, 2o_{0;OD, \text{odd}}^{(1)} \right). \quad (156)$$

LXVII. 3. EXACT DD -SECTOR INSERTION

The DD -block already has the explicit invariant decomposition

$$\mathcal{M}_{DD, \text{inv}}^\mu = a_{\parallel; DD}(\xi) n^\mu \Sigma_3 + a_{E; DD}(\xi) E^\mu{}_a \Sigma_a + a_{O; DD}(\xi) J^\mu{}_a \Sigma_a. \quad (157)$$

Define the reflection-odd parts

$$a_{E; DD}^{\text{odd}}(\xi) = \frac{1}{2} \left(a_{E; DD}(\xi) - a_{E; DD}(-\xi) \right), \quad a_{O; DD}^{\text{odd}}(\xi) = \frac{1}{2} \left(a_{O; DD}(\xi) - a_{O; DD}(-\xi) \right). \quad (158)$$

Then the direct DD -residual coefficients are

$$e_{0; DD, \text{odd}}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} a_{E; DD}^{\text{odd}}, \quad o_{0; DD, \text{odd}}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} a_{O; DD}^{\text{odd}}. \quad (159)$$

Equivalently,

$$e_{0; DD, \text{odd}}^{(1)} = \frac{1}{4} E^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_{DD, \mu, \text{odd}}), \quad o_{0; DD, \text{odd}}^{(1)} = \frac{1}{4} J^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_{DD, \mu, \text{odd}}). \quad (160)$$

Thus

$$r_{DD} = \left(e_{0; DD, \text{odd}}^{(1)}, o_{0; DD, \text{odd}}^{(1)} \right). \quad (161)$$

LXVIII. 4. FINAL COMBINED RESIDUAL VECTOR

Combining (156) and (161), the full residual correction is

$$r_{\text{rem}} = \left(2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)}, 2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)} \right). \quad (162)$$

Therefore the complete transverse coefficient point is

$$\Lambda_E = \Lambda_E^{\text{pr}} + 2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)}, \quad (163)$$

$$\Lambda_O = \Lambda_O^{\text{pr}} + 2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)}. \quad (164)$$

LXIX. 5. FINAL α, β COEFFICIENTS

Using the already fixed branch dictionary

$$\alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O,$$

we obtain

$$\alpha = I_\theta \left(\Lambda_E^{\text{pr}} + 2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)} \right), \quad (165)$$

$$\beta = I_\theta \left(\Lambda_O^{\text{pr}} + 2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)} \right). \quad (166)$$

If the explicit seed profile is used, one may further substitute

$$I_\theta = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa.$$

LXX. 6. CANONICAL SYMMETRIC MIXED-TARGET CONDITION

The canonical mixed odd-active target requires

$$\Lambda_E = \Lambda_O \neq 0.$$

Hence the exact scalar equality condition is

$$2\left(o_{0;OD,\text{odd}}^{(1)} - e_{0;OD,\text{odd}}^{(1)}\right) + \left(o_{0;DD,\text{odd}}^{(1)} - e_{0;DD,\text{odd}}^{(1)}\right) = \Lambda_E^{\text{pr}} - \Lambda_O^{\text{pr}}. \quad (167)$$

The strong sufficient survival conditions are

$$\left| 2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)} \right| < |\Lambda_E^{\text{pr}}|, \quad (168)$$

$$\left| 2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)} \right| < |\Lambda_O^{\text{pr}}|. \quad (169)$$

LXXI. 7. HONEST CLOSURE STATUS

At the present stage, the combined $OD + DD$ problem is fully reduced to the four explicit odd residual numbers

$$e_{0;OD,\text{odd}}^{(1)}, \quad o_{0;OD,\text{odd}}^{(1)}, \quad e_{0;DD,\text{odd}}^{(1)}, \quad o_{0;DD,\text{odd}}^{(1)}.$$

No further structural reduction is needed.

What remains open is not the formula, but the actual evaluation/sign of these four quantities.

LXXII. FINAL SUMMARY

The full residual transverse completion beyond the principal $TD + DT$ pair is

$$r_{\text{rem}} = r_{OD/DO} + r_{DD}.$$

The exchange-type OD/DO correction is obtained from the projected mixed self-energies

$$\Sigma_O^{(OD)}(p), \quad \Sigma_D^{(OD)}(p),$$

through the first momentum derivative at $p = 0$, followed by reflection-odd extraction and transverse projection:

$$e_{0;OD/DO,\text{odd}}^{(1)} = \frac{1}{4} E^\mu_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_{OD/DO,\mu,\text{odd}}), \quad o_{0;OD/DO,\text{odd}}^{(1)} = \frac{1}{4} J^\mu_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_{OD/DO,\mu,\text{odd}}).$$

The intrinsic DD -residual is directly obtained from

$$\mathcal{M}_{DD,\text{inv}}^\mu = a_{||;DD} n^\mu \Sigma_3 + a_{E;DD} E^\mu_a \Sigma_a + a_{O;DD} J^\mu_a \Sigma_a$$

by taking only the reflection-odd parts of $a_{E;DD}$ and $a_{O;DD}$:

$$e_{0;DD,\text{odd}}^{(1)} = a_{E;DD}^{\text{odd}}, \quad o_{0;DD,\text{odd}}^{(1)} = a_{O;DD}^{\text{odd}}.$$

Under leading $O \leftrightarrow D$ exchange symmetry,

$$r_{\text{rem}} = \left(2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)}, \quad 2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)} \right).$$

Therefore the final transverse coefficient point is

$$\Lambda_E = \Lambda_E^{\text{pr}} + 2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)},$$

$$\Lambda_O = \Lambda_O^{\text{pr}} + 2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)}.$$

Hence

$$\alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O.$$

For the canonical symmetric mixed odd-active target, the exact scalar condition is

$$2(o_{0;OD,\text{odd}}^{(1)} - e_{0;OD,\text{odd}}^{(1)}) + (o_{0;DD,\text{odd}}^{(1)} - e_{0;DD,\text{odd}}^{(1)}) = \Lambda_E^{\text{pr}} - \Lambda_O^{\text{pr}}.$$

So the combined $OD + DD$ problem is now fully reduced to four odd residual coefficients and their insertion into the final (α, β) sheet.

LXXIII. COMBINED $OD + DD$ RESIDUAL COMPLETION OF THE TRANSVERSE COEFFICIENT POINT

LXXIV. GOAL

The leading transverse pair is already isolated in the principal mixed blocks

$$TD + DT.$$

Therefore the only remaining transverse completion problem is to insert the residual exchange-type sector OD/DO together with the intrinsic defect-tangent sector DD .

The exact transverse coefficient point must therefore be written in the form

$$(\Lambda_E, \Lambda_O) = (\Lambda_E^{\text{pr}}, \Lambda_O^{\text{pr}}) + r_{OD/DO} + r_{DD}. \quad (170)$$

LXXV. 1. PRINCIPAL PAIR

The already isolated principal pair is

$$\Lambda_E^{\text{pr}} = e_{0;TD}^{(1)} + e_{0;DT}^{(1)}, \quad \Lambda_O^{\text{pr}} = o_{0;TD}^{(1)} + o_{0;DT}^{(1)}. \quad (171)$$

Equivalently, at the conservative working level,

$$\Lambda_E^{\text{pr}} \approx e_{0;K}^{(1)}, \quad \Lambda_O^{\text{pr}} \approx o_{0;K}^{(1)}. \quad (172)$$

LXXVI. 2. UNIVERSAL ODD PROJECTOR RULE

For any residual block $AB \in \{OD, DO, DD\}$, define the tangent-projected first-order kernel

$$\mathcal{M}_{AB}^\mu(\xi) := \Pi_Q \mathcal{V}_{AB}^\mu(\xi) \Pi_Q. \quad (173)$$

Its reflection-odd part is

$$\mathcal{M}_{AB,\text{odd}}^\mu(\xi) = \frac{1}{2} \left(\mathcal{M}_{AB}^\mu(\xi) - \mathcal{M}_{AB}^\mu(-\xi) \right). \quad (174)$$

Then the transverse residual coefficients are extracted by

$$e_{0;AB,\text{odd}}^{(1)} = \frac{1}{4} E^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_{AB,\mu,\text{odd}}), \quad (175)$$

$$o_{0;AB,\text{odd}}^{(1)} = \frac{1}{4} J^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_{AB,\mu,\text{odd}}). \quad (176)$$

So only the odd parts of OD, DO, DD matter.

LXXVII. 3. EXACT OD -SECTOR INSERTION

Choose projected mode bases

$$\mathcal{H}_{OO} u_m = E_m^O u_m, \quad \mathcal{H}_{DD} v_n = E_n^D v_n,$$

with projectors

$$P_O = \sum_m |u_m\rangle\langle u_m|, \quad P_D = \sum_n |v_n\rangle\langle v_n|.$$

The projected propagators are

$$G_O(q; \bar{\Phi}_{\min}) = \left(P_O [\mathcal{H}_{OO} + R_{O,k}] P_O \right)^{-1}, \quad G_D(q; \bar{\Phi}_{\min}) = \left(P_D [\mathcal{H}_{DD} + R_{D,k}] P_D \right)^{-1}. \quad (177)$$

The projected mixed self-energy on the O -sector is

$$\Sigma_O^{(OD)}(p; \bar{\Phi}_{\min}) = - \int \frac{d^d q}{(2\pi)^d} \sum_{m,n} \mathcal{V}_{O;mn}^{(OD)}(p, q) G_D^{(n)}(q) \mathcal{V}_{O;nm}^{(OD)}(-p, q) G_O^{(m)}(p+q) \partial_t R_k(q), \quad (178)$$

with

$$\mathcal{V}_{O;mn}^{(OD)}(p, q) = \left\langle u_m \left| \Gamma_{\nu\nu\zeta}^{(3)}(p, q; \bar{\Phi}_{\min}) \right| v_n \right\rangle = \lambda_{OD} \mathcal{B}_{mn}^x(\xi) \mathcal{P}_O(p, q), \quad (179)$$

where

$$\mathcal{B}_{mn}^x(\xi) \sim \langle u_m, \bar{\chi}_B(\xi) v_n \rangle.$$

Likewise the D -sector companion mixed self-energy is

$$\Sigma_D^{(OD)}(p; \bar{\Phi}_{\min}) = - \int \frac{d^d q}{(2\pi)^d} \sum_{m,n} \tilde{\mathcal{V}}_{D;nm}^{(OD)}(p, q) G_O^{(m)}(q) \tilde{\mathcal{V}}_{D;mn}^{(OD)}(-p, q) G_D^{(n)}(p+q) \partial_t R_k(q), \quad (180)$$

with schematic factorization

$$\tilde{\mathcal{V}}_{D;nm}^{(OD)}(p, q) = \lambda_{OD} \mathcal{B}_{nm}^n(\xi) \mathcal{P}_D(p, q), \quad (181)$$

where

$$\mathcal{B}_{nm}^n(\xi) \sim \langle v_n, \bar{n}_B(\xi) u_m \rangle.$$

The exact odd transport kernel from the exchange-type sector is then

$$\mathcal{M}_{OD/DO, \text{odd}}^\mu(\xi) = \frac{1}{2} \Pi_Q \left[\partial_{p_\mu} \left(\Sigma_O^{(OD)}(p; \xi) + \Sigma_D^{(OD)}(p; \xi) \right)_{p=0} - \partial_{p_\mu} \left(\Sigma_O^{(OD)}(p; -\xi) + \Sigma_D^{(OD)}(p; -\xi) \right)_{p=0} \right] \Pi_Q. \quad (182)$$

Hence

$$e_{0;OD/DO, \text{odd}}^{(1)} = \frac{1}{4} E^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_{OD/DO, \mu, \text{odd}}), \quad (183)$$

$$o_{0;OD/DO, \text{odd}}^{(1)} = \frac{1}{4} J^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_{OD/DO, \mu, \text{odd}}). \quad (184)$$

If the projected background and regulator preserve $O \leftrightarrow D$ exchange symmetry, then

$$e_{0;DO, \text{odd}}^{(1)} = e_{0;OD, \text{odd}}^{(1)}, \quad o_{0;DO, \text{odd}}^{(1)} = o_{0;OD, \text{odd}}^{(1)}. \quad (185)$$

Therefore

$$r_{OD/DO} = \left(2e_{0;OD, \text{odd}}^{(1)}, 2o_{0;OD, \text{odd}}^{(1)} \right). \quad (186)$$

LXXVIII. 4. EXACT DD -SECTOR INSERTION

The projected DD -kernel already has the explicit invariant decomposition

$$\mathcal{M}_{DD,\text{inv}}^\mu = a_{\parallel;DD}(\xi) n^\mu \Sigma_3 + a_{E;DD}(\xi) E^\mu{}_a \Sigma_a + a_{O;DD}(\xi) J^\mu{}_a \Sigma_a. \quad (187)$$

Define the reflection-odd transverse coefficients

$$a_{E;DD}^{\text{odd}}(\xi) = \frac{1}{2} \left(a_{E;DD}(\xi) - a_{E;DD}(-\xi) \right), \quad a_{O;DD}^{\text{odd}}(\xi) = \frac{1}{2} \left(a_{O;DD}(\xi) - a_{O;DD}(-\xi) \right). \quad (188)$$

Then the DD -residual contribution is directly

$$e_{0;DD,\text{odd}}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} a_{E;DD}^{\text{odd}}, \quad o_{0;DD,\text{odd}}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} a_{O;DD}^{\text{odd}}. \quad (189)$$

Equivalently,

$$e_{0;DD,\text{odd}}^{(1)} = \frac{1}{4} E^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_{DD,\mu,\text{odd}}), \quad o_{0;DD,\text{odd}}^{(1)} = \frac{1}{4} J^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_{DD,\mu,\text{odd}}). \quad (190)$$

Therefore

$$r_{DD} = \left(e_{0;DD,\text{odd}}^{(1)}, o_{0;DD,\text{odd}}^{(1)} \right). \quad (191)$$

LXXIX. 5. FINAL COMBINED RESIDUAL VECTOR

Combining (186) and (191), the exact residual correction is

$$r_{\text{rem}} = \left(2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)}, 2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)} \right). \quad (192)$$

Hence the full transverse coefficient point is

$$\Lambda_E = \Lambda_E^{\text{pr}} + 2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)}, \quad (193)$$

$$\Lambda_O = \Lambda_O^{\text{pr}} + 2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)}. \quad (194)$$

LXXX. 6. FINAL α, β INSERTION

Using the exact overlap dictionary

$$\alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O,$$

we obtain

$$\alpha = I_\theta \left(\Lambda_E^{\text{pr}} + 2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)} \right), \quad (195)$$

$$\beta = I_\theta \left(\Lambda_O^{\text{pr}} + 2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)} \right). \quad (196)$$

For the explicit defect profile one may insert

$$I_\theta = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa,$$

so

$$\alpha = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\left(\Lambda_E^{\text{pr}} + 2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)}\right), \quad (197)$$

$$\beta = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\left(\Lambda_O^{\text{pr}} + 2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)}\right). \quad (198)$$

LXXXI. 7. CANONICAL SYMMETRIC MIXED-TARGET CONDITION

The canonical mixed odd-active target requires

$$\Lambda_E = \Lambda_O \neq 0.$$

Thus the exact scalar equality condition is

$$2\left(o_{0;OD,\text{odd}}^{(1)} - e_{0;OD,\text{odd}}^{(1)}\right) + \left(o_{0;DD,\text{odd}}^{(1)} - e_{0;DD,\text{odd}}^{(1)}\right) = \Lambda_E^{\text{pr}} - \Lambda_O^{\text{pr}}. \quad (199)$$

A strong sufficient survival condition is

$$\left|2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)}\right| < |\Lambda_E^{\text{pr}}|, \quad (200)$$

$$\left|2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)}\right| < |\Lambda_O^{\text{pr}}|. \quad (201)$$

LXXXII. 8. HONEST STATUS

At this point the combined $OD + DD$ problem is reduced completely to four explicit odd residual quantities:

$$e_{0;OD,\text{odd}}^{(1)}, \quad o_{0;OD,\text{odd}}^{(1)}, \quad e_{0;DD,\text{odd}}^{(1)}, \quad o_{0;DD,\text{odd}}^{(1)}.$$

No further structural theorem is needed. What remains open is only their actual evaluation/sign.

LXXXIII. FINAL SUMMARY

The full residual transverse completion beyond the principal $TD + DT$ source is

$$r_{\text{rem}} = r_{OD/DO} + r_{DD}.$$

The OD/DO part is obtained from the projected mixed self-energies

$$\Sigma_O^{(OD)}(p), \quad \Sigma_D^{(OD)}(p)$$

by taking the first momentum derivative at $p = 0$, then the reflection-odd part, and finally the transverse projectors:

$$e_{0;OD/DO,\text{odd}}^{(1)} = \frac{1}{4}E^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_{OD/DO,\mu,\text{odd}}), \quad o_{0;OD/DO,\text{odd}}^{(1)} = \frac{1}{4}J^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_{OD/DO,\mu,\text{odd}}).$$

The DD part is directly read from the invariant decomposition

$$\mathcal{M}_{DD,\text{inv}}^\mu = a_{\parallel;DD} n^\mu \Sigma_3 + a_{E;DD} E^\mu{}_a \Sigma_a + a_{O;DD} J^\mu{}_a \Sigma_a$$

by keeping only the odd parts of $a_{E;DD}$ and $a_{O;DD}$:

$$e_{0;DD,\text{odd}}^{(1)} = a_{E;DD}^{\text{odd}}, \quad o_{0;DD,\text{odd}}^{(1)} = a_{O;DD}^{\text{odd}}.$$

Under leading $O \leftrightarrow D$ exchange symmetry,

$$r_{\text{rem}} = \left(2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)}, \quad 2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)} \right).$$

Hence the final transverse coefficients are

$$\Lambda_E = \Lambda_E^{\text{pr}} + 2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)},$$

$$\Lambda_O = \Lambda_O^{\text{pr}} + 2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)}.$$

Therefore

$$\alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O,$$

and with the explicit profile,

$$I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa.$$

The canonical symmetric mixed-target condition is reduced to the single scalar equation

$$2(o_{0;OD,\text{odd}}^{(1)} - e_{0;OD,\text{odd}}^{(1)}) + (o_{0;DD,\text{odd}}^{(1)} - e_{0;DD,\text{odd}}^{(1)}) = \Lambda_E^{\text{pr}} - \Lambda_O^{\text{pr}}.$$

So the combined $OD + DD$ problem is now fully reduced to four explicit odd residual coefficients and their literal insertion into the final (α, β) sheet.

LXXXIV. COMBINED $OD + DD$ RESIDUAL BOUNDS AND A USABLE SUFFICIENT SURVIVAL CRITERION

LXXXV. GOAL

At this stage, the residual transverse completion is no longer a structural question. The exact remaining quantities are

$$e_{0;OD,\text{odd}}^{(1)}, \quad o_{0;OD,\text{odd}}^{(1)}, \quad e_{0;DD,\text{odd}}^{(1)}, \quad o_{0;DD,\text{odd}}^{(1)}.$$

What we now do is reduce these to a finite executable bound package.

LXXXVI. 1. PRINCIPAL-PLUS-RESIDUAL SPLIT

The exact transverse coefficient point is

$$(\Lambda_E, \Lambda_O) = v_{\text{pr}} + r_{\text{rem}}, \tag{202}$$

with

$$v_{\text{pr}} = (\Lambda_E^{\text{pr}}, \Lambda_O^{\text{pr}}) = \left(\frac{1}{2}(A_E + B_E), \frac{1}{2}(A_O + B_O) \right), \tag{203}$$

and

$$r_{\text{rem}} = \left(2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)}, \quad 2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)} \right), \tag{204}$$

assuming the leading $O \leftrightarrow D$ exchange reduction.

LXXXVII. 2. EXACT OVERLAP PACKAGES FOR THE OD -SECTOR

The background overlap matrices are

$$\mathcal{B}_{mn}^x := \langle u_m, \bar{\chi}_B v_n \rangle, \quad \mathcal{B}_{nm}^n := \langle v_n, \bar{n}_B u_m \rangle. \quad (205)$$

Define the Hilbert–Schmidt overlap budgets

$$\mathfrak{B}_\chi := \|P_O \bar{\chi}_B P_D\|_{\text{HS}}, \quad \mathfrak{B}_n := \|P_D \bar{n}_B P_O\|_{\text{HS}}. \quad (206)$$

These satisfy the basis identities

$$\mathfrak{B}_\chi^2 = \sum_{m,n} |\mathcal{B}_{mn}^x|^2, \quad \mathfrak{B}_n^2 = \sum_{m,n} |\mathcal{B}_{nm}^n|^2. \quad (207)$$

LXXXVIII. 3. PROPAGATOR AND PROJECTOR DERIVATIVE BUDGETS

Let

$$\mathfrak{G}_O := \sup_q \|G_O(q)\|, \quad \mathfrak{G}_D := \sup_q \|G_D(q)\|. \quad (208)$$

Let the momentum/projector budgets be

$$\mathfrak{P}_O^{(0)} := \sup_q |\mathcal{P}_O(0, q)|, \quad \mathfrak{P}_D^{(0)} := \sup_q |\mathcal{P}_D(0, q)|, \quad (209)$$

$$\mathfrak{P}_O^{(1)} := \sup_q \left| \partial_{p_\mu} \mathcal{P}_O(p, q) \Big|_{p=0} \right|, \quad \mathfrak{P}_D^{(1)} := \sup_q \left| \partial_{p_\mu} \mathcal{P}_D(p, q) \Big|_{p=0} \right|. \quad (210)$$

Likewise define the propagator momentum-derivative budgets

$$\mathfrak{K}_O^{(1)} := \sup_q \left\| \partial_{p_\mu} G_O(p+q) \Big|_{p=0} \right\|, \quad \mathfrak{K}_D^{(1)} := \sup_q \left\| \partial_{p_\mu} G_D(p+q) \Big|_{p=0} \right\|. \quad (211)$$

These are the exact finite derivative packages needed by the product rule on the projected loop kernel.

LXXXIX. 4. BOUND FOR THE OD -ODD COEFFICIENTS

The projected OD -self-energies have the factorized form

$$\Sigma_O^{(OD)}(p) = -\lambda_{OD}^2 \int_q \sum_{m,n} \mathcal{B}_{mn}^x \mathcal{P}_O(p, q) G_D^{(n)}(q) \mathcal{B}_{nm}^x \mathcal{P}_O(-p, q) G_O^{(m)}(p+q) \partial_t R_k(q), \quad (212)$$

$$\Sigma_D^{(OD)}(p) = -\lambda_{OD}^2 \int_q \sum_{m,n} \mathcal{B}_{nm}^n \mathcal{P}_D(p, q) G_O^{(m)}(q) \mathcal{B}_{mn}^n \mathcal{P}_D(-p, q) G_D^{(n)}(p+q) \partial_t R_k(q). \quad (213)$$

Applying the product rule at $p = 0$, the O -side derivative contributes two projector-derivative terms and one propagator-derivative term. Hence define

$$\mathfrak{L}_O := \int_q \left[2\mathfrak{P}_O^{(0)} \mathfrak{P}_O^{(1)} \mathfrak{G}_D \mathfrak{G}_O + (\mathfrak{P}_O^{(0)})^2 \mathfrak{G}_D \mathfrak{K}_O^{(1)} \right] |\partial_t R_k(q)|. \quad (214)$$

Similarly define

$$\mathfrak{L}_D := \int_q \left[2\mathfrak{P}_D^{(0)} \mathfrak{P}_D^{(1)} \mathfrak{G}_O \mathfrak{G}_D + (\mathfrak{P}_D^{(0)})^2 \mathfrak{G}_O \mathfrak{K}_D^{(1)} \right] |\partial_t R_k(q)|. \quad (215)$$

Then the odd projected transport norm from the exchange-type residual obeys

$$\|\mathcal{M}_{OD/DO,\text{odd}}\| \leq \lambda_{OD}^2 \left(\mathfrak{B}_\chi^2 \mathfrak{L}_O + \mathfrak{B}_n^2 \mathfrak{L}_D \right). \quad (216)$$

Since for any 2×2 tangent endomorphism M ,

$$\left| \frac{1}{4} \text{Tr}(\Sigma_a M) \right| \leq \frac{1}{2} \|M\|_{\text{op}},$$

we obtain

$$|e_{0;OD/DO,\text{odd}}^{(1)}|, |o_{0;OD/DO,\text{odd}}^{(1)}| \leq \frac{1}{2} \lambda_{OD}^2 \left(\mathfrak{B}_\chi^2 \mathfrak{L}_O + \mathfrak{B}_n^2 \mathfrak{L}_D \right). \quad (217)$$

Define

$$\rho_{OD} := \frac{1}{2} \lambda_{OD}^2 \left(\mathfrak{B}_\chi^2 \mathfrak{L}_O + \mathfrak{B}_n^2 \mathfrak{L}_D \right). \quad (218)$$

Then

$$|e_{0;OD,\text{odd}}^{(1)}|, |o_{0;OD,\text{odd}}^{(1)}| \leq \rho_{OD}. \quad (219)$$

XC. 5. EXACT AND BOUNDED DD -SECTOR COEFFICIENTS

The DD -kernel has the invariant decomposition

$$\mathcal{M}_{DD,\text{inv}}^\mu = a_{||;DD} n^\mu \Sigma_3 + a_{E;DD} E^\mu \Sigma_a + a_{O;DD} J^\mu \Sigma_a. \quad (220)$$

Define the odd parts

$$a_{E;DD}^{\text{odd}}(\xi) = \frac{1}{2} (a_{E;DD}(\xi) - a_{E;DD}(-\xi)), \quad a_{O;DD}^{\text{odd}}(\xi) = \frac{1}{2} (a_{O;DD}(\xi) - a_{O;DD}(-\xi)). \quad (221)$$

Then exactly

$$e_{0;DD,\text{odd}}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} a_{E;DD}^{\text{odd}}, \quad o_{0;DD,\text{odd}}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} a_{O;DD}^{\text{odd}}. \quad (222)$$

Introduce the direct DD -budgets

$$\rho_{DD,E} := \sup_{\xi} |a_{E;DD}^{\text{odd}}(\xi)|, \quad \rho_{DD,O} := \sup_{\xi} |a_{O;DD}^{\text{odd}}(\xi)|, \quad \rho_{DD} := \max\{\rho_{DD,E}, \rho_{DD,O}\}. \quad (223)$$

Then

$$|e_{0;DD,\text{odd}}^{(1)}| \leq \rho_{DD,E}, \quad |o_{0;DD,\text{odd}}^{(1)}| \leq \rho_{DD,O}, \quad \max\{|e_{0;DD,\text{odd}}^{(1)}|, |o_{0;DD,\text{odd}}^{(1)}|\} \leq \rho_{DD}. \quad (224)$$

XCI. 6. COMBINED RESIDUAL BOUND

Under the leading exchange reduction,

$$\Delta e_{\text{rem}} = 2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)}, \quad \Delta o_{\text{rem}} = 2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)}. \quad (225)$$

Therefore

$$|\Delta e_{\text{rem}}| \leq 2\rho_{OD} + \rho_{DD,E}, \quad |\Delta o_{\text{rem}}| \leq 2\rho_{OD} + \rho_{DD,O}. \quad (226)$$

In particular,

$$\|r_{\text{rem}}\|_2 \leq \sqrt{2} (2\rho_{OD} + \rho_{DD}). \quad (227)$$

XCII. 7. STRONG SUFFICIENT SURVIVAL CRITERION

The exact transverse coefficient point is

$$\Lambda_E = \Lambda_E^{\text{pr}} + \Delta e_{\text{rem}}, \quad \Lambda_O = \Lambda_O^{\text{pr}} + \Delta o_{\text{rem}}.$$

Hence a strong sufficient survival condition is

$$\boxed{2\rho_{OD} + \rho_{DD,E} < |\Lambda_E^{\text{pr}}|, \quad 2\rho_{OD} + \rho_{DD,O} < |\Lambda_O^{\text{pr}}|.} \quad (228)$$

A weaker Euclidean sufficient condition is

$$\boxed{\sqrt{2}(2\rho_{OD} + \rho_{DD}) < \sqrt{(\Lambda_E^{\text{pr}})^2 + (\Lambda_O^{\text{pr}})^2}.} \quad (229)$$

XCIII. 8. FINAL α, β BOUND SHEET

Using

$$\alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O,$$

we obtain the exact insertion formulas

$$\boxed{\alpha = I_\theta \left(\Lambda_E^{\text{pr}} + 2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)} \right),} \quad (230)$$

$$\boxed{\beta = I_\theta \left(\Lambda_O^{\text{pr}} + 2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)} \right).} \quad (231)$$

For the explicit defect profile,

$$I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa.$$

XCIV. 9. SYMMETRIC-TARGET SUFFICIENT CONDITION

The canonical mixed odd-active target requires

$$\Lambda_E = \Lambda_O \neq 0.$$

An exact equality test still needs the signed residual coefficients. However, a clear sufficient symmetry-preserving condition is the following: if

$$\boxed{\Lambda_E^{\text{pr}} = \Lambda_O^{\text{pr}}, \quad e_{0;OD,\text{odd}}^{(1)} = o_{0;OD,\text{odd}}^{(1)}, \quad e_{0;DD,\text{odd}}^{(1)} = o_{0;DD,\text{odd}}^{(1)},} \quad (232)$$

then automatically

$$\boxed{\Lambda_E = \Lambda_O.} \quad (233)$$

Thus the signed mismatch that must vanish in general is

$$\boxed{\delta_{\text{sym}} := 2(o_{0;OD,\text{odd}}^{(1)} - e_{0;OD,\text{odd}}^{(1)}) + (o_{0;DD,\text{odd}}^{(1)} - e_{0;DD,\text{odd}}^{(1)}) - (\Lambda_E^{\text{pr}} - \Lambda_O^{\text{pr}}).} \quad (234)$$

The exact symmetric target is equivalent to

$$\boxed{\delta_{\text{sym}} = 0.} \quad (235)$$

XCV. HONEST STATUS

At this point the combined $OD + DD$ residual problem is reduced to a finite executable bound package:

$$\mathfrak{B}_\chi, \mathfrak{B}_n, \mathfrak{G}_O, \mathfrak{G}_D, \mathfrak{P}_O^{(0)}, \mathfrak{P}_D^{(0)}, \mathfrak{P}_O^{(1)}, \mathfrak{P}_D^{(1)}, \mathfrak{K}_O^{(1)}, \mathfrak{K}_D^{(1)}, \rho_{DD,E}, \rho_{DD,O}.$$

No additional conceptual reduction is needed. What remains is only the literal evaluation of these finite quantities on the intended branch.

XCVI. FINAL SUMMARY

The exchange-type residual OD/DO can be bounded entirely by the finite packages

$$\mathfrak{B}_\chi = \|P_O \bar{\chi}_B P_D\|_{\text{HS}}, \quad \mathfrak{B}_n = \|P_D \bar{n}_B P_O\|_{\text{HS}},$$

together with projected propagator and momentum/projector derivative budgets

$$\mathfrak{G}_O, \mathfrak{G}_D, \mathfrak{P}_O^{(0)}, \mathfrak{P}_D^{(0)}, \mathfrak{P}_O^{(1)}, \mathfrak{P}_D^{(1)}, \mathfrak{K}_O^{(1)}, \mathfrak{K}_D^{(1)}.$$

Its sharp finite bound is

$$\rho_{OD} = \frac{1}{2} \lambda_{OD}^2 \left(\mathfrak{B}_\chi^2 \mathfrak{L}_O + \mathfrak{B}_n^2 \mathfrak{L}_D \right),$$

with

$$\mathfrak{L}_O = \int_q \left[2\mathfrak{P}_O^{(0)} \mathfrak{P}_O^{(1)} \mathfrak{G}_D \mathfrak{G}_O + (\mathfrak{P}_O^{(0)})^2 \mathfrak{G}_D \mathfrak{K}_O^{(1)} \right] |\partial_t R_k(q)|,$$

$$\mathfrak{L}_D = \int_q \left[2\mathfrak{P}_D^{(0)} \mathfrak{P}_D^{(1)} \mathfrak{G}_O \mathfrak{G}_D + (\mathfrak{P}_D^{(0)})^2 \mathfrak{G}_O \mathfrak{K}_D^{(1)} \right] |\partial_t R_k(q)|.$$

The intrinsic defect-tangent residual DD is directly controlled by the odd parts of its projector-visible transverse coefficients:

$$\rho_{DD,E} := \sup_\xi |a_{E;DD}^{\text{odd}}(\xi)|, \quad \rho_{DD,O} := \sup_\xi |a_{O;DD}^{\text{odd}}(\xi)|, \quad \rho_{DD} := \max\{\rho_{DD,E}, \rho_{DD,O}\}.$$

Thus the combined residual correction obeys

$$|\Delta e_{\text{rem}}| \leq 2\rho_{OD} + \rho_{DD,E}, \quad |\Delta o_{\text{rem}}| \leq 2\rho_{OD} + \rho_{DD,O},$$

and hence

$$\|r_{\text{rem}}\|_2 \leq \sqrt{2} (2\rho_{OD} + \rho_{DD}).$$

A strong sufficient survival criterion is

$$2\rho_{OD} + \rho_{DD,E} < |\Lambda_E^{\text{pr}}|, \quad 2\rho_{OD} + \rho_{DD,O} < |\Lambda_O^{\text{pr}}|.$$

The full transverse coefficient point is therefore

$$\Lambda_E = \Lambda_E^{\text{pr}} + 2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)},$$

$$\Lambda_O = \Lambda_O^{\text{pr}} + 2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)},$$

so

$$\alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O.$$

Finally, the canonical symmetric mixed-target condition is exactly

$$\delta_{\text{sym}} = 2(o_{0;OD,\text{odd}}^{(1)} - e_{0;OD,\text{odd}}^{(1)}) + (o_{0;DD,\text{odd}}^{(1)} - e_{0;DD,\text{odd}}^{(1)}) - (\Lambda_E^{\text{pr}} - \Lambda_O^{\text{pr}}) = 0.$$

So the $OD + DD$ problem is no longer structural. It is reduced to a finite executable budget test plus, if one wants exact symmetry, one final signed residual equality test.

XCVII. OPTIMIZED-REGULATOR REDUCTION OF THE OD -RESIDUAL BUDGET

XCVIII. GOAL

The remaining exchange-type residual budget was

$$\boxed{\rho_{OD} = \frac{1}{2} \lambda_{OD}^2 \left(\mathfrak{B}_\chi^2 \mathfrak{L}_O + \mathfrak{B}_n^2 \mathfrak{L}_D \right)}, \quad (236)$$

with

$$\mathfrak{L}_O, \quad \mathfrak{L}_D$$

still written in terms of abstract propagator/projector derivative packages.

The next nonredundant step is to insert the explicit working regulator choice already fixed in the notes and reduce these packages further.

XCIX. 1. EXPLICIT OPTIMIZED REGULATOR CHOICE

The background-adapted sectoral regulator is taken as

$$\boxed{R_k[\bar{\Phi}_{\min}] = \text{diag}(R_{T,k}, R_{O,k}, R_{D,k}),} \quad (237)$$

with the explicit optimized profile

$$\boxed{R_{I,k} = Z_{I,k} (k^2 - \mathcal{H}_{II}^{(0)}) \Theta(k^2 - \mathcal{H}_{II}^{(0)}), \quad I \in \{T, O, D\}.} \quad (238)$$

In momentum representation this becomes

$$\boxed{R_{I,k}(q) = Z_{I,k} (k^2 - q^2) \Theta(k^2 - q^2).} \quad (239)$$

C. 2. EXPLICIT REGULATOR-DERIVATIVE BOUND

For the optimized momentum profile,

$$R_{I,k}(q) = Z_{I,k} (k^2 - q^2) \Theta(k^2 - q^2),$$

we compute

$$\partial_t R_{I,k}(q) = (\partial_t Z_{I,k})(k^2 - q^2) \Theta(k^2 - q^2) + Z_{I,k} \partial_t [(k^2 - q^2) \Theta(k^2 - q^2)]. \quad (240)$$

Since $t = \ln(k/\Lambda)$, one has $\partial_t = k \partial_k$, and

$$\partial_t [(k^2 - q^2) \Theta(k^2 - q^2)] = 2k^2 \Theta(k^2 - q^2),$$

because the distributional boundary term is multiplied by $k^2 - q^2$ and vanishes.

Therefore

$$\boxed{\partial_t R_{I,k}(q) = \left[(\partial_t Z_{I,k})(k^2 - q^2) + 2Z_{I,k} k^2 \right] \Theta(k^2 - q^2).} \quad (241)$$

Hence the sharp uniform bound is

$$\boxed{\mathfrak{R}_{I,k} := \sup_q |\partial_t R_{I,k}(q)| \leq k^2 (|\partial_t Z_{I,k}| + 2|Z_{I,k}|).} \quad (242)$$

Define

$$\boxed{\mathfrak{R}_{OD} := \max\{\mathfrak{R}_{O,k}, \mathfrak{R}_{D,k}\} \leq k^2 \max\{|\partial_t Z_{O,k}| + 2|Z_{O,k}|, |\partial_t Z_{D,k}| + 2|Z_{D,k}|\}.} \quad (243)$$

CI. 3. PROPAGATOR BOUNDS

The notes already reduce the projected propagator part to the explicit projected spectral gaps:

$$\boxed{\mathfrak{G}_O \leq \Delta_O^{-1}, \quad \mathfrak{G}_D \leq \Delta_D^{-1}.} \quad (244)$$

CII. 4. PROPAGATOR MOMENTUM-DERIVATIVE BOUNDS

Let

$$\boxed{\mathfrak{Y}_O := \sup_q \left\| \partial_{p_\mu} \mathcal{H}_{OO}^{\text{eff}}(p+q) \Big|_{p=0} \right\|, \quad \mathfrak{Y}_D := \sup_q \left\| \partial_{p_\mu} \mathcal{H}_{DD}^{\text{eff}}(p+q) \Big|_{p=0} \right\|.} \quad (245)$$

Since

$$G_I = (P_I[\mathcal{H}_{II} + R_{I,k}]P_I)^{-1},$$

the resolvent identity gives

$$\partial_{p_\mu} G_I = -G_I(\partial_{p_\mu} \mathcal{H}_{II}^{\text{eff}})G_I.$$

Therefore

$$\boxed{\mathfrak{K}_O^{(1)} \leq \mathfrak{G}_O^2 \mathfrak{Y}_O \leq \Delta_O^{-2} \mathfrak{Y}_O, \quad \mathfrak{K}_D^{(1)} \leq \mathfrak{G}_D^2 \mathfrak{Y}_D \leq \Delta_D^{-2} \mathfrak{Y}_D.} \quad (246)$$

CIII. 5. REGULATOR-SUPPORT MOMENTUM VOLUME

For the optimized cutoff, the support of $\partial_t R_{I,k}$ is contained in

$$q^2 \leq k^2.$$

Define the support volume

$$\boxed{\mathfrak{Q}_k := \int_{q^2 \leq k^2} \frac{d^d q}{(2\pi)^d} = \frac{S_{d-1}}{d(2\pi)^d} k^d,} \quad (247)$$

where S_{d-1} is the area of the unit $(d-1)$ -sphere.

CIV. 6. EXPLICIT $\mathfrak{L}_O, \mathfrak{L}_D$ BOUNDS

Recall

$$\mathfrak{L}_O = \int_q \left[2\mathfrak{P}_O^{(0)} \mathfrak{P}_O^{(1)} \mathfrak{G}_D \mathfrak{G}_O + (\mathfrak{P}_O^{(0)})^2 \mathfrak{G}_D \mathfrak{K}_O^{(1)} \right] |\partial_t R_k(q)|, \quad (248)$$

$$\mathfrak{L}_D = \int_q \left[2\mathfrak{P}_D^{(0)} \mathfrak{P}_D^{(1)} \mathfrak{G}_O \mathfrak{G}_D + (\mathfrak{P}_D^{(0)})^2 \mathfrak{G}_O \mathfrak{K}_D^{(1)} \right] |\partial_t R_k(q)|. \quad (249)$$

Using (243), (244), (246), and (247), we obtain

$$\boxed{\mathfrak{L}_O \leq \mathfrak{Q}_k \mathfrak{K}_{OD} \left[2\mathfrak{P}_O^{(0)} \mathfrak{P}_O^{(1)} \Delta_D^{-1} \Delta_O^{-1} + (\mathfrak{P}_O^{(0)})^2 \Delta_D^{-1} \Delta_O^{-2} \mathfrak{Y}_O \right],} \quad (250)$$

$$\boxed{\mathfrak{L}_D \leq \mathfrak{Q}_k \mathfrak{K}_{OD} \left[2\mathfrak{P}_D^{(0)} \mathfrak{P}_D^{(1)} \Delta_O^{-1} \Delta_D^{-1} + (\mathfrak{P}_D^{(0)})^2 \Delta_O^{-1} \Delta_D^{-2} \mathfrak{Y}_D \right].} \quad (251)$$

CV. 7. FULLY EXPLICIT OD -RESIDUAL BUDGET

Substituting (250) and (251) into (236) yields

$$\rho_{OD} \leq \frac{1}{2} \lambda_{OD}^2 \mathfrak{Q}_k \mathfrak{R}_{OD} \left\{ \mathfrak{B}_\chi^2 \left[2\mathfrak{P}_O^{(0)} \mathfrak{P}_O^{(1)} \Delta_D^{-1} \Delta_O^{-1} + (\mathfrak{P}_O^{(0)})^2 \Delta_D^{-1} \Delta_O^{-2} \mathfrak{V}_O \right] + \mathfrak{B}_n^2 \left[2\mathfrak{P}_D^{(0)} \mathfrak{P}_D^{(1)} \Delta_O^{-1} \Delta_D^{-1} + (\mathfrak{P}_D^{(0)})^2 \Delta_O^{-1} \Delta_D^{-2} \mathfrak{V}_D \right] \right\}. \quad (252)$$

This is the sharpest explicit working-branch bound currently licensed by the uploaded notes.

CVI. 8. COMBINED $OD + DD$ SURVIVAL CRITERION

The DD -sector remains directly controlled by

$$\rho_{DD,E} := \sup_{\xi} |a_{E;DD}^{\text{odd}}(\xi)|, \quad \rho_{DD,O} := \sup_{\xi} |a_{O;DD}^{\text{odd}}(\xi)|, \quad \rho_{DD} := \max\{\rho_{DD,E}, \rho_{DD,O}\}. \quad (253)$$

Hence

$$|\Delta e_{\text{rem}}| \leq 2\rho_{OD} + \rho_{DD,E}, \quad |\Delta o_{\text{rem}}| \leq 2\rho_{OD} + \rho_{DD,O}. \quad (254)$$

Therefore the strong sufficient survival condition becomes

$$2\rho_{OD} + \rho_{DD,E} < |\Lambda_E^{\text{pr}}|, \quad 2\rho_{OD} + \rho_{DD,O} < |\Lambda_O^{\text{pr}}|. \quad (255)$$

A Euclidean sufficient condition is

$$\sqrt{2}(2\rho_{OD} + \rho_{DD}) < \sqrt{(\Lambda_E^{\text{pr}})^2 + (\Lambda_O^{\text{pr}})^2}. \quad (256)$$

CVII. 9. FINAL COEFFICIENT SHEET

The full transverse coefficient point is therefore

$$\Lambda_E = \Lambda_E^{\text{pr}} + 2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)}, \quad \Lambda_O = \Lambda_O^{\text{pr}} + 2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)}. \quad (257)$$

Hence

$$\alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O. \quad (258)$$

For the explicit defect profile,

$$I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa.$$

CVIII. 10. HONEST STATUS

The remaining $OD + DD$ completion is no longer a structural problem.

It is reduced to the finite executable package

$$\mathfrak{B}_\chi, \mathfrak{B}_n, \Delta_O, \Delta_D, Z_{O,k}, Z_{D,k}, \partial_t Z_{O,k}, \partial_t Z_{D,k}, \mathfrak{P}_O^{(0)}, \mathfrak{P}_D^{(0)}, \mathfrak{P}_O^{(1)}, \mathfrak{P}_D^{(1)}, \mathfrak{V}_O, \mathfrak{V}_D, \rho_{DD,E}, \rho_{DD,O}.$$

Once these are evaluated on the intended branch, the residual transverse audit is decided.

CIX. FINAL SUMMARY

Using the explicit optimized regulator

$$R_{I,k}(q) = Z_{I,k}(k^2 - q^2)\Theta(k^2 - q^2),$$

the regulator derivative factor is no longer abstract:

$$\mathfrak{R}_{I,k} \leq k^2(|\partial_t Z_{I,k}| + 2|Z_{I,k}|).$$

The projected propagators are bounded by the explicit projected gaps:

$$\mathfrak{G}_O \leq \Delta_O^{-1}, \quad \mathfrak{G}_D \leq \Delta_D^{-1}.$$

The propagator momentum-derivative budgets satisfy

$$\mathfrak{K}_O^{(1)} \leq \Delta_O^{-2} \mathfrak{Y}_O, \quad \mathfrak{K}_D^{(1)} \leq \Delta_D^{-2} \mathfrak{Y}_D.$$

Therefore the exchange-type residual budget becomes

$$\rho_{OD} \leq \frac{1}{2} \lambda_{OD}^2 \mathfrak{Q}_k \mathfrak{R}_{OD} \left\{ \mathfrak{B}_\chi^2 \left[2\mathfrak{P}_O^{(0)} \mathfrak{P}_O^{(1)} \Delta_D^{-1} \Delta_O^{-1} + (\mathfrak{P}_O^{(0)})^2 \Delta_D^{-1} \Delta_O^{-2} \mathfrak{Y}_O \right] + \mathfrak{B}_n^2 \left[2\mathfrak{P}_D^{(0)} \mathfrak{P}_D^{(1)} \Delta_O^{-1} \Delta_D^{-1} + (\mathfrak{P}_D^{(0)})^2 \Delta_O^{-1} \Delta_D^{-2} \mathfrak{Y}_D \right] \right\}.$$

The DD -sector is controlled directly by

$$\rho_{DD,E} = \sup_\xi |a_{E;DD}^{\text{odd}}(\xi)|, \quad \rho_{DD,O} = \sup_\xi |a_{O;DD}^{\text{odd}}(\xi)|.$$

Hence the strong sufficient combined survival criterion is

$$2\rho_{OD} + \rho_{DD,E} < |\Lambda_E^{\text{pr}}|, \quad 2\rho_{OD} + \rho_{DD,O} < |\Lambda_O^{\text{pr}}|.$$

So the $OD + DD$ residual problem is now reduced to a finite explicit budget test on the intended branch.

CX. ACTUAL WORKING PDE WITH THE FULL $OD + DO + DD$ RESIDUAL INSERTION

CXI. SHORT KOREAN EXPLANATION

We now stop rewriting the residual sector as abstract budgets only. Instead, we insert the OD - and DD -residuals directly into the transverse coefficient point and write the actual working PDE in the current reference frame.

At the theorem level currently licensed by the uploaded notes, this is the furthest honest forward form: the residuals are not yet numerically populated, but their exact insertion structure is now complete.

CXII. 1. PRINCIPAL TRANSVERSE PAIR

The leading transverse pair is already isolated as

$$\Lambda_E^{\text{pr}} = e_{0;TD}^{(1)} + e_{0;DT}^{(1)}, \quad \Lambda_O^{\text{pr}} = o_{0;TD}^{(1)} + o_{0;DT}^{(1)}. \quad (259)$$

In the current reference frame, the reduced coefficients satisfy

$$\alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O. \quad (260)$$

CXIII. 2. EXCHANGE-TYPE OD/DO RESIDUAL

The mixed quartic interaction

$$\lambda_{OD} \rho_O \rho_D$$

generates the background-induced cubic vertices

$$\Gamma_{\nu\nu\zeta}^{(3)} \sim \lambda_{OD} \bar{\chi}_B(\xi), \quad \Gamma_{\nu\zeta\zeta}^{(3)} \sim \lambda_{OD} \bar{n}_B(\xi). \quad (261)$$

The corresponding projected mixed self-energies are

$$\Sigma_O^{(OD)}(p) = - \int_q \sum_{m,n} \mathcal{V}_{O;mn}^{(OD)}(p,q) G_D^{(n)}(q) \mathcal{V}_{O;nm}^{(OD)}(-p,q) G_O^{(m)}(p+q) \partial_t R_k(q), \quad (262)$$

with

$$\mathcal{V}_{O;mn}^{(OD)}(p,q) = \lambda_{OD} \mathcal{B}_{mn}^x(\xi) \mathcal{P}_O(p,q), \quad \mathcal{B}_{mn}^x = \langle u_m, \bar{\chi}_B v_n \rangle, \quad (263)$$

and similarly

$$\Sigma_D^{(OD)}(p) = - \int_q \sum_{m,n} \tilde{\mathcal{V}}_{D;nm}^{(OD)}(p,q) G_O^{(m)}(q) \tilde{\mathcal{V}}_{D;mn}^{(OD)}(-p,q) G_D^{(n)}(p+q) \partial_t R_k(q), \quad (264)$$

with

$$\tilde{\mathcal{V}}_{D;nm}^{(OD)}(p,q) = \lambda_{OD} \mathcal{B}_{nm}^n(\xi) \mathcal{P}_D(p,q), \quad \mathcal{B}_{nm}^n = \langle v_n, \bar{n}_B u_m \rangle. \quad (265)$$

The reflection-odd projected transport kernel is

$$\mathcal{M}_{OD/DO,\text{odd}}^\mu(\xi) = \frac{1}{2} \Pi_Q \left[\partial_{p_\mu} (\Sigma_O^{(OD)} + \Sigma_D^{(OD)})_{p=0,\xi} - \partial_{p_\mu} (\Sigma_O^{(OD)} + \Sigma_D^{(OD)})_{p=0,-\xi} \right] \Pi_Q. \quad (266)$$

Thus

$$e_{0;OD/DO,\text{odd}}^{(1)} = \frac{1}{4} E^\mu{}_a \text{Tr}_{Q_B} (\Sigma_a \mathcal{M}_{OD/DO,\mu,\text{odd}}), \quad (267)$$

$$o_{0;OD/DO,\text{odd}}^{(1)} = \frac{1}{4} J^\mu{}_a \text{Tr}_{Q_B} (\Sigma_a \mathcal{M}_{OD/DO,\mu,\text{odd}}). \quad (268)$$

If the projected mixed sector preserves $O \leftrightarrow D$ exchange symmetry, then

$$e_{0;DO,\text{odd}}^{(1)} = e_{0;OD,\text{odd}}^{(1)}, \quad o_{0;DO,\text{odd}}^{(1)} = o_{0;OD,\text{odd}}^{(1)}. \quad (269)$$

Hence

$$r_{OD/DO} = \left(2e_{0;OD,\text{odd}}^{(1)}, 2o_{0;OD,\text{odd}}^{(1)} \right). \quad (270)$$

CXIV. 3. INTRINSIC DD RESIDUAL

The projected DD -kernel is already of the explicit transverse form

$$\mathcal{M}_{DD,\text{inv}}^\mu = a_{\parallel;DD}(\theta) n^\mu \sigma_3 + a_{E;DD}(\theta) E^\mu{}_a \sigma_a + a_{O;DD}(\theta) J^\mu{}_a \sigma_a. \quad (271)$$

Define the odd parts

$$a_{E;DD}^{\text{odd}}(\xi) = \frac{1}{2}(a_{E;DD}(\xi) - a_{E;DD}(-\xi)), \quad a_{O;DD}^{\text{odd}}(\xi) = \frac{1}{2}(a_{O;DD}(\xi) - a_{O;DD}(-\xi)). \quad (272)$$

Then the direct DD -residual coefficients are

$$e_{0;DD,\text{odd}}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} a_{E;DD}^{\text{odd}}, \quad o_{0;DD,\text{odd}}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} a_{O;DD}^{\text{odd}}. \quad (273)$$

Equivalently,

$$e_{0;DD,\text{odd}}^{(1)} = \frac{1}{4} E^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_{DD,\mu,\text{odd}}), \quad o_{0;DD,\text{odd}}^{(1)} = \frac{1}{4} J^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_{DD,\mu,\text{odd}}). \quad (274)$$

So

$$r_{DD} = \left(e_{0;DD,\text{odd}}^{(1)}, o_{0;DD,\text{odd}}^{(1)} \right). \quad (275)$$

CXV. 4. FINAL COMBINED RESIDUAL TRANSVERSE PAIR

Combining (270) and (275),

$$r_{\text{rem}} = \left(2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)}, 2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)} \right). \quad (276)$$

Therefore the full transverse coefficient point is

$$\Lambda_E = \Lambda_E^{\text{pr}} + 2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)}, \quad (277)$$

$$\Lambda_O = \Lambda_O^{\text{pr}} + 2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)}. \quad (278)$$

CXVI. 5. FINAL INSERTED α, β SHEET

Using (260),

$$\alpha = I_\theta \left(\Lambda_E^{\text{pr}} + 2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)} \right), \quad (279)$$

$$\beta = I_\theta \left(\Lambda_O^{\text{pr}} + 2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)} \right). \quad (280)$$

For the explicit defect profile,

$$I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa. \quad (281)$$

So the actual inserted transverse coefficients become

$$\alpha = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left(\Lambda_E^{\text{pr}} + 2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)} \right), \quad (282)$$

$$\beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left(\Lambda_O^{\text{pr}} + 2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)} \right). \quad (283)$$

CXVII. 6. ACTUAL WORKING PDE

The current actual working PDE in the reference frame is therefore

$$\boxed{\left[i\partial_t - \lambda_{\parallel} \sigma_3(-i\partial_{\parallel}) - \alpha \left(\sigma_1(-i\partial_{q_1}) \right) - \beta \left(\sigma_2(-i\partial_{q_2}) \right) \right] \chi = 0,} \quad (284)$$

with

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3,$$

and α, β given by (282)–(283).

Equivalently, in the symmetric canonical target basis where one wants $\alpha = \beta$,

$$\boxed{\left[i\partial_t - \lambda_{\parallel} \sigma_3(-i\partial_{\parallel}) - v_{\perp}^{(1)} \sigma_1(-i\partial_{q_1}) - v_{\perp}^{(2)} \sigma_2(-i\partial_{q_2}) \right] \chi = 0,} \quad (285)$$

where

$$v_{\perp}^{(1)} = \alpha, \quad v_{\perp}^{(2)} = \beta.$$

CXVIII. 7. SYMMETRIC MIXED-TARGET CONDITION

The canonical mixed odd-active target requires

$$\alpha = \beta \quad \Longleftrightarrow \quad \Lambda_E = \Lambda_O.$$

Thus the exact scalar condition is

$$\boxed{2 \left(o_{0;OD,\text{odd}}^{(1)} - e_{0;OD,\text{odd}}^{(1)} \right) + \left(o_{0;DD,\text{odd}}^{(1)} - e_{0;DD,\text{odd}}^{(1)} \right) = \Lambda_E^{\text{pr}} - \Lambda_O^{\text{pr}}.} \quad (286)$$

If this holds, then

$$\boxed{\alpha = \beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left(\Lambda_E^{\text{pr}} + 2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)} \right).} \quad (287)$$

CXIX. 8. STRONG SUFFICIENT SURVIVAL CONDITION

A strong sufficient condition for the transverse carrier to survive is

$$\boxed{\left| 2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)} \right| < |\Lambda_E^{\text{pr}}|,} \quad (288)$$

$$\boxed{\left| 2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)} \right| < |\Lambda_O^{\text{pr}}|.} \quad (289)$$

CXX. 9. HONEST STATUS

At this point the $OD + DD$ sector is not conceptually open any more.

The exact remaining unknowns are only the four odd residual numbers

$$e_{0;OD,\text{odd}}^{(1)}, \quad o_{0;OD,\text{odd}}^{(1)}, \quad e_{0;DD,\text{odd}}^{(1)}, \quad o_{0;DD,\text{odd}}^{(1)}.$$

Everything else is already inserted into the actual working PDE.

CXXI. FINAL SUMMARY

The full residual transverse completion has now been inserted explicitly.
The exchange-type residual is generated by the projected mixed self-energies

$$\Sigma_O^{(OD)}(p), \quad \Sigma_D^{(OD)}(p),$$

with factorized projected vertices

$$\mathcal{V}_{O;mn}^{(OD)} = \lambda_{OD} \mathcal{B}_{mn}^x \mathcal{P}_O, \quad \tilde{\mathcal{V}}_{D;nm}^{(OD)} = \lambda_{OD} \mathcal{B}_{nm}^n \mathcal{P}_D.$$

The intrinsic DD -residual is directly projector-visible through

$$\mathcal{M}_{DD,\text{inv}}^\mu = a_{\parallel;DD} n^\mu \sigma_3 + a_{E;DD} E^\mu{}_a \sigma_a + a_{O;DD} J^\mu{}_a \sigma_a.$$

Under leading $O \leftrightarrow D$ exchange symmetry, the full residual vector is

$$r_{\text{rem}} = \left(2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)}, \quad 2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)} \right).$$

Therefore the full transverse coefficient point is

$$\Lambda_E = \Lambda_E^{\text{pr}} + 2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)},$$

$$\Lambda_O = \Lambda_O^{\text{pr}} + 2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)}.$$

Hence the actual inserted transverse coefficients are

$$\alpha = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left(\Lambda_E^{\text{pr}} + 2e_{0;OD,\text{odd}}^{(1)} + e_{0;DD,\text{odd}}^{(1)} \right),$$

$$\beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left(\Lambda_O^{\text{pr}} + 2o_{0;OD,\text{odd}}^{(1)} + o_{0;DD,\text{odd}}^{(1)} \right).$$

The canonical symmetric mixed-target condition is the single scalar equation

$$2(o_{0;OD,\text{odd}}^{(1)} - e_{0;OD,\text{odd}}^{(1)}) + (o_{0;DD,\text{odd}}^{(1)} - e_{0;DD,\text{odd}}^{(1)}) = \Lambda_E^{\text{pr}} - \Lambda_O^{\text{pr}}.$$

So the actual working PDE is already fully populated except for those four odd residual numbers.

CXXII. ACTUAL FORWARD FORM OF THE TRANSVERSE COEFFICIENT SHEET INCLUDING $OD + DD$

CXXIII. SHORT KOREAN EXPLANATION

We now stop leaving the residual sector as an abstract correction vector only.

Instead, we use the already-closed local invariant algebra to write the DD -sector explicitly at leading and first subleading odd order, and we absorb the OD -sector into the same local odd expansion class.

This produces an actual corrected coefficient sheet for (α, β) , not just a residual placeholder.

CXXIV. 1. GENERAL LOCAL ODD EXPANSION FOR EVERY TRANSVERSE CONTRIBUTION

The complete shell-compatible local transport kernel has the invariant form

$$V^\mu(\xi) = h_{||}(\xi) n^\mu \sigma_3 + h_E(\xi) E^\mu_a \sigma_a + h_O(\xi) J^\mu_a \sigma_a + V_{\text{irr}}^\mu(\xi), \quad (290)$$

and parity implies that every principal transport density must carry one explicit odd factor. Hence the most general local odd expansions are

$$h_E(\xi) = \theta'(\xi) \mathcal{E}_1(\mathcal{X}(\xi)) + \theta'(\xi)^3 \mathcal{E}_3(\mathcal{X}(\xi)) + O_{\text{odd}, \geq 5}, \quad (291)$$

$$h_O(\xi) = \theta'(\xi) \mathcal{O}_1(\mathcal{X}(\xi)) + \theta'(\xi)^3 \mathcal{O}_3(\mathcal{X}(\xi)) + O_{\text{odd}, \geq 5}, \quad (292)$$

where the local invariant list is parity-even:

$$\mathcal{X}(\xi) = (X_0(\xi), X_1(\xi), X_2(\xi), \dots), \quad X_0 = \theta^2, \quad X_1 = \theta'^2, \quad X_2 = \theta\theta'', \dots \quad (293)$$

Therefore every surviving DD - or OD -residual contribution must belong to this same odd local class.

CXXV. 2. LEADING LOCAL EXPANSION OF THE DD -SECTOR

The projected DD -kernel already has the explicit transverse decomposition

$$\mathcal{M}_{DD, \text{inv}}^\mu = a_{||;DD}(\xi) n^\mu \sigma_3 + a_{E;DD}(\xi) E^\mu_a \sigma_a + a_{O;DD}(\xi) J^\mu_a \sigma_a. \quad (294)$$

Because only the odd part survives after graded projection, the transverse coefficients must have the local form

$$a_{E;DD}^{\text{odd}}(\xi) = \theta'(\xi) \left[\Lambda_{E;DD}^{(0)} + \sum_A \Lambda_{E;DD,A}^{(2)} X_A(\xi) \right] + O_{\text{odd}, \geq 5}, \quad (295)$$

$$a_{O;DD}^{\text{odd}}(\xi) = \theta'(\xi) \left[\Lambda_{O;DD}^{(0)} + \sum_A \Lambda_{O;DD,A}^{(2)} X_A(\xi) \right] + O_{\text{odd}, \geq 5}. \quad (296)$$

At the first nontrivial level it is enough to keep

$$X_0 = \theta^2, \quad X_1 = \theta'^2.$$

Then

$$a_{E;DD}^{\text{odd}}(\xi) = \theta'(\xi) \left[\Lambda_{E;DD}^{(0)} + \Lambda_{E;DD,0}^{(2)} \theta(\xi)^2 + \Lambda_{E;DD,1}^{(2)} \theta'(\xi)^2 \right] + O_{\text{odd}, \geq 5}, \quad (297)$$

$$a_{O;DD}^{\text{odd}}(\xi) = \theta'(\xi) \left[\Lambda_{O;DD}^{(0)} + \Lambda_{O;DD,0}^{(2)} \theta(\xi)^2 + \Lambda_{O;DD,1}^{(2)} \theta'(\xi)^2 \right] + O_{\text{odd}, \geq 5}. \quad (298)$$

CXXVI. 3. PROJECTION OF THE DD -SECTOR

The reduced coefficients are

$$\alpha_{DD} = \int d\xi f_+(\xi) f_-(\xi) a_{E;DD}^{\text{odd}}(\xi), \quad \beta_{DD} = \int d\xi f_+(\xi) f_-(\xi) a_{O;DD}^{\text{odd}}(\xi).$$

Substituting (297)–(298) gives

$$\alpha_{DD} = \Lambda_{E;DD}^{(0)} I_\theta + \Lambda_{E;DD,0}^{(2)} I_{\theta^3} + \Lambda_{E;DD,1}^{(2)} I_{\theta'^3} + O(\theta_0^5 \kappa, \theta_0^5 \kappa^3), \quad (299)$$

$$\beta_{DD} = \Lambda_{O;DD}^{(0)} I_\theta + \Lambda_{O;DD,0}^{(2)} I_{\theta^3} + \Lambda_{O;DD,1}^{(2)} I_{\theta'^3} + O(\theta_0^5 \kappa, \theta_0^5 \kappa^3). \quad (300)$$

For the explicit defect profile one has

$$I_\theta = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256}\theta_0^3\kappa^3. \quad (301)$$

Also

$$I_{\theta^3} := \int d\xi f_+ f_- \theta' \theta^2 = -\frac{\sqrt{3}\pi}{32}\theta_0^3\kappa. \quad (302)$$

Therefore

$$\alpha_{DD} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa \left[\Lambda_{E;DD}^{(0)} + \frac{\theta_0^2}{2}\Lambda_{E;DD,0}^{(2)} + \frac{3}{16}\theta_0^2\kappa^2\Lambda_{E;DD,1}^{(2)} \right] + O(\theta_0^5\kappa, \theta_0^5\kappa^3), \quad (303)$$

$$\beta_{DD} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa \left[\Lambda_{O;DD}^{(0)} + \frac{\theta_0^2}{2}\Lambda_{O;DD,0}^{(2)} + \frac{3}{16}\theta_0^2\kappa^2\Lambda_{O;DD,1}^{(2)} \right] + O(\theta_0^5\kappa, \theta_0^5\kappa^3). \quad (304)$$

CXXVII. 4. EFFECTIVE LOCAL EXPANSION OF THE OD -SECTOR

The OD -sector is generated through the projected mixed self-energy derivative, not as a direct tree-level block. However, after the Schur reduction and odd parity projection, it belongs to the same local odd transport class. Hence we may write

$$e_{0;OD,\text{odd}}^{(1)} = \Lambda_{E;OD}^{\text{eff},(0)} + \Lambda_{E;OD,0}^{\text{eff},(2)}\theta^2 + \Lambda_{E;OD,1}^{\text{eff},(2)}\theta'^2 + O_{\geq 4}, \quad (305)$$

$$o_{0;OD,\text{odd}}^{(1)} = \Lambda_{O;OD}^{\text{eff},(0)} + \Lambda_{O;OD,0}^{\text{eff},(2)}\theta^2 + \Lambda_{O;OD,1}^{\text{eff},(2)}\theta'^2 + O_{\geq 4}, \quad (306)$$

so that after projection

$$\alpha_{OD/DO} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa \left[2\Lambda_{E;OD}^{\text{eff},(0)} + \theta_0^2\Lambda_{E;OD,0}^{\text{eff},(2)} + \frac{3}{8}\theta_0^2\kappa^2\Lambda_{E;OD,1}^{\text{eff},(2)} \right] + O(\theta_0^5\kappa, \theta_0^5\kappa^3), \quad (307)$$

$$\beta_{OD/DO} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa \left[2\Lambda_{O;OD}^{\text{eff},(0)} + \theta_0^2\Lambda_{O;OD,0}^{\text{eff},(2)} + \frac{3}{8}\theta_0^2\kappa^2\Lambda_{O;OD,1}^{\text{eff},(2)} \right] + O(\theta_0^5\kappa, \theta_0^5\kappa^3). \quad (308)$$

The factor 2 reflects the $OD + DO$ exchange-reduced insertion.

CXXVIII. 5. FULL $OD + DD$ -CORRECTED TRANSVERSE COEFFICIENTS

The principal pair contributes

$$\alpha_{\text{pr}} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\Lambda_E^{\text{pr}}, \quad \beta_{\text{pr}} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\Lambda_O^{\text{pr}}. \quad (309)$$

Therefore the full transverse coefficients become

$$\alpha = \alpha_{\text{pr}} + \alpha_{OD/DO} + \alpha_{DD} \quad (310)$$

$$\begin{aligned} &= -\frac{\sqrt{3}\pi}{16}\theta_0\kappa \left[\Lambda_E^{\text{pr}} + 2\Lambda_{E;OD}^{\text{eff},(0)} + \Lambda_{E;DD}^{(0)} \right. \\ &\quad \left. + \theta_0^2 \left(\Lambda_{E;OD,0}^{\text{eff},(2)} + \frac{1}{2}\Lambda_{E;DD,0}^{(2)} \right) + \theta_0^2\kappa^2 \left(\frac{3}{8}\Lambda_{E;OD,1}^{\text{eff},(2)} + \frac{3}{16}\Lambda_{E;DD,1}^{(2)} \right) \right] + O(\theta_0^5\kappa, \theta_0^5\kappa^3), \end{aligned} \quad (311)$$

$$\beta = \beta_{\text{pr}} + \beta_{OD/DO} + \beta_{DD} \quad (312)$$

$$\begin{aligned} &= -\frac{\sqrt{3}\pi}{16}\theta_0\kappa \left[\Lambda_O^{\text{pr}} + 2\Lambda_{O;OD}^{\text{eff},(0)} + \Lambda_{O;DD}^{(0)} \right. \\ &\quad \left. + \theta_0^2 \left(\Lambda_{O;OD,0}^{\text{eff},(2)} + \frac{1}{2}\Lambda_{O;DD,0}^{(2)} \right) + \theta_0^2\kappa^2 \left(\frac{3}{8}\Lambda_{O;OD,1}^{\text{eff},(2)} + \frac{3}{16}\Lambda_{O;DD,1}^{(2)} \right) \right] + O(\theta_0^5\kappa, \theta_0^5\kappa^3). \end{aligned} \quad (313)$$

CXXIX. 6. ACTUAL WORKING PDE AFTER $OD + DD$ INSERTION

Thus the actual reference-frame working PDE is

$$\boxed{\left[i\partial_t - \lambda_{\parallel} \sigma_3(-i\partial_{\parallel}) - \alpha \sigma_1(-i\partial_{q_1}) - \beta \sigma_2(-i\partial_{q_2}) \right] \chi = 0,} \quad (314)$$

with

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512}\lambda_c \theta_0^3\kappa^3,$$

and α, β given explicitly by (311) and (313).

CXXX. 7. CANONICAL SYMMETRIC MIXED-TARGET CONDITION

The canonical mixed odd-active target requires

$$\alpha = \beta.$$

At leading local order this is

$$\boxed{\Lambda_E^{\text{pr}} + 2\Lambda_{E;OD}^{\text{eff},(0)} + \Lambda_{E;DD}^{(0)} = \Lambda_O^{\text{pr}} + 2\Lambda_{O;OD}^{\text{eff},(0)} + \Lambda_{O;DD}^{(0)}. \quad (315)}$$

At first subleading order one further needs

$$\boxed{\Lambda_{E;OD,0}^{\text{eff},(2)} + \frac{1}{2}\Lambda_{E;DD,0}^{(2)} = \Lambda_{O;OD,0}^{\text{eff},(2)} + \frac{1}{2}\Lambda_{O;DD,0}^{(2)}, \quad (316)}$$

and

$$\boxed{\frac{3}{8}\Lambda_{E;OD,1}^{\text{eff},(2)} + \frac{3}{16}\Lambda_{E;DD,1}^{(2)} = \frac{3}{8}\Lambda_{O;OD,1}^{\text{eff},(2)} + \frac{3}{16}\Lambda_{O;DD,1}^{(2)}. \quad (317)}$$

CXXXI. 8. HONEST STATUS

At this point the DD -sector is explicitly inserted through its leading and first subleading odd local coefficients, and the OD -sector is inserted through the corresponding effective local odd coefficients generated by the projected self-energy derivative.

So the actual remaining frontier is no longer “what is the form of the corrected PDE?” That form is already (314). What remains is the literal microscopic evaluation of the coefficient set

$$\Lambda_{E;OD}^{\text{eff},(0)}, \Lambda_{O;OD}^{\text{eff},(0)}, \Lambda_{E;DD}^{(0)}, \Lambda_{O;DD}^{(0)},$$

and their first even-invariant corrections.

CXXXII. FINAL SUMMARY

Using the actual-local invariant algebra, every surviving transverse residual contribution from OD or DD must belong to the odd local class

$$\theta'(\xi) \times (\text{parity-even local invariant function}).$$

The DD -sector can therefore be inserted explicitly at leading and first subleading odd order:

$$a_{E;DD}^{\text{odd}}(\xi) = \theta'(\xi) \left[\Lambda_{E;DD}^{(0)} + \Lambda_{E;DD,0}^{(2)} \theta^2 + \Lambda_{E;DD,1}^{(2)} \theta'^2 \right] + O_{\text{odd}, \geq 5},$$

$$a_{O;DD}^{\text{odd}}(\xi) = \theta'(\xi) \left[\Lambda_{O;DD}^{(0)} + \Lambda_{O;DD,0}^{(2)} \theta^2 + \Lambda_{O;DD,1}^{(2)} \theta'^2 \right] + O_{\text{odd}, \geq 5}.$$

After projection, this gives

$$\alpha_{DD} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left[\Lambda_{E;DD}^{(0)} + \frac{\theta_0^2}{2} \Lambda_{E;DD,0}^{(2)} + \frac{3}{16} \theta_0^2 \kappa^2 \Lambda_{E;DD,1}^{(2)} \right] + O(\theta_0^5 \kappa, \theta_0^5 \kappa^3),$$

$$\beta_{DD} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left[\Lambda_{O;DD}^{(0)} + \frac{\theta_0^2}{2} \Lambda_{O;DD,0}^{(2)} + \frac{3}{16} \theta_0^2 \kappa^2 \Lambda_{O;DD,1}^{(2)} \right] + O(\theta_0^5 \kappa, \theta_0^5 \kappa^3).$$

The OD -sector is inserted through effective local odd coefficients generated by the projected self-energy derivative:

$$\alpha_{OD/DO} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left[2\Lambda_{E;OD}^{\text{eff},(0)} + \theta_0^2 \Lambda_{E;OD,0}^{\text{eff},(2)} + \frac{3}{8} \theta_0^2 \kappa^2 \Lambda_{E;OD,1}^{\text{eff},(2)} \right] + O(\theta_0^5 \kappa, \theta_0^5 \kappa^3),$$

$$\beta_{OD/DO} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left[2\Lambda_{O;OD}^{\text{eff},(0)} + \theta_0^2 \Lambda_{O;OD,0}^{\text{eff},(2)} + \frac{3}{8} \theta_0^2 \kappa^2 \Lambda_{O;OD,1}^{\text{eff},(2)} \right] + O(\theta_0^5 \kappa, \theta_0^5 \kappa^3).$$

Therefore the full corrected transverse coefficients are

$$\alpha = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left[\Lambda_E^{\text{pr}} + 2\Lambda_{E;OD}^{\text{eff},(0)} + \Lambda_{E;DD}^{(0)} + \theta_0^2 \left(\Lambda_{E;OD,0}^{\text{eff},(2)} + \frac{1}{2} \Lambda_{E;DD,0}^{(2)} \right) + \theta_0^2 \kappa^2 \left(\frac{3}{8} \Lambda_{E;OD,1}^{\text{eff},(2)} + \frac{3}{16} \Lambda_{E;DD,1}^{(2)} \right) \right] + \dots,$$

$$\beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left[\Lambda_O^{\text{pr}} + 2\Lambda_{O;OD}^{\text{eff},(0)} + \Lambda_{O;DD}^{(0)} + \theta_0^2 \left(\Lambda_{O;OD,0}^{\text{eff},(2)} + \frac{1}{2} \Lambda_{O;DD,0}^{(2)} \right) + \theta_0^2 \kappa^2 \left(\frac{3}{8} \Lambda_{O;OD,1}^{\text{eff},(2)} + \frac{3}{16} \Lambda_{O;DD,1}^{(2)} \right) \right] + \dots.$$

So the actual working PDE is already fixed up to the microscopic evaluation of these local coefficient packages. The problem is no longer structural.

CXXXIII. LEADING DD EQUALITY TEST AND SAME-ORDER $OD + DD$ INSERTION

CXXXIV. GOAL

We now stop postponing the DD -sector behind abstract residual notation. The immediate question is:

Does the current microscopic structure force $\Lambda_{E;DD}^{(0)} = \Lambda_{O;DD}^{(0)}$ at leading local odd order?

(318)

We answer this sharply and then insert the result, together with the same-order OD -correction, into the actual transverse coefficient sheet.

CXXXV. 1. WHAT THE CURRENT NOTES ACTUALLY IMPLY ABOUT DD

The DD -block is already given in the generic invariant form

$$\mathcal{M}_{DD,\text{inv}}^\mu = a_{\parallel;DD}(\theta) n^\mu \sigma_3 + a_{E;DD}(\theta) E^\mu{}_a \sigma_a + a_{O;DD}(\theta) J^\mu{}_a \sigma_a. \quad (319)$$

Therefore the projector-defined leading transverse coefficients are

$$e_{0;DD}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{1}{4} E^\mu{}_a \text{Tr}(\sigma_a \mathcal{M}_\mu^{DD}), \quad o_{0;DD}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{1}{4} J^\mu{}_a \text{Tr}(\sigma_a \mathcal{M}_\mu^{DD}). \quad (320)$$

So the current microscopic formulas treat the E - and O -coefficients as two independent scalar functions. No equality between them follows from the decomposition itself.

Hence the strongest honest theorem-level statement currently supported is:

$$\text{The present notes do not force } \Lambda_{E;DD}^{(0)} = \Lambda_{O;DD}^{(0)}. \quad (321)$$

CXXXVI. 2. LEADING LOCAL ODD FORM OF THE DD -SECTOR

Parity implies that only the odd part survives after graded projection. Therefore the leading local odd expansion must be

$$a_{E;DD}^{\text{odd}}(\xi) = \theta'(\xi) \Lambda_{E;DD}^{(0)} + O(\theta' \theta^2, \theta'^3), \quad (322)$$

$$a_{O;DD}^{\text{odd}}(\xi) = \theta'(\xi) \Lambda_{O;DD}^{(0)} + O(\theta' \theta^2, \theta'^3). \quad (323)$$

Thus after projection one gets

$$\alpha_{DD}^{\text{LO}} = I_\theta \Lambda_{E;DD}^{(0)}, \quad \beta_{DD}^{\text{LO}} = I_\theta \Lambda_{O;DD}^{(0)}. \quad (324)$$

For the explicit defect profile,

$$I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad (325)$$

hence

$$\alpha_{DD}^{\text{LO}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \Lambda_{E;DD}^{(0)}, \quad \beta_{DD}^{\text{LO}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \Lambda_{O;DD}^{(0)}. \quad (326)$$

CXXXVII. 3. EXACT CRITERION FOR LEADING DD -EQUALITY

From (326),

$$\alpha_{DD}^{\text{LO}} = \beta_{DD}^{\text{LO}} \iff \Lambda_{E;DD}^{(0)} = \Lambda_{O;DD}^{(0)}.$$

Therefore the leading DD -sector is symmetric if and only if

$$\Lambda_{E;DD}^{(0)} - \Lambda_{O;DD}^{(0)} = 0. \quad (327)$$

Without an additional microscopic symmetry exchanging the two transverse invariant channels, there is no reason for (327) to hold automatically.

Thus the sharp honest conclusion is:

$$\Lambda_{E;DD}^{(0)} \text{ and } \Lambda_{O;DD}^{(0)} \text{ must be computed separately.} \quad (328)$$

CXXXVIII. 4. SAME-ORDER OD -INSERTION

The OD -sector enters through the projected mixed self-energy derivative and belongs, after parity projection, to the same local odd transport class. At the same leading local order we write

$$\boxed{e_{0;OD,\text{odd}}^{(1)} = \Lambda_{E;OD}^{\text{eff},(0)} + O(\theta^2, \theta'^2), \quad o_{0;OD,\text{odd}}^{(1)} = \Lambda_{O;OD}^{\text{eff},(0)} + O(\theta^2, \theta'^2).} \quad (329)$$

Under the leading $O \leftrightarrow D$ exchange reduction,

$$\boxed{r_{OD/DO}^{\text{LO}} = \left(2\Lambda_{E;OD}^{\text{eff},(0)}, 2\Lambda_{O;OD}^{\text{eff},(0)} \right).} \quad (330)$$

Thus

$$\boxed{\alpha_{OD/DO}^{\text{LO}} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa 2\Lambda_{E;OD}^{\text{eff},(0)}, \quad \beta_{OD/DO}^{\text{LO}} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa 2\Lambda_{O;OD}^{\text{eff},(0)}. } \quad (331)$$

CXXXIX. 5. FULL LEADING-ORDER $OD + DD$ -CORRECTED COEFFICIENT SHEET

Let the principal pair be

$$\boxed{\alpha_{\text{pr}} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa \Lambda_E^{\text{pr}}, \quad \beta_{\text{pr}} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa \Lambda_O^{\text{pr}}.} \quad (332)$$

Then the full leading-order coefficients become

$$\boxed{\alpha^{\text{LO}} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa \left(\Lambda_E^{\text{pr}} + 2\Lambda_{E;OD}^{\text{eff},(0)} + \Lambda_{E;DD}^{(0)} \right),} \quad (333)$$

$$\boxed{\beta^{\text{LO}} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa \left(\Lambda_O^{\text{pr}} + 2\Lambda_{O;OD}^{\text{eff},(0)} + \Lambda_{O;DD}^{(0)} \right).} \quad (334)$$

This is the first genuine forward coefficient sheet beyond the bare principal pair.

CXL. 6. LEADING-ORDER SYMMETRIC MIXED-TARGET CONDITION

The canonical mixed odd-active target requires

$$\alpha = \beta.$$

At leading order this becomes

$$\boxed{\Lambda_E^{\text{pr}} + 2\Lambda_{E;OD}^{\text{eff},(0)} + \Lambda_{E;DD}^{(0)} = \Lambda_O^{\text{pr}} + 2\Lambda_{O;OD}^{\text{eff},(0)} + \Lambda_{O;DD}^{(0)}. } \quad (335)$$

Equivalently,

$$\boxed{(\Lambda_E^{\text{pr}} - \Lambda_O^{\text{pr}}) + 2(\Lambda_{E;OD}^{\text{eff},(0)} - \Lambda_{O;OD}^{\text{eff},(0)}) + (\Lambda_{E;DD}^{(0)} - \Lambda_{O;DD}^{(0)}) = 0.} \quad (336)$$

Thus the leading-order mismatch splits cleanly into:

$$\text{principal mismatch} + OD \text{ mismatch} + DD \text{ mismatch}.$$

CXLI. 7. ACTUAL LEADING-ORDER WORKING PDE

The actual leading-order working PDE is therefore

$$\boxed{\left[i\partial_t - \lambda_{\parallel} \sigma_3(-i\partial_{\parallel}) - \alpha^{\text{LO}} \sigma_1(-i\partial_{q_1}) - \beta^{\text{LO}} \sigma_2(-i\partial_{q_2}) \right] \chi = 0,} \quad (337)$$

with

$$\boxed{\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3,} \quad (338)$$

and $\alpha^{\text{LO}}, \beta^{\text{LO}}$ from (333)–(334).

CXLII. 8. HONEST STATUS

The forward progress achieved here is sharp:

1. the DD -sector is now inserted explicitly at leading order;
2. the OD -sector is inserted at the same order through its effective local coefficients;
3. the full leading-order transverse coefficient sheet is now (333)–(334);
4. the exact leading-order symmetric-target test is (336).

What remains is not another structural reduction. It is the literal microscopic evaluation of the four leading numbers

$$\Lambda_{E;OD}^{\text{eff},(0)}, \quad \Lambda_{O;OD}^{\text{eff},(0)}, \quad \Lambda_{E;DD}^{(0)}, \quad \Lambda_{O;DD}^{(0)}.$$

CXLIII. FINAL SUMMARY

The current notes do *not* imply

$$\Lambda_{E;DD}^{(0)} = \Lambda_{O;DD}^{(0)}$$

automatically. The generic DD -decomposition already treats the two transverse coefficients as independent:

$$\mathcal{M}_{DD,\text{inv}}^{\mu} = a_{\parallel;DD} n^{\mu} \sigma_3 + a_{E;DD} E^{\mu}{}_a \sigma_a + a_{O;DD} J^{\mu}{}_a \sigma_a.$$

Thus the leading odd DD -sector is

$$a_{E;DD}^{\text{odd}}(\xi) = \theta'(\xi) \Lambda_{E;DD}^{(0)} + \cdots, \quad a_{O;DD}^{\text{odd}}(\xi) = \theta'(\xi) \Lambda_{O;DD}^{(0)} + \cdots,$$

so

$$\alpha_{DD}^{\text{LO}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \Lambda_{E;DD}^{(0)}, \quad \beta_{DD}^{\text{LO}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \Lambda_{O;DD}^{(0)}.$$

The OD -sector enters at the same order through effective local coefficients

$$\Lambda_{E;OD}^{\text{eff},(0)}, \quad \Lambda_{O;OD}^{\text{eff},(0)},$$

giving

$$\alpha_{OD/DO}^{\text{LO}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa 2\Lambda_{E;OD}^{\text{eff},(0)}, \quad \beta_{OD/DO}^{\text{LO}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa 2\Lambda_{O;OD}^{\text{eff},(0)}.$$

Therefore the full leading-order transverse coefficients are

$$\alpha^{\text{LO}} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\left(\Lambda_E^{\text{pr}} + 2\Lambda_{E;OD}^{\text{eff},(0)} + \Lambda_{E;DD}^{(0)}\right),$$

$$\beta^{\text{LO}} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\left(\Lambda_O^{\text{pr}} + 2\Lambda_{O;OD}^{\text{eff},(0)} + \Lambda_{O;DD}^{(0)}\right).$$

The exact leading-order canonical mixed-target condition is

$$\left(\Lambda_E^{\text{pr}} - \Lambda_O^{\text{pr}}\right) + 2\left(\Lambda_{E;OD}^{\text{eff},(0)} - \Lambda_{O;OD}^{\text{eff},(0)}\right) + \left(\Lambda_{E;DD}^{(0)} - \Lambda_{O;DD}^{(0)}\right) = 0.$$

So the actual working PDE is already fixed at leading order. What remains is the microscopic evaluation of these four leading transverse correction numbers.

CXLIV. PRINCIPAL-PAIR SYMMETRY TEST AND THE REDUCED LEADING MISMATCH EQUATION

CXLV. GOAL

We now test the shortest possible route: is the principal transverse pair already symmetric before inserting the residual $OD + DD$ correction?

If yes, then the remaining leading mismatch is carried only by the $OD + DD$ sector. If not, then the principal mismatch itself must be included explicitly.

CXLVI. 1. EXACT PRINCIPAL PAIR

The exact principal transverse source is the projected mixed block

$$\mathcal{M}_{TD+DT}^\mu = \Pi_Q(\mathcal{V}_{TD}^\mu + \mathcal{V}_{DT}^\mu)\Pi_Q.$$

Its projector-defined coefficient pair is

$$\Lambda_E^{\text{pr}} = \mathfrak{P}_E[\mathcal{M}_{TD+DT}^{\text{inv}}], \quad \Lambda_O^{\text{pr}} = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}]. \quad (339)$$

At the explicit branch-level reduced realization,

$$V_{TD}^\mu = \theta'(\xi)(A_E E^\mu{}_a \Sigma_a + A_O J^\mu{}_a \Sigma_a), \quad (340)$$

$$V_{DT}^\mu = \theta'(\xi)(B_E E^\mu{}_a \Sigma_a + B_O J^\mu{}_a \Sigma_a). \quad (341)$$

Therefore

$$\Lambda_E^{\text{pr}} = \frac{1}{2}(A_E + B_E), \quad \Lambda_O^{\text{pr}} = \frac{1}{2}(A_O + B_O). \quad (342)$$

CXLVII. 2. EXACT PRINCIPAL SYMMETRY TEST

From (342),

$$\Lambda_E^{\text{pr}} = \Lambda_O^{\text{pr}} \iff A_E + B_E = A_O + B_O. \quad (343)$$

This is the sharpest principal-pair symmetry criterion available at the present stage.

CXLVIII. 3. HONEST STATUS OF THE PRINCIPAL SYMMETRY QUESTION

The uploaded notes do *not* currently provide a theorem forcing

$$A_E + B_E = A_O + B_O.$$

So the strongest honest statement is

$$\boxed{\text{principal-pair symmetry is not yet proved automatically.}} \quad (344)$$

It must be checked by explicit microscopic evaluation of the $TD + DT$ projected Hessian.

CXLIX. 4. RELATION TO THE REFERENCE ROUTE

In the current reference PDE route one imposes

$$\gamma_E = \gamma_O =: \gamma_\perp,$$

which yields

$$\boxed{\alpha = \beta = -\frac{\beta_X \gamma_\perp}{M_X^2} I_c.} \quad (345)$$

Thus the reference route is *constructed* to be symmetric in the transverse pair.

However, this does not by itself prove that the principal microscopic Hessian pair

$$(\Lambda_E^{\text{pr}}, \Lambda_O^{\text{pr}})$$

is already symmetric on the target mixed branch. So the reference-route symmetry and the principal-Hessian symmetry must be kept logically distinct.

CL. 5. LEADING-ORDER FULL MISMATCH EQUATION

The full leading-order transverse coefficient sheet, including the $OD + DD$ correction, is

$$\boxed{\alpha^{\text{LO}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left(\Lambda_E^{\text{pr}} + 2\Lambda_{E;OD}^{\text{eff},(0)} + \Lambda_{E;DD}^{(0)} \right),} \quad (346)$$

$$\boxed{\beta^{\text{LO}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left(\Lambda_O^{\text{pr}} + 2\Lambda_{O;OD}^{\text{eff},(0)} + \Lambda_{O;DD}^{(0)} \right).} \quad (347)$$

Therefore the exact leading-order symmetric-target condition is

$$\boxed{(\Lambda_E^{\text{pr}} - \Lambda_O^{\text{pr}}) + 2(\Lambda_{E;OD}^{\text{eff},(0)} - \Lambda_{O;OD}^{\text{eff},(0)}) + (\Lambda_{E;DD}^{(0)} - \Lambda_{O;DD}^{(0)}) = 0.} \quad (348)$$

Using (342), this becomes

$$\boxed{\frac{1}{2} \left[(A_E + B_E) - (A_O + B_O) \right] + 2(\Lambda_{E;OD}^{\text{eff},(0)} - \Lambda_{O;OD}^{\text{eff},(0)}) + (\Lambda_{E;DD}^{(0)} - \Lambda_{O;DD}^{(0)}) = 0.} \quad (349)$$

CLI. 6. IF THE PRINCIPAL PAIR IS ALREADY SYMMETRIC

If the explicit principal Hessian audit gives

$$A_E + B_E = A_O + B_O,$$

then (349) reduces immediately to

$$\boxed{2(\Lambda_{E;OD}^{\text{eff},(0)} - \Lambda_{O;OD}^{\text{eff},(0)}) + (\Lambda_{E;DD}^{(0)} - \Lambda_{O;DD}^{(0)}) = 0.} \quad (350)$$

So in that case the entire leading-order mismatch is carried only by the $OD + DD$ residual correction.

CLII. 7. IF THE PRINCIPAL PAIR IS NOT SYMMETRIC

If

$$A_E + B_E \neq A_O + B_O,$$

then even before residual insertion there is already a principal mismatch

$$\delta_{\text{pr}} := \frac{1}{2} \left[(A_E + B_E) - (A_O + B_O) \right] \neq 0.$$

Then the residual sector must compensate this:

$$\boxed{2(\Lambda_{E;OD}^{\text{eff},(0)} - \Lambda_{O;OD}^{\text{eff},(0)}) + (\Lambda_{E;DD}^{(0)} - \Lambda_{O;DD}^{(0)}) = -\delta_{\text{pr}}.} \quad (351)$$

CLIII. 8. SHARP CURRENT CONCLUSION

The shortest honest next computation is therefore not a new structural theorem. It is the explicit principal-pair symmetry test

$$\boxed{A_E + B_E \stackrel{?}{=} A_O + B_O.}$$

Once this is known, the leading mixed-target condition collapses either to the reduced residual equation (350) or to the compensation equation (351).

CLIV. FINAL SUMMARY

The exact principal transverse pair is

$$\Lambda_E^{\text{pr}} = \frac{1}{2}(A_E + B_E), \quad \Lambda_O^{\text{pr}} = \frac{1}{2}(A_O + B_O).$$

Therefore the sharp principal symmetry test is

$$\boxed{\Lambda_E^{\text{pr}} = \Lambda_O^{\text{pr}} \iff A_E + B_E = A_O + B_O.}$$

The uploaded notes do not currently prove this automatically. So principal-pair symmetry is still an explicit microscopic question, not a theorem-level consequence of the present formulas.

The full leading-order corrected coefficients are

$$\alpha^{\text{LO}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left(\Lambda_E^{\text{pr}} + 2\Lambda_{E;OD}^{\text{eff},(0)} + \Lambda_{E;DD}^{(0)} \right),$$

$$\beta^{\text{LO}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left(\Lambda_O^{\text{pr}} + 2\Lambda_{O;OD}^{\text{eff},(0)} + \Lambda_{O;DD}^{(0)} \right).$$

So the exact leading-order symmetric-target condition is

$$\boxed{\frac{1}{2} \left[(A_E + B_E) - (A_O + B_O) \right] + 2(\Lambda_{E;OD}^{\text{eff},(0)} - \Lambda_{O;OD}^{\text{eff},(0)}) + (\Lambda_{E;DD}^{(0)} - \Lambda_{O;DD}^{(0)}) = 0.}$$

If the principal pair is already symmetric, this reduces to

$$\boxed{2(\Lambda_{E;OD}^{\text{eff},(0)} - \Lambda_{O;OD}^{\text{eff},(0)}) + (\Lambda_{E;DD}^{(0)} - \Lambda_{O;DD}^{(0)}) = 0.}$$

So the shortest next computation is exactly the principal Hessian symmetry test

$$A_E + B_E \stackrel{?}{=} A_O + B_O.$$

CLV. LEADING-ORDER TRANSVERSE SHEET IN AVERAGE-MISMATCH FORM

CLVI. GOAL

We no longer wait for an a priori proof of

$$\Lambda_E^{\text{pr}} = \Lambda_O^{\text{pr}}.$$

Instead, we absorb the possible principal asymmetry directly into the leading-order coefficient sheet and rewrite the actual PDE in a symmetric-plus-mismatch form.

CLVII. 1. EXACT PRINCIPAL PAIR

The projected mixed $TD + DT$ block gives

$$\Lambda_E^{\text{pr}} = \frac{1}{2}(A_E + B_E), \quad \Lambda_O^{\text{pr}} = \frac{1}{2}(A_O + B_O). \quad (352)$$

Therefore the exact principal symmetry test is

$$\Lambda_E^{\text{pr}} = \Lambda_O^{\text{pr}} \iff A_E + B_E = A_O + B_O. \quad (353)$$

No theorem in the current notes forces this equality automatically.

CLVIII. 2. FULL LEADING-ORDER CORRECTED TRANSVERSE SHEET

Including the leading $OD + DD$ correction, the coefficient pair is

$$\alpha^{\text{LO}} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\left(\Lambda_E^{\text{pr}} + 2\Lambda_{E;OD}^{\text{eff},(0)} + \Lambda_{E;DD}^{(0)}\right), \quad (354)$$

$$\beta^{\text{LO}} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\left(\Lambda_O^{\text{pr}} + 2\Lambda_{O;OD}^{\text{eff},(0)} + \Lambda_{O;DD}^{(0)}\right). \quad (355)$$

CLIX. 3. AVERAGE AND MISMATCH VARIABLES

Define the principal average and principal mismatch

$$\Lambda_\Sigma^{\text{pr}} := \frac{\Lambda_E^{\text{pr}} + \Lambda_O^{\text{pr}}}{2} = \frac{1}{4}\left[(A_E + B_E) + (A_O + B_O)\right], \quad (356)$$

$$\Lambda_\Delta^{\text{pr}} := \frac{\Lambda_E^{\text{pr}} - \Lambda_O^{\text{pr}}}{2} = \frac{1}{4}\left[(A_E + B_E) - (A_O + B_O)\right]. \quad (357)$$

Define likewise the leading residual average and mismatch

$$\Lambda_\Sigma^{\text{rem},(0)} := \frac{2\Lambda_{E;OD}^{\text{eff},(0)} + \Lambda_{E;DD}^{(0)} + 2\Lambda_{O;OD}^{\text{eff},(0)} + \Lambda_{O;DD}^{(0)}}{2}, \quad (358)$$

$$\Lambda_\Delta^{\text{rem},(0)} := \frac{2\Lambda_{E;OD}^{\text{eff},(0)} + \Lambda_{E;DD}^{(0)} - 2\Lambda_{O;OD}^{\text{eff},(0)} - \Lambda_{O;DD}^{(0)}}{2}. \quad (359)$$

Then

$$\Lambda_E^{\text{LO}} = \Lambda_\Sigma^{\text{pr}} + \Lambda_\Delta^{\text{pr}} + \Lambda_\Sigma^{\text{rem},(0)} + \Lambda_\Delta^{\text{rem},(0)}, \quad (360)$$

$$\Lambda_O^{\text{LO}} = \Lambda_\Sigma^{\text{pr}} - \Lambda_\Delta^{\text{pr}} + \Lambda_\Sigma^{\text{rem},(0)} - \Lambda_\Delta^{\text{rem},(0)}. \quad (361)$$

CLX. 4. ACTUAL SYMMETRIC-PLUS-MISMATCH COEFFICIENTS

Introduce

$$v_\perp^{\text{LO}} := -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\left(\Lambda_\Sigma^{\text{pr}} + \Lambda_\Sigma^{\text{rem},(0)}\right), \quad (362)$$

$$\delta_\perp^{\text{LO}} := -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\left(\Lambda_\Delta^{\text{pr}} + \Lambda_\Delta^{\text{rem},(0)}\right). \quad (363)$$

Then the corrected leading-order transverse coefficients take the exact form

$$\alpha^{\text{LO}} = v_\perp^{\text{LO}} + \delta_\perp^{\text{LO}}, \quad \beta^{\text{LO}} = v_\perp^{\text{LO}} - \delta_\perp^{\text{LO}}. \quad (364)$$

This is the cleanest forward form of the current transverse sheet.

CLXI. 5. EXACT LEADING MISMATCH EQUATION

The canonical mixed odd-active target requires

$$\alpha^{\text{LO}} = \beta^{\text{LO}}.$$

By (364), this is equivalent to

$$\delta_\perp^{\text{LO}} = 0. \quad (365)$$

Equivalently,

$$\Lambda_\Delta^{\text{pr}} + \Lambda_\Delta^{\text{rem},(0)} = 0. \quad (366)$$

That is,

$$\frac{1}{4}\left[(A_E + B_E) - (A_O + B_O)\right] + \frac{1}{2}\left[2(\Lambda_{E;OD}^{\text{eff},(0)} - \Lambda_{O;OD}^{\text{eff},(0)}) + (\Lambda_{E;DD}^{(0)} - \Lambda_{O;DD}^{(0)})\right] = 0. \quad (367)$$

Multiplying by 2, one recovers the exact leading-order mismatch equation

$$\frac{1}{2}\left[(A_E + B_E) - (A_O + B_O)\right] + 2(\Lambda_{E;OD}^{\text{eff},(0)} - \Lambda_{O;OD}^{\text{eff},(0)}) + (\Lambda_{E;DD}^{(0)} - \Lambda_{O;DD}^{(0)}) = 0. \quad (368)$$

CLXII. 6. ACTUAL FORWARD PDE

The actual leading-order working PDE is now

$$\left[i\partial_t - \lambda_\parallel \sigma_3(-i\partial_\parallel) - (v_\perp^{\text{LO}} + \delta_\perp^{\text{LO}})\mathcal{D}_E - (v_\perp^{\text{LO}} - \delta_\perp^{\text{LO}})\mathcal{D}_O\right]\chi = 0, \quad (369)$$

where

$$\mathcal{D}_E := \sigma_1(-i\partial_{q_1}), \quad \mathcal{D}_O := \sigma_2(-i\partial_{q_2}), \quad (370)$$

in the simplified directional notation, and in the full reduced shell form one keeps the standard pair

$$\left[i\partial_t - \lambda_\parallel \sigma_3(-i\partial_\parallel) - \alpha^{\text{LO}}(\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) - \beta^{\text{LO}}(-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1))\right]\chi = 0. \quad (371)$$

CLXIII. 7. TWO USEFUL SPECIAL CASES

If the principal pair is already symmetric,

$$A_E + B_E = A_O + B_O,$$

then

$$\Lambda_{\Delta}^{\text{pr}} = 0,$$

and the entire leading mismatch is carried only by the residual correction:

$$\boxed{2(\Lambda_{E;OD}^{\text{eff},(0)} - \Lambda_{O;OD}^{\text{eff},(0)}) + (\Lambda_{E;DD}^{(0)} - \Lambda_{O;DD}^{(0)}) = 0.} \quad (372)$$

If the residual correction is symmetric,

$$2\Lambda_{E;OD}^{\text{eff},(0)} + \Lambda_{E;DD}^{(0)} = 2\Lambda_{O;OD}^{\text{eff},(0)} + \Lambda_{O;DD}^{(0)},$$

then

$$\Lambda_{\Delta}^{\text{rem},(0)} = 0,$$

and the entire leading mismatch is carried by the principal pair:

$$\boxed{A_E + B_E = A_O + B_O.} \quad (373)$$

CLXIV. 8. HONEST STATUS

At this point the forward transverse PDE is already fixed in actual coefficient form.

What remains is not another structural rewrite. It is the microscopic evaluation of

$$(A_E + B_E) - (A_O + B_O), \quad \Lambda_{E;OD}^{\text{eff},(0)} - \Lambda_{O;OD}^{\text{eff},(0)}, \quad \Lambda_{E;DD}^{(0)} - \Lambda_{O;DD}^{(0)}.$$

These three differences completely determine the leading mixed-target mismatch.

CLXV. FINAL SUMMARY

The principal transverse pair is

$$\Lambda_E^{\text{pr}} = \frac{1}{2}(A_E + B_E), \quad \Lambda_O^{\text{pr}} = \frac{1}{2}(A_O + B_O),$$

so principal symmetry is equivalent to

$$A_E + B_E = A_O + B_O.$$

The full leading-order corrected coefficients are

$$\alpha^{\text{LO}} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\left(\Lambda_E^{\text{pr}} + 2\Lambda_{E;OD}^{\text{eff},(0)} + \Lambda_{E;DD}^{(0)}\right),$$

$$\beta^{\text{LO}} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\left(\Lambda_O^{\text{pr}} + 2\Lambda_{O;OD}^{\text{eff},(0)} + \Lambda_{O;DD}^{(0)}\right).$$

Defining average and mismatch variables gives

$$\alpha^{\text{LO}} = v_{\perp}^{\text{LO}} + \delta_{\perp}^{\text{LO}}, \quad \beta^{\text{LO}} = v_{\perp}^{\text{LO}} - \delta_{\perp}^{\text{LO}},$$

with

$$\delta_{\perp}^{\text{LO}} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\left(\Lambda_{\Delta}^{\text{pr}} + \Lambda_{\Delta}^{\text{rem},(0)}\right).$$

Therefore the exact leading-order mixed-target condition is simply

$$\delta_{\perp}^{\text{LO}} = 0,$$

equivalently

$$\frac{1}{2}\left[(A_E + B_E) - (A_O + B_O)\right] + 2(\Lambda_{E;OD}^{\text{eff},(0)} - \Lambda_{O;OD}^{\text{eff},(0)}) + (\Lambda_{E;DD}^{(0)} - \Lambda_{O;DD}^{(0)}) = 0.$$

So the leading-order forward problem is reduced to three microscopic mismatch numbers only.

CLXVI. DIRECT PROJECTOR-TRACE FORMULA FOR THE LEADING DD MISMATCH

CLXVII. GOAL

We now isolate the leading DD -mismatch itself, rather than leaving it inside a residual vector.

The exact question is:

$$\Delta_{DD}^{(0)} := \Lambda_{E;DD}^{(0)} - \Lambda_{O;DD}^{(0)}. \quad (374)$$

Once this is written directly as a projector-trace difference, it can be inserted immediately into the full leading transverse coefficient sheet.

CLXVIII. 1. EXACT DD -PROJECTOR DEFINITIONS

The projected DD -kernel is

$$\mathcal{M}_{DD,\text{inv}}^{\mu} = a_{\parallel;DD}(\theta) n^{\mu} \sigma_3 + a_{E;DD}(\theta) E^{\mu}_a \sigma_a + a_{O;DD}(\theta) J^{\mu}_a \sigma_a. \quad (375)$$

The executable projector definitions are

$$e_{0;DD}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{1}{4} E^{\mu}_a \text{Tr}(\sigma_a \mathcal{M}_{\mu}^{DD}), \quad (376)$$

$$o_{0;DD}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{1}{4} J^{\mu}_a \text{Tr}(\sigma_a \mathcal{M}_{\mu}^{DD}). \quad (377)$$

So the DD -block is not an exchange-type auxiliary piece. It is a direct transverse candidate inside the defect-tangent sector.

CLXIX. 2. LEADING ODD LOCAL FORM

Only the reflection-odd part survives after graded projection. Therefore the leading odd local DD -sector must be

$$\mathcal{M}_{DD,\text{odd}}^{\mu}(\xi) = \theta'(\xi) \left[\Lambda_{E;DD}^{(0)} E^{\mu}_a \sigma_a + \Lambda_{O;DD}^{(0)} J^{\mu}_a \sigma_a \right] + O(\theta'^2, \theta'^3). \quad (378)$$

In the normalization convention already used throughout the notes, this means

$$e_{0;DD}^{(1)} = \Lambda_{E;DD}^{(0)}, \quad o_{0;DD}^{(1)} = \Lambda_{O;DD}^{(0)} \quad \text{at leading local odd order.} \quad (379)$$

Hence the leading DD -mismatch is simply

$$\Delta_{DD}^{(0)} = e_{0;DD}^{(1)} - o_{0;DD}^{(1)}. \quad (380)$$

CLXX. 3. DIRECT PROJECTOR-TRACE FORM OF $\Delta_{DD}^{(0)}$

Subtracting (377) from (376) gives

$$\Delta_{DD}^{(0)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{1}{4} \left(E^\mu{}_a - J^\mu{}_a \right) \text{Tr}(\sigma_a \mathcal{M}_\mu^{DD}). \quad (381)$$

Using only the odd part of the kernel, the same formula may be written more sharply as

$$\Delta_{DD}^{(0)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{1}{4 \theta'(\xi)} \left(E^\mu{}_a - J^\mu{}_a \right) \text{Tr}(\sigma_a \mathcal{M}_{\mu, \text{odd}}^{DD}(\xi)). \quad (382)$$

This is the exact direct projector-trace formula for the leading DD -mismatch.

CLXXI. 4. WHEN DOES THE LEADING DD -MISMATCH VANISH?

The condition

$$\Delta_{DD}^{(0)} = 0$$

is equivalent to

$$\Lambda_{E;DD}^{(0)} = \Lambda_{O;DD}^{(0)}. \quad (383)$$

At the projector level, this means

$$E^\mu{}_a \text{Tr}(\sigma_a \mathcal{M}_{\mu, \text{odd}}^{DD}) = J^\mu{}_a \text{Tr}(\sigma_a \mathcal{M}_{\mu, \text{odd}}^{DD}) \quad \text{at leading local odd order.} \quad (384)$$

The current notes do not provide a theorem forcing (383) or (384). So $\Delta_{DD}^{(0)}$ must be treated as a genuine microscopic observable.

CLXXII. 5. FULL LEADING-ORDER TRANSVERSE COEFFICIENTS INCLUDING $OD + DD$

The principal pair is

$$\Lambda_E^{\text{pr}} = \frac{1}{2}(A_E + B_E), \quad \Lambda_O^{\text{pr}} = \frac{1}{2}(A_O + B_O). \quad (385)$$

The same-order OD -sector contributes

$$\Lambda_{E;OD/DO}^{(0)} = 2\Lambda_{E;OD}^{\text{eff},(0)}, \quad \Lambda_{O;OD/DO}^{(0)} = 2\Lambda_{O;OD}^{\text{eff},(0)}. \quad (386)$$

Therefore the full leading-order transverse coefficients are

$$\alpha^{\text{LO}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left(\Lambda_E^{\text{pr}} + 2\Lambda_{E;OD}^{\text{eff},(0)} + \Lambda_{E;DD}^{(0)} \right), \quad (387)$$

$$\beta^{\text{LO}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left(\Lambda_O^{\text{pr}} + 2\Lambda_{O;OD}^{\text{eff},(0)} + \Lambda_{O;DD}^{(0)} \right). \quad (388)$$

CLXXIII. 6. FULL LEADING MISMATCH EQUATION

Define

$$\Delta_{\text{pr}}^{(0)} := \Lambda_E^{\text{pr}} - \Lambda_O^{\text{pr}} = \frac{1}{2} \left[(A_E + B_E) - (A_O + B_O) \right], \quad (389)$$

$$\Delta_{OD}^{(0)} := 2 \left(\Lambda_{E;OD}^{\text{eff},(0)} - \Lambda_{O;OD}^{\text{eff},(0)} \right), \quad (390)$$

$$\Delta_{DD}^{(0)} := \Lambda_{E;DD}^{(0)} - \Lambda_{O;DD}^{(0)}. \quad (391)$$

Then

$$\alpha^{\text{LO}} - \beta^{\text{LO}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left(\Delta_{\text{pr}}^{(0)} + \Delta_{OD}^{(0)} + \Delta_{DD}^{(0)} \right). \quad (392)$$

So the exact leading-order symmetric mixed-target condition is

$$\Delta_{\text{pr}}^{(0)} + \Delta_{OD}^{(0)} + \Delta_{DD}^{(0)} = 0. \quad (393)$$

CLXXIV. 7. ACTUAL LEADING-ORDER WORKING PDE

The actual leading-order transverse PDE is therefore

$$\left[i\partial_t - \lambda_{\parallel} \sigma_3 (-i\partial_{\parallel}) - \alpha^{\text{LO}} \sigma_1 (-i\partial_{q_1}) - \beta^{\text{LO}} \sigma_2 (-i\partial_{q_2}) \right] \chi = 0, \quad (394)$$

with

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3, \quad (395)$$

and $\alpha^{\text{LO}}, \beta^{\text{LO}}$ given by (387)–(388).

CLXXV. 8. HONEST STATUS

The forward reduction is now complete at leading order.

The DD -mismatch is no longer a vague residual phrase. It is the explicit projector-trace observable (381)–(382). Therefore the whole leading mixed-target problem is reduced to evaluating three microscopic differences:

$$\Delta_{\text{pr}}^{(0)}, \quad \Delta_{OD}^{(0)}, \quad \Delta_{DD}^{(0)}.$$

CLXXVI. FINAL SUMMARY

The DD -block acts directly inside the defect-tangent doublet sector and has the explicit projected decomposition

$$\mathcal{M}_{DD,\text{inv}}^{\mu} = a_{\parallel;DD} n^{\mu} \sigma_3 + a_{E;DD} E^{\mu}{}_a \sigma_a + a_{O;DD} J^{\mu}{}_a \sigma_a.$$

Its leading odd local transverse part is

$$\mathcal{M}_{DD,\text{odd}}^{\mu} = \theta'(\xi) \left[\Lambda_{E;DD}^{(0)} E^{\mu}{}_a \sigma_a + \Lambda_{O;DD}^{(0)} J^{\mu}{}_a \sigma_a \right] + \cdots,$$

so

$$\Delta_{DD}^{(0)} = \Lambda_{E;DD}^{(0)} - \Lambda_{O;DD}^{(0)} = e_{0;DD}^{(1)} - o_{0;DD}^{(1)}.$$

The exact direct projector-trace formula is

$$\Delta_{DD}^{(0)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{1}{4} (E^\mu{}_a - J^\mu{}_a) \text{Tr}(\sigma_a \mathcal{M}_\mu^{DD}),$$

or more sharply, using only the odd part,

$$\Delta_{DD}^{(0)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{1}{4\theta'(\xi)} (E^\mu{}_a - J^\mu{}_a) \text{Tr}(\sigma_a \mathcal{M}_{\mu, \text{odd}}^{DD}(\xi)).$$

The full leading-order transverse coefficients are

$$\alpha^{\text{LO}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left(\Lambda_E^{\text{pr}} + 2\Lambda_{E;OD}^{\text{eff},(0)} + \Lambda_{E;DD}^{(0)} \right),$$

$$\beta^{\text{LO}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left(\Lambda_O^{\text{pr}} + 2\Lambda_{O;OD}^{\text{eff},(0)} + \Lambda_{O;DD}^{(0)} \right).$$

Therefore the exact leading mixed-target condition is

$$\Delta_{\text{pr}}^{(0)} + \Delta_{OD}^{(0)} + \Delta_{DD}^{(0)} = 0,$$

where

$$\Delta_{\text{pr}}^{(0)} = \frac{1}{2} [(A_E + B_E) - (A_O + B_O)], \quad \Delta_{OD}^{(0)} = 2(\Lambda_{E;OD}^{\text{eff},(0)} - \Lambda_{O;OD}^{\text{eff},(0)}), \quad \Delta_{DD}^{(0)} = \Lambda_{E;DD}^{(0)} - \Lambda_{O;DD}^{(0)}.$$

So the leading-order forward problem is now reduced to three explicit microscopic mismatch numbers only.

CLXXVII. DIRECT PROJECTOR-TRACE FORMULA FOR THE PRINCIPAL LEADING MISMATCH

CLXXVIII. SHORT KOREAN EXPLANATION

We now stop asking whether the principal pair might be symmetric. Instead, we write the principal mismatch itself as a direct observable extracted from the projected $TD + DT$ block.

This turns the leading mixed-target condition into a literal sum of three microscopic mismatch observables:

$$\Delta_{\text{pr}}^{(0)}, \quad \Delta_{OD}^{(0)}, \quad \Delta_{DD}^{(0)}.$$

CLXXIX. 1. EXACT PROJECTED PRINCIPAL BLOCK

The principal transverse source is the mixed shell-defect block

$$\mathcal{M}_{TD+DT}^\mu := \Pi_Q (\mathcal{V}_{TD}^\mu + \mathcal{V}_{DT}^\mu) \Pi_Q. \quad (396)$$

The exact invariant projectors are

$$\mathfrak{P}_E[\mathcal{M}] = \frac{1}{4} E^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_\mu), \quad \mathfrak{P}_O[\mathcal{M}] = \frac{1}{4} J^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_\mu). \quad (397)$$

Therefore the principal coefficient pair is

$$e_0^{(1)} = \mathfrak{P}_E[\mathcal{M}_{TD+DT}^{\text{inv}}], \quad o_0^{(1)} = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}], \quad (398)$$

and at the minimal mixed-block level

$$\Lambda_E^{\text{pr}} = e_0^{(1)}, \quad \Lambda_O^{\text{pr}} = o_0^{(1)}. \quad (399)$$

CLXXX. 2. EXPLICIT $TD + DT$ BLOCK FORM

The two mixed blocks have the branch-level reduced form

$$V_{TD}^\mu = \theta'(\xi) \left(A_E E^\mu{}_a \Sigma_a + A_O J^\mu{}_a \Sigma_a \right), \quad (400)$$

$$V_{DT}^\mu = \theta'(\xi) \left(B_E E^\mu{}_a \Sigma_a + B_O J^\mu{}_a \Sigma_a \right). \quad (401)$$

Hence

$$\mathcal{M}_{TD+DT}^\mu = \theta'(\xi) \left[(A_E + B_E) E^\mu{}_a \Sigma_a + (A_O + B_O) J^\mu{}_a \Sigma_a \right]. \quad (402)$$

Under the canonical invariant normalization, the two channels decouple, so

$$e_0^{(1)} = \frac{1}{2}(A_E + B_E), \quad o_0^{(1)} = \frac{1}{2}(A_O + B_O). \quad (403)$$

Equivalently,

$$\Lambda_E^{\text{pr}} = \frac{1}{2}(A_E + B_E), \quad \Lambda_O^{\text{pr}} = \frac{1}{2}(A_O + B_O). \quad (404)$$

CLXXXI. 3. PRINCIPAL LEADING MISMATCH AS AN EXACT OBSERVABLE

Define the principal leading mismatch

$$\Delta_{\text{pr}}^{(0)} := \Lambda_E^{\text{pr}} - \Lambda_O^{\text{pr}}. \quad (405)$$

Then by (404),

$$\Delta_{\text{pr}}^{(0)} = \frac{1}{2} \left[(A_E + B_E) - (A_O + B_O) \right]. \quad (406)$$

This is already the exact finite principal mismatch number.

CLXXXII. 4. DIRECT PROJECTOR-TRACE FORMULA FOR $\Delta_{\text{pr}}^{(0)}$

Subtracting the two projector definitions gives

$$\Delta_{\text{pr}}^{(0)} = \mathfrak{P}_E[\mathcal{M}_{TD+DT}^{\text{inv}}] - \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}]. \quad (407)$$

Using (397), this is

$$\Delta_{\text{pr}}^{(0)} = \frac{1}{4} \left(E^\mu{}_a - J^\mu{}_a \right) \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_\mu^{TD+DT}). \quad (408)$$

Since $\mathcal{M}_{TD+DT}^\mu$ contains one explicit factor of $\theta'(\xi)$, one may also write the normalized odd-coefficient form

$$\Delta_{\text{pr}}^{(0)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{1}{4 \theta'(\xi)} \left(E^\mu{}_a - J^\mu{}_a \right) \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_\mu^{TD+DT}(\xi)). \quad (409)$$

Thus the principal mismatch is not a vague branch label. It is a direct projector-trace observable.

CLXXXIII. 5. FULL LEADING MISMATCH EQUATION

The OD - and DD -sector leading mismatches are

$$\Delta_{OD}^{(0)} = 2 \left(\Lambda_{E;OD}^{\text{eff},(0)} - \Lambda_{O;OD}^{\text{eff},(0)} \right), \quad (410)$$

$$\Delta_{DD}^{(0)} = \Lambda_{E;DD}^{(0)} - \Lambda_{O;DD}^{(0)}. \quad (411)$$

Hence the exact leading mixed-target condition is

$$\Delta_{\text{pr}}^{(0)} + \Delta_{OD}^{(0)} + \Delta_{DD}^{(0)} = 0. \quad (412)$$

Substituting (406) gives

$$\frac{1}{2} \left[(A_E + B_E) - (A_O + B_O) \right] + 2 \left(\Lambda_{E;OD}^{\text{eff},(0)} - \Lambda_{O;OD}^{\text{eff},(0)} \right) + \left(\Lambda_{E;DD}^{(0)} - \Lambda_{O;DD}^{(0)} \right) = 0. \quad (413)$$

CLXXXIV. 6. LEADING-ORDER TRANSVERSE COEFFICIENT SHEET

With

$$\alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O, \quad I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa,$$

the principal pair alone gives

$$\alpha_{\text{pr}} = -\frac{\sqrt{3}\pi}{32} \theta_0 \kappa (A_E + B_E), \quad \beta_{\text{pr}} = -\frac{\sqrt{3}\pi}{32} \theta_0 \kappa (A_O + B_O). \quad (414)$$

Including the same-order $OD + DD$ correction,

$$\alpha^{\text{LO}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left(\Lambda_E^{\text{pr}} + 2\Lambda_{E;OD}^{\text{eff},(0)} + \Lambda_{E;DD}^{(0)} \right), \quad (415)$$

$$\beta^{\text{LO}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left(\Lambda_O^{\text{pr}} + 2\Lambda_{O;OD}^{\text{eff},(0)} + \Lambda_{O;DD}^{(0)} \right). \quad (416)$$

Hence

$$\alpha^{\text{LO}} - \beta^{\text{LO}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left(\Delta_{\text{pr}}^{(0)} + \Delta_{OD}^{(0)} + \Delta_{DD}^{(0)} \right). \quad (417)$$

CLXXXV. 7. ACTUAL LEADING-ORDER WORKING PDE

The actual leading-order working PDE is therefore

$$\left[i\partial_t - \lambda_{\parallel} \sigma_3 (-i\partial_{\parallel}) - \alpha^{\text{LO}} \sigma_1 (-i\partial_{q_1}) - \beta^{\text{LO}} \sigma_2 (-i\partial_{q_2}) \right] \chi = 0, \quad (418)$$

with

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3. \quad (419)$$

CLXXXVI. 8. HONEST STATUS

The principal mismatch is now fully reduced to the direct observable

$$\Delta_{\text{pr}}^{(0)} = \frac{1}{4}(E^\mu{}_a - J^\mu{}_a) \text{Tr}(\Sigma_a \mathcal{M}_\mu^{TD+DT}),$$

equivalently

$$\Delta_{\text{pr}}^{(0)} = \frac{1}{2}[(A_E + B_E) - (A_O + B_O)].$$

So the whole leading-order mixed-target problem is reduced to three explicit microscopic mismatch observables:

$$\Delta_{\text{pr}}^{(0)}, \quad \Delta_{OD}^{(0)}, \quad \Delta_{DD}^{(0)}.$$

CLXXXVII. FINAL SUMMARY

The principal $TD + DT$ pair is already an exact finite observable:

$$\Lambda_E^{\text{pr}} = \frac{1}{2}(A_E + B_E), \quad \Lambda_O^{\text{pr}} = \frac{1}{2}(A_O + B_O).$$

Therefore the principal leading mismatch is

$$\Delta_{\text{pr}}^{(0)} = \Lambda_E^{\text{pr}} - \Lambda_O^{\text{pr}} = \frac{1}{2}[(A_E + B_E) - (A_O + B_O)].$$

It is also given directly by the projector-trace formula

$$\Delta_{\text{pr}}^{(0)} = \frac{1}{4}(E^\mu{}_a - J^\mu{}_a) \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_\mu^{TD+DT}).$$

The full leading mixed-target condition is

$$\Delta_{\text{pr}}^{(0)} + \Delta_{OD}^{(0)} + \Delta_{DD}^{(0)} = 0,$$

where

$$\Delta_{OD}^{(0)} = 2(\Lambda_{E;OD}^{\text{eff},(0)} - \Lambda_{O;OD}^{\text{eff},(0)}), \quad \Delta_{DD}^{(0)} = \Lambda_{E;DD}^{(0)} - \Lambda_{O;DD}^{(0)}.$$

Hence

$$\alpha^{\text{LO}} - \beta^{\text{LO}} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\left(\Delta_{\text{pr}}^{(0)} + \Delta_{OD}^{(0)} + \Delta_{DD}^{(0)}\right).$$

So the leading-order problem is now reduced exactly to three microscopic mismatch observables only.

CLXXXVIII. PRINCIPAL LEADING MISMATCH AS A DIRECT OBSERVABLE

CLXXXIX. GOAL

We now freeze the principal $TD + DT$ contribution itself as an explicit observable, rather than leaving it as a qualitative symmetry question.

CXC. 1. EXACT PROJECTED PRINCIPAL BLOCK

The principal mixed shell–defect transverse carrier is

$$\mathcal{M}_{TD+DT}^\mu := \Pi_Q (\mathcal{V}_{TD}^\mu + \mathcal{V}_{DT}^\mu) \Pi_Q. \quad (420)$$

The exact invariant projector functionals are

$$\mathfrak{P}_E[\mathcal{M}] = \frac{1}{4} E^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_\mu), \quad \mathfrak{P}_O[\mathcal{M}] = \frac{1}{4} J^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_\mu). \quad (421)$$

Therefore

$$\Lambda_E^{\text{pr}} = \mathfrak{P}_E[\mathcal{M}_{TD+DT}^{\text{inv}}], \quad \Lambda_O^{\text{pr}} = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}]. \quad (422)$$

CXCI. 2. EXPLICIT BRANCH-LEVEL FORM

At the current branch-level reduced realization,

$$V_{TD}^\mu = \theta'(\xi) \left(A_E E^\mu{}_a \Sigma_a + A_O J^\mu{}_a \Sigma_a \right), \quad (423)$$

$$V_{DT}^\mu = \theta'(\xi) \left(B_E E^\mu{}_a \Sigma_a + B_O J^\mu{}_a \Sigma_a \right). \quad (424)$$

Hence

$$\mathcal{M}_{TD+DT}^\mu = \theta'(\xi) \left[(A_E + B_E) E^\mu{}_a \Sigma_a + (A_O + B_O) J^\mu{}_a \Sigma_a \right]. \quad (425)$$

By canonical invariant normalization,

$$\Lambda_E^{\text{pr}} = \frac{1}{2} (A_E + B_E), \quad \Lambda_O^{\text{pr}} = \frac{1}{2} (A_O + B_O). \quad (426)$$

CXCII. 3. PRINCIPAL MISMATCH

Define

$$\Delta_{\text{pr}}^{(0)} := \Lambda_E^{\text{pr}} - \Lambda_O^{\text{pr}}. \quad (427)$$

Then

$$\Delta_{\text{pr}}^{(0)} = \frac{1}{2} \left[(A_E + B_E) - (A_O + B_O) \right]. \quad (428)$$

Equivalently, directly at the projector level,

$$\Delta_{\text{pr}}^{(0)} = \mathfrak{P}_E[\mathcal{M}_{TD+DT}^{\text{inv}}] - \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}] = \frac{1}{4} (E^\mu{}_a - J^\mu{}_a) \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_\mu^{TD+DT}). \quad (429)$$

So the principal transverse mismatch is already a direct finite observable.

CXCIII. 4. PRINCIPAL CONTRIBUTION TO THE REDUCED PDE COEFFICIENTS

The reduced shell dictionary is

$$\boxed{\alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O, \quad I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa.} \quad (430)$$

Therefore the principal pair contributes

$$\boxed{\alpha_{\text{pr}} = -\frac{\sqrt{3}\pi}{32} \theta_0 \kappa (A_E + B_E), \quad \beta_{\text{pr}} = -\frac{\sqrt{3}\pi}{32} \theta_0 \kappa (A_O + B_O).} \quad (431)$$

and hence

$$\boxed{\alpha_{\text{pr}} - \beta_{\text{pr}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \Delta_{\text{pr}}^{(0)}.} \quad (432)$$

CXCIV. 5. FULL LEADING-ORDER MISMATCH EQUATION

The same-order residual mismatches are

$$\boxed{\Delta_{OD}^{(0)} = 2\left(\Lambda_{E;OD}^{\text{eff},(0)} - \Lambda_{O;OD}^{\text{eff},(0)}\right),} \quad (433)$$

$$\boxed{\Delta_{DD}^{(0)} = \Lambda_{E;DD}^{(0)} - \Lambda_{O;DD}^{(0)}.} \quad (434)$$

Hence the exact leading mixed-target condition is

$$\boxed{\Delta_{\text{pr}}^{(0)} + \Delta_{OD}^{(0)} + \Delta_{DD}^{(0)} = 0.} \quad (435)$$

Equivalently,

$$\boxed{\frac{1}{2} \left[(A_E + B_E) - (A_O + B_O) \right] + 2\left(\Lambda_{E;OD}^{\text{eff},(0)} - \Lambda_{O;OD}^{\text{eff},(0)}\right) + \left(\Lambda_{E;DD}^{(0)} - \Lambda_{O;DD}^{(0)}\right) = 0.} \quad (436)$$

CXCV. 6. ACTUAL LEADING-ORDER WORKING PDE

The leading-order transverse coefficients are

$$\boxed{\alpha^{\text{LO}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left(\Lambda_E^{\text{pr}} + 2\Lambda_{E;OD}^{\text{eff},(0)} + \Lambda_{E;DD}^{(0)} \right),} \quad (437)$$

$$\boxed{\beta^{\text{LO}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left(\Lambda_O^{\text{pr}} + 2\Lambda_{O;OD}^{\text{eff},(0)} + \Lambda_{O;DD}^{(0)} \right).} \quad (438)$$

Therefore

$$\boxed{\alpha^{\text{LO}} - \beta^{\text{LO}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left(\Delta_{\text{pr}}^{(0)} + \Delta_{OD}^{(0)} + \Delta_{DD}^{(0)} \right).} \quad (439)$$

This is the actual leading-order mismatch equation inside the reduced PDE.

CXCVI. FINAL SUMMARY

The principal $TD + DT$ audit is already reduced to the exact finite pair

$$\Lambda_E^{\text{pr}} = \frac{1}{2}(A_E + B_E), \quad \Lambda_O^{\text{pr}} = \frac{1}{2}(A_O + B_O).$$

Therefore the principal leading mismatch is the direct observable

$$\Delta_{\text{pr}}^{(0)} = \frac{1}{2} \left[(A_E + B_E) - (A_O + B_O) \right] = \frac{1}{4} (E^\mu_a - J^\mu_a) \text{Tr}_{Q_B} (\Sigma_a \mathcal{M}_\mu^{TD+DT}).$$

The full leading mixed-target condition is

$$\Delta_{\text{pr}}^{(0)} + \Delta_{OD}^{(0)} + \Delta_{DD}^{(0)} = 0.$$

Hence

$$\alpha^{\text{LO}} - \beta^{\text{LO}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left(\Delta_{\text{pr}}^{(0)} + \Delta_{OD}^{(0)} + \Delta_{DD}^{(0)} \right).$$

So the entire leading-order transverse problem is now reduced exactly to three microscopic mismatch observables:

$$\Delta_{\text{pr}}^{(0)}, \quad \Delta_{OD}^{(0)}, \quad \Delta_{DD}^{(0)}.$$

CXCVII. FORWARD REORGANIZATION: FROM THE TARGET-AUDIT DECOMPOSITION BACK TO THE REFERENCE PDE

CXCVIII. GOAL

We now stop further decomposing the target-branch audits and reorganize the program according to the latest fixed master contents:

$$\text{current forward route} = \text{explicit odd extra-carrier reference PDE},$$

$$\text{current target route} = \text{canonical mixed odd-active branch}.$$

The honest next step is therefore not another audit decomposition, but:

$$\text{populate the reference PDE coefficients explicitly and use them as the bridge to the target branch.} \quad (440)$$

CXCIX. 1. REFERENCE-ROUTE COEFFICIENT DICTIONARY

The explicit odd-route reduced coefficient dictionary is already fixed:

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = I_c \tilde{\Lambda}_E, \quad \beta = I_c \tilde{\Lambda}_O. \quad (441)$$

The overlap integrals are

$$I_3 = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3, \quad I_c = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) \cos \theta(\xi). \quad (442)$$

At one-mediator level,

$$\tilde{\Lambda}_E^{(1)} = -\frac{\beta_X \gamma_E}{M_X^2}, \quad \tilde{\Lambda}_O^{(1)} = -\frac{\beta_X \gamma_O}{M_X^2}. \quad (443)$$

Under the fixed reference choice

$$\gamma_E = \gamma_O =: \gamma_\perp,$$

this becomes

$$\lambda_\parallel = \frac{\lambda_c}{2} I_3, \quad \alpha = \beta = -\frac{\beta_X \gamma_\perp}{M_X^2} I_c. \quad (444)$$

CC. 2. CURRENT REFERENCE PDE

The current reference reduced PDE is

$$\left[i\partial_t - \frac{\lambda_c}{2} I_3 \sigma_3 (-i\partial_\parallel) + \frac{\beta_X \gamma_\perp}{M_X^2} I_c (\sigma_1 (-i\partial_1) + \sigma_2 (-i\partial_2)) + \frac{\beta_X \gamma_\perp}{M_X^2} I_c (-\sigma_1 (-i\partial_2) + \sigma_2 (-i\partial_1)) \right] \chi = 0. \quad (445)$$

Introducing the rotated coordinates

$$q_1 := \frac{p_1 - p_2}{\sqrt{2}}, \quad q_2 := \frac{p_1 + p_2}{\sqrt{2}}, \quad (446)$$

the same PDE becomes

$$\left[i\partial_t - \frac{\lambda_c}{2} I_3 \sigma_3 (-i\partial_\parallel) - v_\perp^{\text{ref}} (\sigma_1 (-i\partial_{q_1}) + \sigma_2 (-i\partial_{q_2})) \right] \chi = 0, \quad (447)$$

with

$$v_\perp^{\text{ref}} = -\sqrt{2} \frac{\beta_X \gamma_\perp}{M_X^2} I_c. \quad (448)$$

CCI. 3. TARGET BRANCH AS A COEFFICIENT-MATCHING PROBLEM

The canonical mixed odd-active target is already structurally fixed as

$$\lambda_\parallel = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad \alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa. \quad (449)$$

Therefore the reference route matches the target iff

$$\frac{\lambda_c}{2} I_3 = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad (450)$$

and

$$\frac{\gamma_\perp}{M_X^2} I_c = \frac{\sqrt{3}\pi}{16} \theta_0 \kappa. \quad (451)$$

Thus the whole target problem is now reduced to:

$$\text{explicitly evaluate } I_3, I_c, \lambda_c, \gamma_\perp \text{ and choose their signs compatibly.} \quad (452)$$

CCII. 4. WHAT HAPPENS TO THE TARGET-BRANCH MISMATCH ANALYSIS?

The target-branch leading-order mismatch analysis is still valid:

$$\boxed{\Delta_{\text{pr}}^{(0)} + \Delta_{OD}^{(0)} + \Delta_{DD}^{(0)} = 0.} \quad (453)$$

But after the latest master update, this should be interpreted as a *secondary diagnostic* for the target branch, not as the primary forward route.

The primary forward route is the explicit reference PDE (447) and the matching conditions (450)–(451).

CCIII. 5. EXPLICIT STATUS SENTENCE

So the correct current status is:

$$\boxed{\text{reference PDE fixed, target branch structurally closed, remaining work} = \text{finite coefficient insertion and matching.}} \quad (454)$$

CCIV. 6. ACTUAL NEXT COMPUTATION

The shortest nonredundant next computation is therefore:

$$\boxed{\text{compute } I_3 \text{ exactly on the explicit seed, compute } I_c \text{ to the same order, and solve for } \lambda_c, \gamma_\perp.} \quad (455)$$

Once that is done, the reference route either lands exactly on the target branch or produces a controlled mismatch that can then be compared against the target-audit diagnostics.

CCV. FINAL SUMMARY

The latest fixed master contents imply that the active forward route is no longer the direct target-branch audit chain, but the explicit odd extra-carrier reference PDE.

Its exact coefficient dictionary is

$$\lambda_\parallel = \frac{\lambda_c}{2} I_3, \quad \alpha = I_c \tilde{\Lambda}_E, \quad \beta = I_c \tilde{\Lambda}_O,$$

and under the fixed symmetric reference choice

$$\gamma_E = \gamma_O =: \gamma_\perp,$$

this becomes

$$\lambda_\parallel = \frac{\lambda_c}{2} I_3, \quad \alpha = \beta = -\frac{\beta_X \gamma_\perp}{M_X^2} I_c.$$

The current reference PDE is therefore

$$\left[i\partial_t - \frac{\lambda_c}{2} I_3 \sigma_3 (-i\partial_\parallel) - v_\perp^{\text{ref}} (\sigma_1 (-i\partial_{q_1}) + \sigma_2 (-i\partial_{q_2})) \right] \chi = 0,$$

with

$$v_\perp^{\text{ref}} = -\sqrt{2} \frac{\beta_X \gamma_\perp}{M_X^2} I_c.$$

The target mixed odd-active branch remains

$$\lambda_\parallel = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad \alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa.$$

So the exact bridge conditions are

$$\frac{\lambda_c}{2} I_3 = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad \frac{\gamma_\perp}{M_X^2} I_c = \frac{\sqrt{3}\pi}{16} \theta_0 \kappa.$$

The target-branch mismatch equation

$$\Delta_{\text{pr}}^{(0)} + \Delta_{OD}^{(0)} + \Delta_{DD}^{(0)} = 0$$

remains correct, but it is now secondary. The primary forward problem is simply the finite extraction of

$$I_3, I_c, \lambda_c, \gamma_\perp.$$

CCVI. ACTUAL NONLINEAR PDE BRANCH CONSTRUCTION

CCVII. GOAL

The true bottleneck is no longer the reduced projector algebra. It is the construction of an actual nonlinear stationary branch

$$\Psi_{\text{actual}}(x)$$

solving the microscopic Euler–Lagrange equation

$$\boxed{\mathcal{E}'[\Psi] = 0.} \quad (456)$$

The correct next step is therefore to formulate the nonlinear branch problem in a way that is simultaneously:

1. variationally well-defined,
2. numerically solvable,
3. compatible with the intended BCC shell structure,
4. directly usable for later Bloch linearization.

CCVIII. 1. MINIMAL STATIONARY MICROSCOPIC EQUATION

We use the current strongest working microscopic stationary equation in the generic form

$$\boxed{-\mu^2 \Psi + Z \nabla^2 \Psi - Y \nabla^4 \Psi - \lambda (\Psi^\dagger \Psi) \Psi + \mathcal{N}_{\text{ref}}[\Psi, P, X, \sigma] = 0,} \quad (457)$$

where $\mathcal{N}_{\text{ref}}$ denotes the remaining refined microscopic nonlinear terms already admitted by the working branch.

For the actual nonlinear branch construction, the correct viewpoint is:

$$\boxed{\text{first solve the full stationary branch problem, then linearize around that solution.}} \quad (458)$$

CCIX. 2. ENERGY FUNCTIONAL FORMULATION

Assume the stationary equation is the Euler–Lagrange equation of an energy functional

$$\boxed{\mathcal{E}[\Psi] = \int d^3x \left[-\mu^2 \Psi^\dagger \Psi + Z \partial_i \Psi^\dagger \partial_i \Psi + Y (\nabla^2 \Psi)^\dagger (\nabla^2 \Psi) + \frac{\lambda}{2} (\Psi^\dagger \Psi)^2 + \mathcal{E}_{\text{ref}}[\Psi, P, X, \sigma] \right].} \quad (459)$$

Then

$$\boxed{\frac{\delta \mathcal{E}}{\delta \Psi^\dagger} = 0 \iff \mathcal{E}'[\Psi] = 0.} \quad (460)$$

So the actual branch problem is equivalent to finding a nontrivial critical point of $\mathcal{E}$.

CCX. 3. INTENDED BRANCH ANSATZ

To avoid drifting into irrelevant solution sectors, we restrict the nonlinear search to the intended BCC-shell branch. Write the field as

$$\Psi(x) = \sum_{A \in S_{\text{BCC}}} \phi_A(x) e^{iG_A \cdot x} z_A(x), \quad (461)$$

where

- S_{BCC} is the chosen BCC reciprocal shell,
- G_A are the shell wavevectors,
- $\phi_A(x)$ are slowly varying complex amplitudes,
- $z_A(x)$ are internal orientation vectors.

For the first actual branch construction, we impose the minimal locked branch reduction

$$\phi_A(x) = \phi(x) \eta_A, \quad z_A(x) = z(x), \quad (462)$$

with fixed shell weights η_A specifying the intended BCC sector.

Then

$$\Psi(x) = \phi(x) z(x) \sum_{A \in S_{\text{BCC}}} \eta_A e^{iG_A \cdot x}. \quad (463)$$

This is the first nontrivial nonlinear search manifold.

CCXI. 4. DEFECT-COMPATIBLE DECOMPOSITION

Because the intended branch must later support the carrier construction, we further resolve

$$z(x) = z_B(\xi) + \delta z(x), \quad \phi(x) = \phi_B(\xi) + \delta \phi(x), \quad (464)$$

where ξ is the transverse defect coordinate and z_B, ϕ_B are the defect-profile background fields.

At the first construction level, the actual branch search is reduced to solving for

$$\phi_B(\xi), \quad z_B(\xi), \quad (465)$$

subject to asymptotic and regularity conditions.

CCXII. 5. REDUCED ONE-DIMENSIONAL STATIONARY PROBLEM

Under the defect-compatible locked reduction, the stationary problem collapses to an effective one-dimensional variational problem

$$\mathcal{E}_{\text{red}}[\phi_B, z_B] = \int d\xi \mathcal{L}_{\text{red}}(\phi_B, \phi'_B, \phi''_B, z_B, z'_B, z''_B, \dots), \quad (466)$$

with Euler–Lagrange equations

$$\frac{\delta \mathcal{E}_{\text{red}}}{\delta \phi_B^\dagger} = 0, \quad \frac{\delta \mathcal{E}_{\text{red}}}{\delta z_B^\dagger} = 0, \quad z_B^\dagger z_B = 1. \quad (467)$$

This is the first actual nonlinear branch equation we must solve.

CCXIII. 6. RESIDUAL MAP FOR THE ACTUAL BRANCH SOLVER

Introduce the nonlinear residual map

$$\mathcal{R}[\phi_B, z_B] := \begin{pmatrix} R_\phi[\phi_B, z_B] \\ R_z[\phi_B, z_B] \\ z_B^\dagger z_B - 1 \end{pmatrix}, \quad (468)$$

where R_ϕ and R_z are the reduced Euler–Lagrange residuals.

The actual nonlinear branch problem is then

$$\mathcal{R}[\phi_B, z_B] = 0. \quad (469)$$

This is the correct object for Newton, gradient flow, or continuation methods.

CCXIV. 7. NEWTON–KANTOROVICH STEP

Given an approximate branch $(\phi_B^{(n)}, z_B^{(n)})$, define the linearized Hessian/Jacobian operator

$$\mathcal{J}^{(n)} := D\mathcal{R}[\phi_B^{(n)}, z_B^{(n)}]. \quad (470)$$

Then the Newton step is

$$\mathcal{J}^{(n)} \begin{pmatrix} \delta\phi_B^{(n)} \\ \delta z_B^{(n)} \\ \delta\lambda^{(n)} \end{pmatrix} = -\mathcal{R}[\phi_B^{(n)}, z_B^{(n)}], \quad (471)$$

where $\delta\lambda^{(n)}$ enforces the normalization constraint on z_B .

The update is

$$\phi_B^{(n+1)} = \phi_B^{(n)} + \delta\phi_B^{(n)}, \quad z_B^{(n+1)} = \frac{z_B^{(n)} + \delta z_B^{(n)}}{\|z_B^{(n)} + \delta z_B^{(n)}\|}. \quad (472)$$

CCXV. 8. GRADIENT-FLOW ALTERNATIVE

If the Newton step is numerically stiff, use the relaxation flow

$$\partial_s \phi_B = -\frac{\delta\mathcal{E}_{\text{red}}}{\delta\phi_B^\dagger}, \quad \partial_s z_B = -\Pi_{z_B^\perp} \frac{\delta\mathcal{E}_{\text{red}}}{\delta z_B^\dagger}, \quad (473)$$

where $\Pi_{z_B^\perp}$ projects orthogonally to preserve $z_B^\dagger z_B = 1$.

A stationary branch is then a fixed point:

$$\partial_s \phi_B = 0, \quad \partial_s z_B = 0. \quad (474)$$

CCXVI. 9. ACCEPTANCE CRITERIA FOR Ψ_{actual}

A candidate branch Ψ_{actual} is accepted only if all of the following hold:

$$\|\mathcal{R}[\phi_B, z_B]\| < \varepsilon_{\text{NL}}, \quad (475)$$

$$\boxed{\text{grid refinement changes observables by } < \varepsilon_{\text{grid}},} \quad (476)$$

$$\boxed{\text{flow/Newton convergence is stable under initialization perturbations,}} \quad (477)$$

$$\boxed{\text{the shell content is BCC-dominant,}} \quad (478)$$

$$\boxed{\text{the Hessian around the branch has no forbidden unstable mode.}} \quad (479)$$

Only after these pass do we call the result

$$\Psi_{\text{actual}}(x).$$

CCXVII. 10. WHY THIS ENDS THE REPETITION

All later objects are then derived from the same branch:

$$\boxed{\Psi_{\text{actual}}(x) \implies L_{\text{Bloch}}(k) \implies K_i = \partial_{k_i} L|_{k_*} \implies P_* K_i P_* \implies (\lambda_{\parallel}, \alpha, \beta).} \quad (480)$$

Thus the carrier coefficients, CKM-type reduced quantities, and all branch audits are no longer computed from different fronts. They all come from one common nonlinear branch.

CCXVIII. 11. IMMEDIATE NEXT TASK

The immediate next task is not another decomposition of reduced coefficients. It is to write the first explicit reduced residual map

$$R_{\phi}[\phi_B, z_B], \quad R_z[\phi_B, z_B]$$

for the locked BCC defect ansatz and to choose the first numerical scheme.

That is the first real construction step toward

$$\Psi_{\text{actual}}(x).$$

CCXIX. FINAL SUMMARY

The actual bottleneck is the construction of the nonlinear stationary branch

$$\Psi_{\text{actual}}(x),$$

not further decomposition of reduced audit quantities.

The correct formulation is:

$$\mathcal{E}'[\Psi] = 0 \implies \Psi_{\text{actual}}(x),$$

with the intended BCC locked defect ansatz

$$\Psi(x) = \phi(x) z(x) \sum_{A \in S_{\text{BCC}}} \eta_A e^{i G_A \cdot x},$$

further reduced to the defect-profile unknowns

$$\phi_B(\xi), \quad z_B(\xi).$$

This gives the reduced nonlinear residual map

$$\mathcal{R}[\phi_B, z_B] = 0,$$

which can be solved either by Newton–Kantorovich iteration or by constrained gradient flow.

A candidate branch is accepted only if:

$$\|\mathcal{R}\| < \varepsilon_{\text{NL}},$$

the branch is grid-converged, the convergence is stable, the shell content is BCC-dominant, and the Hessian has no forbidden unstable mode.

Once $\Psi_{\text{actual}}(x)$ is obtained, all later quantities follow from the same chain:

$$\Psi_{\text{actual}}(x) \rightarrow L_{\text{Bloch}}(k) \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow (\lambda_{\parallel}, \alpha, \beta).$$

So from here on, there is one route only: construct the actual nonlinear branch first, then derive everything else from it.

CCXX. EXPLICIT CLASS II CALLBACK AND PHASE-I ACTUAL BRANCH SEARCH

CCXXI. GOAL

After reviewing the accumulated materials, the most forward nonredundant step is clear:

freeze the explicit solver-ready Class II callback $\mathcal{G}^{(II)}[\Psi]$ and run the first accepted actual-BCC branch search.

(481)

The point is not to further decompose reduced audits. It is to construct a stationary nonlinear branch

$$\Psi_{\text{actual}}(x)$$

of the strongest currently explicit working microscopic PDE.

CCXXII. 1. FROZEN STATIONARY WORKING PDE

We take the Phase-I stationary microscopic PDE in the form

$-\mu^2 \Psi + Z \nabla^2 \Psi - Y \nabla^4 \Psi - \lambda |\Psi|^2 \Psi - \gamma |\Psi|^4 \Psi + \mathcal{G}^{(II)}[\Psi] = 0.$

(482)

The pseudo-time relaxation flow is

$\partial_\tau \Psi = \mu^2 \Psi - Z \nabla^2 \Psi + Y \nabla^4 \Psi + \lambda |\Psi|^2 \Psi + \gamma |\Psi|^4 \Psi - \mathcal{G}^{(II)}[\Psi].$

(483)

CCXXIII. 2. EXPLICIT WORKING CLASS II BRANCH

The strongest explicit working microscopic branch is

$L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[P, \Psi] + \beta_X X_i^a K_i^a[P, \Psi].$

(484)

The working Ψ -only promotion used by the solver is

$\rho_\varepsilon := \Psi^\dagger \Psi + \varepsilon, \quad m^a := \Psi^\dagger T^a \Psi, \quad n^a := \frac{m^a}{\rho_\varepsilon}.$

(485)

Then the current sector is realized as

$J_i^a = \partial_i m^a, \quad K_i^a = \rho_\varepsilon \partial_i n^a = \partial_i m^a - n^a \partial_i \rho_\varepsilon.$

(486)

CCXXIV. 3. HEAVY-MEDIATOR ELIMINATION

The algebraic heavy-field equation is

$$X_i^a = -\frac{1}{M_X^2} (\alpha_X J_i^a + \beta_X K_i^a). \quad (487)$$

Substituting (486) gives

$$X_i^a = -\frac{1}{M_X^2} [(\alpha_X + \beta_X) \partial_i m^a - \beta_X n^a \partial_i \rho_\varepsilon]. \quad (488)$$

Define

$$F_i^a := (\alpha_X + \beta_X) \partial_i m^a - \beta_X n^a \partial_i \rho_\varepsilon. \quad (489)$$

Then the induced energy correction is

$$\delta \mathcal{E}_{II} = -\frac{1}{2M_X^2} \sum_{i,a} F_i^a F_i^a. \quad (490)$$

CCXXV. 4. EXPLICIT SOLVER-READY CALLBACK

Define the auxiliary fields

$$A^a = (\alpha_X + \beta_X) T^a \Psi - \beta_X n^a \Psi, \quad (491)$$

$$B_i^a = (\alpha_X + \beta_X) T^a (\partial_i \Psi) - \beta_X n^a (\partial_i \Psi) - \beta_X \frac{(T^a - n^a \mathbf{1}) \Psi}{\rho_\varepsilon} \partial_i \rho_\varepsilon. \quad (492)$$

The explicit solver-ready working callback is therefore

$$\mathcal{G}^{(II)}[\Psi] = \frac{1}{M_X^2} \sum_{i,a} [\partial_i (F_i^a A^a) - F_i^a B_i^a]. \quad (493)$$

This is the actual model-specific ingredient that the solver was missing.

CCXXVI. 5. SEMI-IMPLICIT FOURIER STEP

Split the flow into

$$L = \mu^2 - Z \nabla^2 + Y \nabla^4, \quad N(\Psi) = \lambda |\Psi|^2 \Psi + \gamma |\Psi|^4 \Psi - \mathcal{G}^{(II)}[\Psi].$$

Then one semi-implicit step is

$$(1 - \Delta \tau L) \Psi^{n+1} = \Psi^n + \Delta \tau N(\Psi^n). \quad (494)$$

In Fourier space,

$$\widehat{\Psi}^{n+1}(k) = \frac{\widehat{\Psi}^n(k) + \Delta \tau \widehat{N(\Psi^n)}(k)}{1 + \Delta \tau (\mu^2 - Z(-|k|^2) + Y|k|^4)}. \quad (495)$$

CCXXVII. 6. COMMENSURATE BCC BRANCH SETUP

To remove box-induced shell ambiguity, freeze the first commensurate BCC setup

$$L = 4\pi m, \quad q_0 = q_* = \frac{1}{\sqrt{2}}, \quad m \in \mathbb{Z}_{>0}. \quad (496)$$

Initialize from the equal-amplitude 12-mode BCC shell seed

$$\phi_0(x) = A_{\text{BCC}} \sum_{Q \in \mathcal{S}_{12}} e^{iQ \cdot x}, \quad (497)$$

plus a small defect-seeded perturbation.

CCXXVIII. 7. ORBIT-SYMMETRIC STABILIZATION LAYER

Because the actual observed bottleneck is orbit-level instability inside the retained BCC shell, add the weak orbit-balancing map after each PDE step:

$$\widehat{\Psi}_\alpha \mapsto \widehat{\Psi}_\alpha + \varepsilon_{\text{bal}}(\widehat{\Psi} - \widehat{\Psi}_\alpha), \quad \widehat{\Psi} = \frac{1}{12} \sum_{\beta=1}^{12} \widehat{\Psi}_\beta. \quad (498)$$

At the current working level one may take

$$\varepsilon_{\text{bal}} = 0.08 \quad (499)$$

as the first test value.

This is not yet promoted to a constitutive physical term. It is a numerical stabilization layer used only to test whether an orbit-symmetric BCC stationary branch exists nearby.

CCXXIX. 8. RESIDUAL MAP AND ACCEPTANCE CRITERIA

Define the nonlinear PDE residual

$$R_{\text{PDE}}[\Psi] := -\mu^2 \Psi + Z \nabla^2 \Psi - Y \nabla^4 \Psi - \lambda |\Psi|^2 \Psi - \gamma |\Psi|^4 \Psi + \mathcal{G}^{(II)}[\Psi]. \quad (500)$$

The branch is accepted only if:

$$\text{rel_residual} := \frac{\|R_{\text{PDE}}[\Psi]\|}{\|\Psi\|} < \varepsilon_{\text{res}}, \quad (501)$$

$$\text{shell_fraction} = \frac{\sum_{Q \in \mathcal{S}_{12}} |\widehat{\Psi}(Q)|^2}{\sum_k |\widehat{\Psi}(k)|^2} > \varepsilon_{\text{shell}}, \quad (502)$$

$$\text{orbit_balance} = \frac{\min_{\alpha \in \mathcal{S}_{12}} P_\alpha}{\max_{\alpha \in \mathcal{S}_{12}} P_\alpha} > \varepsilon_{\text{orb}}, \quad (503)$$

$$\text{no-one-patch-collapse indicator passes}, \quad (504)$$

and finally

$$\text{the accepted branch is stable under grid refinement and seed perturbation.} \quad (505)$$

CCXXX. 9. OUTPUT MAP

Once an accepted branch Ψ_{actual} is found, all later extractions come from that same branch:

$$\boxed{\Psi_{\text{actual}} \rightarrow L_{\text{Bloch}}(k) \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow (\lambda_{\parallel}, \alpha, \beta),} \quad (506)$$

and in parallel

$$\boxed{\Psi_{\text{actual}} \rightarrow \text{carrier gates, CKM finite tables, } (W_2, m_*, g_{\text{eff}}, Z_{\text{cub}}).} \quad (507)$$

CCXXXI. 10. IMMEDIATE NEXT CALCULATION

Therefore the actual next calculation is not another reduced audit.

It is:

$$\boxed{\text{run the Phase-I relaxation with } (493) + (496) + (498), \text{ and test acceptance.}} \quad (508)$$

CCXXXII. FINAL SUMMARY

After reviewing the accumulated materials carefully, the most forward nonredundant move is now unambiguous:

$$\boxed{\text{freeze the explicit working Class II callback } \mathcal{G}^{(II)}[\Psi] \text{ and run the first accepted actual-BCC branch search.}} \quad (509)$$

The correct stationary PDE is

$$-\mu^2 \Psi + Z \nabla^2 \Psi - Y \nabla^4 \Psi - \lambda |\Psi|^2 \Psi - \gamma |\Psi|^4 \Psi + \mathcal{G}^{(II)}[\Psi] = 0,$$

with pseudo-time relaxation

$$\partial_\tau \Psi = \mu^2 \Psi - Z \nabla^2 \Psi + Y \nabla^4 \Psi + \lambda |\Psi|^2 \Psi + \gamma |\Psi|^4 \Psi - \mathcal{G}^{(II)}[\Psi].$$

The explicit working callback is

$$\mathcal{G}^{(II)}[\Psi] = \frac{1}{M_X^2} \sum_{i,a} [\partial_i (F_i^a A^a) - F_i^a B_i^a],$$

with

$$\rho_\varepsilon = \Psi^\dagger \Psi + \varepsilon, \quad m^a = \Psi^\dagger T^a \Psi, \quad n^a = \frac{m^a}{\rho_\varepsilon},$$

$$F_i^a = (\alpha_X + \beta_X) \partial_i m^a - \beta_X n^a \partial_i \rho_\varepsilon,$$

$$A^a = (\alpha_X + \beta_X) T^a \Psi - \beta_X n^a \Psi,$$

$$B_i^a = (\alpha_X + \beta_X) T^a (\partial_i \Psi) - \beta_X n^a (\partial_i \Psi) - \beta_X \frac{(T^a - n^a \mathbf{1}) \Psi}{\rho_\varepsilon} \partial_i \rho_\varepsilon.$$

The branch-preserving numerical setup is:

$$L = 4\pi m, \quad q_0 = q_* = \frac{1}{\sqrt{2}},$$

equal-amplitude 12-mode BCC initial data, and the weak orbit-balancing step

$$\widehat{\Psi}_\alpha \mapsto \widehat{\Psi}_\alpha + \varepsilon_{\text{bal}} (\widehat{\Psi} - \widehat{\Psi}_\alpha), \quad \varepsilon_{\text{bal}} = 0.08.$$

A branch is accepted only if the PDE residual is small, the shell fraction remains high, the orbit balance remains high, no one-patch collapse occurs, and the result is stable under grid/seed perturbations.

So the project has now been reduced to one executable next step:

$$\boxed{\text{run the Phase-I actual nonlinear branch search with the frozen callback and stabilization protocol.}}$$

CCXXXIII. PHASE-I ACTUAL BRANCH EXECUTION PROTOCOL

CCXXXIV. GOAL

The solver infrastructure is already frozen. The true remaining issue is not another reduction of audits, but the execution protocol for the actual nonlinear branch search.

We therefore fix one concrete Phase-I protocol:

same solver backbone, same commensurate BCC box, same working callback, fixed acceptance ladder.

CCXXXV. 1. FROZEN BRANCH GEOMETRY AND SOLVER BACKBONE

Keep the commensurate BCC setup fixed:

$$L = 4\pi, \quad q_0 = q_* = \frac{1}{\sqrt{2}}. \quad (510)$$

Keep the same explicit working Class II callback

$$\mathcal{G}^{(II)}[\Psi]$$

and the same semi-implicit pseudo-time relaxation scheme already fixed in the solver backbone.

Keep the orbit-balancing map

$$\widehat{\Psi}_\alpha \mapsto \widehat{\Psi}_\alpha + \varepsilon_{\text{bal}}(\widehat{\Psi} - \widehat{\Psi}_\alpha), \quad \varepsilon_{\text{bal}} = 0.08, \quad (511)$$

as the default stabilization layer for the first accepted-branch search.

CCXXXVI. 2. GRID LADDER AND TIME-STEP LADDER

Use the following fixed refinement ladder:

$$N \in \{32, 48, 64\}, \quad \Delta\tau \in \{5 \times 10^{-3}, 2.5 \times 10^{-3}, 1.25 \times 10^{-3}\}. \quad (512)$$

The branch search always proceeds in this order:

$$(N, \Delta\tau) = (32, 5 \times 10^{-3}) \rightarrow (48, 2.5 \times 10^{-3}) \rightarrow (64, 1.25 \times 10^{-3}).$$

No new seed family is introduced after this point. Only the same intended BCC seed is refined.

CCXXXVII. 3. MEASURED DIAGNOSTICS

At every checkpoint, record exactly the four solver diagnostics already implemented:

$$\text{rel_step} = \frac{\|\Psi^{n+1} - \Psi^n\|_{L^2}}{\|\Psi^n\|_{L^2}}, \quad (513)$$

$$\text{rel_residual} = \frac{\|R[\Psi]\|_{L^2}}{\|\Psi\|_{L^2}}, \quad (514)$$

$$f_{\text{shell}} = \frac{\sum_{||k|-q_0| \leq \delta} |\widehat{\Psi}(k)|^2}{\sum_k |\widehat{\Psi}(k)|^2}, \quad (515)$$

$$b_{\text{orb}} = \frac{\min_{\alpha \in \mathcal{S}_{12}} P_\alpha}{\max_{\alpha \in \mathcal{S}_{12}} P_\alpha}. \quad (516)$$

These are the only primary phase-I diagnostics.

CCXXXVIII. 4. THREE-LEVEL ACCEPTANCE LADDER

Because recent runs show that shell/orbit geometry can already become excellent while the residual stays large, we separate the protocol into three levels.

A. Level 1: geometric pre-acceptance

A run passes the geometric gate if

$$\boxed{f_{\text{shell}} > 0.90, \quad b_{\text{orb}} > 0.95.} \quad (517)$$

This is chosen because one recent run already achieved

$$f_{\text{shell}} \approx 0.9184, \quad b_{\text{orb}} \approx 0.9963,$$

while still failing stationarity. So (517) is demanding but already known to be reachable.

B. Level 2: stationary candidate

A run is called a stationary candidate if, in addition to (517),

$$\boxed{\text{rel_step} < 10^{-3}, \quad \text{rel_residual} < 10^{-1}.} \quad (518)$$

This is not the final acceptance level. It is only the first point at which it is meaningful to perform refinement and stability checks.

C. Level 3: accepted actual branch

A run is accepted as Ψ_{actual} only if, after refinement,

$$\boxed{\text{rel_step} < 10^{-4}, \quad \text{rel_residual} < 10^{-2}, \quad f_{\text{shell}} > 0.90, \quad b_{\text{orb}} > 0.95.} \quad (519)$$

In addition, the extracted observables must satisfy

$$\boxed{\frac{|O_{64,1.25 \times 10^{-3}} - O_{48,2.5 \times 10^{-3}}|}{|O_{64,1.25 \times 10^{-3}}|} < 10^{-2}} \quad (520)$$

for every monitored observable O in the set

$$\{f_{\text{shell}}, b_{\text{orb}}, W_0, W_2, \|\Psi\|_{L^2}\}.$$

CCXXXIX. 5. EARLY REJECTION AND BOTTLENECK DIAGNOSIS

To prevent further repetition, impose the following hard decision rules.

A. Rule A: shell-loss rejection

If after 100 pseudo-time steps,

$$\boxed{f_{\text{shell}} < 0.80,} \quad (521)$$

reject the run as leaving the intended BCC sector.

B. Rule B: orbit-collapse rejection

If after 100 pseudo-time steps,

$$\boxed{b_{\text{orb}} < 0.20,} \quad (522)$$

reject the run as orbit-collapsed.

C. Rule C: callback-mismatch diagnosis

If a run satisfies the geometric gate (517) but, over three successive halvings of $\Delta\tau$,

$$\boxed{\text{rel_residual} > 10^{-1} \quad \text{and} \quad \text{rel_residual} \text{ does not decrease by at least a factor 2,}} \quad (523)$$

then diagnose the failure as

$$\boxed{\text{constitutive/callback mismatch, not geometric branch failure.}} \quad (524)$$

This rule is directly motivated by the already observed pattern: geometry can be nearly perfect while the residual remains $O(10^{-1}-10^0)$.

CCXL. 6. PHASE-I STOP CONDITION

The phase-I search stops immediately when either:

$$\boxed{\text{an accepted actual branch satisfying (519)–(520) is found,}} \quad (525)$$

or

$$\boxed{\text{the callback-mismatch diagnosis (523)–(524) is triggered.}} \quad (526)$$

In the second case, the correct next step is not a new seed or a new box. It is a revision of the constitutive callback $\mathcal{G}^{(II)}[\Psi]$.

CCXLI. 7. WHAT THIS PROTOCOL ACCOMPLISHES

This protocol prevents the project from drifting.
It cleanly distinguishes:

1. geometric failure of the intended BCC branch,
2. orbit-collapse failure,
3. and constitutive mismatch of the working callback.

So Phase I now has an exact executable outcome map:

$$\boxed{\text{accept branch} \quad \text{or} \quad \text{reject geometry} \quad \text{or} \quad \text{diagnose callback mismatch.}}$$

CCXLII. FINAL SUMMARY

The actual nonlinear branch search is now fixed by one execution protocol.
The frozen setup is:

$$L = 4\pi, \quad q_0 = \frac{1}{\sqrt{2}}, \quad \varepsilon_{\text{bal}} = 0.08,$$

with refinement ladder

$$N \in \{32, 48, 64\}, \quad \Delta\tau \in \{5 \times 10^{-3}, 2.5 \times 10^{-3}, 1.25 \times 10^{-3}\}.$$

The only primary diagnostics are:

$$\text{rel_step}, \quad \text{rel_residual}, \quad f_{\text{shell}}, \quad b_{\text{orb}}.$$

The three-level acceptance ladder is:

$$f_{\text{shell}} > 0.90, \quad b_{\text{orb}} > 0.95 \quad (\text{geometric pre-acceptance}),$$

$$\text{rel_step} < 10^{-3}, \quad \text{rel_residual} < 10^{-1} \quad (\text{stationary candidate}),$$

$$\text{rel_step} < 10^{-4}, \quad \text{rel_residual} < 10^{-2}, \quad f_{\text{shell}} > 0.90, \quad b_{\text{orb}} > 0.95,$$

plus $< 1\%$ grid/time refinement stability

(accepted actual branch).

The hard early decisions are:

$$f_{\text{shell}} < 0.80 \Rightarrow \text{reject as shell-loss},$$

$$b_{\text{orb}} < 0.20 \Rightarrow \text{reject as orbit-collapse},$$

and if geometry is good but the residual stays $> 10^{-1}$ without a factor-2 decrease across three halvings of $\Delta\tau$,

diagnose callback mismatch, not branch failure.

So Phase I is no longer vague. It now has one exact executable outcome map:

accept branch or reject geometry or diagnose constitutive mismatch.

CCXLIII. STAGE IA: ORIENTATION-FROZEN ACTUAL NONLINEAR BRANCH

CCXLIV. GOAL

The first real construction step of the actual nonlinear branch is not the full coupled (ϕ_B, z_B) problem yet. Instead, we first freeze the internal orientation and solve the scalar amplitude branch:

$\Psi(x) = \Phi(x) z_0, \quad z_0^\dagger z_0 = 1, \quad z_0 = \text{fixed.}$	(527)
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This produces the first actual nonlinear stationary branch candidate. Only after that do we release the tangent orientation sector.

CCXLV. 1. FROZEN-ORIENTATION REDUCTION OF THE CLASS II DICTIONARY

Let

$$\boxed{\rho_\varepsilon = \Psi^\dagger \Psi + \varepsilon, \quad m^a = \Psi^\dagger T^a \Psi, \quad n^a = \frac{m^a}{\rho_\varepsilon}.} \quad (528)$$

For the frozen ansatz $\Psi = \Phi z_0$, define

$$\boxed{n_0^a := z_0^\dagger T^a z_0.} \quad (529)$$

Then

$$\boxed{m^a = |\Phi|^2 n_0^a, \quad n^a = n_0^a \quad (\text{constant in } x).} \quad (530)$$

Hence

$$\boxed{J_i^a = \partial_i m^a = n_0^a \partial_i |\Phi|^2,} \quad (531)$$

while

$$\boxed{K_i^a = \partial_i m^a - n^a \partial_i \rho_\varepsilon = 0} \quad (532)$$

at this frozen-orientation level.

Therefore the heavy-field elimination simplifies to

$$\boxed{F_i^a = (\alpha_X + \beta_X) \partial_i m^a - \beta_X n^a \partial_i \rho_\varepsilon = \alpha_X n_0^a \partial_i |\Phi|^2.} \quad (533)$$

CCXLVI. 2. SCALAR PROJECTION OF THE WORKING CALLBACK

The solver-ready callback is

$$\mathcal{G}^{(II)}[\Psi] = \frac{1}{M_X^2} \sum_{i,a} \left[\partial_i (F_i^a A^a) - F_i^a B_i^a \right]. \quad (534)$$

Projecting along z_0 , only the z_0 -parallel component matters for the Stage-IA scalar branch. Using

$$z_0^\dagger T^a z_0 = n_0^a, \quad z_0^\dagger (T^a - n_0^a \mathbf{1}) z_0 = 0,$$

one obtains

$$\boxed{z_0^\dagger A^a = \alpha_X n_0^a \Phi, \quad z_0^\dagger B_i^a = \alpha_X n_0^a \partial_i \Phi.} \quad (535)$$

Substituting (533) and (535) into (534),

$$z_0^\dagger \mathcal{G}^{(II)}[\Phi z_0] = \frac{1}{M_X^2} \sum_{i,a} \left[\partial_i (\alpha_X n_0^a \partial_i |\Phi|^2 \cdot \alpha_X n_0^a \Phi) - \alpha_X n_0^a \partial_i |\Phi|^2 \cdot \alpha_X n_0^a \partial_i \Phi \right] \quad (536)$$

$$= \frac{\alpha_X^2}{M_X^2} \sum_a (n_0^a)^2 \left[\partial_i ((\partial_i |\Phi|^2) \Phi) - (\partial_i |\Phi|^2) (\partial_i \Phi) \right] \quad (537)$$

$$= \frac{\alpha_X^2}{M_X^2} \sum_a (n_0^a)^2 [\nabla^2 (|\Phi|^2)] \Phi. \quad (538)$$

Define

$$\boxed{\nu_0 := \sum_a (n_0^a)^2, \quad \chi_{II} := \frac{\alpha_X^2 \nu_0}{M_X^2}.} \quad (539)$$

Then

$$\boxed{z_0^\dagger \mathcal{G}^{(II)}[\Phi z_0] = \chi_{II} \nabla^2 (|\Phi|^2) \Phi.} \quad (540)$$

CCXLVII. 3. STAGE-IA SCALAR STATIONARY PDE

Projecting the full stationary microscopic PDE along z_0 , the frozen-orientation branch satisfies

$$\boxed{-\mu^2 \Phi + Z \nabla^2 \Phi - Y \nabla^4 \Phi - \lambda |\Phi|^2 \Phi - \gamma |\Phi|^4 \Phi + \chi_{II} \nabla^2 (|\Phi|^2) \Phi = 0.} \quad (541)$$

This is the first actual nonlinear PDE we must solve.

CCXLVIII. 4. STAGE-IA PSEUDO-TIME RELAXATION

The corresponding relaxation flow is

$$\boxed{\partial_\tau \Phi = \mu^2 \Phi - Z \nabla^2 \Phi + Y \nabla^4 \Phi + \lambda |\Phi|^2 \Phi + \gamma |\Phi|^4 \Phi - \chi_{II} \nabla^2 (|\Phi|^2) \Phi.} \quad (542)$$

A semi-implicit step may be written as

$$\boxed{(1 - \Delta \tau L) \Phi^{n+1} = \Phi^n + \Delta \tau N_{IA}(\Phi^n),} \quad (543)$$

with

$$L = \mu^2 - Z \nabla^2 + Y \nabla^4, \quad N_{IA}(\Phi) = \lambda |\Phi|^2 \Phi + \gamma |\Phi|^4 \Phi - \chi_{II} \nabla^2 (|\Phi|^2) \Phi. \quad (544)$$

CCXLIX. 5. BCC-LOCKED AMPLITUDE REDUCTION

Now impose the commensurate BCC locked ansatz

$$\boxed{\Phi(x) = \phi(\xi) S_{\text{BCC}}(x), \quad S_{\text{BCC}}(x) := \sum_{A \in S_{12}} \eta_A e^{i G_A \cdot x}.} \quad (545)$$

Define the cell averages

$$\boxed{C_2 := \langle |S_{\text{BCC}}|^2 \rangle_{\text{cell}}, \quad C_4 := \langle |S_{\text{BCC}}|^4 \rangle_{\text{cell}}, \quad C_6 := \langle |S_{\text{BCC}}|^6 \rangle_{\text{cell}}.} \quad (546)$$

Then the first shell-averaged reduced residual is

$$\boxed{R_\phi[\phi] = -\mu^2 C_2 \phi + Z \mathcal{L}_2[\phi] - Y \mathcal{L}_4[\phi] - \lambda C_4 |\phi|^2 \phi - \gamma C_6 |\phi|^4 \phi + \chi_{II} \mathcal{C}_{II}[\phi],} \quad (547)$$

where

$$\boxed{\mathcal{L}_2[\phi] := \langle \overline{S_{\text{BCC}}} \nabla^2 (\phi S_{\text{BCC}}) \rangle_{\text{cell}},} \quad (548)$$

$$\boxed{\mathcal{L}_4[\phi] := \langle \overline{S_{\text{BCC}}} \nabla^4 (\phi S_{\text{BCC}}) \rangle_{\text{cell}},} \quad (549)$$

$$\boxed{\mathcal{C}_{II}[\phi] := \langle \overline{S_{\text{BCC}}} \nabla^2 (|\phi S_{\text{BCC}}|^2) S_{\text{BCC}} \rangle_{\text{cell}} \phi.} \quad (550)$$

Thus the first actual branch search is reduced to the scalar one-dimensional residual equation

$$\boxed{R_\phi[\phi] = 0.} \quad (551)$$

CCL. 6. TANGENT CONSISTENCY RESIDUAL

The frozen-orientation branch is acceptable only if the orthogonal tangent forcing is small. Define

$$\boxed{R_{\perp}[\Phi; z_0] := \Pi_{z_0^{\perp}} \mathcal{E}'[\Phi z_0].} \quad (552)$$

At leading Class II order, a sufficient consistency condition is

$$\boxed{\Pi_{z_0^{\perp}}(n_0^a T^a z_0) = 0.} \quad (553)$$

More generally, Stage IA is accepted only if

$$\boxed{\frac{\|R_{\perp}[\Phi; z_0]\|}{\|\Phi\|} < \varepsilon_{\perp},} \quad (554)$$

with $\varepsilon_{\perp}$ chosen smaller than the PDE residual tolerance.

If (554) fails, the Stage-IA branch is still useful as the seed for Stage IB, where $z(\xi)$ is released dynamically.

CCLI. 7. WHAT THIS ACCOMPLISHES

Stage IA already removes the biggest ambiguity.

The actual nonlinear branch construction is no longer a vague search over the full Ψ -space. It is first reduced to:

1. a scalar fourth-order nonlinear PDE (541),
2. then a shell-averaged defect-amplitude residual equation (551),
3. together with the tangent consistency check (552).

So the immediate next executable problem is:

$$\boxed{\text{solve } R_{\phi}[\phi] = 0 \text{ on the commensurate BCC box, then test } R_{\perp}.} \quad (555)$$

CCLII. FINAL SUMMARY

The first real forward step of the actual nonlinear branch construction is the orientation-frozen Stage-IA ansatz

$$\Psi(x) = \Phi(x) z_0.$$

In this sector,

$$m^a = |\Phi|^2 n_0^a, \quad n^a = n_0^a, \quad J_i^a = n_0^a \partial_i |\Phi|^2, \quad K_i^a = 0,$$

so the heavy-mediator Class II callback simplifies drastically.

Projecting along z_0 , one gets the exact scalar callback

$$\boxed{z_0^{\dagger} \mathcal{G}^{(II)}[\Phi z_0] = \chi_{II} \nabla^2(|\Phi|^2) \Phi, \quad \chi_{II} = \frac{\alpha_X^2 \nu_0}{M_X^2}, \quad \nu_0 = \sum_a (n_0^a)^2.}$$

Therefore the Stage-IA stationary PDE is

$$\boxed{-\mu^2 \Phi + Z \nabla^2 \Phi - Y \nabla^4 \Phi - \lambda |\Phi|^2 \Phi - \gamma |\Phi|^4 \Phi + \chi_{II} \nabla^2(|\Phi|^2) \Phi = 0.}$$

With the locked BCC shell ansatz

$$\Phi(x) = \phi(\xi) S_{\text{BCC}}(x),$$

this reduces to the scalar residual equation

$$R_\phi[\phi] = -\mu^2 C_2 \phi + Z \mathcal{L}_2[\phi] - Y \mathcal{L}_4[\phi] - \lambda C_4 |\phi|^2 \phi - \gamma C_6 |\phi|^4 \phi + \chi_{II} \mathcal{C}_{II}[\phi] = 0.$$

The frozen-orientation branch is accepted only if the tangent consistency residual

$$R_\perp[\Phi; z_0] = \Pi_{z_0^\perp} \mathcal{E}'[\Phi z_0]$$

is also sufficiently small.

So the immediate next executable problem is no longer vague:

$$\text{solve } R_\phi[\phi] = 0 \text{ first, then test } R_\perp.$$

CCLIII. STAGE IA: BCC CELL-AVERAGED SCALAR AMPLITUDE ODE

CCLIV. GOAL

Starting from the Stage-IA scalar stationary PDE

$$-\mu^2 \Phi + Z \nabla^2 \Phi - Y \nabla^4 \Phi - \lambda |\Phi|^2 \Phi - \gamma |\Phi|^4 \Phi + \chi_{II} \nabla^2 (|\Phi|^2) \Phi = 0, \quad (556)$$

we now insert the commensurate locked BCC ansatz

$$\Phi(x) = \phi(\xi) S_{\text{BCC}}(x), \quad \xi = \hat{n} \cdot x, \quad (557)$$

and reduce the PDE to a one-dimensional amplitude equation for $\phi(\xi)$.

CCLV. 1. BASIC SHELL IDENTITIES

Let

$$S_{\text{BCC}}(x) = \sum_{A \in \mathcal{S}_{12}} \eta_A e^{i G_A \cdot x}, \quad |G_A| = q_*. \quad (558)$$

Define the cell averages

$$C_2 := \langle |S_{\text{BCC}}|^2 \rangle_{\text{cell}}, \quad C_4 := \langle |S_{\text{BCC}}|^4 \rangle_{\text{cell}}, \quad C_6 := \langle |S_{\text{BCC}}|^6 \rangle_{\text{cell}}. \quad (559)$$

We also define the fourth-order shell-gradient constant

$$C_\Delta := \langle |S_{\text{BCC}}|^2 \nabla^2 |S_{\text{BCC}}|^2 \rangle_{\text{cell}}. \quad (560)$$

For the orbit-symmetric BCC shell,

$$\langle \overline{S_{\text{BCC}}} \partial_\xi S_{\text{BCC}} \rangle_{\text{cell}} = 0, \quad \langle (\hat{n} \cdot G_A)^2 \rangle_{\mathcal{S}_{12}} = \frac{q_*^2}{3}. \quad (561)$$

CCLVI. 2. QUADRATIC OPERATOR REDUCTION

We define

$$\mathcal{L}_2[\phi] := \langle \overline{S_{\text{BCC}}} \nabla^2 (\phi S_{\text{BCC}}) \rangle_{\text{cell}}. \quad (562)$$

Now

$$\nabla^2 (\phi e^{i G_A \cdot x}) = (\phi'' + 2i(\hat{n} \cdot G_A) \phi' - q_*^2 \phi) e^{i G_A \cdot x}. \quad (563)$$

Summing over the shell and averaging over the cell, the odd term cancels by symmetry, hence

$$\mathcal{L}_2[\phi] = C_2 (\phi'' - q_*^2 \phi). \quad (564)$$

CCLVII. 3. QUARTIC DERIVATIVE OPERATOR REDUCTION

Define

$$\boxed{\mathcal{L}_4[\phi] := \langle \overline{S_{\text{BCC}}} \nabla^4(\phi S_{\text{BCC}}) \rangle_{\text{cell}}.} \quad (565)$$

For each mode, with $g_A := \hat{n} \cdot G_A$,

$$\nabla^2(\phi e^{iG_A \cdot x}) = (D^2 + 2ig_AD - q_*^2)\phi e^{iG_A \cdot x}, \quad D := \frac{d}{d\xi}. \quad (566)$$

Therefore

$$\nabla^4(\phi e^{iG_A \cdot x}) = (D^2 + 2ig_AD - q_*^2)^2 \phi e^{iG_A \cdot x}, \quad (567)$$

that is,

$$(D^2 + 2ig_AD - q_*^2)^2 = D^4 + 4ig_AD^3 - (4g_A^2 + 2q_*^2)D^2 - 4ig_Aq_*^2D + q_*^4. \quad (568)$$

After shell averaging, the odd D^3 and D terms vanish, while

$$\langle g_A^2 \rangle = q_*^2/3.$$

Hence

$$\boxed{\mathcal{L}_4[\phi] = C_2 \left(\phi'''' - \frac{10}{3} q_*^2 \phi'' + q_*^4 \phi \right).} \quad (569)$$

CCLVIII. 4. CLASS-II NONLINEAR CALLBACK REDUCTION

Define

$$\boxed{\mathcal{C}_{II}[\phi] := \langle \overline{S_{\text{BCC}}} \nabla^2(|\phi S_{\text{BCC}}|^2) S_{\text{BCC}} \rangle_{\text{cell}} \phi.} \quad (570)$$

Since

$$|\phi S_{\text{BCC}}|^2 = |\phi|^2 |S_{\text{BCC}}|^2,$$

we have

$$\nabla^2(|\phi S_{\text{BCC}}|^2) = (|\phi|^2)'' |S_{\text{BCC}}|^2 + 2(|\phi|^2)' \partial_\xi |S_{\text{BCC}}|^2 + |\phi|^2 \nabla^2 |S_{\text{BCC}}|^2. \quad (571)$$

Multiplying by $\overline{S_{\text{BCC}}} S_{\text{BCC}} = |S_{\text{BCC}}|^2$ and averaging over the cell, the odd middle term vanishes by symmetry:

$$\langle |S_{\text{BCC}}|^2 \partial_\xi |S_{\text{BCC}}|^2 \rangle_{\text{cell}} = 0. \quad (572)$$

Therefore

$$\boxed{\mathcal{C}_{II}[\phi] = C_4 (|\phi|^2)'' \phi + C_\Delta |\phi|^2 \phi.} \quad (573)$$

For the real branch $\phi \in \mathbb{R}$,

$$(|\phi|^2)'' = (\phi^2)'' = 2(\phi')^2 + 2\phi\phi'',$$

so

$$\boxed{\mathcal{C}_{II}[\phi] = 2C_4 \phi^2 \phi'' + 2C_4 \phi (\phi')^2 + C_\Delta \phi^3.} \quad (574)$$

CCLIX. 5. REDUCED SCALAR AMPLITUDE RESIDUAL

The shell-averaged scalar residual is

$$\boxed{R_\phi[\phi] = -\mu^2 C_2 \phi + Z \mathcal{L}_2[\phi] - Y \mathcal{L}_4[\phi] - \lambda C_4 |\phi|^2 \phi - \gamma C_6 |\phi|^4 \phi + \chi_{II} \mathcal{C}_{II}[\phi].} \quad (575)$$

Substituting (564), (569), and (574) yields the explicit real scalar ODE

$$\boxed{-Y C_2 \phi'''' + C_2 \left(Z + \frac{10}{3} Y q_*^2 \right) \phi'' - C_2 (\mu^2 + Z q_*^2 + Y q_*^4) \phi + 2\chi_{II} C_4 \phi^2 \phi'' + 2\chi_{II} C_4 \phi (\phi')^2 + (\chi_{II} C_\Delta - \lambda C_4) \phi^3 - \gamma C_6 \phi^5 = 0.} \quad (576)$$

This is the first actual one-dimensional nonlinear branch equation.

CCLX. 6. EFFECTIVE COEFFICIENT INTERPRETATION

Equation (576) shows that the frozen-orientation actual branch has the following effective structure:

$$\boxed{\text{effective linear mass} = C_2 (\mu^2 + Z q_*^2 + Y q_*^4),} \quad (577)$$

$$\boxed{\text{effective second-derivative stiffness} = C_2 \left(Z + \frac{10}{3} Y q_*^2 \right),} \quad (578)$$

$$\boxed{\text{effective cubic coefficient} = \chi_{II} C_\Delta - \lambda C_4,} \quad (579)$$

and two genuinely new derivative-cubic terms

$$2\chi_{II} C_4 \phi^2 \phi'', \quad 2\chi_{II} C_4 \phi (\phi')^2.$$

So the Class-II callback survives the scalar reduction as a derivative-cubic nonlinear stiffness correction.

CCLXI. 7. BOUNDARY CONDITIONS

For the defect branch search, impose the standard localized/heteroclinic conditions

$$\boxed{\phi'(\xi) \rightarrow 0, \quad \phi'''(\xi) \rightarrow 0, \quad \phi(\xi) \rightarrow \phi_\infty \quad \text{as } |\xi| \rightarrow \infty,} \quad (580)$$

together with center symmetry

$$\boxed{\phi'(0) = 0, \quad \phi'''(0) = 0,} \quad (581)$$

for the even branch.

If a node-type branch is desired, replace (581) by the appropriate odd conditions.

CCLXII. 8. IMMEDIATE NEXT STEP

The actual next calculation is now completely concrete:

$$\boxed{\text{solve the scalar fourth-order ODE (576) with (580)–(581), then evaluate } R_\perp.} \quad (582)$$

Only after that do we release the internal orientation and move to Stage IB.

CCLXIII. FINAL SUMMARY

Under the locked BCC ansatz

$$\Phi(x) = \phi(\xi) S_{\text{BCC}}(x),$$

the shell-averaged operators reduce to

$$\boxed{\mathcal{L}_2[\phi] = C_2(\phi'' - q_*^2 \phi),}$$

$$\boxed{\mathcal{L}_4[\phi] = C_2 \left(\phi'''' - \frac{10}{3} q_*^2 \phi'' + q_*^4 \phi \right),}$$

and the Class-II nonlinear callback becomes

$$\boxed{\mathcal{C}_{II}[\phi] = 2C_4 \phi^2 \phi'' + 2C_4 \phi (\phi')^2 + C_\Delta \phi^3}$$

for the real branch.

Therefore the Stage-IA actual scalar amplitude equation is

$$\begin{aligned} & -YC_2 \phi'''' + C_2 \left(Z + \frac{10}{3} Y q_*^2 \right) \phi'' - C_2 (\mu^2 + Z q_*^2 + Y q_*^4) \phi \\ & + 2\chi_{II} C_4 \phi^2 \phi'' + 2\chi_{II} C_4 \phi (\phi')^2 + (\chi_{II} C_\Delta - \lambda C_4) \phi^3 - \gamma C_6 \phi^5 = 0. \end{aligned}$$

This is the first real one-dimensional nonlinear branch equation for the actual branch search. So the project is now past abstract solver scaffolding. The immediate next problem is simply:

$$\boxed{\text{solve this fourth-order scalar ODE, then test the tangent residual } R_\perp.}$$

CCLXIV. STAGE IA SOLVER-READY FORM: TWO SECOND-ORDER EQUATIONS AND A FIRST-ORDER SYSTEM

CCLXV. STARTING POINT

From the BCC cell-averaged Stage-IA reduction, the real scalar amplitude equation is

$$\begin{aligned} & -YC_2 \phi'''' + C_2 \left(Z + \frac{10}{3} Y q_*^2 \right) \phi'' - C_2 (\mu^2 + Z q_*^2 + Y q_*^4) \phi \\ & + 2\chi_{II} C_4 \phi^2 \phi'' + 2\chi_{II} C_4 \phi (\phi')^2 + (\chi_{II} C_\Delta - \lambda C_4) \phi^3 - \gamma C_6 \phi^5 = 0. \end{aligned}$$

For compactness, define the frozen coefficients

$$\boxed{A_2 := C_2 \left(Z + \frac{10}{3} Y q_*^2 \right), \quad M_2 := C_2 (\mu^2 + Z q_*^2 + Y q_*^4), \quad U_3 := \chi_{II} C_\Delta - \lambda C_4.} \quad (583)$$

Then (CCLXV) becomes

$$\boxed{-YC_2 \phi'''' + A_2 \phi'' - M_2 \phi + 2\chi_{II} C_4 \phi^2 \phi'' + 2\chi_{II} C_4 \phi (\phi')^2 + U_3 \phi^3 - \gamma C_6 \phi^5 = 0.} \quad (584)$$

CCLXVI. 1. REDUCTION TO TWO SECOND-ORDER EQUATIONS

Introduce

$$\boxed{u(\xi) := \phi(\xi), \quad w(\xi) := \phi''(\xi).} \quad (585)$$

Then the first relation is simply

$$\boxed{u'' - w = 0.} \quad (586)$$

Since $\phi'''' = w''$, the second equation is

$$\boxed{-YC_2 w'' + A_2 w - M_2 u + 2\chi_{II} C_4 u^2 w + 2\chi_{II} C_4 u(u')^2 + U_3 u^3 - \gamma C_6 u^5 = 0.} \quad (587)$$

Therefore the Stage-IA branch equation is exactly equivalent to the coupled two-field second-order system

$$\boxed{u'' = w,} \quad (588)$$

$$\boxed{w'' = \frac{1}{YC_2} \left[A_2 w - M_2 u + 2\chi_{II} C_4 u^2 w + 2\chi_{II} C_4 u(u')^2 + U_3 u^3 - \gamma C_6 u^5 \right].} \quad (589)$$

This is the first solver-ready BVP form.

CCLXVII. 2. FIRST-ORDER SYSTEM FORM

For shooting or general ODE integrators, introduce

$$\boxed{y_1 := u, \quad y_2 := u', \quad y_3 := w, \quad y_4 := w'.} \quad (590)$$

Then

$$\boxed{y_1' = y_2,} \quad (591)$$

$$\boxed{y_2' = y_3,} \quad (592)$$

$$\boxed{y_3' = y_4,} \quad (593)$$

and

$$\boxed{y_4' = \frac{1}{YC_2} \left[A_2 y_3 - M_2 y_1 + 2\chi_{II} C_4 y_1^2 y_3 + 2\chi_{II} C_4 y_1 y_2^2 + U_3 y_1^3 - \gamma C_6 y_1^5 \right].} \quad (594)$$

So the scalar actual branch search is now reduced to a four-dimensional first-order autonomous system.

CCLXVIII. 3. BOUNDARY CONDITIONS FOR THE EVEN LOCALIZED BRANCH

For the even branch centered at $\xi = 0$, the symmetry conditions are

$$\boxed{u'(0) = 0, \quad w'(0) = 0.} \quad (595)$$

At infinity, a stationary defect branch must approach a constant vacuum:

$$\boxed{u(\xi) \rightarrow u_\infty, \quad u'(\xi) \rightarrow 0, \quad w(\xi) \rightarrow 0, \quad w'(\xi) \rightarrow 0 \quad (\xi \rightarrow \infty).} \quad (596)$$

Thus on a finite computational interval $[0, L_\xi]$, the practical BVP conditions are

$$\boxed{u'(0) = 0, \quad w'(0) = 0, \quad u(L_\xi) = u_\infty, \quad w(L_\xi) = 0.} \quad (597)$$

CCLXIX. 4. ASYMPTOTIC VACUUM EQUATION

The asymptotic constant u_∞ is not free. Setting

$$u' = u'' = u''' = u'''' = 0$$

in (584) gives the vacuum equation

$$\boxed{-M_2 u_\infty + U_3 u_\infty^3 - \gamma C_6 u_\infty^5 = 0.} \quad (598)$$

Therefore either

$$\boxed{u_\infty = 0,} \quad (599)$$

or, for a nontrivial asymptotic branch, $s := u_\infty^2$ must satisfy

$$\boxed{\gamma C_6 s^2 - U_3 s + M_2 = 0.} \quad (600)$$

Thus

$$\boxed{u_\infty^2 = \frac{U_3 \pm \sqrt{U_3^2 - 4\gamma C_6 M_2}}{2\gamma C_6},} \quad (601)$$

provided the discriminant is nonnegative and the selected root is positive.

CCLXX. 5. RESIDUAL OPERATOR FOR NEWTON/COLLOCATION

For a two-field BVP solver, define

$$\boxed{R_1[u, w] := u'' - w,} \quad (602)$$

$$\boxed{R_2[u, w] := -YC_2 w'' + A_2 w - M_2 u + 2\chi_{II} C_4 u^2 w + 2\chi_{II} C_4 u (u')^2 + U_3 u^3 - \gamma C_6 u^5.} \quad (603)$$

Then the Stage-IA BVP is exactly

$$\boxed{R_1[u, w] = 0, \quad R_2[u, w] = 0,} \quad (604)$$

with boundary data (597).

CCLXXI. 6. LINEARIZATION FOR NEWTON-KANTOROVICH

Let

$$u \mapsto u + \delta u, \quad w \mapsto w + \delta w.$$

Then

$$\boxed{\delta R_1 = (\delta u)'' - \delta w.} \quad (605)$$

For R_2 , one gets

$$\begin{aligned} \delta R_2 = & -YC_2 (\delta w)'' + (A_2 + 2\chi_{II} C_4 u^2) \delta w \\ & + \left[-M_2 + 4\chi_{II} C_4 u w + 2\chi_{II} C_4 (u')^2 + 3U_3 u^2 - 5\gamma C_6 u^4 \right] \delta u \\ & + 4\chi_{II} C_4 u u' (\delta u)'. \end{aligned} \quad (606)$$

So the Newton step solves

$$\boxed{\delta R_1 = -R_1, \quad \delta R_2 = -R_2} \quad (607)$$

with linearized boundary conditions.

CCLXXII. 7. RELAXATION FLOW FORM

If one wants a geometry-preserving residual flow rather than direct Newton, use

$$\boxed{\partial_s u = -R_1[u, w], \quad \partial_s w = -R_2[u, w].} \quad (608)$$

Explicitly,

$$\boxed{\partial_s u = -(u'' - w),} \quad (609)$$

$$\boxed{\partial_s w = Y C_2 w'' - A_2 w + M_2 u - 2\chi_{II} C_4 u^2 w - 2\chi_{II} C_4 u (u')^2 - U_3 u^3 + \gamma C_6 u^5.} \quad (610)$$

A stationary point of this s -flow is exactly a solution of the Stage-IA branch equation.

CCLXXIII. 8. ACCEPTANCE OBSERVABLES AT THE SCALAR LEVEL

At Stage IA, before tangent release, the scalar branch is accepted only if

$$\boxed{\|R_1\|_{L^2} + \|R_2\|_{L^2} < \varepsilon_{\text{IA}},} \quad (611)$$

$$\boxed{u(\xi) \text{ remains in the intended BCC-shell basin through } S_{\text{BCC}},} \quad (612)$$

$$\boxed{\frac{\|R_{\perp}[\Phi; z_0]\|}{\|\Phi\|} < \varepsilon_{\perp}.} \quad (613)$$

Only then is the result promoted to the Stage-IA actual branch candidate.

CCLXXIV. 9. IMMEDIATE NEXT STEP

The actual next step is therefore no longer conceptual. It is exactly:

$$\boxed{\text{solve the BVP } (R_1, R_2) = 0 \text{ with (597), using either (607) or (608).}} \quad (614)$$

CCLXXV. FINAL SUMMARY

The Stage-IA scalar amplitude equation has now been reduced from one fourth-order ODE to two coupled second-order equations:

$$u'' = w,$$

$$w'' = \frac{1}{Y C_2} \left[A_2 w - M_2 u + 2\chi_{II} C_4 u^2 w + 2\chi_{II} C_4 u (u')^2 + U_3 u^3 - \gamma C_6 u^5 \right].$$

Equivalently, in first-order form:

$$y'_1 = y_2, \quad y'_2 = y_3, \quad y'_3 = y_4,$$

$$y_4' = \frac{1}{YC_2} \left[A_2 y_3 - M_2 y_1 + 2\chi_{II} C_4 y_1^2 y_3 + 2\chi_{II} C_4 y_1 y_2^2 + U_3 y_1^3 - \gamma C_6 y_1^5 \right].$$

The finite-interval even-branch boundary conditions are:

$$u'(0) = 0, \quad w'(0) = 0, \quad u(L_\xi) = u_\infty, \quad w(L_\xi) = 0,$$

where u_∞ is determined by

$$-M_2 u_\infty + U_3 u_\infty^3 - \gamma C_6 u_\infty^5 = 0.$$

The Newton residual system is

$$R_1[u, w] = u'' - w,$$

$$R_2[u, w] = -YC_2 w'' + A_2 w - M_2 u + 2\chi_{II} C_4 u^2 w + 2\chi_{II} C_4 u (u')^2 + U_3 u^3 - \gamma C_6 u^5.$$

So the project has now advanced from abstract branch construction to a concrete solver-ready Stage-IA BVP. The next step is simply to solve $(R_1, R_2) = 0$ and then test the tangent residual $R_\perp$.

CCLXXVI. STAGE IA NUMERICAL REALIZATION: TAIL, SEED, AND CHEBYSHEV-NEWTON COLLOCATION

CCLXXVII. GOAL

The Stage-IA scalar branch has already been reduced to the two-field second-order system

$$u'' = w, \tag{615}$$

$$w'' = \frac{1}{YC_2} \left[A_2 w - M_2 u + 2\chi_{II} C_4 u^2 w + 2\chi_{II} C_4 u (u')^2 + U_3 u^3 - \gamma C_6 u^5 \right]. \tag{616}$$

We now convert this into a concrete numerical boundary-value problem.

CCLXXVIII. 1. VACUUM SELECTION

The asymptotic vacuum u_∞ is determined by

$$\boxed{-M_2 u_\infty + U_3 u_\infty^3 - \gamma C_6 u_\infty^5 = 0.} \tag{617}$$

Hence either

$$u_\infty = 0,$$

or $s = u_\infty^2$ satisfies

$$\boxed{\gamma C_6 s^2 - U_3 s + M_2 = 0.} \tag{618}$$

For the nontrivial branch we choose the positive root

$$\boxed{u_\infty^2 = \frac{U_3 + \sqrt{U_3^2 - 4\gamma C_6 M_2}}{2\gamma C_6}}, \tag{619}$$

provided the discriminant is nonnegative and the result is positive. If both roots are positive, the larger one is the default first probe for the stable large-amplitude vacuum.

CCLXXIX. 2. TAIL LINEARIZATION AND DECAY SCALE

Write

$$u(\xi) = u_\infty + \delta(\xi), \quad |\delta| \ll 1.$$

Since $u'_\infty = w_\infty = 0$, linearization of (615)–(616) yields

$$\delta'' = \omega, \tag{620}$$

$$-YC_2 \omega'' + (A_2 + 2\chi_{II} C_4 u_\infty^2) \omega + V_\infty \delta = 0, \tag{621}$$

where

$$\boxed{V_\infty := -M_2 + 3U_3 u_\infty^2 - 5\gamma C_6 u_\infty^4.} \tag{622}$$

Eliminating $\omega = \delta''$, the tail satisfies

$$\boxed{-YC_2 \delta'''' + (A_2 + 2\chi_{II} C_4 u_\infty^2) \delta'' + V_\infty \delta = 0.} \tag{623}$$

Using $\delta(\xi) \sim e^{-m\xi}$, we obtain the characteristic equation

$$\boxed{-YC_2 m^4 + (A_2 + 2\chi_{II} C_4 u_\infty^2) m^2 + V_\infty = 0.} \tag{624}$$

Setting $s = m^2$, this is

$$\boxed{YC_2 s^2 - (A_2 + 2\chi_{II} C_4 u_\infty^2) s - V_\infty = 0.} \tag{625}$$

The admissible tail branch requires at least one positive root $s_* > 0$, giving decay rate

$$\boxed{m_* = \sqrt{s_*} > 0.} \tag{626}$$

CCLXXX. 3. FINITE-INTERVAL TRUNCATION

Choose the computational interval $[0, L_\xi]$ by requiring

$$\boxed{e^{-m_* L_\xi} \leq \varepsilon_{\text{tail}}.} \tag{627}$$

Equivalently,

$$\boxed{L_\xi \geq \frac{|\log \varepsilon_{\text{tail}}|}{m_*}.} \tag{628}$$

For the first run, take

$$\boxed{\varepsilon_{\text{tail}} = 10^{-8}, \quad L_\xi = \frac{18}{m_*}} \tag{629}$$

as the default practical choice.

CCLXXXI. 4. EVEN-BRANCH SEED PROFILE

For the even localized branch around u_∞ , use the first seed

$$u^{(0)}(\xi) = u_\infty - a_0 \operatorname{sech}^2\left(\frac{\xi}{\ell_0}\right), \quad (630)$$

with

$$w^{(0)}(\xi) = (u^{(0)})''(\xi). \quad (631)$$

A practical initial amplitude is

$$a_0 = \frac{u_\infty}{2}, \quad (632)$$

and the initial width is chosen from the tail rate:

$$\ell_0 = \frac{1}{m_*}. \quad (633)$$

This seed matches the correct asymptotic scale and preserves the even symmetry:

$$u'(0) = 0, \quad w'(0) = 0.$$

CCLXXXII. 5. CHEBYSHEV-LOBATTO GRID

Map the finite interval $[0, L_\xi]$ to $x \in [-1, 1]$ by

$$\xi = \frac{L_\xi}{2}(1 + x). \quad (634)$$

Take the Chebyshev-Lobatto nodes

$$x_j = \cos \frac{\pi j}{N}, \quad j = 0, \dots, N. \quad (635)$$

Let $D^{(1)}$ and $D^{(2)}$ be the usual first and second Chebyshev differentiation matrices on $[-1, 1]$. Then on $[0, L_\xi]$,

$$\partial_\xi = \frac{2}{L_\xi} D^{(1)}, \quad \partial_\xi^2 = \frac{4}{L_\xi^2} D^{(2)}. \quad (636)$$

CCLXXXIII. 6. DISCRETE RESIDUAL VECTOR

Let

$$u_j \approx u(\xi_j), \quad w_j \approx w(\xi_j).$$

Define the discrete derivative vectors

$$u_\xi := \frac{2}{L_\xi} D^{(1)} u, \quad u_{\xi\xi} := \frac{4}{L_\xi^2} D^{(2)} u, \quad w_{\xi\xi} := \frac{4}{L_\xi^2} D^{(2)} w.$$

For the interior nodes $j = 1, \dots, N-1$, define

$$(R_1)_j = (u_{\xi\xi})_j - w_j, \quad (637)$$

$$(R_2)_j = -YC_2(w_{\xi\xi})_j + A_2 w_j - M_2 u_j + 2\chi_{II} C_4 u_j^2 w_j + 2\chi_{II} C_4 u_j (u_\xi)_j^2 + U_3 u_j^3 - \gamma C_6 u_j^5. \quad (638)$$

CCLXXXIV. 7. BOUNDARY RESIDUALS

The even-branch boundary conditions are imposed by replacing four rows of the residual vector with:

$$\boxed{B_1 = (u_\xi)_0, \quad B_2 = (w_\xi)_0,} \quad (639)$$

$$\boxed{B_3 = u_N - u_\infty, \quad B_4 = w_N.} \quad (640)$$

Thus the full discrete nonlinear system is

$$\boxed{\mathcal{R}_{\text{disc}}(u, w) = (B_1, B_2, (R_1)_1, \dots, (R_1)_{N-1}, (R_2)_1, \dots, (R_2)_{N-1}, B_3, B_4) = 0.} \quad (641)$$

CCLXXXV. 8. NEWTON STEP

Let

$$U := (u_0, \dots, u_N, w_0, \dots, w_N)^T.$$

Then the Newton update is

$$\boxed{J_{\text{disc}}(U^{(n)}) \delta U^{(n)} = -\mathcal{R}_{\text{disc}}(U^{(n)}),} \quad (642)$$

with

$$\boxed{U^{(n+1)} = U^{(n)} + \delta U^{(n)}.} \quad (643)$$

Because the continuous Jacobian was already computed,

$$\delta R_1 = (\delta u)'' - \delta w, \quad (644)$$

$$\delta R_2 = -Y C_2 (\delta w)'' + (A_2 + 2\chi_{II} C_4 u^2) \delta w + \left[-M_2 + 4\chi_{II} C_4 u w + 2\chi_{II} C_4 (u')^2 + 3U_3 u^2 - 5\gamma C_6 u^4 \right] \delta u + 4\chi_{II} C_4 u u' (\delta u)', \quad (645)$$

the discrete Jacobian is obtained by direct matrix substitution:

$$\boxed{(\delta u)' \mapsto \frac{2}{L_\xi} D^{(1)} \delta u, \quad (\delta u)'' \mapsto \frac{4}{L_\xi^2} D^{(2)} \delta u, \quad (\delta w)'' \mapsto \frac{4}{L_\xi^2} D^{(2)} \delta w.} \quad (646)$$

CCLXXXVI. 9. DAMPED NEWTON AND CONTINUATION

Because the branch can be stiff, use a damping parameter $\eta_n \in (0, 1]$:

$$\boxed{U^{(n+1)} = U^{(n)} + \eta_n \delta U^{(n)}.} \quad (647)$$

Start with $\eta_n = 1$. If

$$\|\mathcal{R}_{\text{disc}}(U^{(n+1)})\| > \|\mathcal{R}_{\text{disc}}(U^{(n)})\|,$$

halve η_n until descent occurs.

The preferred continuation order is:

$$\boxed{\chi_{II} = 0 \rightarrow \text{solve the simpler branch} \rightarrow \chi_{II} > 0 \text{ continuation.}} \quad (648)$$

That is the cheapest robust route.

CCLXXXVII. 10. STAGE IA ACCEPTANCE TEST

The scalar branch is accepted when

$$\boxed{\|\mathcal{R}_{\text{disc}}\|_2 < \varepsilon_{\text{IA}},} \quad (649)$$

with default

$$\boxed{\varepsilon_{\text{IA}} = 10^{-8},} \quad (650)$$

and, in addition,

$$\boxed{\frac{\|R_{\perp}[\Phi; z_0]\|}{\|\Phi\|} < \varepsilon_{\perp}, \quad \varepsilon_{\perp} = 10^{-9}.} \quad (651)$$

Finally, repeat at

$$N, 2N$$

and accept only if

$$\boxed{\frac{\|u_{2N} - u_N\|}{\|u_{2N}\|} < 10^{-4}.} \quad (652)$$

CCLXXXVIII. 11. IMMEDIATE NEXT COMPUTATION

The next actual computation is therefore fully explicit:

$$\boxed{\text{construct } u_{\infty}, m_*, L_{\xi}, u^{(0)}, w^{(0)}, \text{ assemble } \mathcal{R}_{\text{disc}}, \text{ and run damped Newton with continuation in } \chi_{II}.} \quad (653)$$

Once this converges, Stage IA is done. Then we test the tangent residual and release $z(\xi)$ in Stage IB.

CCLXXXIX. FINAL SUMMARY

The Stage-IA scalar actual branch is now a fully explicit numerical BVP.

First, choose the asymptotic vacuum u_{∞} from

$$-M_2 u_{\infty} + U_3 u_{\infty}^3 - \gamma C_6 u_{\infty}^5 = 0.$$

Next, linearize around u_{∞} and solve the tail characteristic equation

$$YC_2 s^2 - (A_2 + 2\chi_{II} C_4 u_{\infty}^2) s - V_{\infty} = 0, \quad s = m^2,$$

to get the decay scale m_* , then choose

$$L_{\xi} = \frac{18}{m_*}, \quad u^{(0)}(\xi) = u_{\infty} - \frac{u_{\infty}}{2} \text{sech}^2(m_* \xi), \quad w^{(0)} = (u^{(0)})''.$$

Then solve the discrete collocation system

$$(R_1)_j = (u_{\xi\xi})_j - w_j = 0,$$

$$(R_2)_j = -YC_2(w_{\xi\xi})_j + A_2 w_j - M_2 u_j + 2\chi_{II} C_4 u_j^2 w_j + 2\chi_{II} C_4 u_j (u_{\xi})_j^2 + U_3 u_j^3 - \gamma C_6 u_j^5 = 0,$$

with boundary data

$$(u_{\xi})_0 = 0, \quad (w_{\xi})_0 = 0, \quad u_N = u_{\infty}, \quad w_N = 0.$$

Use damped Newton and continuation from $\chi_{II} = 0$ to the desired $\chi_{II} > 0$.

So the project is now beyond structural setup: the next step is literally to run this Stage-IA collocation/Newton solve.

CCXC. STAGE IA BASE BRANCH AT $\chi_{II} = 0$: VACUUM, TAIL, FIRST INTEGRAL, AND ANALYTIC SEED

CCXCI. SHORT KOREAN EXPLANATION

We now freeze the continuation base point by turning off the Class-II derivative-cubic callback:

$$\chi_{II} = 0.$$

This does not yet solve the full actual branch problem, but it gives the cleanest base equation for continuation.

At this base level we can do four things exactly:

1. determine the asymptotic vacuum u_∞ ,
2. determine the tail decay rate m_* ,
3. derive an exact first integral,
4. construct a closed-form analytic seed family consistent with the vacuum and the tail.

CCXCII. 1. BASE ODE AT $\chi_{II} = 0$

The Stage-IA scalar ODE was

$$-YC_2 u'''' + A_2 u'' - M_2 u + 2\chi_{II} C_4 u^2 u'' + 2\chi_{II} C_4 u(u')^2 + U_3 u^3 - \gamma C_6 u^5 = 0,$$

with

$$A_2 = C_2 \left(Z + \frac{10}{3} Y q_*^2 \right), \quad M_2 = C_2 (\mu^2 + Z q_*^2 + Y q_*^4), \quad U_3 = \chi_{II} C_\Delta - \lambda C_4.$$

At $\chi_{II} = 0$, this reduces to

$$\boxed{-YC_2 u'''' + A_2 u'' - M_2 u - \lambda C_4 u^3 - \gamma C_6 u^5 = 0.} \quad (654)$$

This is the clean continuation base equation.

CCXCIII. 2. EXACT ASYMPTOTIC VACUUM

A constant vacuum u_∞ satisfies

$$\boxed{-M_2 u_\infty - \lambda C_4 u_\infty^3 - \gamma C_6 u_\infty^5 = 0.} \quad (655)$$

Thus either

$$\boxed{u_\infty = 0,} \quad (656)$$

or $s = u_\infty^2$ solves

$$\boxed{\gamma C_6 s^2 + \lambda C_4 s + M_2 = 0.} \quad (657)$$

Hence the nontrivial vacuum branches are

$$\boxed{u_\infty^2 = \frac{-\lambda C_4 \pm \sqrt{\lambda^2 C_4^2 - 4\gamma C_6 M_2}}{2\gamma C_6}.} \quad (658)$$

A real nonzero vacuum exists iff

$$\boxed{\lambda^2 C_4^2 - 4\gamma C_6 M_2 \geq 0} \quad (659)$$

and at least one root in (658) is positive.

CCXCIV. 3. TAIL LINEARIZATION AND EXACT DECAY RATE

Linearize around the selected vacuum:

$$u(\xi) = u_\infty + \delta(\xi), \quad |\delta| \ll 1.$$

The linearized equation is

$$-YC_2 \delta'''' + A_2 \delta'' + V_\infty^{(0)} \delta = 0, \quad (660)$$

where

$$V_\infty^{(0)} = -M_2 - 3\lambda C_4 u_\infty^2 - 5\gamma C_6 u_\infty^4. \quad (661)$$

Using the vacuum equation (655), this simplifies to

$$V_\infty^{(0)} = -2\lambda C_4 u_\infty^2 - 4\gamma C_6 u_\infty^4 = -2u_\infty^2(\lambda C_4 + 2\gamma C_6 u_\infty^2). \quad (662)$$

With the tail ansatz

$$\delta(\xi) \sim e^{-m\xi},$$

we obtain

$$-YC_2 m^4 + A_2 m^2 + V_\infty^{(0)} = 0. \quad (663)$$

Setting $s = m^2$, the admissible tail root satisfies

$$YC_2 s^2 - A_2 s - V_\infty^{(0)} = 0. \quad (664)$$

Thus the positive decay rate is

$$m_*^2 = \frac{A_2 + \sqrt{A_2^2 + 4YC_2 V_\infty^{(0)}}}{2YC_2}, \quad (665)$$

provided the numerator is positive and the discriminant is nonnegative.

CCXCV. 4. EXACT FIRST INTEGRAL

Multiply (654) by u' . Then

$$-YC_2 u'''' u' = \frac{d}{d\xi} \left(-YC_2 u''' u' + \frac{YC_2}{2} (u'')^2 \right),$$

and

$$A_2 u'' u' = \frac{d}{d\xi} \left(\frac{A_2}{2} (u')^2 \right).$$

Therefore the ODE has the exact first integral

$$-YC_2 u''' u' + \frac{YC_2}{2} (u'')^2 + \frac{A_2}{2} (u')^2 - \frac{M_2}{2} u^2 - \frac{\lambda C_4}{4} u^4 - \frac{\gamma C_6}{6} u^6 = \mathcal{C}_0. \quad (666)$$

For a localized defect branch with

$$u \rightarrow u_\infty, \quad u' \rightarrow 0, \quad u'' \rightarrow 0, \quad u''' \rightarrow 0 \quad (|\xi| \rightarrow \infty),$$

the constant is

$$\mathcal{C}_0 = -\frac{M_2}{2} u_\infty^2 - \frac{\lambda C_4}{4} u_\infty^4 - \frac{\gamma C_6}{6} u_\infty^6. \quad (667)$$

This first integral is exact and should be used as a solver diagnostic:

$$\mathcal{I}[u](\xi) := -YC_2 u''' u' + \frac{YC_2}{2} (u'')^2 + \frac{A_2}{2} (u')^2 - \frac{M_2}{2} u^2 - \frac{\lambda C_4}{4} u^4 - \frac{\gamma C_6}{6} u^6 - \mathcal{C}_0 \equiv 0. \quad (668)$$

CCXCVI. 5. CLOSED-FORM ANALYTIC SEED FAMILY

A generic exact elementary solution of (654) is not available for generic parameters. So the correct base strategy is not an exact closed-form solution, but an exact vacuum/tail matched analytic seed family.

The default even localized dip seed is

$$u^{(0)}(\xi) = u_\infty - a \operatorname{sech}^2(b\xi), \quad (669)$$

with

$$u^{(0)'}(0) = 0, \quad u^{(0)'''}(0) = 0. \quad (670)$$

Its derivatives at the center are

$$u^{(0)}(0) = u_\infty - a, \quad u^{(0)''}(0) = 2ab^2, \quad u^{(0)''''}(0) = -16ab^4. \quad (671)$$

To match the exact tail, choose

$$b = m_*. \quad (672)$$

Then the center-amplitude parameter a can be fixed by the center equation (654) evaluated at $\xi = 0$:

$$16YC_2 am_*^4 + 2A_2 am_*^2 - M_2(u_\infty - a) - \lambda C_4(u_\infty - a)^3 - \gamma C_6(u_\infty - a)^5 = 0. \quad (673)$$

This is the natural algebraic amplitude-fixing condition for the base seed.

CCXCVII. 6. TWO PRACTICAL SEED CHOICES

There are two especially useful seed subclasses.

A. Dark-defect seed

Impose a core dip to zero:

$$u^{(0)}(0) = 0 \iff a = u_\infty. \quad (674)$$

Then

$$u_{\text{dark}}^{(0)}(\xi) = u_\infty (1 - \operatorname{sech}^2(m_*\xi)) = u_\infty \tanh^2(m_*\xi). \quad (675)$$

B. Algebraically matched shallow-dip seed

Solve (673) for a with $b = m_*$. This gives the most faithful one-parameter analytic seed near the actual branch.

CCXCVIII. 7. NEWTON CONTINUATION ORDER

The actual continuation order is now fixed:

$$\chi_{II} = 0 \text{ base branch} \rightarrow \text{solve (654)} \rightarrow \text{obtain } u_0(\xi) \rightarrow \chi_{II} > 0 \text{ continuation.} \quad (676)$$

At the $\chi_{II} = 0$ stage, the solver should monitor simultaneously:

$$\|R_1\| + \|R_2\|, \quad \|\mathcal{I}[u]\|, \quad \text{grid refinement error.} \quad (677)$$

CCXCIX. 8. HONEST STATUS

At $\chi_{II} = 0$, we do *not* have a generic exact closed-form branch. But we now have:

1. the exact vacuum equation,
2. the exact tail equation,
3. the exact first integral,
4. and an exact vacuum/tail matched analytic seed family.

That is sufficient to fix a robust Newton/continuation starting point for the actual nonlinear branch search.

CCC. FINAL SUMMARY

The $\chi_{II} = 0$ Stage-IA base equation is

$$-YC_2 u'''' + A_2 u'' - M_2 u - \lambda C_4 u^3 - \gamma C_6 u^5 = 0.$$

Its nontrivial vacuum is determined exactly by

$$\gamma C_6 u_\infty^4 + \lambda C_4 u_\infty^2 + M_2 = 0,$$

so

$$u_\infty^2 = \frac{-\lambda C_4 \pm \sqrt{\lambda^2 C_4^2 - 4\gamma C_6 M_2}}{2\gamma C_6}.$$

Its exact tail decay rate is determined by

$$-YC_2 m^4 + A_2 m^2 + V_\infty^{(0)} = 0, \quad V_\infty^{(0)} = -2\lambda C_4 u_\infty^2 - 4\gamma C_6 u_\infty^4,$$

so

$$m_*^2 = \frac{A_2 + \sqrt{A_2^2 + 4YC_2 V_\infty^{(0)}}}{2YC_2}.$$

The equation also has the exact first integral

$$-YC_2 u''' u' + \frac{YC_2}{2} (u'')^2 + \frac{A_2}{2} (u')^2 - \frac{M_2}{2} u^2 - \frac{\lambda C_4}{4} u^4 - \frac{\gamma C_6}{6} u^6 = C_0.$$

A robust analytic continuation seed is

$$u^{(0)}(\xi) = u_\infty - a \operatorname{sech}^2(m_* \xi),$$

with either the dark-defect choice $a = u_\infty$, or a fixed by the center equation

$$16YC_2 a m_*^4 + 2A_2 a m_*^2 - M_2(u_\infty - a) - \lambda C_4(u_\infty - a)^3 - \gamma C_6(u_\infty - a)^5 = 0.$$

So the $\chi_{II} = 0$ base branch is now fixed strongly enough to serve as the continuation start for the full $\chi_{II} > 0$ actual branch search.

CCCI. STAGE IA NEWTON START: INITIAL UNKNOWN VECTOR AND BLOCK JACOBIAN

CCCII. GOAL

We now freeze the actual Newton input for the Stage-IA scalar branch. The purpose is to make the $\chi_{II} = 0$ base branch the explicit starting point for continuation to $\chi_{II} > 0$.

CCCI. 1. BASE BRANCH EQUATION

At $\chi_{II} = 0$, the scalar amplitude equation is

$$\boxed{-YC_2 u'''' + A_2 u'' - M_2 u - \lambda C_4 u^3 - \gamma C_6 u^5 = 0,} \quad (678)$$

with

$$\boxed{A_2 = C_2 \left(Z + \frac{10}{3} Y q_*^2 \right), \quad M_2 = C_2 (\mu^2 + Z q_*^2 + Y q_*^4).} \quad (679)$$

Introduce

$$\boxed{u(\xi) := \phi(\xi), \quad w(\xi) := u''(\xi).} \quad (680)$$

Then the coupled second-order system is

$$\boxed{u'' - w = 0,} \quad (681)$$

$$\boxed{-YC_2 w'' + A_2 w - M_2 u - \lambda C_4 u^3 - \gamma C_6 u^5 = 0.} \quad (682)$$

CCCI. 2. CHEBYSHEV GRID AND DISCRETE UNKNOWN VECTOR

On $[0, L_\xi]$, map from $x \in [-1, 1]$ by

$$\boxed{\xi = \frac{L_\xi}{2} (1 + x).} \quad (683)$$

Take the Chebyshev–Lobatto nodes

$$\boxed{x_j = \cos \frac{\pi j}{N}, \quad j = 0, \dots, N.} \quad (684)$$

Let

$$D^{(1)}, \quad D^{(2)}$$

be the standard Chebyshev differentiation matrices on $[-1, 1]$. Then

$$\boxed{D_\xi^{(1)} := \frac{2}{L_\xi} D^{(1)}, \quad D_\xi^{(2)} := \frac{4}{L_\xi^2} D^{(2)}.} \quad (685)$$

The Newton unknown vector is

$$\boxed{U := \begin{pmatrix} u_0 \\ \vdots \\ u_N \\ w_0 \\ \vdots \\ w_N \end{pmatrix} \in \mathbb{R}^{2N+2}.} \quad (686)$$

CCCV. 3. ANALYTIC INITIAL SEED

Choose the asymptotic vacuum u_∞ from

$$\boxed{-M_2 u_\infty - \lambda C_4 u_\infty^3 - \gamma C_6 u_\infty^5 = 0.} \quad (687)$$

Let the admissible tail decay be $m_* > 0$. Then the default even seed is

$$\boxed{u^{(0)}(\xi) = u_\infty - a \operatorname{sech}^2(m_* \xi),} \quad (688)$$

with

$$\boxed{w^{(0)}(\xi) = u^{(0)''}(\xi) = 2am_*^2 \left(3 \operatorname{sech}^4(m_* \xi) - 2 \operatorname{sech}^2(m_* \xi) \right).} \quad (689)$$

Hence the initial Newton vector is

$$\boxed{U^{(0)} = \begin{pmatrix} u^{(0)}(\xi_0) \\ \vdots \\ u^{(0)}(\xi_N) \\ w^{(0)}(\xi_0) \\ \vdots \\ w^{(0)}(\xi_N) \end{pmatrix}.} \quad (690)$$

For the dark-defect seed one sets

$$\boxed{a = u_\infty.} \quad (691)$$

CCCVI. 4. DISCRETE RESIDUAL VECTOR

Define the derivative vectors

$$u_\xi = D_\xi^{(1)} u, \quad u_{\xi\xi} = D_\xi^{(2)} u, \quad w_\xi = D_\xi^{(1)} w, \quad w_{\xi\xi} = D_\xi^{(2)} w.$$

For the interior nodes $j = 1, \dots, N-1$, set

$$\boxed{(R_1)_j = (u_{\xi\xi})_j - w_j,} \quad (692)$$

$$\boxed{(R_2)_j = -Y C_2 (w_{\xi\xi})_j + A_2 w_j - M_2 u_j - \lambda C_4 u_j^3 - \gamma C_6 u_j^5.} \quad (693)$$

The boundary rows are

$$\boxed{B_1 = (u_\xi)_0, \quad B_2 = (w_\xi)_0, \quad B_3 = u_N - u_\infty, \quad B_4 = w_N.} \quad (694)$$

Thus the full discrete nonlinear system is

$$\boxed{\mathcal{R}_{\text{disc}}(U) = (B_1, B_2, (R_1)_1, \dots, (R_1)_{N-1}, (R_2)_1, \dots, (R_2)_{N-1}, B_3, B_4)^T = 0.} \quad (695)$$

CCCVII. 5. BLOCK JACOBIAN STRUCTURE

Write

$$J_{\text{disc}}(U) = \begin{pmatrix} J_{uu} & J_{uw} \\ J_{wu} & J_{ww} \end{pmatrix}, \quad (696)$$

where each block is $(N+1) \times (N+1)$.

For the interior rows corresponding to R_1 ,

$$J_{uu}^{(R_1)} = D_\xi^{(2)}, \quad J_{uw}^{(R_1)} = -I. \quad (697)$$

For the interior rows corresponding to R_2 ,

$$J_{wu}^{(R_2)} = \text{diag}(-M_2 - 3\lambda C_4 u^2 - 5\gamma C_6 u^4), \quad (698)$$

$$J_{ww}^{(R_2)} = -Y C_2 D_\xi^{(2)} + A_2 I. \quad (699)$$

So for $\chi_{II} = 0$, the Newton Jacobian is especially simple: there are no derivative-cubic mixed terms yet.

CCCVIII. 6. BOUNDARY ROWS OF THE JACOBIAN

The boundary rows are inserted by replacing the corresponding rows of the block matrix with:

$$B_1 = (u_\xi)_0 \implies \text{row}(B_1) = ((D_\xi^{(1)})_{0,:} \mid 0), \quad (700)$$

$$B_2 = (w_\xi)_0 \implies \text{row}(B_2) = (0 \mid (D_\xi^{(1)})_{0,:}), \quad (701)$$

$$B_3 = u_N - u_\infty \implies \text{row}(B_3) = (e_N^T \mid 0), \quad (702)$$

$$B_4 = w_N \implies \text{row}(B_4) = (0 \mid e_N^T), \quad (703)$$

where e_N is the standard basis vector selecting the last node.

CCCIX. 7. NEWTON STEP

The damped Newton step is

$$J_{\text{disc}}(U^{(n)}) \delta U^{(n)} = -\mathcal{R}_{\text{disc}}(U^{(n)}), \quad (704)$$

$$U^{(n+1)} = U^{(n)} + \eta_n \delta U^{(n)}, \quad 0 < \eta_n \leq 1. \quad (705)$$

Use the backtracking rule: start with $\eta_n = 1$, and halve until

$$\|\mathcal{R}_{\text{disc}}(U^{(n+1)})\|_2 < \|\mathcal{R}_{\text{disc}}(U^{(n)})\|_2. \quad (706)$$

CCCX. 8. CONTINUATION IN χ_{II}

Once the $\chi_{II} = 0$ base branch converges, continue in small increments

$$\boxed{\chi_{II}^{(k+1)} = \chi_{II}^{(k)} + \Delta\chi,} \quad (707)$$

using the converged solution at $\chi_{II}^{(k)}$ as the initial vector for $\chi_{II}^{(k+1)}$.

At $\chi_{II} > 0$, the second residual becomes

$$\boxed{(R_2)_j = -YC_2(w_{\xi\xi})_j + A_2w_j - M_2u_j + 2\chi_{II}C_4u_j^2w_j + 2\chi_{II}C_4u_j(u_\xi)_j^2 + U_3u_j^3 - \gamma C_6u_j^5,} \quad (708)$$

and the Jacobian picks up the extra terms

$$\boxed{\Delta J_{wu} = \text{diag}\left(4\chi_{II}C_4uw + 2\chi_{II}C_4(u_\xi)^2 + 3\chi_{II}C_\Delta u^2\right) + \text{diag}(4\chi_{II}C_4u u_\xi)D_\xi^{(1)},} \quad (709)$$

$$\boxed{\Delta J_{ww} = \text{diag}(2\chi_{II}C_4u^2).} \quad (710)$$

Thus the $\chi_{II} = 0$ Jacobian is the clean exact base block, and the full $\chi_{II} > 0$ Jacobian is obtained by explicit low-order corrections.

CCCXI. 9. IMMEDIATE EXECUTION ORDER

The exact next execution order is now:

$$\boxed{\text{(i) build } U^{(0)} \text{ from (688)–(689),}} \quad (711)$$

$$\boxed{\text{(ii) solve } \mathcal{R}_{\text{disc}}(U) = 0 \text{ at } \chi_{II} = 0,} \quad (712)$$

$$\boxed{\text{(iii) continue in } \chi_{II} > 0,} \quad (713)$$

$$\boxed{\text{(iv) test } R_\perp[\Phi; z_0].} \quad (714)$$

At this point the actual nonlinear branch problem is fully solver-ready.

CCCXII. FINAL SUMMARY

The Stage-IA $\chi_{II} = 0$ base branch is now in fully explicit Newton form.

The discrete unknown vector is

$$U = (u_0, \dots, u_N, w_0, \dots, w_N)^T,$$

with analytic initial seed

$$u^{(0)}(\xi) = u_\infty - a \operatorname{sech}^2(m_*\xi), \quad w^{(0)}(\xi) = 2am_*^2(3 \operatorname{sech}^4(m_*\xi) - 2 \operatorname{sech}^2(m_*\xi)).$$

The residual system is

$$(R_1)_j = (u_{\xi\xi})_j - w_j,$$

$$(R_2)_j = -YC_2(w_{\xi\xi})_j + A_2w_j - M_2u_j - \lambda C_4u_j^3 - \gamma C_6u_j^5,$$

with boundary rows

$$(u_\xi)_0 = 0, \quad (w_\xi)_0 = 0, \quad u_N = u_\infty, \quad w_N = 0.$$

The $\chi_{II} = 0$ Jacobian block structure is

$$J_{\text{disc}} = \begin{pmatrix} D_\xi^{(2)} & -I \\ \text{diag}(-M_2 - 3\lambda C_4u^2 - 5\gamma C_6u^4) & -YC_2D_\xi^{(2)} + A_2I \end{pmatrix},$$

with the appropriate boundary-row replacements.

Once this base branch converges, continuation to $\chi_{II} > 0$ is performed by reusing the converged solution as the next initial seed and adding the explicit derivative-cubic Jacobian corrections.

So the problem is now fully in executable Newton/continuation form.

CCCXIII. STAGE IA TANGENT CONSISTENCY RESIDUAL

CCCXIV. GOAL

After solving the scalar frozen-orientation branch

$$\Psi(x) = \Phi(x) z_0, \quad z_0^\dagger z_0 = 1,$$

the next nonredundant question is whether this branch is also stationary in the internal tangent directions.

Therefore we must compute the exact tangent residual

$$\boxed{R_\perp[\Phi; z_0] := \Pi_{z_0^\perp} \mathcal{E}'[\Phi z_0].} \quad (715)$$

If $R_\perp = 0$, Stage IA is already an exact actual branch inside the current working microscopic theory. If $R_\perp \neq 0$, it is only a scalar precursor and we must release the orientation in Stage IB.

CCCXV. 1. FROZEN-ORIENTATION IDENTITIES

Recall

$$\boxed{\rho_\varepsilon = \Psi^\dagger \Psi + \varepsilon, \quad m^a = \Psi^\dagger T^a \Psi, \quad n^a = \frac{m^a}{\rho_\varepsilon}.} \quad (716)$$

For

$$\Psi = \Phi z_0, \quad n_0^a := z_0^\dagger T^a z_0,$$

we have

$$\boxed{m^a = |\Phi|^2 n_0^a, \quad n^a = n_0^a, \quad \rho_\varepsilon = |\Phi|^2 + \varepsilon.} \quad (717)$$

Hence

$$\boxed{J_i^a = \partial_i m^a = n_0^a \partial_i |\Phi|^2, \quad K_i^a = 0,} \quad (718)$$

and therefore

$$\boxed{F_i^a = (\alpha_X + \beta_X) \partial_i m^a - \beta_X n^a \partial_i \rho_\varepsilon = \alpha_X n_0^a \partial_i |\Phi|^2.} \quad (719)$$

CCCXVI. 2. EXACT PERPENDICULAR PROJECTION OF A^a AND B_i^a

The explicit working callback was

$$\mathcal{G}^{(II)}[\Psi] = \frac{1}{M_X^2} \sum_{i,a} \left[\partial_i (F_i^a A^a) - F_i^a B_i^a \right],$$

with

$$A^a = (\alpha_X + \beta_X) T^a \Psi - \beta_X n^a \Psi, \quad (720)$$

$$B_i^a = (\alpha_X + \beta_X) T^a (\partial_i \Psi) - \beta_X n^a (\partial_i \Psi) - \beta_X \frac{(T^a - n^a \mathbf{1}) \Psi}{\rho_\varepsilon} \partial_i \rho_\varepsilon. \quad (721)$$

Define the tangent vectors

$$\boxed{Q^a := \Pi_{z_0^\perp}(T^a z_0).} \quad (722)$$

Since

$$\Pi_{z_0^\perp}(n_0^a z_0) = 0, \quad \Psi = \Phi z_0, \quad \partial_i \Psi = (\partial_i \Phi) z_0,$$

we get

$$\boxed{\Pi_{z_0^\perp} A^a = (\alpha_X + \beta_X) \Phi Q^a,} \quad (723)$$

and

$$\Pi_{z_0^\perp} B_i^a = (\alpha_X + \beta_X) (\partial_i \Phi) Q^a - \beta_X \frac{\Phi}{|\Phi|^2 + \varepsilon} \partial_i |\Phi|^2 Q^a. \quad (724)$$

CCCXVII. 3. EXACT TANGENT RESIDUAL FORMULA

Projecting the callback orthogonally to z_0 , we obtain

$$\boxed{R_\perp[\Phi; z_0] = \Pi_{z_0^\perp} \mathcal{G}^{(II)}[\Phi z_0] = \frac{1}{M_X^2} \sum_{i,a} \left[\partial_i (F_i^a \Pi_{z_0^\perp} A^a) - F_i^a \Pi_{z_0^\perp} B_i^a \right].} \quad (725)$$

Insert (719), (723), (724). Since Q^a is constant in space,

$$\partial_i (F_i^a \Pi_{z_0^\perp} A^a) = (\alpha_X + \beta_X) \partial_i (F_i^a \Phi) Q^a, \quad (726)$$

$$F_i^a \Pi_{z_0^\perp} B_i^a = F_i^a \left[(\alpha_X + \beta_X) (\partial_i \Phi) - \beta_X \frac{\Phi}{|\Phi|^2 + \varepsilon} \partial_i |\Phi|^2 \right] Q^a. \quad (727)$$

Therefore

$$R_\perp[\Phi; z_0] = \frac{1}{M_X^2} \sum_{i,a} \left[(\alpha_X + \beta_X) \partial_i (F_i^a \Phi) - (\alpha_X + \beta_X) F_i^a \partial_i \Phi + \beta_X F_i^a \Phi \frac{\partial_i |\Phi|^2}{|\Phi|^2 + \varepsilon} \right] Q^a \quad (728)$$

$$= \frac{1}{M_X^2} \sum_{i,a} \left[(\alpha_X + \beta_X) (\partial_i F_i^a) \Phi + \beta_X F_i^a \Phi \frac{\partial_i |\Phi|^2}{|\Phi|^2 + \varepsilon} \right] Q^a. \quad (729)$$

Now use

$$F_i^a = \alpha_X n_0^a \partial_i |\Phi|^2, \quad \partial_i F_i^a = \alpha_X n_0^a \nabla^2 |\Phi|^2.$$

Then

$$R_{\perp}[\Phi; z_0] = \frac{\alpha_X}{M_X^2} \left[(\alpha_X + \beta_X) \Phi \nabla^2 |\Phi|^2 + \beta_X \Phi \frac{|\nabla |\Phi|^2|^2}{|\Phi|^2 + \varepsilon} \right] \sum_a n_0^a Q^a. \quad (730)$$

Define the internal tangent mismatch vector

$$\Xi_0 := \sum_a n_0^a Q^a = \Pi_{z_0^\perp} \left(\sum_a n_0^a T^a \right) z_0. \quad (731)$$

Then the tangent residual takes the compact exact form

$$R_{\perp}[\Phi; z_0] = \frac{\alpha_X}{M_X^2} \Xi_0 \left[(\alpha_X + \beta_X) \Phi \nabla^2 |\Phi|^2 + \beta_X \Phi \frac{|\nabla |\Phi|^2|^2}{|\Phi|^2 + \varepsilon} \right]. \quad (732)$$

This is the exact Stage-IA tangent consistency residual.

CCCXVIII. 4. EXACT VANISHING CRITERION

From (732), the strongest exact sufficient condition for perfect tangent consistency is

$$\Xi_0 = 0. \quad (733)$$

Equivalently,

$$\Pi_{z_0^\perp} \left(\sum_a n_0^a T^a \right) z_0 = 0. \quad (734)$$

Thus Stage IA is exactly tangent-consistent if z_0 is an eigenvector of the effective internal operator

$$H_0 := \sum_a n_0^a T^a. \quad (735)$$

So the frozen orientation is not arbitrary: the natural exact Stage-IA orientations are precisely the eigenvectors of H_0 .

CCCXIX. 5. QUANTITATIVE TANGENT-CONSISTENCY BOUND

Even when $\Xi_0 \neq 0$, (732) gives a direct bound. Using any consistent L^2 -based norm,

$$\frac{\|R_{\perp}[\Phi; z_0]\|}{\|\Phi\|} \leq \frac{|\alpha_X| \|\Xi_0\|}{M_X^2} \left[|\alpha_X + \beta_X| \|\nabla^2 |\Phi|^2\|_{\infty} + |\beta_X| \left\| \frac{|\nabla |\Phi|^2|^2}{|\Phi|^2 + \varepsilon} \right\|_{\infty} \right]. \quad (736)$$

This is the exact diagnostic inequality to decide whether Stage IA is good enough as an actual branch or only as a seed for Stage IB.

CCCXX. 6. SIMPLIFICATION ON THE SCALAR STAGE-IA BRANCH

For the locked BCC ansatz

$$\Phi(x) = \phi(\xi) S_{\text{BCC}}(x),$$

the leading scalar branch is already concentrated on one defect coordinate ξ . At this level,

$$|\nabla|\Phi|^2|^2 \sim ((|\phi|^2)')^2 \cdot (\text{cell factor}), \quad \nabla^2|\Phi|^2 \sim (|\phi|^2)'' \cdot (\text{cell factor}).$$

Therefore

$$R_\perp[\phi; z_0] \propto \Xi_0 \left[(\alpha_X + \beta_X) \phi (|\phi|^2)'' + \beta_X \phi \frac{((|\phi|^2)')^2}{|\phi|^2 + \varepsilon} \right] \quad (737)$$

up to frozen BCC cell factors.

So the tangent failure is completely controlled by two things only:

1. the internal mismatch vector Ξ_0 ,
2. the scalar branch curvature/gradient profile.

CCCXXI. 7. ACCEPTANCE RULE FOR STAGE IA

The Stage-IA scalar branch is promoted to an actual branch if, in addition to the scalar residual acceptance,

$$\frac{\|R_\perp[\Phi; z_0]\|}{\|\Phi\|} < \varepsilon_\perp. \quad (738)$$

If (738) fails but the scalar branch residual is already small, then the correct next move is not to discard the branch. Instead, use it as the initial data for Stage IB, where the internal orientation $z(\xi)$ is released dynamically.

CCCXXII. 8. IMMEDIATE NEXT STEP

The actual next step is now completely concrete:

$$\boxed{\text{solve the Stage-IA scalar BVP for } \phi, \quad \text{compute } \Xi_0, \quad \text{evaluate } R_\perp[\phi; z_0].} \quad (739)$$

If $\Xi_0 = 0$, Stage IA is already tangent-consistent. If not, Stage IA is the exact continuation seed for Stage IB.

CCCXXIII. FINAL SUMMARY

For the frozen-orientation branch

$$\Psi(x) = \Phi(x) z_0,$$

the Stage-IA tangent consistency residual can be computed exactly.

One has

$$m^a = |\Phi|^2 n_0^a, \quad n^a = n_0^a, \quad J_i^a = n_0^a \partial_i |\Phi|^2, \quad K_i^a = 0,$$

so

$$F_i^a = \alpha_X n_0^a \partial_i |\Phi|^2.$$

Projecting the Class II callback orthogonally to z_0 gives the exact formula

$$R_\perp[\Phi; z_0] = \frac{\alpha_X}{M_X^2} \Xi_0 \left[(\alpha_X + \beta_X) \Phi \nabla^2 |\Phi|^2 + \beta_X \Phi \frac{|\nabla |\Phi|^2|^2}{|\Phi|^2 + \varepsilon} \right],$$

where

$$\Xi_0 = \Pi_{z_0^\perp} \left(\sum_a n_0^a T^a \right) z_0.$$

Therefore the strongest exact sufficient condition for perfect Stage-IA tangent consistency is

$$\Xi_0 = 0.$$

Equivalently, z_0 must be an eigenvector of

$$H_0 = \sum_a n_0^a T^a.$$

More generally, the tangent residual satisfies the explicit bound

$$\frac{\|R_\perp[\Phi; z_0]\|}{\|\Phi\|} \leq \frac{|\alpha_X| \|\Xi_0\|}{M_X^2} \left[|\alpha_X + \beta_X| \|\nabla^2 |\Phi|^2\|_\infty + |\beta_X| \left\| \frac{|\nabla |\Phi|^2|^2}{|\Phi|^2 + \varepsilon} \right\|_\infty \right].$$

So the Stage-IA branch is now fully split into two exact checks:

$$\text{scalar BVP } R_\phi[\phi] = 0, \quad \text{tangent consistency } R_\perp[\phi; z_0].$$

If $R_\perp$ is small enough, Stage IA is already an actual branch. If not, it becomes the exact initial seed for Stage IB.

CCCXXIV. CANONICAL INTERNAL CHOICE FOR EXACT STAGE-IA TANGENT CONSISTENCY

CCCXXV. GOAL

We now choose the internal locked direction z_0 so that the Stage-IA tangent residual vanishes exactly at the current working Class II level.

The exact condition derived previously was

$$\Xi_0 := \Pi_{z_0^\perp} \left(\sum_a n_0^a T^a \right) z_0 = 0, \quad n_0^a := z_0^\dagger T^a z_0. \quad (740)$$

CCCXXVI. 1. CANONICAL LOCKED BASIS

Use the canonical locked basis already fixed in the notes:

$$z_0 = e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad u_1 = e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad u_2 = e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}. \quad (741)$$

The tangent space is

$$Q_{z_0} = \text{span}_{\mathbb{C}}\{u_1, u_2\}. \quad (742)$$

Let

$$T^a = \frac{\lambda^a}{2} \quad (743)$$

be the standard Hermitian $SU(3)$ generators.

CCCXXVII. 2. EXACT VALUES OF n_0^a

Since $z_0 = e_1$,

$$n_0^a = e_1^\dagger T^a e_1 = (T^a)_{11}.$$

For the standard Gell-Mann basis,

$$\boxed{n_0^3 = \frac{1}{2}, \quad n_0^8 = \frac{1}{2\sqrt{3}}, \quad n_0^a = 0 \quad (a \neq 3, 8).} \quad (744)$$

CCCXXVIII. 3. EXACT EFFECTIVE INTERNAL OPERATOR

Define

$$\boxed{H_0 := \sum_a n_0^a T^a.} \quad (745)$$

Using (744),

$$\boxed{H_0 = \frac{1}{2} T^3 + \frac{1}{2\sqrt{3}} T^8.} \quad (746)$$

Now

$$T^3 = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad T^8 = \frac{1}{2\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}.$$

Therefore

$$\boxed{H_0 = \begin{pmatrix} \frac{1}{3} & 0 & 0 \\ 0 & -\frac{1}{6} & 0 \\ 0 & 0 & -\frac{1}{6} \end{pmatrix}.} \quad (747)$$

Hence

$$\boxed{H_0 z_0 = \frac{1}{3} z_0.} \quad (748)$$

So $z_0 = e_1$ is an exact eigenvector of H_0 .

CCCXXIX. 4. EXACT VANISHING OF THE TANGENT MISMATCH VECTOR

By definition,

$$\boxed{\Xi_0 = \Pi_{z_0^\perp} H_0 z_0.} \quad (749)$$

Using (748),

$$H_0 z_0 = \frac{1}{3} z_0$$

is purely longitudinal. Therefore its orthogonal projection to Q_{z_0} vanishes:

$$\boxed{\Xi_0 = 0.} \quad (750)$$

This is the exact canonical solution of the Stage-IA tangent-consistency condition.

CCCXXX. 5. CONSEQUENCE FOR THE STAGE-IA TANGENT RESIDUAL

The exact tangent residual formula derived previously was

$$R_{\perp}[\Phi; z_0] = \frac{\alpha_X}{M_X^2} \Xi_0 \left[(\alpha_X + \beta_X) \Phi \nabla^2 |\Phi|^2 + \beta_X \Phi \frac{|\nabla |\Phi|^2|^2}{|\Phi|^2 + \varepsilon} \right]. \quad (751)$$

Substituting (750) gives

$$\boxed{R_{\perp}[\Phi; z_0] = 0} \quad (752)$$

identically, for every scalar Stage-IA profile Φ , at the current working Class II level.

CCCXXXI. 6. SHARP CONSEQUENCE

Therefore the canonical internal choice

$$\boxed{z_0 = e_1} \quad (753)$$

does not merely simplify the scalar branch. It makes the frozen-orientation Stage-IA branch exactly tangent-consistent inside the current working microscopic model.

Hence the actual nonlinear branch construction is now reduced to:

$$\boxed{\text{solve the Stage-IA scalar BVP only.}} \quad (754)$$

There is no Stage-IA obstruction from internal tangent forcing in this canonical choice.

CCCXXXII. 7. HONEST STATUS

This does *not* yet prove full unconditional closure of the complete microscopic TECT action. But at the current explicit working Class II level, it proves the stronger practical statement:

$$\boxed{\text{the canonical frozen-orientation Stage-IA branch is internally self-consistent.}} \quad (755)$$

So Stage IB is no longer required as an immediate next step. The shortest path is now to solve the scalar Stage-IA BVP and extract the accepted branch.

CCCXXXIII. FINAL SUMMARY

Using the canonical locked basis

$$z_0 = e_1, \quad u_1 = e_2, \quad u_2 = e_3,$$

with standard $SU(3)$ generators $T^a = \lambda^a/2$, one gets

$$n_0^3 = \frac{1}{2}, \quad n_0^8 = \frac{1}{2\sqrt{3}}, \quad n_0^a = 0 \quad (a \neq 3, 8).$$

Therefore

$$H_0 = \sum_a n_0^a T^a = \frac{1}{2} T^3 + \frac{1}{2\sqrt{3}} T^8 = \begin{pmatrix} \frac{1}{3} & 0 & 0 \\ 0 & -\frac{1}{6} & 0 \\ 0 & 0 & -\frac{1}{6} \end{pmatrix},$$

so

$$H_0 z_0 = \frac{1}{3} z_0.$$

Hence

$$\Xi_0 = \Pi_{z_0^\perp} H_0 z_0 = 0.$$

Since the exact Stage-IA tangent residual is

$$R_\perp[\Phi; z_0] = \frac{\alpha_X}{M_X^2} \Xi_0 \left[(\alpha_X + \beta_X) \Phi \nabla^2 |\Phi|^2 + \beta_X \Phi \frac{|\nabla |\Phi|^2|^2}{|\Phi|^2 + \varepsilon} \right],$$

it follows that

$$\boxed{R_\perp[\Phi; z_0] = 0}$$

identically for the canonical choice $z_0 = e_1$, at the current working Class II level.

So Stage IA is already tangent-consistent. The immediate next problem is now only:

$$\boxed{\text{solve the scalar Stage-IA BVP.}}$$

CCCXXXIV. STAGE IA ACTUAL SOLVE PROTOCOL

CCCXXXV. SHORT KOREAN EXPLANATION

Because the canonical frozen orientation $z_0 = e_1$ makes

$$R_\perp[\Phi; z_0] = 0$$

at the current working Class II level, the immediate actual branch problem is reduced exactly to the scalar Stage-IA BVP.

So the right next step is no longer structural decomposition. It is the actual numerical solve of the scalar branch, first at $\chi_{II} = 0$, then by continuation to $\chi_{II} > 0$.

CCCXXXVI. 1. FINAL STAGE-IA BVP

The scalar Stage-IA system is

$$\boxed{u'' = w,} \tag{756}$$

$$\boxed{-YC_2 w'' + A_2 w - M_2 u + 2\chi_{II} C_4 u^2 w + 2\chi_{II} C_4 u (u')^2 + U_3 u^3 - \gamma C_6 u^5 = 0,} \tag{757}$$

with

$$\boxed{A_2 = C_2 \left(Z + \frac{10}{3} Y q_*^2 \right), \quad M_2 = C_2 (\mu^2 + Z q_*^2 + Y q_*^4), \quad U_3 = \chi_{II} C_\Delta - \lambda C_4.} \tag{758}$$

The finite-interval even-branch boundary conditions are

$$\boxed{u'(0) = 0, \quad w'(0) = 0, \quad u(L_\xi) = u_\infty, \quad w(L_\xi) = 0.} \tag{759}$$

CCCXXXVII. 2. EXECUTION ORDER

The exact execution order is:

$$\boxed{\text{Step A: solve the base branch at } \chi_{II} = 0,} \tag{760}$$

$$\boxed{\text{Step B: continue the converged branch in } \chi_{II} > 0,} \tag{761}$$

$$\boxed{\text{Step C: refine } N \text{ and } L_\xi \text{ until the branch is grid-stable.}} \tag{762}$$

Because $R_\perp = 0$ already holds for the canonical choice, there is no immediate need to release the internal orientation.

CCXXXVIII. 3. BASE BRANCH AT $\chi_{II} = 0$

At the base point,

$$\boxed{-YC_2 u'''' + A_2 u'' - M_2 u - \lambda C_4 u^3 - \gamma C_6 u^5 = 0.} \quad (763)$$

The nontrivial vacuum is determined by

$$\boxed{-M_2 u_\infty - \lambda C_4 u_\infty^3 - \gamma C_6 u_\infty^5 = 0,} \quad (764)$$

equivalently

$$\boxed{\gamma C_6 u_\infty^4 + \lambda C_4 u_\infty^2 + M_2 = 0.} \quad (765)$$

The admissible tail rate m_* is determined from

$$\boxed{-YC_2 m^4 + A_2 m^2 + V_\infty^{(0)} = 0, \quad V_\infty^{(0)} = -2\lambda C_4 u_\infty^2 - 4\gamma C_6 u_\infty^4.} \quad (766)$$

CCXXXIX. 4. PRACTICAL FIRST SEED

Use the analytic vacuum/tail-matched seed

$$\boxed{u^{(0)}(\xi) = u_\infty - a \operatorname{sech}^2(m_* \xi),} \quad (767)$$

$$\boxed{w^{(0)}(\xi) = u^{(0)''}(\xi).} \quad (768)$$

Default first choice:

$$\boxed{a = u_\infty} \quad (769)$$

(dark-defect seed), unless the center equation indicates a shallower defect.

CCCXL. 5. DOMAIN AND COLLOCATION DEFAULTS

Choose the computational interval by the tail rule

$$\boxed{L_\xi = \frac{18}{m_*}.} \quad (770)$$

Use the Chebyshev–Lobatto grid

$$\boxed{x_j = \cos \frac{\pi j}{N}, \quad \xi_j = \frac{L_\xi}{2}(1 + x_j), \quad j = 0, \dots, N.} \quad (771)$$

Recommended refinement ladder:

$$\boxed{N \in \{64, 96, 128\}.} \quad (772)$$

Start with $N = 64$, converge the branch, then push to 96, then 128.

CCCXLI. 6. DISCRETE RESIDUAL SYSTEM

Let

$$U = (u_0, \dots, u_N, w_0, \dots, w_N)^T.$$

Then the interior residuals are

$$(R_1)_j = (u_{\xi\xi})_j - w_j, \quad (773)$$

$$(R_2)_j = -YC_2(w_{\xi\xi})_j + A_2w_j - M_2u_j + 2\chi_{II}C_4u_j^2w_j + 2\chi_{II}C_4u_j(u_{\xi})_j^2 + U_3u_j^3 - \gamma C_6u_j^5. \quad (774)$$

The boundary rows are

$$(u_{\xi})_0 = 0, \quad (w_{\xi})_0 = 0, \quad u_N - u_{\infty} = 0, \quad w_N = 0. \quad (775)$$

Thus the nonlinear system is

$$\mathcal{R}_{\text{disc}}(U) = 0. \quad (776)$$

CCCXLII. 7. NEWTON SOLVE AND DAMPING

At fixed χ_{II} , solve

$$J_{\text{disc}}(U^{(n)}) \delta U^{(n)} = -\mathcal{R}_{\text{disc}}(U^{(n)}), \quad (777)$$

$$U^{(n+1)} = U^{(n)} + \eta_n \delta U^{(n)}, \quad 0 < \eta_n \leq 1. \quad (778)$$

Use backtracking: start with $\eta_n = 1$, and halve until

$$\|\mathcal{R}_{\text{disc}}(U^{(n+1)})\|_2 < \|\mathcal{R}_{\text{disc}}(U^{(n)})\|_2.$$

CCCXLIII. 8. CONTINUATION IN χ_{II}

Once the $\chi_{II} = 0$ base branch converges, continue in small increments:

$$\chi_{II}^{(k+1)} = \chi_{II}^{(k)} + \Delta\chi. \quad (779)$$

A safe first ladder is

$$\Delta\chi = 0.05 \chi_{II}^{\text{target}}, \quad (780)$$

for 20 steps, or smaller if Newton loses monotonicity.

At each continuation step, reuse the previously converged solution as the new initial guess.

CCCXLIV. 9. STAGE-IA ACCEPTANCE CRITERIA

The Stage-IA scalar branch is accepted if

$$\|\mathcal{R}_{\text{disc}}\|_2 < 10^{-8}, \quad (781)$$

$$\boxed{\frac{\|u_{2N} - u_N\|_2}{\|u_{2N}\|_2} < 10^{-4}}, \quad (782)$$

$$\boxed{\left\| -YC_2 u''' u' + \frac{YC_2}{2} (u'')^2 + \frac{A_2}{2} (u')^2 - \frac{M_2}{2} u^2 - \frac{\lambda C_4}{4} u^4 - \frac{\gamma C_6}{6} u^6 - \mathcal{C}_0 \right\|_\infty < 10^{-7}}, \quad (783)$$

and, because $z_0 = e_1$ gives $\Xi_0 = 0$,

$$\boxed{R_\perp[\Phi; z_0] = 0} \quad (784)$$

at the current working Class II level.

CCCXLV. 10. OUTPUT OF STAGE IA

When the Stage-IA branch is accepted, denote it

$$\boxed{u_{\text{IA}}(\xi), \quad \Phi_{\text{IA}}(x) = u_{\text{IA}}(\xi) S_{\text{BCC}}(x), \quad \Psi_{\text{IA}}(x) = \Phi_{\text{IA}}(x) e_1.} \quad (785)$$

This is the first actual nonlinear branch candidate.

Only after this is obtained do we proceed to Bloch linearization

$$L_{\text{Bloch}}(k) \rightarrow K_i \rightarrow P_* K_i P_*$$

and extract $(\lambda_\parallel, \alpha, \beta)$.

CCCXLVI. 11. IMMEDIATE NEXT MOVE

Thus the immediate next move is not another reduction. It is exactly:

$$\boxed{\text{run the Stage-IA Newton/collocation solve at } \chi_{II} = 0, \text{ then continue to } \chi_{II} > 0.} \quad (786)$$

CCCXLVII. FINAL SUMMARY

Because the canonical internal choice $z_0 = e_1$ makes

$$R_\perp[\Phi; z_0] = 0$$

at the current working Class II level, the actual nonlinear branch problem is immediately reduced to the scalar Stage-IA BVP.

The practical execution protocol is:

1. solve the $\chi_{II} = 0$ base branch first,
2. use the vacuum/tail-matched seed

$$u^{(0)}(\xi) = u_\infty - a \operatorname{sech}^2(m_* \xi), \quad a = u_\infty,$$

3. take

$$L_\xi = \frac{18}{m_*}, \quad N = 64 \rightarrow 96 \rightarrow 128,$$

4. solve the Chebyshev collocation Newton system

$$\mathcal{R}_{\text{disc}}(U) = 0,$$

5. continue in small steps from $\chi_{II} = 0$ to $\chi_{II} > 0$,
6. accept only when the residual, grid convergence, and first-integral defect all pass.

At that point the resulting branch

$$\Psi_{\text{IA}}(x) = u_{\text{IA}}(\xi) S_{\text{BCC}}(x) e_1$$

is the first actual nonlinear branch candidate, and the next stage is Bloch linearization and coefficient extraction.

CCCXLVIII. STAGE IA AS A TWO-PARAMETER SHOOTING PROBLEM

CCCXLIX. GOAL

The Stage-IA scalar branch has already been reduced to the solver-ready system

$$u'' = w, \tag{787}$$

$$w'' = \frac{1}{YC_2} \left[A_2 w - M_2 u + 2\chi_{II} C_4 u^2 w + 2\chi_{II} C_4 u (u')^2 + U_3 u^3 - \gamma C_6 u^5 \right]. \tag{788}$$

We now reduce the even branch to a two-parameter shooting problem.

CCCL. 1. EVEN-BRANCH CENTER DATA

For the even branch centered at $\xi = 0$, the symmetry conditions are

$$\boxed{u'(0) = 0, \quad w'(0) = 0.} \tag{789}$$

Therefore the only free center data are

$$\boxed{u_c := u(0), \quad w_c := w(0).} \tag{790}$$

Since $w = u''$, this means the entire local branch is determined by

$$u(0), \quad u''(0),$$

with odd derivatives fixed to zero by symmetry.

CCCL. 2. EXACT CENTER ACCELERATION

Evaluating (788) at $\xi = 0$, and using $u'(0) = 0$, we obtain

$$\boxed{w''(0) = \frac{1}{YC_2} \left[A_2 w_c - M_2 u_c + 2\chi_{II} C_4 u_c^2 w_c + U_3 u_c^3 - \gamma C_6 u_c^5 \right].} \tag{791}$$

Since $u'' = w$, we also have

$$\boxed{u''(0) = w_c, \quad u'''(0) = w''(0).} \tag{792}$$

Thus the full local Taylor data are fixed by (u_c, w_c) .

CCCLII. 3. CENTER TAYLOR EXPANSION

The even branch therefore has the local expansions

$$\boxed{u(\xi) = u_c + \frac{w_c}{2}\xi^2 + \frac{w''(0)}{24}\xi^4 + O(\xi^6),} \quad (793)$$

$$\boxed{w(\xi) = w_c + \frac{w''(0)}{2}\xi^2 + O(\xi^4).} \quad (794)$$

This is the correct regular seed for outward integration from the center.

CCCLIII. 4. FIRST-ORDER SHOOTING SYSTEM

Introduce

$$\boxed{y_1 := u, \quad y_2 := u', \quad y_3 := w, \quad y_4 := w'.} \quad (795)$$

Then the autonomous first-order system is

$$\boxed{y_1' = y_2, \quad y_2' = y_3, \quad y_3' = y_4,} \quad (796)$$

$$\boxed{y_4' = \frac{1}{YC_2} \left[A_2 y_3 - M_2 y_1 + 2\chi_{II} C_4 y_1^2 y_3 + 2\chi_{II} C_4 y_1 y_2^2 + U_3 y_1^3 - \gamma C_6 y_1^5 \right].} \quad (797)$$

The center initial data are

$$\boxed{y_1(0) = u_c, \quad y_2(0) = 0, \quad y_3(0) = w_c, \quad y_4(0) = 0.} \quad (798)$$

CCCLIV. 5. SHOOTING TARGET MAP

Fix the finite truncation length L_ξ , determined from the tail scale as before. The desired asymptotic boundary conditions are

$$\boxed{u(L_\xi) = u_\infty, \quad w(L_\xi) = 0.} \quad (799)$$

Therefore define the two-component shooting map

$$\boxed{\mathcal{S}(u_c, w_c) := \begin{pmatrix} u(L_\xi; u_c, w_c) - u_\infty \\ w(L_\xi; u_c, w_c) \end{pmatrix}.} \quad (800)$$

The Stage-IA branch problem is exactly

$$\boxed{\mathcal{S}(u_c, w_c) = 0.} \quad (801)$$

Thus the full scalar BVP has now been reduced to a two-dimensional root-finding problem.

CCCLV. 6. VARIATIONAL EQUATIONS FOR THE SHOOTING JACOBIAN

To apply Newton in (u_c, w_c) , define the two variational solutions

$$\delta^{(u)}y := \frac{\partial y}{\partial u_c}, \quad \delta^{(w)}y := \frac{\partial y}{\partial w_c}.$$

Linearizing (796)–(797), one gets

$$\boxed{\delta y'_1 = \delta y_2, \quad \delta y'_2 = \delta y_3, \quad \delta y'_3 = \delta y_4,} \quad (802)$$

and

$$\begin{aligned} \delta y'_4 = \frac{1}{YC_2} & \left[(-M_2 + 4\chi_{II}C_4y_1y_3 + 2\chi_{II}C_4y_2^2 + 3U_3y_1^2 - 5\gamma C_6y_1^4)\delta y_1 \right. \\ & \left. + 4\chi_{II}C_4y_1y_2\delta y_2 + (A_2 + 2\chi_{II}C_4y_1^2)\delta y_3 \right]. \end{aligned} \quad (803)$$

For the u_c -variation, the initial data are

$$\boxed{\delta^{(u)}y(0) = (1, 0, 0, 0)^T.} \quad (804)$$

For the w_c -variation,

$$\boxed{\delta^{(w)}y(0) = (0, 0, 1, 0)^T.} \quad (805)$$

Hence the exact shooting Jacobian is

$$\boxed{DS(u_c, w_c) = \begin{pmatrix} \delta^{(u)}y_1(L_\xi) & \delta^{(w)}y_1(L_\xi) \\ \delta^{(u)}y_3(L_\xi) & \delta^{(w)}y_3(L_\xi) \end{pmatrix}.} \quad (806)$$

CCCLVI. 7. TWO-DIMENSIONAL NEWTON SHOOTING

The Newton correction for the center data is therefore

$$\boxed{DS(u_c^{(n)}, w_c^{(n)}) \begin{pmatrix} \delta u_c^{(n)} \\ \delta w_c^{(n)} \end{pmatrix} = -\mathcal{S}(u_c^{(n)}, w_c^{(n)})}.} \quad (807)$$

Then update

$$\boxed{u_c^{(n+1)} = u_c^{(n)} + \eta_n \delta u_c^{(n)}, \quad w_c^{(n+1)} = w_c^{(n)} + \eta_n \delta w_c^{(n)},} \quad (808)$$

with damping $0 < \eta_n \leq 1$ chosen by backtracking.

CCCLVII. 8. NATURAL ADMISSIBLE REGION FOR THE CENTER DATA

The even branch must remain in the basin of the chosen vacuum u_∞ . So the first admissible search window is

$$\boxed{0 \leq u_c \leq u_\infty,} \quad (809)$$

for the dip branch, and

$$\boxed{w_c \geq 0,} \quad (810)$$

for a center minimum.

The dark-defect seed corresponds to

$$\boxed{u_c = 0, \quad w_c = 2u_\infty m_*^2} \quad (811)$$

at leading seed level.

CCCLVIII. 9. ACCEPTANCE AT THE SHOOTING LEVEL

The Stage-IA branch is accepted if

$$\|\mathcal{S}(u_c, w_c)\|_2 < \varepsilon_{\text{shoot}}, \quad (812)$$

with a default choice

$$\varepsilon_{\text{shoot}} = 10^{-10}, \quad (813)$$

and, in addition, the first-integral defect satisfies

$$\|\mathcal{I}[u]\|_\infty < 10^{-8}. \quad (814)$$

Because the canonical $z_0 = e_1$ choice gives $R_\perp = 0$, these conditions are sufficient for Stage IA acceptance at the current working Class II level.

CCCLIX. 10. IMMEDIATE NEXT MOVE

The actual next move is now completely explicit:

$$\text{choose } (u_c^{(0)}, w_c^{(0)}) \text{ from the dark seed, integrate outward, construct } \mathcal{S}, \text{ and solve } \mathcal{S}(u_c, w_c) = 0 \quad (815)$$

first at $\chi_{II} = 0$, then by continuation to $\chi_{II} > 0$.

CCCLX. FINAL SUMMARY

The Stage-IA scalar branch has now been reduced from a fourth-order ODE to a two-parameter shooting problem. For the even branch, the only free center data are

$$u_c = u(0), \quad w_c = u''(0),$$

since

$$u'(0) = 0, \quad w'(0) = 0$$

are fixed by symmetry.

The exact shooting target map is

$$\mathcal{S}(u_c, w_c) = \begin{pmatrix} u(L_\xi; u_c, w_c) - u_\infty \\ w(L_\xi; u_c, w_c) \end{pmatrix},$$

and the branch equation is simply

$$\mathcal{S}(u_c, w_c) = 0.$$

The Jacobian of this map is obtained from the variational system and is

$$D\mathcal{S}(u_c, w_c) = \begin{pmatrix} \partial_{u_c} u(L_\xi) & \partial_{w_c} u(L_\xi) \\ \partial_{u_c} w(L_\xi) & \partial_{w_c} w(L_\xi) \end{pmatrix}.$$

So the Stage-IA actual branch solve is now a literal two-dimensional Newton shooting problem:

$$D\mathcal{S} \begin{pmatrix} \delta u_c \\ \delta w_c \end{pmatrix} = -\mathcal{S}.$$

With the canonical choice $z_0 = e_1$, the tangent residual already vanishes, so solving this shooting problem is enough to construct the Stage-IA actual branch at the current working Class II level.

CCCLXI. STAGE IA SHOOTING NEWTON: DARK-SEED INITIALIZATION, LOCAL START, AND CONTINUATION

CCCLXII. GOAL

The Stage-IA branch has already been reduced to the two-parameter even shooting problem

$$\mathcal{S}(u_c, w_c) = 0, \quad u_c := u(0), \quad w_c := u''(0).$$

The next real step is to convert this into a numerically stable integration-and-Newton procedure. We now fix:

1. the dark-seed initial center data,
2. the exact local Taylor start away from $\xi = 0$,
3. the actual continuation rule in χ_{II} .

CCCLXIII. 1. FULL STAGE-IA CENTER EQUATION

The Stage-IA scalar system is

$$u'' = w, \tag{816}$$

$$-YC_2 w'' + A_2 w - M_2 u + 2\chi_{II} C_4 u^2 w + 2\chi_{II} C_4 u(u')^2 + U_3 u^3 - \gamma C_6 u^5 = 0. \tag{817}$$

At the center of the even branch,

$$u'(0) = 0, \quad w'(0) = 0.$$

Therefore the exact center acceleration is

$$w''(0) = \frac{1}{YC_2} \left[A_2 w_c - M_2 u_c + 2\chi_{II} C_4 u_c^2 w_c + U_3 u_c^3 - \gamma C_6 u_c^5 \right]. \tag{818}$$

This is the key local datum needed for a stable outward integration.

CCCLXIV. 2. DARK-DEFECT CENTER SEED

For the $\chi_{II} = 0$ base branch, the natural analytic seed was

$$u^{(0)}(\xi) = u_\infty - a \operatorname{sech}^2(m_* \xi).$$

For the dark branch we take

$$a = u_\infty, \tag{819}$$

so

$$u_{\text{dark}}^{(0)}(\xi) = u_\infty \tanh^2(m_* \xi). \tag{820}$$

Hence the corresponding center seed values are

$$u_c^{(0)} = 0, \quad w_c^{(0)} = u''(0) = 2u_\infty m_*^2. \tag{821}$$

This is the default initial shooting point for the $\chi_{II} = 0$ solve.

CCCLXV. 3. EXACT LOCAL TAYLOR START

Direct integration from $\xi = 0$ is numerically awkward because of symmetry locking. So introduce a small positive start radius δ , and begin integration at $\xi = \delta$.

For the even branch,

$$u'(0) = 0, \quad u'''(0) = 0, \quad w'(0) = 0.$$

Using (818), the local Taylor expansions are

$$u(\delta) = u_c + \frac{w_c}{2}\delta^2 + \frac{w''(0)}{24}\delta^4 + O(\delta^6), \quad (822)$$

$$u'(\delta) = w_c\delta + \frac{w''(0)}{6}\delta^3 + O(\delta^5), \quad (823)$$

$$w(\delta) = w_c + \frac{w''(0)}{2}\delta^2 + O(\delta^4), \quad (824)$$

$$w'(\delta) = w''(0)\delta + O(\delta^3). \quad (825)$$

Thus the exact local start vector for the first-order system

$$y_1 = u, \quad y_2 = u', \quad y_3 = w, \quad y_4 = w'$$

is

$$y(\delta; u_c, w_c) = \begin{pmatrix} u_c + \frac{w_c}{2}\delta^2 + \frac{w''(0)}{24}\delta^4 \\ w_c\delta + \frac{w''(0)}{6}\delta^3 \\ w_c + \frac{w''(0)}{2}\delta^2 \\ w''(0)\delta \end{pmatrix} + O(\delta^5), \quad (826)$$

with $w''(0)$ given by (818).

CCCLXVI. 4. PRACTICAL CHOICE OF THE START RADIUS

The start radius must be small compared to the tail scale but not so small that roundoff dominates. The natural choice is

$$\delta = \frac{10^{-3}}{m_*}. \quad (827)$$

If the Newton map becomes noisy, one may increase to

$$\delta = \frac{5 \times 10^{-3}}{m_*},$$

but never larger than $10^{-2}/m_*$ in the first pass.

CCCLXVII. 5. SHOOTING MAP WITH LOCAL START

Let $y(\xi; u_c, w_c, \chi_{II})$ be the unique solution of the Stage-IA first-order system starting from (826) at $\xi = \delta$. Then the shooting map is

$$\mathcal{S}(u_c, w_c; \chi_{II}) = \begin{pmatrix} u(L_\xi; u_c, w_c, \chi_{II}) - u_\infty(\chi_{II}) \\ w(L_\xi; u_c, w_c, \chi_{II}) \end{pmatrix}. \quad (828)$$

The branch solve is

$$\mathcal{S}(u_c, w_c; \chi_{II}) = 0. \quad (829)$$

CCCLXVIII. 6. VARIATIONAL INITIAL DATA FOR THE SHOOTING JACOBIAN

To compute the Jacobian

$$D_{(u_c, w_c)} \mathcal{S},$$

we integrate the variational system together with the main ODE.

Because the local start data depend on u_c, w_c through $w''(0)$, the initial variational vectors are not simply $(1, 0, 0, 0)$ and $(0, 0, 1, 0)$ at $\xi = \delta$. They must include the derivatives of $w''(0)$.

Define

$$\Omega(u_c, w_c; \chi_{II}) := \frac{1}{YC_2} \left[A_2 w_c - M_2 u_c + 2\chi_{II} C_4 u_c^2 w_c + U_3 u_c^3 - \gamma C_6 u_c^5 \right]. \quad (830)$$

Then

$$w''(0) = \Omega(u_c, w_c; \chi_{II}).$$

Hence

$$\frac{\partial \Omega}{\partial u_c} = \frac{1}{YC_2} \left[-M_2 + 4\chi_{II} C_4 u_c w_c + 3U_3 u_c^2 - 5\gamma C_6 u_c^4 \right], \quad (831)$$

$$\frac{\partial \Omega}{\partial w_c} = \frac{1}{YC_2} \left[A_2 + 2\chi_{II} C_4 u_c^2 \right]. \quad (832)$$

Therefore the u_c -variation initial vector at $\xi = \delta$ is

$$\partial_{u_c} y(\delta) = \begin{pmatrix} 1 + \frac{1}{24} \Omega_{u_c} \delta^4 \\ \frac{1}{6} \Omega_{u_c} \delta^3 \\ \frac{1}{2} \Omega_{u_c} \delta^2 \\ \Omega_{u_c} \delta \end{pmatrix} + O(\delta^5), \quad (833)$$

and the w_c -variation initial vector is

$$\partial_{w_c} y(\delta) = \begin{pmatrix} \frac{1}{2} \delta^2 + \frac{1}{24} \Omega_{w_c} \delta^4 \\ \delta + \frac{1}{6} \Omega_{w_c} \delta^3 \\ 1 + \frac{1}{2} \Omega_{w_c} \delta^2 \\ \Omega_{w_c} \delta \end{pmatrix} + O(\delta^5). \quad (834)$$

This is the correct exact local initialization for the shooting Jacobian.

CCCLXIX. 7. TWO-DIMENSIONAL NEWTON SHOOTING

At fixed χ_{II} , solve

$$D\mathcal{S}(u_c^{(n)}, w_c^{(n)}; \chi_{II}) \begin{pmatrix} \delta u_c^{(n)} \\ \delta w_c^{(n)} \end{pmatrix} = -\mathcal{S}(u_c^{(n)}, w_c^{(n)}; \chi_{II}). \quad (835)$$

Update by

$$u_c^{(n+1)} = u_c^{(n)} + \eta_n \delta u_c^{(n)}, \quad w_c^{(n+1)} = w_c^{(n)} + \eta_n \delta w_c^{(n)}, \quad (836)$$

with backtracking damping $0 < \eta_n \leq 1$.

CCCLXX. 8. CONTINUATION SCHEDULE IN χ_{II}

After solving the base branch at $\chi_{II} = 0$, use continuation

$$\chi_{II}^{(k+1)} = \chi_{II}^{(k)} + \Delta\chi. \quad (837)$$

The robust first schedule is

$$\Delta\chi = 0.02 \chi_{II}^{\text{target}} \quad (838)$$

for the first 10 steps, then

$$\Delta\chi = 0.05 \chi_{II}^{\text{target}} \quad (839)$$

once Newton converges in 1 or 2 damped steps.

Use the converged center data $(u_c^{(k)}, w_c^{(k)})$ as the predictor for the next step.

CCCLXXI. 9. ACCEPTANCE AT THE SHOOTING LEVEL

At each continuation step, accept the center pair if

$$\|\mathcal{S}(u_c, w_c; \chi_{II})\|_2 < 10^{-10}, \quad (840)$$

and the first-integral defect satisfies

$$\|\mathcal{I}[u]\|_\infty < 10^{-8}. \quad (841)$$

At the final target χ_{II} , also require refinement stability:

$$\frac{|u_c^{(2N)} - u_c^{(N)}|}{|u_c^{(2N)}|} < 10^{-4}, \quad \frac{|w_c^{(2N)} - w_c^{(N)}|}{|w_c^{(2N)}|} < 10^{-4}. \quad (842)$$

CCCLXXII. 10. IMMEDIATE EXECUTABLE ORDER

So the exact immediate executable order is:

$$(i) \text{ solve } \mathcal{S}(u_c, w_c; 0) = 0 \text{ from the dark seed } (u_c^{(0)}, w_c^{(0)}) = (0, 2u_\infty m_*^2), \quad (843)$$

$$(ii) \text{ continue the root in } \chi_{II} > 0, \quad (844)$$

$$(iii) \text{ reconstruct } u(\xi) \text{ and hence } \Psi_{IA}(x) = u(\xi) S_{BCC}(x) e_1. \quad (845)$$

At this point Stage IA is fully executable as a two-parameter Newton shooting solve.

CCCLXXIII. FINAL SUMMARY

The Stage-IA scalar branch is now a fully explicit two-parameter Newton shooting problem. The free center data are

$$u_c = u(0), \quad w_c = u''(0),$$

with symmetry fixing

$$u'(0) = 0, \quad w'(0) = 0.$$

The dark base seed is

$$u_c^{(0)} = 0, \quad w_c^{(0)} = 2u_\infty m_*^2.$$

To avoid numerical issues at $\xi = 0$, start integration at

$$\delta = \frac{10^{-3}}{m_*},$$

using the exact local Taylor start

$$u(\delta) = u_c + \frac{w_c}{2}\delta^2 + \frac{\Omega}{24}\delta^4, \quad u'(\delta) = w_c\delta + \frac{\Omega}{6}\delta^3,$$

$$w(\delta) = w_c + \frac{\Omega}{2}\delta^2, \quad w'(\delta) = \Omega\delta,$$

where

$$\Omega = \frac{1}{YC_2} \left[A_2 w_c - M_2 u_c + 2\chi_{II} C_4 u_c^2 w_c + U_3 u_c^3 - \gamma C_6 u_c^5 \right].$$

The exact shooting map is

$$\mathcal{S}(u_c, w_c; \chi_{II}) = \begin{pmatrix} u(L_\xi; u_c, w_c, \chi_{II}) - u_\infty(\chi_{II}) \\ w(L_\xi; u_c, w_c, \chi_{II}) \end{pmatrix},$$

and the branch is found by solving

$$\mathcal{S}(u_c, w_c; \chi_{II}) = 0$$

with two-dimensional damped Newton.

After solving at $\chi_{II} = 0$, continue in small χ_{II} -steps to the target value.

So Stage IA is now no longer a general BVP description. It is a literal two-variable root-finding problem with an explicit seed, explicit local start, and explicit continuation rule.

CCCLXXIV. STAGE IA CONTINUATION PREDICTOR AROUND $\chi_{II} = 0$

CCCLXXV. GOAL

Assume the base branch $u_0(\xi)$ at $\chi_{II} = 0$ has already been obtained from

$$\boxed{-YC_2 u_0'''' + A_2 u_0'' - M_2 u_0 - \lambda C_4 u_0^3 - \gamma C_6 u_0^5 = 0.} \quad (846)$$

The next exact step is to derive the first continuation predictor for $\chi_{II} > 0$.

CCCLXXVI. 1. FULL STAGE-IA EQUATION

The full Stage-IA scalar equation is

$$-Y C_2 u'''' + A_2 u'' - M_2 u + 2\chi_{II} C_4 u^2 u'' + 2\chi_{II} C_4 u (u')^2 + (\chi_{II} C_\Delta - \lambda C_4) u^3 - \gamma C_6 u^5 = 0. \quad (847)$$

We expand

$$u(\xi; \chi_{II}) = u_0(\xi) + \chi_{II} u_1(\xi) + O(\chi_{II}^2). \quad (848)$$

CCCLXXVII. 2. FIRST-ORDER CONTINUATION EQUATION

Substitute (848) into (847) and use (846). At order $O(\chi_{II})$, one obtains

$$-Y C_2 u_1'''' + A_2 u_1'' + V_0(\xi) u_1 + S_0(\xi) = 0, \quad (849)$$

where the linear potential is

$$V_0(\xi) = -M_2 - 3\lambda C_4 u_0(\xi)^2 - 5\gamma C_6 u_0(\xi)^4, \quad (850)$$

and the explicit source is

$$S_0(\xi) = 2C_4 u_0(\xi)^2 u_0''(\xi) + 2C_4 u_0(\xi) (u_0'(\xi))^2 + C_\Delta u_0(\xi)^3. \quad (851)$$

Thus the first continuation predictor is governed by a linear inhomogeneous fourth-order ODE on the known background u_0 .

CCCLXXVIII. 3. VACUUM CONTINUATION

Let

$$u_\infty(\chi_{II}) = u_{\infty,0} + \chi_{II} u_{\infty,1} + O(\chi_{II}^2).$$

The full vacuum equation is

$$-M_2 u_\infty + (\chi_{II} C_\Delta - \lambda C_4) u_\infty^3 - \gamma C_6 u_\infty^5 = 0. \quad (852)$$

Differentiating at $\chi_{II} = 0$ gives

$$V_{\infty,0} u_{\infty,1} + C_\Delta u_{\infty,0}^3 = 0, \quad (853)$$

with

$$V_{\infty,0} = -M_2 - 3\lambda C_4 u_{\infty,0}^2 - 5\gamma C_6 u_{\infty,0}^4. \quad (854)$$

Hence

$$u_{\infty,1} = -\frac{C_\Delta u_{\infty,0}^3}{V_{\infty,0}}. \quad (855)$$

CCCLXXIX. 4. BOUNDARY CONDITIONS FOR THE PREDICTOR

For the even branch,

$$u_1'(0) = 0, \quad u_1'''(0) = 0. \quad (856)$$

At infinity,

$$u_1(\infty) = u_{\infty,1}, \quad u_1''(\infty) = 0. \quad (857)$$

Thus u_1 is a well-posed linear BVP once u_0 is known.

CCCLXXX. 5. TWO-FIELD FORM FOR THE PREDICTOR

Introduce

$$\boxed{v := u_1, \quad z := u_1''} \quad (858)$$

Then the continuation predictor system becomes

$$\boxed{v'' = z,} \quad (859)$$

$$\boxed{-YC_2 z'' + A_2 z + V_0(\xi) v + S_0(\xi) = 0.} \quad (860)$$

This is the clean solver form for the first predictor.

CCCLXXXI. 6. PREDICTOR FOR THE CENTER DATA

Write the shooting data as functions of χ_{II} :

$$\boxed{u_c(\chi_{II}) = u_{c,0} + \chi_{II} \dot{u}_c + O(\chi_{II}^2), \quad w_c(\chi_{II}) = w_{c,0} + \chi_{II} \dot{w}_c + O(\chi_{II}^2).} \quad (861)$$

The full shooting map is

$$\boxed{\mathcal{S}(u_c, w_c; \chi_{II}) = \begin{pmatrix} u(L_\xi; u_c, w_c, \chi_{II}) - u_\infty(\chi_{II}) \\ w(L_\xi; u_c, w_c, \chi_{II}) \end{pmatrix}.} \quad (862)$$

Differentiating the identity

$$\mathcal{S}(u_c(\chi_{II}), w_c(\chi_{II}); \chi_{II}) = 0$$

at $\chi_{II} = 0$, we get the exact predictor equation

$$\boxed{D_{(u_c, w_c)} \mathcal{S}(u_{c,0}, w_{c,0}; 0) \begin{pmatrix} \dot{u}_c \\ \dot{w}_c \end{pmatrix} + \partial_{\chi_{II}} \mathcal{S}(u_{c,0}, w_{c,0}; 0) = 0.} \quad (863)$$

Therefore

$$\boxed{\begin{pmatrix} \dot{u}_c \\ \dot{w}_c \end{pmatrix} = - \left[D_{(u_c, w_c)} \mathcal{S}(u_{c,0}, w_{c,0}; 0) \right]^{-1} \partial_{\chi_{II}} \mathcal{S}(u_{c,0}, w_{c,0}; 0).} \quad (864)$$

This is the exact first-order center-data predictor for continuation.

CCCLXXXII. 7. PREDICTOR-CORRECTOR STEP

Given the converged base solution at $\chi_{II} = 0$, define the first predictor

$$\boxed{u_c^{\text{pred}} = u_{c,0} + \Delta\chi \dot{u}_c, \quad w_c^{\text{pred}} = w_{c,0} + \Delta\chi \dot{w}_c.} \quad (865)$$

Then use $(u_c^{\text{pred}}, w_c^{\text{pred}})$ as the initial guess for the full Newton corrector at $\chi_{II} = \Delta\chi$:

$$\boxed{\mathcal{S}(u_c, w_c; \Delta\chi) = 0.} \quad (866)$$

CCCLXXXIII. 8. HONEST STATUS

At this point the Stage-IA route is fully explicit:

1. solve the base branch u_0 at $\chi_{II} = 0$;
2. solve the linear predictor problem (849) or equivalently (859)–(860);
3. extract $(\dot{u}_c, \dot{w}_c)$ from (864);
4. run predictor-corrector continuation in χ_{II} .

Nothing conceptual remains open at the Stage-IA scalar level.

CCCLXXXIV. FINAL SUMMARY

The first continuation step from $\chi_{II} = 0$ is now fixed exactly.

Expanding

$$u(\xi; \chi_{II}) = u_0(\xi) + \chi_{II} u_1(\xi) + O(\chi_{II}^2),$$

the first-order response u_1 satisfies the linear inhomogeneous ODE

$$\boxed{-Y C_2 u_1'''' + A_2 u_1'' + V_0(\xi) u_1 + S_0(\xi) = 0,}$$

with

$$V_0(\xi) = -M_2 - 3\lambda C_4 u_0(\xi)^2 - 5\gamma C_6 u_0(\xi)^4,$$

$$S_0(\xi) = 2C_4 u_0(\xi)^2 u_0''(\xi) + 2C_4 u_0(\xi) (u_0'(\xi))^2 + C_\Delta u_0(\xi)^3.$$

The vacuum also continues explicitly:

$$u_\infty(\chi_{II}) = u_{\infty,0} + \chi_{II} u_{\infty,1} + O(\chi_{II}^2), \quad u_{\infty,1} = -\frac{C_\Delta u_{\infty,0}^3}{V_{\infty,0}}.$$

At the shooting level, the center data predictor is

$$\boxed{\begin{pmatrix} \dot{u}_c \\ \dot{w}_c \end{pmatrix} = -\left[D_{(u_c, w_c)} \mathcal{S}(u_{c,0}, w_{c,0}; 0) \right]^{-1} \partial_{\chi_{II}} \mathcal{S}(u_{c,0}, w_{c,0}; 0).}$$

So the Stage-IA continuation is now a standard predictor-corrector problem: solve the base branch, solve the linear predictor equation, predict (u_c, w_c) , then correct by full Newton at the next χ_{II} -value.

CCCLXXXV. STAGE IA AUGMENTED VARIATIONAL INTEGRATION FOR THE CONTINUATION PREDICTOR

CCCLXXXVI. GOAL

We now derive the explicit executable formula for

$$\partial_{\chi_{II}} \mathcal{S}(u_{c,0}, w_{c,0}; 0),$$

so that the continuation predictor

$$\begin{pmatrix} \dot{u}_c \\ \dot{w}_c \end{pmatrix} = -\left[D_{(u_c, w_c)} \mathcal{S}(u_{c,0}, w_{c,0}; 0) \right]^{-1} \partial_{\chi_{II}} \mathcal{S}(u_{c,0}, w_{c,0}; 0)$$

can be computed directly by one augmented variational integration.

CCCLXXXVII. 1. STAGE-IA FIRST-ORDER SYSTEM

Recall the first-order system

$$\boxed{y'_1 = y_2, \quad y'_2 = y_3, \quad y'_3 = y_4,} \quad (867)$$

$$\boxed{y'_4 = f(y; \chi_{II}) := \frac{1}{YC_2} \left[A_2 y_3 - M_2 y_1 + 2\chi_{II} C_4 y_1^2 y_3 + 2\chi_{II} C_4 y_1 y_2^2 + U_3 y_1^3 - \gamma C_6 y_1^5 \right],} \quad (868)$$

with

$$\boxed{U_3 = \chi_{II} C_\Delta - \lambda C_4.} \quad (869)$$

The shooting map is

$$\boxed{\mathcal{S}(u_c, w_c; \chi_{II}) = \begin{pmatrix} y_1(L_\xi; u_c, w_c, \chi_{II}) - u_\infty(\chi_{II}) \\ y_3(L_\xi; u_c, w_c, \chi_{II}) \end{pmatrix}.} \quad (870)$$

CCCLXXXVIII. 2. BASE-CENTER INITIALIZATION AT $\xi = \delta$

At $\xi = \delta$, the local Taylor start is

$$\boxed{y(\delta; u_c, w_c, \chi_{II}) = \begin{pmatrix} u_c + \frac{w_c}{2} \delta^2 + \frac{\Omega}{24} \delta^4 \\ w_c \delta + \frac{\Omega}{6} \delta^3 \\ w_c + \frac{\Omega}{2} \delta^2 \\ \Omega \delta \end{pmatrix} + O(\delta^5),} \quad (871)$$

where

$$\boxed{\Omega(u_c, w_c; \chi_{II}) = \frac{1}{YC_2} \left[A_2 w_c - M_2 u_c + 2\chi_{II} C_4 u_c^2 w_c + U_3 u_c^3 - \gamma C_6 u_c^5 \right].} \quad (872)$$

CCCLXXXIX. 3. TWO CENTER-DATA VARIATIONS

Define the two variational fields

$$a(\xi) := \partial_{u_c} y(\xi), \quad b(\xi) := \partial_{w_c} y(\xi).$$

They satisfy

$$\boxed{a'_1 = a_2, \quad a'_2 = a_3, \quad a'_3 = a_4,} \quad (873)$$

$$\boxed{a'_4 = f_{y_1} a_1 + f_{y_2} a_2 + f_{y_3} a_3,} \quad (874)$$

and similarly

$$\boxed{b'_1 = b_2, \quad b'_2 = b_3, \quad b'_3 = b_4,} \quad (875)$$

$$\boxed{b'_4 = f_{y_1} b_1 + f_{y_2} b_2 + f_{y_3} b_3.} \quad (876)$$

The coefficients are

$$f_{y_1} = \frac{1}{YC_2} \left[-M_2 + 4\chi_{II} C_4 y_1 y_3 + 2\chi_{II} C_4 y_2^2 + 3U_3 y_1^2 - 5\gamma C_6 y_1^4 \right], \quad (877)$$

$$f_{y_2} = \frac{1}{YC_2} \left[4\chi_{II} C_4 y_1 y_2 \right], \quad (878)$$

$$f_{y_3} = \frac{1}{YC_2} \left[A_2 + 2\chi_{II} C_4 y_1^2 \right]. \quad (879)$$

At $\xi = \delta$, the exact local initial data are

$$a(\delta) = \begin{pmatrix} 1 + \frac{1}{24}\Omega_{u_c}\delta^4 \\ \frac{1}{6}\Omega_{u_c}\delta^3 \\ \frac{1}{2}\Omega_{u_c}\delta^2 \\ \Omega_{u_c}\delta \end{pmatrix} + O(\delta^5), \quad (880)$$

$$b(\delta) = \begin{pmatrix} \frac{1}{2}\delta^2 + \frac{1}{24}\Omega_{w_c}\delta^4 \\ \delta + \frac{1}{6}\Omega_{w_c}\delta^3 \\ 1 + \frac{1}{2}\Omega_{w_c}\delta^2 \\ \Omega_{w_c}\delta \end{pmatrix} + O(\delta^5), \quad (881)$$

with

$$\Omega_{u_c} = \frac{1}{YC_2} \left[-M_2 + 4\chi_{II} C_4 u_c w_c + 3U_3 u_c^2 - 5\gamma C_6 u_c^4 \right], \quad (882)$$

$$\Omega_{w_c} = \frac{1}{YC_2} \left[A_2 + 2\chi_{II} C_4 u_c^2 \right]. \quad (883)$$

CCCXC. 4. EXPLICIT χ_{II} -SENSITIVITY FIELD

Now define the genuine continuation-sensitivity field

$$p(\xi) := \partial_{\chi_{II}} y(\xi).$$

Then

$$p'_1 = p_2, \quad p'_2 = p_3, \quad p'_3 = p_4, \quad (884)$$

and

$$p'_4 = f_{y_1} p_1 + f_{y_2} p_2 + f_{y_3} p_3 + f_\chi, \quad (885)$$

where the explicit forcing is

$$f_\chi = \frac{1}{YC_2} \left[2C_4 y_1^2 y_3 + 2C_4 y_1 y_2^2 + C_\Delta y_1^3 \right]. \quad (886)$$

So p is obtained by integrating one inhomogeneous linear system on top of the base branch y .

CCCXCI. 5. INITIAL DATA FOR THE χ_{II} -SENSITIVITY

At fixed center data (u_c, w_c) , the only χ_{II} -dependence of the local start enters through Ω . Hence

$$\Omega_\chi = \frac{1}{YC_2} \left[2C_4 u_c^2 w_c + C_\Delta u_c^3 \right]. \quad (887)$$

Therefore the exact local initial data for p are

$$p(\delta) = \begin{pmatrix} \frac{1}{24} \Omega_\chi \delta^4 \\ \frac{1}{6} \Omega_\chi \delta^3 \\ \frac{1}{2} \Omega_\chi \delta^2 \\ \Omega_\chi \delta \end{pmatrix} + O(\delta^5). \quad (888)$$

CCCXCII. 6. EXACT FORMULAS FOR THE SHOOTING JACOBIAN AND $\partial_{\chi_{II}} \mathcal{S}$

Once y, a, b, p are integrated from $\xi = \delta$ to $\xi = L_\xi$, we obtain

$$D_{(u_c, w_c)} \mathcal{S} = \begin{pmatrix} a_1(L_\xi) & b_1(L_\xi) \\ a_3(L_\xi) & b_3(L_\xi) \end{pmatrix}, \quad (889)$$

and

$$\partial_{\chi_{II}} \mathcal{S} = \begin{pmatrix} p_1(L_\xi) - u_{\infty,1} \\ p_3(L_\xi) \end{pmatrix}, \quad (890)$$

where

$$u_{\infty,1} = -\frac{C_\Delta u_{\infty,0}^3}{V_{\infty,0}}, \quad V_{\infty,0} = -M_2 - 3\lambda C_4 u_{\infty,0}^2 - 5\gamma C_6 u_{\infty,0}^4. \quad (891)$$

This is the explicit executable formula we needed.

CCCXCIII. 7. EXACT CENTER-DATA PREDICTOR

The continuation predictor is therefore

$$\begin{pmatrix} \dot{u}_c \\ \dot{w}_c \end{pmatrix} = - \begin{pmatrix} a_1(L_\xi) & b_1(L_\xi) \\ a_3(L_\xi) & b_3(L_\xi) \end{pmatrix}^{-1} \begin{pmatrix} p_1(L_\xi) - u_{\infty,1} \\ p_3(L_\xi) \end{pmatrix}. \quad (892)$$

Then for a continuation step $\Delta\chi$,

$$u_c^{\text{pred}} = u_{c,0} + \Delta\chi \dot{u}_c, \quad w_c^{\text{pred}} = w_{c,0} + \Delta\chi \dot{w}_c. \quad (893)$$

CCCXCIV. 8. MINIMAL AUGMENTED IVP

Thus the entire Stage-IA predictor-corrector can be implemented by integrating the following augmented IVP on $[\delta, L_\xi]$:

$$\underbrace{y}_4 \oplus \underbrace{a}_4 \oplus \underbrace{b}_4 \oplus \underbrace{p}_4$$

that is, a 16-dimensional first-order system:

$$(y, a, b, p)' = \mathcal{F}_{\text{aug}}(y, a, b, p; \chi_{II}). \quad (894)$$

No further formal reduction is needed.

CCCXCV. 9. IMMEDIATE NEXT MOVE

The immediate executable order is now exactly:

$$\boxed{\text{(i) solve the base branch } y_0,} \quad (895)$$

$$\boxed{\text{(ii) integrate } (a, b, p) \text{ on top of } y_0,} \quad (896)$$

$$\boxed{\text{(iii) compute } \dot{u}_c, \dot{w}_c \text{ from (892),} \quad (897)$$

$$\boxed{\text{(iv) use } (u_c^{\text{pred}}, w_c^{\text{pred}}) \text{ as the Newton corrector seed at } \chi_{II} = \Delta\chi.} \quad (898)$$

CCCXCVI. FINAL SUMMARY

The missing continuation ingredient is now fully explicit.

Let $y = (u, u', w, w')$ solve the Stage-IA first-order system. Then the two center-data variations

$$a = \partial_{u_c} y, \quad b = \partial_{w_c} y$$

satisfy the homogeneous variational system, while the true continuation sensitivity

$$p = \partial_{\chi_{II}} y$$

satisfies the inhomogeneous system

$$\begin{aligned} p'_1 &= p_2, & p'_2 &= p_3, & p'_3 &= p_4, \\ p'_4 &= f_{y_1} p_1 + f_{y_2} p_2 + f_{y_3} p_3 + f_\chi, \end{aligned}$$

with

$$f_\chi = \frac{1}{YC_2} \left[2C_4 y_1^2 y_3 + 2C_4 y_1 y_2^2 + C_\Delta y_1^3 \right].$$

Its local initial data at $\xi = \delta$ are

$$p(\delta) = \begin{pmatrix} \frac{1}{24} \Omega_\chi \delta^4 \\ \frac{1}{6} \Omega_\chi \delta^3 \\ \frac{1}{2} \Omega_\chi \delta^2 \\ \Omega_\chi \delta \end{pmatrix}, \quad \Omega_\chi = \frac{1}{YC_2} \left[2C_4 u_c^2 w_c + C_\Delta u_c^3 \right].$$

After integrating y, a, b, p to L_ξ , the exact formulas are

$$\begin{aligned} D_{(u_c, w_c)} \mathcal{S} &= \begin{pmatrix} a_1(L_\xi) & b_1(L_\xi) \\ a_3(L_\xi) & b_3(L_\xi) \end{pmatrix}, \\ \partial_{\chi_{II}} \mathcal{S} &= \begin{pmatrix} p_1(L_\xi) - u_{\infty,1} \\ p_3(L_\xi) \end{pmatrix}. \end{aligned}$$

Therefore the exact continuation predictor is

$$\boxed{\begin{pmatrix} \dot{u}_c \\ \dot{w}_c \end{pmatrix} = - \left[D_{(u_c, w_c)} \mathcal{S} \right]^{-1} \partial_{\chi_{II}} \mathcal{S}.}$$

So the full Stage-IA predictor-corrector is now reduced to a single augmented 16-dimensional first-order integration plus a 2×2 linear solve.

CCCXCVII. STAGE IA LOCAL CONTINUATION THEOREM AT $\chi_{II} = 0$

CCCXCVIII. GOAL

The Stage-IA branch has already been reduced to the two-parameter shooting equation

$$\mathcal{S}(u_c, w_c; \chi_{II}) = 0.$$

The remaining nonredundant step is to close the local continuation structure around the base solution at $\chi_{II} = 0$.

We now make the following explicit:

1. the fully simplified augmented IVP at $\chi_{II} = 0$,
2. the exact 2×2 invertibility condition for the shooting Jacobian,
3. the resulting local continuation theorem,
4. the special simplification for the dark-core branch.

CCCXCIX. 1. BASE FIRST-ORDER SYSTEM AT $\chi_{II} = 0$

At $\chi_{II} = 0$, the Stage-IA first-order system is

$$\boxed{y'_1 = y_2, \quad y'_2 = y_3, \quad y'_3 = y_4,} \quad (899)$$

$$\boxed{y'_4 = f_0(y) = \frac{1}{YC_2} \left[A_2 y_3 - M_2 y_1 - \lambda C_4 y_1^3 - \gamma C_6 y_1^5 \right].} \quad (900)$$

The local Taylor start at $\xi = \delta$ is

$$\boxed{y(\delta; u_c, w_c) = \begin{pmatrix} u_c + \frac{w_c}{2} \delta^2 + \frac{\Omega_0}{24} \delta^4 \\ w_c \delta + \frac{\Omega_0}{6} \delta^3 \\ w_c + \frac{\Omega_0}{2} \delta^2 \\ \Omega_0 \delta \end{pmatrix} + O(\delta^5),} \quad (901)$$

with

$$\boxed{\Omega_0(u_c, w_c) = \frac{1}{YC_2} \left[A_2 w_c - M_2 u_c - \lambda C_4 u_c^3 - \gamma C_6 u_c^5 \right].} \quad (902)$$

CD. 2. HOMOGENEOUS VARIATIONAL PAIR AT THE BASE POINT

Define the two center-data variations

$$a(\xi) = \partial_{u_c} y(\xi), \quad b(\xi) = \partial_{w_c} y(\xi).$$

At $\chi_{II} = 0$, they satisfy

$$\boxed{a'_1 = a_2, \quad a'_2 = a_3, \quad a'_3 = a_4,} \quad (903)$$

$$\boxed{a'_4 = V_0(\xi) a_1 + \frac{A_2}{YC_2} a_3,} \quad (904)$$

and similarly

$$\boxed{b'_1 = b_2, \quad b'_2 = b_3, \quad b'_3 = b_4,} \quad (905)$$

$$\boxed{b'_4 = V_0(\xi) b_1 + \frac{A_2}{YC_2} b_3,} \quad (906)$$

where the base linear potential is

$$\boxed{V_0(\xi) = \frac{1}{YC_2} \left[-M_2 - 3\lambda C_4 y_{1,0}(\xi)^2 - 5\gamma C_6 y_{1,0}(\xi)^4 \right].} \quad (907)$$

Their local starts at $\xi = \delta$ are

$$\boxed{a(\delta) = \begin{pmatrix} 1 + \frac{1}{24}\Omega_{0,u}\delta^4 \\ \frac{1}{6}\Omega_{0,u}\delta^3 \\ \frac{1}{2}\Omega_{0,u}\delta^2 \\ \Omega_{0,u}\delta \end{pmatrix} + O(\delta^5),} \quad (908)$$

$$\boxed{b(\delta) = \begin{pmatrix} \frac{1}{2}\delta^2 + \frac{1}{24}\Omega_{0,w}\delta^4 \\ \delta + \frac{1}{6}\Omega_{0,w}\delta^3 \\ 1 + \frac{1}{2}\Omega_{0,w}\delta^2 \\ \Omega_{0,w}\delta \end{pmatrix} + O(\delta^5),} \quad (909)$$

with

$$\boxed{\Omega_{0,u} = \frac{1}{YC_2} \left[-M_2 - 3\lambda C_4 u_c^2 - 5\gamma C_6 u_c^4 \right],} \quad (910)$$

$$\boxed{\Omega_{0,w} = \frac{A_2}{YC_2}.} \quad (911)$$

CDI. 3. INHOMOGENEOUS χ_{II} -SENSITIVITY AT THE BASE POINT

Define

$$p(\xi) = \partial_{\chi_{II}} y(\xi) \Big|_{\chi_{II}=0}.$$

Then p satisfies

$$\boxed{p'_1 = p_2, \quad p'_2 = p_3, \quad p'_3 = p_4,} \quad (912)$$

$$\boxed{p'_4 = V_0(\xi) p_1 + \frac{A_2}{YC_2} p_3 + s_0(\xi),} \quad (913)$$

with explicit source

$$\boxed{s_0(\xi) = \frac{1}{YC_2} \left[2C_4 y_{1,0}(\xi)^2 y_{3,0}(\xi) + 2C_4 y_{1,0}(\xi) y_{2,0}(\xi)^2 + C_\Delta y_{1,0}(\xi)^3 \right].} \quad (914)$$

The local initial data are

$$p(\delta) = \begin{pmatrix} \frac{1}{24}\Omega_{\chi,0}\delta^4 \\ \frac{1}{6}\Omega_{\chi,0}\delta^3 \\ \frac{1}{2}\Omega_{\chi,0}\delta^2 \\ \Omega_{\chi,0}\delta \end{pmatrix} + O(\delta^5), \quad (915)$$

with

$$\Omega_{\chi,0} = \frac{1}{YC_2} \left[2C_4 u_c^2 w_c + C_\Delta u_c^3 \right]. \quad (916)$$

CDII. 4. EXACT SHOOTING JACOBIAN AND PARAMETER SENSITIVITY

Let $(u_{c,0}, w_{c,0})$ be the converged base branch at $\chi_{II} = 0$. Then the shooting Jacobian is

$$D\mathcal{S}_0 = \begin{pmatrix} a_1(L_\xi) & b_1(L_\xi) \\ a_3(L_\xi) & b_3(L_\xi) \end{pmatrix}, \quad (917)$$

and the χ_{II} -sensitivity is

$$\partial_{\chi_{II}} \mathcal{S}_0 = \begin{pmatrix} p_1(L_\xi) - u_{\infty,1} \\ p_3(L_\xi) \end{pmatrix}, \quad (918)$$

where

$$u_{\infty,1} = -\frac{C_\Delta u_{\infty,0}^3}{V_{\infty,0}}, \quad V_{\infty,0} = -M_2 - 3\lambda C_4 u_{\infty,0}^2 - 5\gamma C_6 u_{\infty,0}^4. \quad (919)$$

CDIII. 5. INVERTIBILITY CRITERION AND LOCAL CONTINUATION

Define the determinant

$$\Delta_{\text{shoot}} := \det D\mathcal{S}_0 = a_1(L_\xi)b_3(L_\xi) - b_1(L_\xi)a_3(L_\xi). \quad (920)$$

If

$$\Delta_{\text{shoot}} \neq 0, \quad (921)$$

then by the implicit function theorem there exists a unique local continuation branch

$$(u_c(\chi_{II}), w_c(\chi_{II})) \quad (922)$$

in a neighborhood of $\chi_{II} = 0$, satisfying

$$\mathcal{S}(u_c(\chi_{II}), w_c(\chi_{II}); \chi_{II}) = 0.$$

Moreover,

$$\begin{pmatrix} \dot{u}_c \\ \dot{w}_c \end{pmatrix} = -(D\mathcal{S}_0)^{-1} \partial_{\chi_{II}} \mathcal{S}_0. \quad (923)$$

This is the exact first predictor.

CDIV. 6. DARK-CORE SIMPLIFICATION

If the converged base branch is the dark-core branch with

$$\boxed{u_{c,0} = 0}, \quad (924)$$

then the local χ_{II} -start simplifies sharply:

$$\boxed{\Omega_{\chi,0} = 0}. \quad (925)$$

So in the dark-core case,

$$\boxed{p(\delta) = 0 + O(\delta^5)}, \quad (926)$$

and the entire χ_{II} -sensitivity is generated by the bulk forcing $s_0(\xi)$ and the vacuum shift $u_{\infty,1}$, not by the local center Taylor start.

This is numerically favorable.

CDV. 7. PREDICTOR-CORRECTOR STEP

For a small continuation step $\Delta\chi$, define

$$\boxed{u_c^{\text{pred}} = u_{c,0} + \Delta\chi \dot{u}_c, \quad w_c^{\text{pred}} = w_{c,0} + \Delta\chi \dot{w}_c}. \quad (927)$$

Then solve the full nonlinear shooting equation at $\chi_{II} = \Delta\chi$:

$$\boxed{\mathcal{S}(u_c, w_c; \Delta\chi) = 0} \quad (928)$$

using $(u_c^{\text{pred}}, w_c^{\text{pred}})$ as the initial Newton guess.

CDVI. 8. SHARP STATUS

The Stage-IA scalar branch is now closed at the local-continuation level.

What remains is no longer formal derivation. It is the actual numerical integration of:

$$y_0, \quad a, \quad b, \quad p$$

on $[\delta, L_\xi]$, the evaluation of Δ_{shoot} , and the construction of the first predictor step (927).

CDVII. FINAL SUMMARY

At $\chi_{II} = 0$, the Stage-IA base branch y_0 satisfies the autonomous first-order system

$$y'_1 = y_2, \quad y'_2 = y_3, \quad y'_3 = y_4, \quad y'_4 = \frac{1}{YC_2}(A_2 y_3 - M_2 y_1 - \lambda C_4 y_1^3 - \gamma C_6 y_1^5).$$

On top of this base branch, the two center-data variations a, b satisfy homogeneous linear systems, while the χ_{II} -sensitivity p satisfies the inhomogeneous linear system

$$p'_1 = p_2, \quad p'_2 = p_3, \quad p'_3 = p_4,$$

$$p'_4 = V_0(\xi)p_1 + \frac{A_2}{YC_2}p_3 + s_0(\xi),$$

with

$$V_0(\xi) = \frac{1}{YC_2} [-M_2 - 3\lambda C_4 y_{1,0}^2 - 5\gamma C_6 y_{1,0}^4],$$

$$s_0(\xi) = \frac{1}{YC_2} [2C_4 y_{1,0}^2 y_{3,0} + 2C_4 y_{1,0} y_{2,0}^2 + C_\Delta y_{1,0}^3].$$

After integrating to L_ξ , one gets

$$D\mathcal{S}_0 = \begin{pmatrix} a_1(L_\xi) & b_1(L_\xi) \\ a_3(L_\xi) & b_3(L_\xi) \end{pmatrix}, \quad \partial_{\chi_{II}} \mathcal{S}_0 = \begin{pmatrix} p_1(L_\xi) - u_{\infty,1} \\ p_3(L_\xi) \end{pmatrix}.$$

Therefore the exact first continuation predictor is

$$\begin{pmatrix} \dot{u}_c \\ \dot{w}_c \end{pmatrix} = -(D\mathcal{S}_0)^{-1} \partial_{\chi_{II}} \mathcal{S}_0.$$

If

$$\det D\mathcal{S}_0 \neq 0,$$

then a unique local continuation branch $(u_c(\chi_{II}), w_c(\chi_{II}))$ exists near $\chi_{II} = 0$.

So the Stage-IA problem is now reduced to one augmented integration and one 2×2 inversion.

CDVIII. FROM THE ACCEPTED STAGE-IA BRANCH TO THE COMMON BLOCH EXTRACTOR

CDIX. SHORT KOREAN EXPLANATION

The goal is not to keep refining a working PDE forever. The goal is to obtain one accepted actual branch and read all physical quantities from that same branch.

At the current working Class II level, the canonical Stage-IA branch is already tangent-consistent. Therefore, once the scalar Stage-IA branch is numerically accepted, the next immediate step is not another branch split. It is the common Bloch extractor.

CDX. 1. ACCEPTED STAGE-IA BRANCH

Assume the scalar Stage-IA solve has converged and produced

$$u_{IA}(\xi). \tag{929}$$

Then the corresponding actual branch candidate is

$$\Psi_{IA}(x) = u_{IA}(\xi) S_{BCC}(x) e_1. \tag{930}$$

At the current working Class II level with the canonical internal choice $z_0 = e_1$, this branch is already tangent-consistent, so it is the first legitimate input to the common extractor.

CDXI. 2. BLOCH LINEARIZATION AROUND THE ACCEPTED BRANCH

Linearize the full microscopic stationary PDE around Ψ_{IA} :

$$\Psi(x, t) = \Psi_{IA}(x) + \eta(x, t). \tag{931}$$

The exact linearized Bloch operator is then

$$L_{\text{Bloch}}(k) = \mathcal{H}_{\Psi\Psi}^{\text{lin}}[\Psi_{\text{IA}}; k], \quad (932)$$

where k is the Bloch momentum around the chosen Bragg point.

Choose the physical expansion point

$$k_* = \frac{G_*}{2}, \quad k = k_* + p, \quad |p| \ll |G_*|. \quad (933)$$

CDXII. 3. FIRST-DERIVATIVE TRANSPORT OPERATORS

Define the exact Bloch-derivative operators

$$K_i := \partial_{k_i} L_{\text{Bloch}}(k) \Big|_{k=k_*}. \quad (934)$$

These are the actual branch-dependent transport operators.

Next choose the isolated low carrier projector

$$P_* \quad (935)$$

onto the accepted doubled low block.

Then the effective first-order symbol is

$$A_1(p) = \sum_i p_i P_* K_i P_*. \quad (936)$$

CDXIII. 4. CANONICAL TRANSVERSE DECOMPOSITION

Introduce the longitudinal and rotated transverse coordinates

$$p_{\parallel}, \quad q_1 := \frac{p_1 - p_2}{\sqrt{2}}, \quad q_2 := \frac{p_1 + p_2}{\sqrt{2}}. \quad (937)$$

The effective Dirac-type decomposition is then

$$P_* K_{\parallel} P_* = \lambda_c \sigma_3, \quad (938)$$

$$P_* K_{q_1} P_* = \Lambda_E \sigma_1 + \Lambda_O \sigma_2, \quad (939)$$

$$P_* K_{q_2} P_* = -\Lambda_O \sigma_1 + \Lambda_E \sigma_2. \quad (940)$$

Equivalently, the scalar extractor formulas are

$$\lambda_c = \frac{1}{2} \text{Tr}(\sigma_3 P_* K_{\parallel} P_*), \quad (941)$$

$$\Lambda_E = \frac{1}{2} \text{Tr}(\sigma_1 P_* K_{q_1} P_*) = \frac{1}{2} \text{Tr}(\sigma_2 P_* K_{q_2} P_*), \quad (942)$$

$$\Lambda_O = \frac{1}{2} \text{Tr}(\sigma_2 P_* K_{q_1} P_*) = -\frac{1}{2} \text{Tr}(\sigma_1 P_* K_{q_2} P_*). \quad (943)$$

CDXIV. 5. REDUCED COEFFICIENT DICTIONARY

Once $(\lambda_c, \Lambda_E, \Lambda_O)$ are extracted from the accepted branch, the reduced carrier coefficients are given by the exact overlap dictionary

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O. \quad (944)$$

Using the explicit defect overlaps,

$$I_{\theta} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3. \quad (945)$$

Hence

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3, \quad (946)$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \Lambda_E, \quad \beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \Lambda_O. \quad (947)$$

CDXV. 6. ACTUAL MICROSCOPIC CARRIER PDE ON THE ACCEPTED BRANCH

Therefore the actual reduced carrier PDE extracted from the accepted Stage-IA branch is

$$\left[i\partial_t - \lambda_{\parallel} \sigma_3 (-i\partial_{\parallel}) - \alpha (\sigma_1 (-i\partial_{q_1}) + \sigma_2 (-i\partial_{q_2})) - \beta (\sigma_1 (-i\partial_{q_2}) + \sigma_2 (-i\partial_{q_1})) \right] \chi = 0. \quad (948)$$

This is no longer a symbolic target. It is the actual branch-level PDE obtained from the accepted branch through the common extractor.

CDXVI. 7. CONNECTION TO THE FULL PROJECT GOAL

Once the accepted branch Ψ_{IA} is frozen, the same common extractor pipeline feeds:

$$(\lambda_{\parallel}, \alpha, \beta) \quad (949)$$

for the carrier PDE,

$$\Delta_{\text{bench}, \rho}^{\text{fin}}, \eta_{R, \rho}, \Delta_{00}, L_*, (c_E, c_O), (C_E, C_O) \quad (950)$$

for the carrier gates,
and

$$W_2, m_*, g_{\text{eff}}, Z_{\text{cub}} \quad (951)$$

for the anisotropy/hierarchy side.

So the project goal is still exactly the same:

$$\text{one accepted actual branch} \rightarrow \text{one common extractor} \rightarrow \text{full certificate.}$$

CDXVII. 8. HONEST STATUS

At the current point, the path is correct and aligned with the final goal.

What is not yet done is the actual numerical acceptance of the Stage-IA branch and the subsequent Bloch extraction. But the route is no longer drifting:

$$\boxed{\text{solve Stage IA} \rightarrow \text{freeze } \Psi_{\text{IA}} \rightarrow \text{compute } L_{\text{Bloch}}(k), K_i, P_* K_i P_*}.$$

CDXVIII. FINAL SUMMARY

Yes: the goal is still to obtain an accepted actual branch and extract the actual microscopic carrier PDE from it. The exact route is

$$\boxed{\Psi_{\text{actual}}(x) \rightarrow L_{\text{Bloch}}(k) \rightarrow K_i = \partial_{k_i} L_{\text{Bloch}}|_{k_*} \rightarrow P_* K_i P_* \rightarrow (\lambda_c, \Lambda_E, \Lambda_O) \rightarrow (\lambda_{\parallel}, \alpha, \beta)}.$$

For the accepted Stage-IA branch

$$\Psi_{\text{IA}}(x) = u_{\text{IA}}(\xi) S_{\text{BCC}}(x) e_1,$$

the extractor is

$$\lambda_c = \frac{1}{2} \text{Tr}(\sigma_3 P_* K_{\parallel} P_*),$$

$$\Lambda_E = \frac{1}{2} \text{Tr}(\sigma_1 P_* K_{q_1} P_*) = \frac{1}{2} \text{Tr}(\sigma_2 P_* K_{q_2} P_*),$$

$$\Lambda_O = \frac{1}{2} \text{Tr}(\sigma_2 P_* K_{q_1} P_*) = -\frac{1}{2} \text{Tr}(\sigma_1 P_* K_{q_2} P_*).$$

Then

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3, \quad \alpha = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \Lambda_E, \quad \beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \Lambda_O.$$

So we are going toward the correct goal. The remaining immediate task is not another conceptual rewrite, but the accepted numerical solve of Stage IA and the common Bloch extractor built on top of it.

CDXIX. BLOCH COMMON EXTRACTOR ON THE ACCEPTED STAGE-IA BRANCH

CDXX. SHORT KOREAN EXPLANATION

The accepted Stage-IA branch is no longer just a nonlinear solve target. It is now the common microscopic background from which the carrier PDE coefficients must be extracted.

So the next legitimate step is not another branch split. It is the explicit Bloch-fiber construction and the projected derivative audit on that same accepted branch.

CDXXI. 1. ACCEPTED BACKGROUND AND FLUCTUATION SPACE

Assume the Stage-IA solve has converged and produced the accepted branch

$$\boxed{\Psi_{\text{IA}}(x) = u_{\text{IA}}(\xi) S_{\text{BCC}}(x) e_1.} \tag{952}$$

We linearize the working refined-locked + Class II microscopic branch

$$\boxed{L_{\text{work}} = L_{\Psi}^{\text{ref}}[\Psi] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \Psi]} \tag{953}$$

around Ψ_{IA} , using the fluctuation pair

$$(\delta\Psi, \delta X).$$

CDXXII. 2. BLOCH FIBER HESSIAN

On the accepted periodic background, the Bloch fiber quadratic operator is

$$\mathcal{H}_{\text{Bloch}}(k) = \begin{pmatrix} H_{\Psi\Psi}(k) & H_{\Psi X}(k) \\ H_{X\Psi}(k) & H_{XX}(k) \end{pmatrix}. \quad (954)$$

At the current working Class II level, the heavy block is

$$H_{XX}(k) = M_X^2 \mathbf{1}_X \quad (955)$$

up to subleading background-induced algebraic corrections, and the mixed block is

$$H_{\Psi X}(k) = \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k), \quad (956)$$

where $\mathcal{J}(k)$ and $\mathcal{K}(k)$ are the Bloch-space linearized current vertices on the accepted branch. :contentReference[oaicite:2]index=2 :contentReference[oaicite:3]index=3

CDXXIII. 3. EXACT EFFECTIVE BLOCH OPERATOR

Eliminating the heavy mediator gives the exact quadratic effective Bloch operator

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - H_{\Psi X}(k) H_{XX}(k)^{-1} H_{X\Psi}(k). \quad (957)$$

At the strict working level $H_{XX}(k) = M_X^2 \mathbf{1}_X$, this becomes

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - \frac{1}{M_X^2} (\alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k)) (\alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k))^\dagger. \quad (958)$$

CDXXIV. 4. FOLDED BRAGG CONVENTION AND SHELL PROJECTION

We work in the folded Bragg convention

$$k_* = \frac{G_*}{2}, \quad k = k_* + p, \quad |p| \ll |G_*|. \quad (959)$$

Let Π_{S_*} be the projector onto the near-resonant shell pair S_* . Then the shell operator is

$$H_{\text{shell}}(p) = \Pi_{S_*} H_{\text{eff}}(k_* + p) \Pi_{S_*}. \quad (960)$$

Let Γ be a contour around only the isolated low valley doublet of $H_{\text{shell}}(0)$. Then the exact valley projector is the Riesz projector

$$P_* = \frac{1}{2\pi i} \oint_{\Gamma} (z - H_{\text{shell}}(0))^{-1} dz. \quad (961)$$

This is the correct basis-independent definition of the local doubled carrier block. :contentReference[oaicite:4]index=4

CDXXV. 5. FIRST DERIVATIVE OPERATORS AND THEIR EXACT SPLIT

Define the Bloch derivatives

$$K_i := \partial_{k_i} H_{\text{eff}}(k) \Big|_{k=k_*}. \quad (962)$$

At the exact operator level, the derivative splits as

$$K_i = K_i^{(\text{bare})} + K_i^{(\text{II})} + K_i^{(\text{spec})}, \quad (963)$$

with

$$K_i^{(\text{bare})} = \partial_{k_i} H_{\Psi\Psi}(k) \Big|_{k_*}, \quad (964)$$

$$K_i^{(\text{II})} = - \left[(\partial_{k_i} H_{\Psi X}) H_{XX}^{-1} H_{X\Psi} + H_{\Psi X} H_{XX}^{-1} (\partial_{k_i} H_{X\Psi}) \right]_{k_*}, \quad (965)$$

$$K_i^{(\text{spec})} = \left[H_{\Psi X} H_{XX}^{-1} (\partial_{k_i} H_{XX}) H_{XX}^{-1} H_{X\Psi} \right]_{k_*}. \quad (966)$$

At the strict working level $H_{XX} = M_X^2 \mathbf{1}_X$, one has

$$K_i^{(\text{spec})} = 0, \quad (967)$$

so the load-bearing terms are reduced to

$$K_i = K_i^{(\text{bare})} + K_i^{(\text{II})}.$$

This is exactly the current honest decomposition of the full-PDE transport source. :contentReference[oaicite:5]index=5 :contentReference[oaicite:6]index=6

CDXXVI. 6. SHELL-ADAPTED PROJECTED MATRICES

Choose the shell-adapted orthonormal frame

$$\hat{n} = \frac{G_*}{|G_*|}, \quad \{e_1, e_2, \hat{n}\}. \quad (968)$$

Define the projected derivative matrices

$$M_i := P_* K_i P_*, \quad M_{\parallel} := \hat{n}_i M_i, \quad M_1 := e_{1i} M_i, \quad M_2 := e_{2i} M_i. \quad (969)$$

Then the first-order projected symbol is

$$A_1(p) = p_i M_i + O(|p|^2). \quad (970)$$

CDXXVII. 7. PAULI EXTRACTION OF THE MICROSCOPIC COEFFICIENTS

On $\text{Ran}(P_*) \cong \mathbb{C}^2$, choose the Pauli basis $\sigma_1, \sigma_2, \sigma_3$. Then the microscopic Bloch-extracted coefficients are

$$\lambda_c = \frac{1}{2} \text{Tr}(\sigma_3 M_{\parallel}), \quad (971)$$

$$\Lambda_E = \frac{1}{2} \text{Tr}(\sigma_1 M_1) = \frac{1}{2} \text{Tr}(\sigma_2 M_2), \quad (972)$$

$$\Lambda_O = \frac{1}{2} \text{Tr}(\sigma_2 M_1) = -\frac{1}{2} \text{Tr}(\sigma_1 M_2). \quad (973)$$

Equivalently, the reduced first-order symbol is

$$A_1(p) = \lambda_c p_{\parallel} \sigma_3 + \Lambda_E (p_1 \sigma_1 + p_2 \sigma_2) + \Lambda_O (-p_2 \sigma_1 + p_1 \sigma_2). \quad (974)$$

This is the finite shell-projected coefficient-matching stage in explicit form. :contentReference[oaicite:7]index=7 :contentReference[oaicite:8]index=8

CDXXVIII. 8. REDUCED CARRIER PDE COEFFICIENTS

After the Bloch extraction, the reduced carrier coefficients are obtained by the already-fixed overlap dictionary

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O. \quad (975)$$

Using the explicit overlaps,

$$I_{\theta} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3, \quad (976)$$

we obtain

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3, \quad (977)$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \Lambda_E, \quad \beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \Lambda_O. \quad (978)$$

So the actual reduced microscopic carrier PDE is fixed once $(\lambda_c, \Lambda_E, \Lambda_O)$ are read from the accepted branch. :contentReference[oaicite:9]index=9

CDXXIX. 9. EXACT IMMEDIATE COMPUTATIONAL TARGETS

Therefore the next explicit computational targets on the accepted Stage-IA branch are the four shell quantities

$$\widehat{U}_{\text{eff}}(G_*), \quad \widehat{T}_{\text{eff}}^1(G_*), \quad \widehat{T}_{\text{eff}}^2(G_*), \quad v_{\parallel}. \quad (979)$$

Equivalently, one computes the branch-specific reciprocal-harmonic matrix elements of

$$\mathcal{J}(k), \quad \mathcal{K}(k), \quad H_{\text{rem,rem}}(k)$$

at the intervalley harmonic $Q = G_*$, and then assembles K_i , M_i , and finally $(\lambda_c, \Lambda_E, \Lambda_O)$. :contentReference[oaicite:10]index=10

CDXXX. 10. HONEST STATUS

At this point the Bloch/operator pipeline is not conceptually open.

What remains is not another theorem, but the actual branch-specific derivative-and-projection computation:

$$H_{\Psi\Psi}(k), \quad H_{\Psi X}(k), \quad H_{XX}(k) \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow (\lambda_c, \Lambda_E, \Lambda_O).$$

So the path is correct, and the immediate next move is concrete: freeze the accepted Stage-IA branch and compute the Bloch derivative matrices on that branch.

CDXXXI. FINAL SUMMARY

Once the accepted Stage-IA branch

$$\Psi_{\text{IA}}(x) = u_{\text{IA}}(\xi) S_{\text{BCC}}(x) e_1$$

is available, the correct next step is the common Bloch extractor.

The exact effective Bloch operator is

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - H_{\Psi X}(k) H_{XX}(k)^{-1} H_{X\Psi}(k),$$

with

$$H_{\Psi X}(k) = \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k), \quad H_{XX}(k) \approx M_X^2 \mathbf{1}_X.$$

Its derivative at the folded Bragg point $k_* = G_*/2$ is

$$K_i = \partial_{k_i} H_{\text{eff}}(k) \big|_{k=k_*} = K_i^{(\text{bare})} + K_i^{(\text{II})} + K_i^{(\text{spec})},$$

and at the strict working level $K_i^{(\text{spec})} = 0$.

Projecting to the isolated valley block,

$$M_i = P_* K_i P_*,$$

the microscopic coefficients are

$$\lambda_c = \frac{1}{2} \text{Tr}(\sigma_3 M_{\parallel}), \quad \Lambda_E = \frac{1}{2} \text{Tr}(\sigma_1 M_1) = \frac{1}{2} \text{Tr}(\sigma_2 M_2), \quad \Lambda_O = \frac{1}{2} \text{Tr}(\sigma_2 M_1) = -\frac{1}{2} \text{Tr}(\sigma_1 M_2).$$

Then the reduced carrier coefficients are

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3, \quad \alpha = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \Lambda_E, \quad \beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \Lambda_O.$$

So the immediate forward problem is now completely concrete:

$$\boxed{\Psi_{\text{IA}} \rightarrow H_{\text{eff}}(k) \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow (\lambda_c, \Lambda_E, \Lambda_O).}$$

CDXXXII. EXPLICIT BLOCH-HARMONIC MATRIX ELEMENTS OF $H_{\Psi X}(k)$ ON THE ACCEPTED STAGE-IA BRANCH

CDXXXIII. SHORT KOREAN EXPLANATION

The accepted Stage-IA branch is

$$\Psi_{\text{IA}}(x) = \Phi_{\text{IA}}(x) e_1, \quad \Phi_{\text{IA}}(x) = u_{\text{IA}}(\xi) S_{\text{BCC}}(x).$$

The next real microscopic step is to build the mixed Bloch block

$$H_{\Psi X}(k) = \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k)$$

on this branch.

At the canonical internal choice $z_0 = e_1$, the J -vertex is explicit in both the longitudinal and tangent sectors, while the K -vertex vanishes on the pure longitudinal scalar channel and survives only in the tangent channel through the profile-gradient term.

CDXXXIV. 1. ACCEPTED BACKGROUND AND INTERNAL FRAME

Fix the accepted Stage-IA branch

$$\boxed{\Psi_{\text{IA}}(x) = \Phi_{\text{IA}}(x) e_1, \quad \Phi_{\text{IA}}(x) = u_{\text{IA}}(\xi) S_{\text{BCC}}(x).} \quad (980)$$

Use the canonical internal frame

$$\boxed{e_1, \quad e_2, \quad e_3,} \quad (981)$$

and define the tangent basis

$$\boxed{u_1 := e_2, \quad u_2 := e_3.} \quad (982)$$

Let

$$T^a = \frac{\lambda^a}{2} \quad (983)$$

be the Hermitian $SU(3)$ generators, and recall

$$n_0^a = e_1^\dagger T^a e_1. \quad (984)$$

Define also the tangent couplings

$$Q_\mu^a := u_\mu^\dagger T^a e_1, \quad \mu = 1, 2. \quad (985)$$

For the canonical choice e_1 , one has

$$n_0^3 = \frac{1}{2}, \quad n_0^8 = \frac{1}{2\sqrt{3}}, \quad n_0^a = 0 \quad (a \neq 3, 8),$$

while Q_μ^a is nonzero only for the off-diagonal generators that connect e_1 to e_2, e_3 .

CDXXXV. 2. BLOCH HARMONIC CONVENTIONS

Write the fluctuation in Bloch form

$$\delta\Psi(x) = e^{ik \cdot x} \sum_Q \psi_Q e^{iQ \cdot x}, \quad \delta X_i^a(x) = e^{ik \cdot x} \sum_Q x_{i,Q}^a e^{iQ \cdot x}. \quad (986)$$

For any background field $F(x)$, define the reciprocal-harmonic coefficient

$$\widehat{F}(G) := \frac{1}{|\Omega|} \int_\Omega e^{-iG \cdot x} F(x) d^3x. \quad (987)$$

Then multiplication by $F(x)$ in real space becomes convolution in shell/Bloch space:

$$\langle Q|F|Q' \rangle = \widehat{F}(Q - Q'). \quad (988)$$

CDXXXVI. 3. MIXED BLOCK FROM THE J -VERTEX

At the working Class II level, the mixed interaction is

$$L_{\text{int}}^{(J)} = \alpha_X X_i^a J_i^a[\Psi], \quad J_i^a = \partial_i(\Psi^\dagger T^a \Psi). \quad (989)$$

Linearizing around $\Psi_{\text{IA}} = \Phi_{\text{IA}} e_1$, the variation of J_i^a with respect to $\delta\Psi$ gives the mixed block.

A. 3A. Longitudinal scalar channel

Restrict first to the longitudinal fluctuation

$$\delta\Psi_\parallel = \varphi e_1.$$

Then

$$\delta J_i^a|_\parallel = n_0^a \partial_i (\overline{\Phi_{\text{IA}}} \varphi + \Phi_{\text{IA}} \overline{\varphi}). \quad (990)$$

Equivalently, the block from δX to the longitudinal Ψ -sector is

$$(H_{\Psi X}^{(J,\parallel)}(k) x)_Q = -\alpha_X \sum_{Q', i, a} i(k_i + Q'_i) n_0^a \widehat{\Phi_{\text{IA}}}(Q - Q') x_{i, Q'}^a. \quad (991)$$

Thus the explicit Bloch-harmonic matrix element is

$$\langle Q, e_1 | H_{\Psi X}^{(J)}(k) | Q', i, a \rangle = -i\alpha_X (k_i + Q'_i) n_0^a \widehat{\Phi_{\text{IA}}}(Q - Q'). \quad (992)$$

This is the exact longitudinal mixed matrix element on the accepted Stage-IA branch.

B. 3B. Tangent channel

Now take the tangent fluctuation

$$\delta\Psi_\perp = \sum_{\mu=1}^2 \eta_\mu u_\mu.$$

Then

$$\delta J_i^a|_\perp = \partial_i \left(\overline{\Phi_{\text{IA}}} \eta_\mu Q_\mu^a + \Phi_{\text{IA}} \overline{\eta_\mu} \overline{Q_\mu^a} \right). \quad (993)$$

Hence the tangent mixed matrix elements are

$$\langle Q, u_\mu | H_{\Psi X}^{(J)}(k) | Q', i, a \rangle = -i\alpha_X (k_i + Q'_i) Q_\mu^a \widehat{\Phi_{\text{IA}}}(Q - Q'). \quad (994)$$

So the J -vertex is completely explicit on the accepted Stage-IA background.

CDXXXVII. 4. MIXED BLOCK FROM THE K -VERTEX

The second Class II interaction is

$$L_{\text{int}}^{(K)} = \beta_X X_i^a K_i^a, \quad K_i^a = \rho_\varepsilon \partial_i n^a. \quad (995)$$

On the accepted Stage-IA background,

$$n^a = n_0^a = \text{constant}, \quad K_i^a[\Psi_{\text{IA}}] = 0.$$

So the K -block has no background value, but its linearization survives in the tangent sector.

Define the profile-gradient field

$$\Xi_i(x) := \frac{\Phi_{\text{IA}}(x) \partial_i |\Phi_{\text{IA}}(x)|^2}{|\Phi_{\text{IA}}(x)|^2 + \varepsilon}. \quad (996)$$

Then the canonical tangent linearization gives

$$\langle Q, u_\mu | H_{\Psi X}^{(K)}(k) | Q', i, a \rangle = -\beta_X Q_\mu^a \widehat{\Xi}_i(Q - Q'). \quad (997)$$

By contrast, on the pure longitudinal scalar channel,

$$\langle Q, e_1 | H_{\Psi X}^{(K)}(k) | Q', i, a \rangle = 0 \quad (998)$$

at the current frozen-orientation Stage-IA level.

Therefore the exact working mixed block splits as

$$H_{\Psi X}(k) = H_{\Psi X}^{(J,\parallel)}(k) \oplus (H_{\Psi X}^{(J,\perp)}(k) + H_{\Psi X}^{(K,\perp)}(k)). \quad (999)$$

CDXXXVIII. 5. BRANCH-SPECIFIC HARMONIC DATA ACTUALLY NEEDED

Thus all branch dependence is reduced to the following three harmonic families on the accepted Stage-IA branch:

$$\boxed{\widehat{\Phi}_{\text{IA}}(G), \quad \widehat{\partial_i \Phi}_{\text{IA}}(G), \quad \widehat{\Xi}_i(G).} \quad (1000)$$

For the locked BCC branch

$$\Phi_{\text{IA}}(x) = u_{\text{IA}}(\xi) S_{\text{BCC}}(x),$$

the shell harmonics satisfy

$$\boxed{\widehat{\Phi}_{\text{IA}}(G) = \sum_{Q \in \mathcal{S}_{12}} \eta_Q \widehat{u}_{\text{IA}}(G - Q),} \quad (1001)$$

and similarly for $\widehat{\partial_i \Phi}_{\text{IA}}(G)$ and $\widehat{\Xi}_i(G)$.

So the intervalley extractor only needs those reciprocal coefficients at the physically relevant harmonics $G = G_*, 0, \pm 2G_*, \dots$, and in practice the load-bearing pieces are

$$\boxed{\widehat{\Phi}_{\text{IA}}(G_*), \quad \widehat{\Xi}_1(G_*), \quad \widehat{\Xi}_2(G_*).} \quad (1002)$$

CDXXXIX. 6. EFFECTIVE BLOCH OPERATOR AND DERIVATIVE AUDIT

Once the matrix elements above are assembled, the exact effective Bloch operator is

$$\boxed{H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - H_{\Psi X}(k) H_{XX}(k)^{-1} H_{X\Psi}(k).} \quad (1003)$$

At the strict working level,

$$\boxed{H_{XX}(k) \approx M_X^2 \mathbf{1}_X,} \quad (1004)$$

so

$$\boxed{H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - \frac{1}{M_X^2} H_{\Psi X}(k) H_{X\Psi}(k).} \quad (1005)$$

Then define

$$\boxed{K_i = \partial_{k_i} H_{\text{eff}}(k) \Big|_{k=k_*}, \quad M_i = P_* K_i P_*} \quad (1006)$$

Finally the microscopic coefficients are extracted by

$$\boxed{\lambda_c = \frac{1}{2} \text{Tr}(\sigma_3 M_{\parallel}), \quad \Lambda_E = \frac{1}{2} \text{Tr}(\sigma_1 M_1) = \frac{1}{2} \text{Tr}(\sigma_2 M_2), \quad \Lambda_O = \frac{1}{2} \text{Tr}(\sigma_2 M_1) = -\frac{1}{2} \text{Tr}(\sigma_1 M_2).} \quad (1007)$$

CDXL. 7. REDUCED CARRIER PDE AND BENCHMARK BRIDGE

The reduced dictionary remains

$$\boxed{\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = I_c \Lambda_E, \quad \beta = I_c \Lambda_O,} \quad (1008)$$

and the current explicit reference route is already fixed by

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = \beta = -\frac{\beta_X \gamma_{\perp}}{M_X^2} I_c,$$

with the matching conditions to the canonical mixed odd-active target also fixed. :contentReference[oaicite:1]index=1

Therefore the accepted-branch extractor is now aligned with the project goal:

$$\boxed{\Psi_{\text{IA}} \rightarrow H_{\Psi X}(k) \rightarrow H_{\text{eff}}(k) \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow (\lambda_c, \Lambda_E, \Lambda_O) \rightarrow (\lambda_{\parallel}, \alpha, \beta).} \quad (1009)$$

CDXLI. 8. HONEST STATUS

At this point the only thing not yet done is the literal branch-specific harmonic evaluation

$$\widehat{\Phi}_{\text{IA}}(G), \quad \widehat{\Xi}_i(G)$$

and the subsequent derivative/projection computation.

No new branch logic is needed. No new audit decomposition is needed. The remaining work is finite harmonic extraction on the accepted Stage-IA branch.

CDXLII. FINAL SUMMARY

On the accepted Stage-IA branch

$$\Psi_{\text{IA}}(x) = \Phi_{\text{IA}}(x)e_1, \quad \Phi_{\text{IA}}(x) = u_{\text{IA}}(\xi)S_{\text{BCC}}(x),$$

the mixed Bloch block $H_{\Psi X}(k)$ becomes explicit.

For the J -vertex,

$$L_{\text{int}}^{(J)} = \alpha_X X_i^a J_i^a,$$

the longitudinal scalar matrix element is

$$\langle Q, e_1 | H_{\Psi X}^{(J)}(k) | Q', i, a \rangle = -i\alpha_X (k_i + Q'_i) n_0^a \widehat{\Phi}_{\text{IA}}(Q - Q').$$

The tangent matrix element is

$$\langle Q, u_\mu | H_{\Psi X}^{(J)}(k) | Q', i, a \rangle = -i\alpha_X (k_i + Q'_i) Q_\mu^a \widehat{\Phi}_{\text{IA}}(Q - Q').$$

For the K -vertex,

$$L_{\text{int}}^{(K)} = \beta_X X_i^a K_i^a,$$

the longitudinal scalar channel vanishes at the frozen-orientation Stage-IA level,

$$\langle Q, e_1 | H_{\Psi X}^{(K)}(k) | Q', i, a \rangle = 0,$$

while the tangent matrix element is

$$\langle Q, u_\mu | H_{\Psi X}^{(K)}(k) | Q', i, a \rangle = -\beta_X Q_\mu^a \widehat{\Xi}_i(Q - Q'), \quad \Xi_i = \frac{\Phi_{\text{IA}} \partial_i |\Phi_{\text{IA}}|^2}{|\Phi_{\text{IA}}|^2 + \varepsilon}.$$

So the branch-specific data needed by the common extractor are reduced to the reciprocal harmonics

$$\widehat{\Phi}_{\text{IA}}(G), \quad \widehat{\Xi}_i(G).$$

From these one constructs

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - H_{\Psi X}(k)H_{XX}(k)^{-1}H_{X\Psi}(k), \quad K_i = \partial_{k_i} H_{\text{eff}}(k)|_{k_*}, \quad M_i = P_* K_i P_*,$$

and then extracts

$$\lambda_c, \quad \Lambda_E, \quad \Lambda_O.$$

So the immediate next problem is now completely concrete:

$$\text{evaluate } \widehat{\Phi}_{\text{IA}}(G) \text{ and } \widehat{\Xi}_i(G) \text{ on the accepted Stage-IA branch.}$$

CDXLIII. RECIPROCAL-HARMONIC EXTRACTION ON THE ACCEPTED STAGE-IA BRANCH

CDXLIV. 1. ACCEPTED BRANCH AND LONGITUDINAL HARMONIC SUPPORT

Take the accepted Stage-IA branch

$$\boxed{\Psi_{\text{IA}}(x) = \Phi_{\text{IA}}(x)e_1, \quad \Phi_{\text{IA}}(x) = u_{\text{IA}}(\xi) S_{\text{BCC}}(x), \quad \xi = \hat{n} \cdot x.} \quad (1010)$$

Write the locked BCC shell factor as

$$\boxed{S_{\text{BCC}}(x) = \sum_{Q \in \mathcal{S}_{12}} \eta_Q e^{iQ \cdot x}.} \quad (1011)$$

For any reciprocal vector K , define the branch-profile harmonic

$$\boxed{U(K) := \widehat{u_{\text{IA}}}(K) = \frac{1}{|\Omega|} \int_{\Omega} e^{-iK \cdot x} u_{\text{IA}}(\hat{n} \cdot x) d^3x.} \quad (1012)$$

Because u_{IA} depends only on $\xi = \hat{n} \cdot x$, its Fourier support is purely longitudinal:

$$\boxed{U(K) = 0 \quad \text{unless} \quad K_{\perp} = 0.} \quad (1013)$$

Hence the exact reciprocal-harmonic coefficient of the accepted branch is

$$\boxed{\widehat{\Phi_{\text{IA}}}(G) = \sum_{Q \in \mathcal{S}_{12}} \eta_Q U(G - Q).} \quad (1014)$$

By (1013), only those Q with

$$(G - Q)_{\perp} = 0$$

contribute. Therefore the actual support is the Minkowski sum

$$\boxed{\text{supp } \widehat{\Phi_{\text{IA}}} \subset \mathcal{S}_{12} + \Lambda_{\parallel},} \quad (1015)$$

where $\Lambda_{\parallel}$ is the one-dimensional longitudinal reciprocal set.

For the load-bearing intervalley harmonic G_* , define

$$\boxed{\mathcal{I}_{\Phi}(G_*) := \{Q \in \mathcal{S}_{12} : (G_* - Q)_{\perp} = 0\}.} \quad (1016)$$

Then

$$\boxed{\widehat{\Phi_{\text{IA}}}(G_*) = \sum_{Q \in \mathcal{I}_{\Phi}(G_*)} \eta_Q U((G_* - Q)_{\parallel}).} \quad (1017)$$

CDXLV. 2. EXACT HARMONIC FORMULA FOR $|\Phi_{\text{IA}}|^2$

The exact density harmonics are

$$\boxed{|\widehat{\Phi_{\text{IA}}}|^2(G) = \sum_H \widehat{\Phi_{\text{IA}}}(H) \overline{\widehat{\Phi_{\text{IA}}}(H - G)}.} \quad (1018)$$

Equivalently, using (1014),

$$|\widehat{\Phi_{\text{IA}}}|^2(G) = \sum_{Q, Q' \in \mathcal{S}_{12}} \eta_Q \overline{\eta_{Q'}} \sum_H U(H - Q) \overline{U(H - G - Q')}. \quad (1019)$$

This shows that the density harmonics lie on

$$\text{supp } \widehat{|\Phi_{\text{IA}}|^2} \subset (\mathcal{S}_{12} - \mathcal{S}_{12}) + \Lambda_{\parallel}. \quad (1020)$$

Define the shell structure factors

$$B_{\Delta} := \sum_{\substack{Q, Q' \in \mathcal{S}_{12} \\ Q - Q' = \Delta}} \eta_Q \overline{\eta_{Q'}}. \quad (1021)$$

Then

$$|S_{\text{BCC}}(x)|^2 = \sum_{\Delta \in \Delta \mathcal{S}} B_{\Delta} e^{i\Delta \cdot x}, \quad \Delta \mathcal{S} := \mathcal{S}_{12} - \mathcal{S}_{12}. \quad (1022)$$

CDXLVI. 3. EXACT AND PRACTICAL FORMULAS FOR $\widehat{\Xi}_i(G)$

Recall

$$\Xi_i(x) = \frac{\Phi_{\text{IA}}(x) \partial_i |\Phi_{\text{IA}}(x)|^2}{|\Phi_{\text{IA}}(x)|^2 + \varepsilon}. \quad (1023)$$

Define

$$R(x) := \frac{1}{|\Phi_{\text{IA}}(x)|^2 + \varepsilon}, \quad \widehat{R}(G) := \frac{1}{|\Omega|} \int_{\Omega} e^{-iG \cdot x} R(x) d^3x. \quad (1024)$$

Since

$$\partial_i |\Phi_{\text{IA}}|^2 \wedge (G) = iG_i \widehat{|\Phi_{\text{IA}}|^2}(G),$$

the exact reciprocal-harmonic coefficient of Ξ_i is the triple convolution

$$\widehat{\Xi}_i(G) = \sum_{G_1 + G_2 + G_3 = G} \widehat{\Phi_{\text{IA}}}(G_1) i(G_2)_i \widehat{|\Phi_{\text{IA}}|^2}(G_2) \widehat{R}(G_3). \quad (1025)$$

This is the exact branch-specific harmonic formula.

CDXLVII. 4. STAGE-IA AVERAGED-DENOMINATOR APPROXIMATION

For the first explicit extractor pass, it is convenient to use the longitudinal averaged-denominator approximation

$$R(x) \approx R_0(\xi) := \frac{1}{C_2 u_{\text{IA}}(\xi)^2 + \varepsilon}. \quad (1026)$$

Then

$$|\Phi_{\text{IA}}|^2 = u_{\text{IA}}(\xi)^2 |S_{\text{BCC}}(x)|^2,$$

so

$$\partial_i |\Phi_{\text{IA}}|^2 = n_i (u_{\text{IA}}^2)'(\xi) |S_{\text{BCC}}|^2 + u_{\text{IA}}(\xi)^2 \partial_i |S_{\text{BCC}}|^2. \quad (1027)$$

Hence

$$\Xi_i(x) \approx n_i r_1(\xi) |S_{\text{BCC}}(x)|^2 S_{\text{BCC}}(x) + r_2(\xi) S_{\text{BCC}}(x) \partial_i |S_{\text{BCC}}(x)|^2, \quad (1028)$$

where

$$r_1(\xi) := \frac{u_{\text{IA}}(\xi) (u_{\text{IA}}^2)'(\xi)}{C_2 u_{\text{IA}}(\xi)^2 + \varepsilon}, \quad r_2(\xi) := \frac{u_{\text{IA}}(\xi)^3}{C_2 u_{\text{IA}}(\xi)^2 + \varepsilon}. \quad (1029)$$

Write the longitudinal harmonics

$$R_1(K) := \widehat{r}_1(K), \quad R_2(K) := \widehat{r}_2(K), \quad (1030)$$

again supported only on $K_\perp = 0$.

Using (1022) and

$$\partial_i |S_{\text{BCC}}|^2 = \sum_{\Delta \in \Delta\mathcal{S}} i\Delta_i B_\Delta e^{i\Delta \cdot x},$$

we obtain the practical explicit approximation

$$\widehat{\Xi}_i(G) \approx \sum_{Q \in \mathcal{S}_{12}} \sum_{\Delta \in \Delta\mathcal{S}} \eta_Q B_\Delta \left[n_i R_1(G - Q - \Delta) + i\Delta_i R_2(G - Q - \Delta) \right]. \quad (1031)$$

Because R_1, R_2 are longitudinally supported, only those pairs (Q, Δ) with

$$(G - Q - \Delta)_\perp = 0$$

actually contribute.

Define the corresponding admissible set

$$\mathcal{I}_\Xi(G) := \{(Q, \Delta) \in \mathcal{S}_{12} \times \Delta\mathcal{S} : (G - Q - \Delta)_\perp = 0\}. \quad (1032)$$

Then the load-bearing harmonic is

$$\widehat{\Xi}_i(G_*) \approx \sum_{(Q, \Delta) \in \mathcal{I}_\Xi(G_*)} \eta_Q B_\Delta \left[n_i R_1((G_* - Q - \Delta)_\parallel) + i\Delta_i R_2((G_* - Q - \Delta)_\parallel) \right]. \quad (1033)$$

CDXLVIII. 5. MIXED BLOCH BLOCK IN HARMONIC FORM

On the canonical Stage-IA branch, the mixed J -vertex matrix elements are

$$\langle G, e_1 | H_{\Psi X}^{(J)}(k) | G', i, a \rangle = -i\alpha_X (k_i + G'_i) n_0^a \widehat{\Phi}_{\text{IA}}(G - G'), \quad (1034)$$

$$\langle G, u_\mu | H_{\Psi X}^{(J)}(k) | G', i, a \rangle = -i\alpha_X (k_i + G'_i) Q_\mu^a \widehat{\Phi}_{\text{IA}}(G - G'). \quad (1035)$$

The K -vertex vanishes on the pure longitudinal scalar channel at the frozen-orientation Stage-IA level,

$$\langle G, e_1 | H_{\Psi X}^{(K)}(k) | G', i, a \rangle = 0, \quad (1036)$$

while on the tangent channel it is

$$\langle G, u_\mu | H_{\Psi X}^{(K)}(k) | G', i, a \rangle = -\beta_X Q_\mu^a \widehat{\Xi}_i(G - G'). \quad (1037)$$

Substituting (1014) and (1031) gives the fully branch-resolved mixed block

$$H_{\Psi X}(k) = H_{\Psi X}^{(J, \parallel)}(k) \oplus (H_{\Psi X}^{(J, \perp)}(k) + H_{\Psi X}^{(K, \perp)}(k)). \quad (1038)$$

CDXLIX. 6. IMMEDIATE LOAD-BEARING HARMONICS

Therefore the immediate branch-specific harmonic data that must be extracted numerically are

$$\boxed{\widehat{\Phi}_{\text{IA}}(G_*), \quad \widehat{\Xi}_1(G_*), \quad \widehat{\Xi}_2(G_*),} \quad (1039)$$

together with the corresponding near-shell harmonics needed to differentiate $H_{\text{eff}}(k)$ at $k_* = G_*/2$.

Once these are inserted, the remaining extractor chain is standard:

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - H_{\Psi X}(k)H_{XX}(k)^{-1}H_{X\Psi}(k), \quad K_i = \partial_{k_i} H_{\text{eff}}(k)|_{k_*}, \quad M_i = P_* K_i P_*.$$

Then

$$\lambda_c = \frac{1}{2} \text{Tr}(\sigma_3 M_{\parallel}), \quad \Lambda_E = \frac{1}{2} \text{Tr}(\sigma_1 M_1) = \frac{1}{2} \text{Tr}(\sigma_2 M_2), \quad \Lambda_O = \frac{1}{2} \text{Tr}(\sigma_2 M_1) = -\frac{1}{2} \text{Tr}(\sigma_1 M_2).$$

So the accepted-branch extractor has now been reduced to a finite reciprocal-harmonic evaluation problem.

CDL. FINAL SUMMARY

For the accepted Stage-IA branch

$$\Phi_{\text{IA}}(x) = u_{\text{IA}}(\xi) S_{\text{BCC}}(x),$$

the exact branch harmonic is

$$\boxed{\widehat{\Phi}_{\text{IA}}(G) = \sum_{Q \in \mathcal{S}_{12}} \eta_Q U(G - Q), \quad U(K) = \widehat{u}_{\text{IA}}(K),}$$

with the selection rule

$$U(K) = 0 \quad \text{unless} \quad K_{\perp} = 0.$$

The exact Ξ_i -harmonic is the triple convolution

$$\boxed{\widehat{\Xi}_i(G) = \sum_{G_1 + G_2 + G_3 = G} \widehat{\Phi}_{\text{IA}}(G_1) i(G_2)_i |\widehat{\Phi}_{\text{IA}}|^2(G_2) \widehat{R}(G_3),}$$

where

$$R = (|\Phi_{\text{IA}}|^2 + \varepsilon)^{-1}.$$

For the first practical extractor pass, the averaged-denominator approximation gives

$$\boxed{\widehat{\Xi}_i(G) \approx \sum_{Q \in \mathcal{S}_{12}} \sum_{\Delta \in \Delta \mathcal{S}} \eta_Q B_{\Delta} \left[n_i R_1(G - Q - \Delta) + i \Delta_i R_2(G - Q - \Delta) \right],}$$

where

$$R_1 = \frac{\widehat{u_{\text{IA}}(u_{\text{IA}}^2)}'}{C_2 u_{\text{IA}}^2 + \varepsilon}, \quad R_2 = \frac{\widehat{u_{\text{IA}}^3}}{C_2 u_{\text{IA}}^2 + \varepsilon},$$

again with purely longitudinal support.

Therefore the mixed Bloch block is now explicit:

$$\langle G, e_1 | H_{\Psi X}^{(J)}(k) | G', i, a \rangle = -i \alpha_X(k_i + G'_i) n_0^a \widehat{\Phi}_{\text{IA}}(G - G'),$$

$$\langle G, u_{\mu} | H_{\Psi X}^{(J)}(k) | G', i, a \rangle = -i \alpha_X(k_i + G'_i) Q_{\mu}^a \widehat{\Phi}_{\text{IA}}(G - G'),$$

$$\langle G, u_{\mu} | H_{\Psi X}^{(K)}(k) | G', i, a \rangle = -\beta_X Q_{\mu}^a \widehat{\Xi}_i(G - G'),$$

while the longitudinal K -channel vanishes at frozen-orientation Stage IA.

So the immediate next finite task is exactly:

$$\boxed{\text{evaluate } \widehat{\Phi}_{\text{IA}}(G_*), \widehat{\Xi}_1(G_*), \widehat{\Xi}_2(G_*) \text{ on the accepted Stage-IA branch.}}$$

CDLI. STAGE-IA BRANCH-SPECIFIC FOURIER OVERLAPS FOR THE COMMON EXTRACTOR

CDLII. GOAL

We now reduce the accepted Stage-IA branch to the finite Fourier overlaps that directly feed the common Bloch extractor.

The branch is

$$\boxed{\Psi_{\text{IA}}(x) = \Phi_{\text{IA}}(x)e_1, \quad \Phi_{\text{IA}}(x) = u_{\text{IA}}(\xi) S_{\text{BCC}}(x).} \quad (1040)$$

Define the density harmonic

$$\boxed{\rho(x) := |\Phi_{\text{IA}}(x)|^2, \quad \widehat{\rho}(Q) := |\widehat{\Phi_{\text{IA}}}^2(Q).} \quad (1041)$$

CDLIII. 1. STAGE-IA J -HARMONICS

On the canonical frozen-orientation Stage-IA branch, one has

$$J_i^A(x) = n_0^A \partial_i \rho(x),$$

so the exact reciprocal harmonic is

$$\boxed{\widehat{J}_i^A(Q) = iQ_i n_0^A \widehat{\rho}(Q).} \quad (1042)$$

Thus the J -sector is completely determined by the density harmonics $\widehat{\rho}(Q)$.

CDLIV. 2. STAGE-IA TANGENT K -HARMONICS

From the tangent-channel linearization on the accepted Stage-IA branch, the K -sector is encoded by the profile-gradient field $\Xi_i(x)$. Write the tangent projection coefficient as

$$\boxed{\widehat{K}_{i,\alpha}^A(Q) = \mathcal{Q}_\alpha^A \widehat{\Xi}_i(Q),} \quad (1043)$$

where $\alpha = 1, 2$ labels the two transverse tangent channels and $\mathcal{Q}_\alpha^A$ is the fixed internal shell-tangent coupling on the canonical e_1 -branch.

Therefore the entire transverse Class II overlap is determined by the pair

$$\widehat{\rho}(Q), \quad \widehat{\Xi}_i(Q).$$

CDLIV. 3. EXACT STAGE-IA CLASS II TRANSVERSE OVERLAP

The explicit folded-Bragg Class II overlap formula is

$$\boxed{\widehat{T}_{(II)}^\alpha(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \sum_A \sum_{Q+Q'=G_*} \widehat{J}_i^A(Q) \widehat{K}_{i,\alpha}^A(Q').} \quad (1044)$$

Substituting (1042) and (1043), we obtain

$$\boxed{\widehat{T}_{(II)}^\alpha(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \sum_A n_0^A \mathcal{Q}_\alpha^A \sum_{Q+Q'=G_*} iQ_i \widehat{\rho}(Q) \widehat{\Xi}_i(Q').} \quad (1045)$$

Define the fixed internal contraction

$$\Gamma_\alpha := \sum_A n_0^A \mathcal{Q}_\alpha^A. \quad (1046)$$

Then

$$\widehat{T}_{(II)}^\alpha(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \Gamma_\alpha \sum_{Q+Q'=G_*} i Q_i \widehat{\rho}(Q) \widehat{\Xi}_i(Q'). \quad (1047)$$

This is the exact Stage-IA branch-specific transverse overlap formula.

CDLVI. 4. PRACTICAL EXPLICIT FORM USING THE $\widehat{\Xi}_i$ -REDUCTION

From the previous Stage-IA harmonic reduction,

$$\widehat{\Xi}_i(Q') \approx \sum_{(Q_1, \Delta) \in \mathcal{I}_\Xi(Q')} \eta_{Q_1} B_\Delta \left[n_i R_1((Q' - Q_1 - \Delta)_\parallel) + i \Delta_i R_2((Q' - Q_1 - \Delta)_\parallel) \right],$$

where

$$R_1 = \frac{\widehat{u_{IA}(u_{IA}^2)'} }{C_2 u_{IA}^2 + \varepsilon}, \quad R_2 = \frac{\widehat{u_{IA}^3}}{C_2 u_{IA}^2 + \varepsilon}.$$

Substituting into (1047) gives

$$\widehat{T}_{(II)}^\alpha(G_*) \approx -\frac{\alpha_X \beta_X}{M_X^2} \Gamma_\alpha \sum_{Q+Q'=G_*} \widehat{\rho}(Q) \sum_{(Q_1, \Delta) \in \mathcal{I}_\Xi(Q')} \eta_{Q_1} B_\Delta \left[i(Q \cdot \hat{n}) R_1(L) - (Q \cdot \Delta) R_2(L) \right], \quad (1048)$$

where

$$L := (Q' - Q_1 - \Delta)_\parallel. \quad (1049)$$

So the transverse witness is now reduced to a finite shell sum dressed by one-dimensional longitudinal profile harmonics.

CDLVII. 5. EXACT STAGE-IA SCALAR BRAGG BLOCK

The scalar shell-pair block has the form

$$M_0 = \widehat{U}_{\text{eff}}(G_*) \mathbf{1}_2.$$

On the Stage-IA branch, the Class II contribution comes from the JJ -type elimination because the longitudinal K -channel vanishes at frozen orientation.

Thus

$$\widehat{U}_{(II)}(G_*) = -\frac{\alpha_X^2}{M_X^2} \sum_A \sum_{Q+Q'=G_*} \widehat{J}_i^A(Q) \widehat{J}_i^A(Q'). \quad (1050)$$

Using (1042),

$$\widehat{U}_{(II)}(G_*) = \frac{\alpha_X^2}{M_X^2} \left(\sum_A (n_0^A)^2 \right) \sum_{Q+Q'=G_*} (Q \cdot Q') \widehat{\rho}(Q) \widehat{\rho}(Q'). \quad (1051)$$

Define

$$\boxed{\nu_0 := \sum_A (n_0^A)^2.} \quad (1052)$$

Then

$$\boxed{\widehat{U}_{(II)}(G_*) = \frac{\alpha_X^2 \nu_0}{M_X^2} \sum_{Q+Q'=G_*} (Q \cdot Q') \widehat{\rho}(Q) \widehat{\rho}(Q').} \quad (1053)$$

Hence the scalar Bragg mass obstruction/survival test is also reduced to the same density harmonics $\widehat{\rho}(Q)$.

CDLVIII. 6. LOAD-BEARING STAGE-IA HARMONIC DATA

Therefore the accepted Stage-IA branch feeds the common extractor through the finite data set

$$\boxed{\widehat{\rho}(Q), \quad R_1(L), \quad R_2(L),} \quad (1054)$$

with the principal intervalley outputs

$$\boxed{\widehat{U}_{(II)}(G_*), \quad \widehat{T}_{(II)}^1(G_*), \quad \widehat{T}_{(II)}^2(G_*).} \quad (1055)$$

At the practical first-shell level, the load-bearing pieces are the harmonics with

$$Q, \quad Q' \text{ near } 0, \quad G_*,$$

because the intervalley selection rule enforces

$$Q + Q' = G_*.$$

CDLIX. 7. IMMEDIATE NEXT COMPUTATION

Thus the next literal computation on the accepted Stage-IA branch is:

$$\boxed{\text{(i) compute } \widehat{\rho}(Q), \quad \text{(ii) compute } R_1(L), R_2(L), \quad \text{(iii) assemble } \widehat{U}_{(II)}(G_*), \widehat{T}_{(II)}^1(G_*), \widehat{T}_{(II)}^2(G_*).} \quad (1056)$$

Once these are known, the rest of the common extractor is standard:

$$H_{\text{eff}}(k) \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow (\lambda_c, \Lambda_E, \Lambda_O) \rightarrow (\lambda_{\parallel}, \alpha, \beta).$$

CDLX. ONE-DIMENSIONAL PROFILE REDUCTION OF THE ACCEPTED STAGE-IA HARMONICS

CDLXI. SHORT KOREAN EXPLANATION

The accepted Stage-IA branch has the separated form

$$\Phi_{\text{IA}}(x) = u_{\text{IA}}(\xi) S_{\text{BCC}}(x), \quad \xi = \hat{n} \cdot x.$$

Therefore every branch-specific harmonic needed by the common extractor reduces to a finite BCC geometry sum dressed by one-dimensional Fourier coefficients of the profile $u_{\text{IA}}(\xi)$.

This is the exact point at which the accepted-branch extractor becomes a one-dimensional profile-data problem.

CDLXII. 1. ACCEPTED BRANCH AND ONE-DIMENSIONAL PROFILE TRANSFORMS

Take

$$\Phi_{\text{IA}}(x) = u(\xi) S(x), \quad \xi = \hat{n} \cdot x, \quad S(x) = \sum_{Q \in \mathcal{S}_{12}} \eta_Q e^{iQ \cdot x}. \quad (1057)$$

For any longitudinal reciprocal number $\ell \in \Lambda_{\parallel}$, define the one-dimensional profile transforms

$$\mathcal{U}_1(\ell) := \frac{1}{L_{\parallel}} \int_0^{L_{\parallel}} e^{-i\ell\xi} u(\xi) d\xi, \quad (1058)$$

$$\mathcal{U}_2(\ell) := \frac{1}{L_{\parallel}} \int_0^{L_{\parallel}} e^{-i\ell\xi} u(\xi)^2 d\xi, \quad (1059)$$

and

$$r_1(\xi) := \frac{u(\xi) (u(\xi)^2)'}{C_2 u(\xi)^2 + \varepsilon}, \quad r_2(\xi) := \frac{u(\xi)^3}{C_2 u(\xi)^2 + \varepsilon}, \quad (1060)$$

with transforms

$$\mathcal{R}_1(\ell) := \frac{1}{L_{\parallel}} \int_0^{L_{\parallel}} e^{-i\ell\xi} r_1(\xi) d\xi, \quad \mathcal{R}_2(\ell) := \frac{1}{L_{\parallel}} \int_0^{L_{\parallel}} e^{-i\ell\xi} r_2(\xi) d\xi. \quad (1061)$$

Because u , r_1 , and r_2 depend only on ξ , their full reciprocal-space coefficients have purely longitudinal support:

$$\widehat{u}(K) = \mathbf{1}_{K_{\perp}=0} \mathcal{U}_1(K_{\parallel}), \quad \widehat{u^2}(K) = \mathbf{1}_{K_{\perp}=0} \mathcal{U}_2(K_{\parallel}), \quad (1062)$$

$$\widehat{r_1}(K) = \mathbf{1}_{K_{\perp}=0} \mathcal{R}_1(K_{\parallel}), \quad \widehat{r_2}(K) = \mathbf{1}_{K_{\perp}=0} \mathcal{R}_2(K_{\parallel}). \quad (1063)$$

CDLXIII. 2. EXACT REDUCTION OF $\widehat{\Phi}(G)$

The accepted branch harmonic is

$$\widehat{\Phi}(G) = \sum_{Q \in \mathcal{S}_{12}} \eta_Q \widehat{u}(G - Q). \quad (1064)$$

Using (1062),

$$\widehat{\Phi}(G) = \sum_{Q \in \mathcal{S}_{12}} \eta_Q \mathbf{1}_{(G-Q)_{\perp}=0} \mathcal{U}_1((G-Q)_{\parallel}). \quad (1065)$$

So the branch harmonic is already a finite BCC shell sum dressed by the one-dimensional profile transform $\mathcal{U}_1$. In particular, at the load-bearing intervalley harmonic G_* ,

$$\widehat{\Phi}(G_*) = \sum_{Q \in \mathcal{I}_{\Phi}(G_*)} \eta_Q \mathcal{U}_1((G_* - Q)_{\parallel}), \quad (1066)$$

where

$$\mathcal{I}_{\Phi}(G_*) := \{Q \in \mathcal{S}_{12} : (G_* - Q)_{\perp} = 0\}. \quad (1067)$$

CDLXIV. 3. EXACT REDUCTION OF $\widehat{\rho}(G)$

Define

$$\rho(x) = |\Phi(x)|^2 = u(\xi)^2 |S(x)|^2. \quad (1068)$$

Write the BCC density shell factor as

$$|S(x)|^2 = \sum_{\Delta \in \Delta\mathcal{S}} B_{\Delta} e^{i\Delta \cdot x}, \quad \Delta\mathcal{S} := \mathcal{S}_{12} - \mathcal{S}_{12}, \quad (1069)$$

with structure coefficients

$$B_{\Delta} := \sum_{\substack{Q, Q' \in \mathcal{S}_{12} \\ Q - Q' = \Delta}} \eta_Q \overline{\eta_{Q'}}. \quad (1070)$$

Then

$$\rho(x) = \sum_{\Delta \in \Delta\mathcal{S}} B_{\Delta} u(\xi)^2 e^{i\Delta \cdot x}. \quad (1071)$$

Therefore

$$\widehat{\rho}(G) = \sum_{\Delta \in \Delta\mathcal{S}} B_{\Delta} \widehat{u^2}(G - \Delta). \quad (1072)$$

Using (1062),

$$\widehat{\rho}(G) = \sum_{\Delta \in \Delta\mathcal{S}} B_{\Delta} \mathbf{1}_{(G - \Delta)_{\perp} = 0} \mathcal{U}_2((G - \Delta)_{\parallel}). \quad (1073)$$

So $\widehat{\rho}(G)$ is fully reduced to the one-dimensional profile transform $\mathcal{U}_2$ plus pure BCC geometry. In particular,

$$\widehat{\rho}(G_*) = \sum_{\Delta \in \mathcal{I}_{\rho}(G_*)} B_{\Delta} \mathcal{U}_2((G_* - \Delta)_{\parallel}), \quad (1074)$$

where

$$\mathcal{I}_{\rho}(G_*) := \{\Delta \in \Delta\mathcal{S} : (G_* - \Delta)_{\perp} = 0\}. \quad (1075)$$

CDLXV. 4. EXACT REDUCTION OF $\widehat{\Xi}_i(G)$ TO ONE-DIMENSIONAL PROFILE DATA

In the averaged-denominator Stage-IA reduction, one has

$$\Xi_i(x) \approx n_i r_1(\xi) |S(x)|^2 S(x) + r_2(\xi) S(x) \partial_i |S(x)|^2. \quad (1076)$$

First define the two pure BCC geometry tensors

$$C_M^{(0)} := \sum_{\substack{Q \in \mathcal{S}_{12}, \Delta \in \Delta\mathcal{S} \\ Q + \Delta = M}} \eta_Q B_{\Delta}, \quad (1077)$$

$$C_{M,i}^{(1)} := \sum_{\substack{Q \in \mathcal{S}_{12}, \Delta \in \Delta \mathcal{S} \\ Q+\Delta=M}} i\Delta_i \eta_Q B_\Delta. \quad (1078)$$

Then the shell identities are

$$|S(x)|^2 S(x) = \sum_M C_M^{(0)} e^{iM \cdot x}, \quad (1079)$$

$$S(x) \partial_i |S(x)|^2 = \sum_M C_{M,i}^{(1)} e^{iM \cdot x}. \quad (1080)$$

Therefore (1076) becomes

$$\Xi_i(x) \approx \sum_M \left[n_i C_M^{(0)} r_1(\xi) + C_{M,i}^{(1)} r_2(\xi) \right] e^{iM \cdot x}. \quad (1081)$$

Taking the reciprocal harmonic and using (1063),

$$\widehat{\Xi}_i(G) \approx \sum_M \left[n_i C_M^{(0)} \mathbf{1}_{(G-M)_\perp=0} \mathcal{R}_1((G-M)_\parallel) + C_{M,i}^{(1)} \mathbf{1}_{(G-M)_\perp=0} \mathcal{R}_2((G-M)_\parallel) \right]. \quad (1082)$$

This is the desired one-dimensional profile reduction.

At the intervalley harmonic G_* ,

$$\widehat{\Xi}_i(G_*) \approx \sum_{M \in \mathcal{I}_\Xi(G_*)} \left[n_i C_M^{(0)} \mathcal{R}_1((G_* - M)_\parallel) + C_{M,i}^{(1)} \mathcal{R}_2((G_* - M)_\parallel) \right], \quad (1083)$$

where

$$\mathcal{I}_\Xi(G_*) := \{M : (G_* - M)_\perp = 0\}. \quad (1084)$$

So $\widehat{\Xi}_i(G_*)$ is now completely reduced to the one-dimensional profile transforms $\mathcal{R}_1, \mathcal{R}_2$ plus pure BCC geometry.

CDLXVI. 5. CLASS II INTERVALLEY OVERLAPS REDUCED TO ONE-DIMENSIONAL PROFILE DATA

The scalar intervalley overlap is

$$\widehat{U}_{(II)}(G_*) = \frac{\alpha_X^2 \nu_0}{M_X^2} \sum_{Q+Q'=G_*} (Q \cdot Q') \widehat{\rho}(Q) \widehat{\rho}(Q'), \quad \nu_0 := \sum_A (n_0^A)^2. \quad (1085)$$

Substituting (1073), $\widehat{U}_{(II)}(G_*)$ becomes a finite BCC geometry sum dressed only by $\mathcal{U}_2$. Likewise the transverse overlaps are

$$\widehat{T}_{(II)}^\alpha(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \Gamma_\alpha \sum_{Q+Q'=G_*} iQ_i \widehat{\rho}(Q) \widehat{\Xi}_i(Q'), \quad (1086)$$

with

$$\Gamma_\alpha := \sum_A n_0^A \mathcal{Q}_\alpha^A. \quad (1087)$$

Substituting (1073) and (1082), each $\widehat{T}_{(II)}^\alpha(G_*)$ becomes a finite BCC shell sum dressed only by

$$\mathcal{U}_2(\ell), \quad \mathcal{R}_1(\ell), \quad \mathcal{R}_2(\ell).$$

CDLXVII. 6. FINAL ONE-DIMENSIONAL PROFILE-DATA STATEMENT

Therefore the entire accepted-branch common extractor is reduced to the one-dimensional profile data set

$$\boxed{\mathcal{U}_1(\ell), \quad \mathcal{U}_2(\ell), \quad \mathcal{R}_1(\ell), \quad \mathcal{R}_2(\ell), \quad \ell \in \Lambda_{\parallel}}, \quad (1088)$$

together with fixed pure BCC geometry tensors

$$\eta_Q, \quad B_{\Delta}, \quad C_M^{(0)}, \quad C_{M,i}^{(1)}.$$

Then

$$\hat{\Phi}(G), \quad \hat{\rho}(G), \quad \hat{\Xi}_i(G)$$

are explicit finite sums, and so are

$$\hat{U}_{(II)}(G_*), \quad \hat{T}_{(II)}^1(G_*), \quad \hat{T}_{(II)}^2(G_*).$$

Once these are assembled, the remaining extractor chain is standard:

$$H_{\text{eff}}(k) \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow (\lambda_c, \Lambda_E, \Lambda_O) \rightarrow (\lambda_{\parallel}, \alpha, \beta).$$

CDLXVIII. FINAL SUMMARY

The accepted Stage-IA branch

$$\Phi_{\text{IA}}(x) = u(\xi) S_{\text{BCC}}(x)$$

reduces the branch-specific harmonic data to one-dimensional profile Fourier transforms.

The exact branch harmonic is

$$\hat{\Phi}(G) = \sum_{Q \in \mathcal{S}_{12}} \eta_Q \mathbf{1}_{(G-Q)_{\perp}=0} \mathcal{U}_1((G-Q)_{\parallel}).$$

The density harmonic is

$$\hat{\rho}(G) = \sum_{\Delta \in \Delta \mathcal{S}} B_{\Delta} \mathbf{1}_{(G-\Delta)_{\perp}=0} \mathcal{U}_2((G-\Delta)_{\parallel}).$$

The profile-gradient harmonic is

$$\hat{\Xi}_i(G) \approx \sum_M \left[n_i C_M^{(0)} \mathbf{1}_{(G-M)_{\perp}=0} \mathcal{R}_1((G-M)_{\parallel}) + C_{M,i}^{(1)} \mathbf{1}_{(G-M)_{\perp}=0} \mathcal{R}_2((G-M)_{\parallel}) \right].$$

Therefore the load-bearing Class II overlaps

$$\hat{U}_{(II)}(G_*), \quad \hat{T}_{(II)}^1(G_*), \quad \hat{T}_{(II)}^2(G_*)$$

are finite BCC geometry sums dressed only by the one-dimensional profile data

$$\mathcal{U}_1, \quad \mathcal{U}_2, \quad \mathcal{R}_1, \quad \mathcal{R}_2.$$

So the common extractor has now been reduced to a pure one-dimensional profile-data problem on the accepted Stage-IA branch.

CDLXIX. ONE-DIMENSIONAL FOURIER DATA OF THE ACCEPTED STAGE-IA SHOOTING SOLUTION

CDLXX. SHORT KOREAN EXPLANATION

We now reduce the accepted Stage-IA branch completely to one-dimensional Fourier data extracted from the shooting solution $u_{\text{IA}}(\xi)$.

Because the accepted branch is even in ξ , the required profile transforms split into cosine transforms for the even quantities and a sine transform for the odd derivative-current quantity.

This is the last reduction before actual numerical extraction.

CDLXXI. 1. EVEN EXTENSION AND LONGITUDINAL RECIPROCAL LATTICE

Let the accepted Stage-IA profile on $[0, L_\xi]$ be

$$u_{\text{IA}}(\xi).$$

Extend it evenly to $[-L_\xi, L_\xi]$, and then periodically with period $2L_\xi$.

The longitudinal reciprocal lattice is therefore

$$\ell_m = \frac{m\pi}{L_\xi}, \quad m \in \mathbb{Z}. \quad (1089)$$

For any even profile $f(\xi)$, the longitudinal Fourier coefficient is

$$\widehat{f}(\ell_m) = \frac{1}{2L_\xi} \int_{-L_\xi}^{L_\xi} e^{-i\ell_m \xi} f(\xi) d\xi = \frac{1}{L_\xi} \int_0^{L_\xi} f(\xi) \cos(\ell_m \xi) d\xi. \quad (1090)$$

For any odd profile $g(\xi)$,

$$\widehat{g}(\ell_m) = \frac{1}{2L_\xi} \int_{-L_\xi}^{L_\xi} e^{-i\ell_m \xi} g(\xi) d\xi = -\frac{i}{L_\xi} \int_0^{L_\xi} g(\xi) \sin(\ell_m \xi) d\xi. \quad (1091)$$

CDLXXII. 2. THE FOUR ONE-DIMENSIONAL PROFILE TRANSFORMS

Define the Stage-IA one-dimensional data by

$$\mathcal{U}_1^{(m)} := \frac{1}{L_\xi} \int_0^{L_\xi} u_{\text{IA}}(\xi) \cos(\ell_m \xi) d\xi, \quad (1092)$$

$$\mathcal{U}_2^{(m)} := \frac{1}{L_\xi} \int_0^{L_\xi} u_{\text{IA}}(\xi)^2 \cos(\ell_m \xi) d\xi, \quad (1093)$$

$$\mathcal{R}_2^{(m)} := \frac{1}{L_\xi} \int_0^{L_\xi} \frac{u_{\text{IA}}(\xi)^3}{C_2 u_{\text{IA}}(\xi)^2 + \varepsilon} \cos(\ell_m \xi) d\xi, \quad (1094)$$

and

$$\mathcal{R}_1^{(m)} := -\frac{i}{L_\xi} \int_0^{L_\xi} \frac{u_{\text{IA}}(\xi) (u_{\text{IA}}(\xi)^2)'}{C_2 u_{\text{IA}}(\xi)^2 + \varepsilon} \sin(\ell_m \xi) d\xi. \quad (1095)$$

These are exactly the four one-dimensional profile inputs required by the accepted-branch common extractor.

CDLXXIII. 3. STABLE REFORMULATION OF $\mathcal{R}_1^{(m)}$

The quantity $r_1(\xi)$ contains a derivative and is numerically less stable than the others if evaluated directly. So define

$$F(u) := \frac{2u}{C_2} - \frac{2\sqrt{\varepsilon}}{C_2^{3/2}} \arctan\left(\sqrt{\frac{C_2}{\varepsilon}} u\right). \quad (1096)$$

Then

$$\boxed{\frac{d}{d\xi} F(u_{\text{IA}}(\xi)) = \frac{u_{\text{IA}}(\xi)(u_{\text{IA}}(\xi)^2)'}{C_2 u_{\text{IA}}(\xi)^2 + \varepsilon} =: r_1(\xi).} \quad (1097)$$

Since $u_{\text{IA}}(\xi)$ is even, $F(u_{\text{IA}}(\xi))$ is also even, and the boundary term vanishes on the periodic-even extension. Therefore

$$\boxed{\mathcal{R}_1^{(m)} = i\ell_m \mathcal{F}^{(m)},} \quad (1098)$$

where

$$\boxed{\mathcal{F}^{(m)} := \frac{1}{L_\xi} \int_0^{L_\xi} F(u_{\text{IA}}(\xi)) \cos(\ell_m \xi) d\xi.} \quad (1099)$$

This is the preferred stable formula for $\mathcal{R}_1^{(m)}$.

CDLXXIV. 4. DISCRETE FORMULAS FROM THE SHOOTING SOLUTION

Suppose the accepted shooting solution is available at quadrature nodes $\xi_j \in [0, L_\xi]$ with weights ω_j . Then the numerical extraction formulas are

$$\boxed{\mathcal{U}_1^{(m)} \approx \frac{1}{L_\xi} \sum_j \omega_j u_j \cos(\ell_m \xi_j),} \quad (1100)$$

$$\boxed{\mathcal{U}_2^{(m)} \approx \frac{1}{L_\xi} \sum_j \omega_j u_j^2 \cos(\ell_m \xi_j),} \quad (1101)$$

$$\boxed{\mathcal{R}_2^{(m)} \approx \frac{1}{L_\xi} \sum_j \omega_j \frac{u_j^3}{C_2 u_j^2 + \varepsilon} \cos(\ell_m \xi_j),} \quad (1102)$$

and in the stable form,

$$\boxed{\mathcal{R}_1^{(m)} \approx i\ell_m \frac{1}{L_\xi} \sum_j \omega_j F(u_j) \cos(\ell_m \xi_j).} \quad (1103)$$

For the present Stage-IA pipeline, these are the actual practical formulas to use.

CDLXXV. 5. REDUCTION OF $\hat{\Phi}(G_*)$ TO ONE-DIMENSIONAL DATA

The accepted-branch harmonic was already reduced to

$$\boxed{\hat{\Phi}(G_*) = \sum_{Q \in \mathcal{I}_\Phi(G_*)} \eta_Q \mathcal{U}_1^{(m_Q)},} \quad (1104)$$

where

$$\boxed{m_Q := \frac{L_\xi}{\pi} (G_* - Q)_\parallel, \quad Q \in \mathcal{I}_\Phi(G_*),} \quad (1105)$$

and $\mathcal{I}_\Phi(G_*)$ is the admissible BCC shell subset satisfying $(G_* - Q)_\perp = 0$.

So $\hat{\Phi}(G_*)$ depends only on the one-dimensional profile coefficients $\mathcal{U}_1^{(m)}$.

CDLXXVI. 6. REDUCTION OF $\widehat{\rho}(G_*)$

The density harmonic is

$$\widehat{\rho}(G_*) = \sum_{\Delta \in \mathcal{I}_\rho(G_*)} B_\Delta \mathcal{U}_2^{(m_\Delta)}, \quad (1106)$$

with

$$m_\Delta := \frac{L_\xi}{\pi} (G_* - \Delta)_\parallel, \quad \Delta \in \mathcal{I}_\rho(G_*). \quad (1107)$$

Thus $\widehat{\rho}(G_*)$ is determined entirely by $\mathcal{U}_2^{(m)}$ and fixed BCC geometry.

CDLXXVII. 7. REDUCTION OF $\widehat{\Xi}_i(G_*)$

The profile-gradient harmonic was reduced to

$$\widehat{\Xi}_i(G_*) \approx \sum_{M \in \mathcal{I}_\Xi(G_*)} \left[n_i C_M^{(0)} \mathcal{R}_1^{(m_M)} + C_{M,i}^{(1)} \mathcal{R}_2^{(m_M)} \right], \quad (1108)$$

where

$$m_M := \frac{L_\xi}{\pi} (G_* - M)_\parallel, \quad M \in \mathcal{I}_\Xi(G_*). \quad (1109)$$

Therefore $\widehat{\Xi}_i(G_*)$ is determined only by the one-dimensional profile coefficients $\mathcal{R}_1^{(m)}$ and $\mathcal{R}_2^{(m)}$.

CDLXXVIII. 8. FINAL ACCEPTED-BRANCH OVERLAP FORMULAS

The accepted Stage-IA branch feeds the Class II intervalley overlaps through

$$\widehat{U}_{(II)}(G_*) = \frac{\alpha_X^2 \nu_0}{M_X^2} \sum_{Q+Q'=G_*} (Q \cdot Q') \widehat{\rho}(Q) \widehat{\rho}(Q'), \quad \nu_0 := \sum_A (n_0^A)^2, \quad (1110)$$

and

$$\widehat{T}_{(II)}^\alpha(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \Gamma_\alpha \sum_{Q+Q'=G_*} iQ_i \widehat{\rho}(Q) \widehat{\Xi}_i(Q'), \quad (1111)$$

with

$$\Gamma_\alpha := \sum_A n_0^A \mathcal{Q}_\alpha^A. \quad (1112)$$

Substituting (1106) and (1108), both $\widehat{U}_{(II)}$ and $\widehat{T}_{(II)}^\alpha$ become finite BCC geometry sums depending only on

$$\mathcal{U}_2^{(m)}, \quad \mathcal{R}_1^{(m)}, \quad \mathcal{R}_2^{(m)}.$$

Thus the accepted-branch common extractor is now completely reduced to one-dimensional Fourier data of the Stage-IA shooting solution.

CDLXXIX. 9. IMMEDIATE NEXT MOVE

The exact next finite task is:

$$\boxed{\text{compute } \mathcal{U}_1^{(m)}, \mathcal{U}_2^{(m)}, \mathcal{R}_1^{(m)}, \mathcal{R}_2^{(m)} \text{ from } u_{\text{IA}}(\xi), \text{ for the finitely many } m \text{ selected by } G_*}. \quad (1113)$$

After that, the remainder is mechanical:

$$\widehat{\Phi}(G_*), \widehat{\rho}(G_*), \widehat{\Xi}_i(G_*) \rightarrow \widehat{U}_{(II)}(G_*), \widehat{T}_{(II)}^1(G_*), \widehat{T}_{(II)}^2(G_*) \rightarrow H_{\text{eff}}(k), K_i, P_* K_i P_*.$$

CDLXXX. FINAL SUMMARY

The accepted Stage-IA branch extractor is now reduced completely to one-dimensional Fourier data of the shooting profile $u_{\text{IA}}(\xi)$.

The four required profile transforms are:

$$\mathcal{U}_1^{(m)} = \frac{1}{L_\xi} \int_0^{L_\xi} u_{\text{IA}}(\xi) \cos(\ell_m \xi) d\xi, \quad \ell_m = \frac{m\pi}{L_\xi},$$

$$\mathcal{U}_2^{(m)} = \frac{1}{L_\xi} \int_0^{L_\xi} u_{\text{IA}}(\xi)^2 \cos(\ell_m \xi) d\xi,$$

$$\mathcal{R}_2^{(m)} = \frac{1}{L_\xi} \int_0^{L_\xi} \frac{u_{\text{IA}}(\xi)^3}{C_2 u_{\text{IA}}(\xi)^2 + \varepsilon} \cos(\ell_m \xi) d\xi,$$

and the stable odd-current transform

$$\mathcal{R}_1^{(m)} = i\ell_m \frac{1}{L_\xi} \int_0^{L_\xi} F(u_{\text{IA}}(\xi)) \cos(\ell_m \xi) d\xi,$$

with

$$F(u) = \frac{2u}{C_2} - \frac{2\sqrt{\varepsilon}}{C_2^{3/2}} \arctan\left(\sqrt{\frac{C_2}{\varepsilon}} u\right).$$

From these, the branch harmonics become

$$\widehat{\Phi}(G_*) = \sum_{Q \in \mathcal{I}_\Phi(G_*)} \eta_Q \mathcal{U}_1^{(m_Q)},$$

$$\widehat{\rho}(G_*) = \sum_{\Delta \in \mathcal{I}_\rho(G_*)} B_\Delta \mathcal{U}_2^{(m_\Delta)},$$

$$\widehat{\Xi}_i(G_*) \approx \sum_{M \in \mathcal{I}_\Xi(G_*)} \left[n_i C_M^{(0)} \mathcal{R}_1^{(m_M)} + C_{M,i}^{(1)} \mathcal{R}_2^{(m_M)} \right].$$

Therefore the accepted-branch Class II overlaps

$$\widehat{U}_{(II)}(G_*), \quad \widehat{T}_{(II)}^1(G_*), \quad \widehat{T}_{(II)}^2(G_*)$$

are now finite BCC geometry sums depending only on the one-dimensional Fourier data of $u_{\text{IA}}(\xi)$.

So the common extractor has been reduced all the way to a pure one-dimensional profile-data computation.

CDLXXXI. FINITE LONGITUDINAL SAMPLE SET AT THE SELECTED BCC HARMONIC G_*

CDLXXXII. SHORT KOREAN EXPLANATION

We now stop speaking about generic harmonic sums and fix the actual finite longitudinal sample set required by the accepted Stage-IA branch at the selected BCC vector

$$G_* = \frac{q_*}{\sqrt{2}}(1, 1, 0).$$

The result is:

$$\boxed{\widehat{\Phi}(G_*) \text{ needs only } \mathcal{U}_1(0), \mathcal{U}_1(2q_*),} \quad (1114)$$

$$\boxed{\widehat{\rho}(G_*) \text{ needs only } \mathcal{U}_2(0), \mathcal{U}_2(q_*), \mathcal{U}_2(2q_*), \mathcal{U}_2(3q_*),} \quad (1115)$$

$$\boxed{\widehat{\Xi}_i(G_*) \text{ needs only } \mathcal{R}_1(\ell), \mathcal{R}_2(\ell) \text{ at } \ell \in \{0, q_*, 2q_*, 3q_*, 4q_*\},} \quad (1116)$$

with $\mathcal{R}_1(0) = 0$ automatically by parity.

So the accepted-branch common extractor is now reduced to a very small one-dimensional data problem.

CDLXXXIII. 1. SELECTED SHELL GEOMETRY

Fix

$$\boxed{G_* = \frac{q_*}{\sqrt{2}}(1, 1, 0), \quad \hat{n} = \frac{G_*}{|G_*|} = \frac{1}{\sqrt{2}}(1, 1, 0).} \quad (1117)$$

The first BCC shell is

$$\boxed{S_{\text{BCC}} = \frac{q_*}{\sqrt{2}} \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}.} \quad (1118)$$

All longitudinal sample frequencies below are multiples of

$$q_* = |G_*|.$$

CDLXXXIV. 2. EXACT FINITE REDUCTION OF $\widehat{\Phi}(G_*)$

The accepted branch harmonic is

$$\widehat{\Phi}(G_*) = \sum_{Q \in S_{\text{BCC}}} \eta_Q \mathbf{1}_{(G_* - Q)_\perp = 0} \mathcal{U}_1((G_* - Q)_\parallel). \quad (1119)$$

For $G_* = \frac{q_*}{\sqrt{2}}(1, 1, 0)$, the condition

$$(G_* - Q)_\perp = 0$$

is equivalent to

$$Q = \pm G_*.$$

Indeed:

- if $Q = +G_*$, then $G_* - Q = 0$;
- if $Q = -G_*$, then $G_* - Q = 2G_*$;
- every other first-shell vector produces a difference with nonzero perpendicular part.

Therefore

$$\widehat{\Phi}(G_*) = \eta_+ \mathcal{U}_1(0) + \eta_- \mathcal{U}_1(2q_*), \quad (1120)$$

where

$$\eta_+ := \eta_{G_*}, \quad \eta_- := \eta_{-G_*}.$$

So $\widehat{\Phi}(G_*)$ depends only on the two longitudinal samples

$$\mathcal{U}_1(0), \quad \mathcal{U}_1(2q_*).$$

CDLXXXV. 3. EXACT FINITE REDUCTION OF $\widehat{\rho}(G_*)$

Write

$$|S_{\text{BCC}}|^2 = \sum_{\Delta \in \Delta S} B_{\Delta} e^{i\Delta \cdot x}, \quad \Delta S := S_{\text{BCC}} - S_{\text{BCC}}.$$

Then

$$\widehat{\rho}(G_*) = \sum_{\Delta \in \Delta S} B_{\Delta} \mathbf{1}_{(G_* - \Delta)_{\perp} = 0} \mathcal{U}_2((G_* - \Delta)_{\parallel}). \quad (1121)$$

Now $G_* - \Delta$ is parallel to $\hat{n}$ if and only if Δ itself is parallel to G_* . Within the BCC difference set, the parallel possibilities are

$$\Delta \in \{-2G_*, -G_*, 0, G_*, 2G_*\}. \quad (1122)$$

Hence

$$\widehat{\rho}(G_*) = B_{G_*} \mathcal{U}_2(0) + B_0 \mathcal{U}_2(q_*) + B_{-G_*} \mathcal{U}_2(2q_*) + B_{2G_*} \mathcal{U}_2(-q_*) + B_{-2G_*} \mathcal{U}_2(3q_*). \quad (1123)$$

Since $\mathcal{U}_2(\ell)$ is even,

$$\mathcal{U}_2(-q_*) = \mathcal{U}_2(q_*).$$

Therefore

$$\widehat{\rho}(G_*) = B_{G_*} \mathcal{U}_2(0) + (B_0 + B_{2G_*}) \mathcal{U}_2(q_*) + B_{-G_*} \mathcal{U}_2(2q_*) + B_{-2G_*} \mathcal{U}_2(3q_*). \quad (1124)$$

So $\widehat{\rho}(G_*)$ needs only the four longitudinal samples

$$\mathcal{U}_2(0), \quad \mathcal{U}_2(q_*), \quad \mathcal{U}_2(2q_*), \quad \mathcal{U}_2(3q_*). \quad (1125)$$

CDLXXXVI. 4. EXACT FINITE REDUCTION OF $\widehat{\Xi}_i(G_*)$

In the Stage-IA averaged-denominator reduction,

$$\widehat{\Xi}_i(G_*) \approx \sum_M \left[n_i C_M^{(0)} \mathbf{1}_{(G_* - M)_{\perp} = 0} \mathcal{R}_1((G_* - M)_{\parallel}) + C_{M,i}^{(1)} \mathbf{1}_{(G_* - M)_{\perp} = 0} \mathcal{R}_2((G_* - M)_{\parallel}) \right]. \quad (1126)$$

Here M runs over the shell-plus-density support

$$M \in S_{\text{BCC}} + \Delta S.$$

The condition $(G_* - M)_\perp = 0$ means that M must also lie on the G_* -axis. Inside $S_{\text{BCC}} + \Delta S$, the parallel possibilities are

$$M \in \{-3G_*, -2G_*, -G_*, 0, G_*, 2G_*, 3G_*\}. \quad (1127)$$

Therefore

$$\widehat{\Xi}_i(G_*) \approx \sum_{m=-3}^3 \left[n_i C_{mG_*}^{(0)} \mathcal{R}_1((1-m)q_*) + C_{mG_*,i}^{(1)} \mathcal{R}_2((1-m)q_*) \right]. \quad (1128)$$

Using parity:

- $\mathcal{R}_2(\ell)$ is even,
- $\mathcal{R}_1(\ell) = i\ell \mathcal{F}(\ell)$ and hence $\mathcal{R}_1(0) = 0$.

So the only independent longitudinal samples needed are

$$\mathcal{R}_1(q_*), \quad \mathcal{R}_1(2q_*), \quad \mathcal{R}_1(3q_*), \quad \mathcal{R}_1(4q_*), \quad (1129)$$

and

$$\mathcal{R}_2(0), \quad \mathcal{R}_2(q_*), \quad \mathcal{R}_2(2q_*), \quad \mathcal{R}_2(3q_*), \quad \mathcal{R}_2(4q_*). \quad (1130)$$

Thus $\widehat{\Xi}_i(G_*)$ is reduced to a finite one-dimensional data set of size $4 + 5$.

CDLXXXVII. 5. FINAL FINITE ONE-DIMENSIONAL DATA SET

Collecting the above results, the accepted Stage-IA branch requires only the following one-dimensional Fourier samples:

$$\mathcal{U}_1(0), \quad \mathcal{U}_1(2q_*), \quad (1131)$$

$$\mathcal{U}_2(0), \quad \mathcal{U}_2(q_*), \quad \mathcal{U}_2(2q_*), \quad \mathcal{U}_2(3q_*), \quad (1132)$$

$$\mathcal{R}_1(q_*), \quad \mathcal{R}_1(2q_*), \quad \mathcal{R}_1(3q_*), \quad \mathcal{R}_1(4q_*), \quad (1133)$$

$$\mathcal{R}_2(0), \quad \mathcal{R}_2(q_*), \quad \mathcal{R}_2(2q_*), \quad \mathcal{R}_2(3q_*), \quad \mathcal{R}_2(4q_*). \quad (1134)$$

So the entire accepted-branch common extractor is reduced to

$$2 + 4 + 4 + 5 = 15 \quad (1135)$$

independent one-dimensional profile samples.

CDLXXXVIII. 6. IMMEDIATE NEXT FORMULAS

Once those 15 numbers are known, one inserts them into

$$\boxed{\hat{\Phi}(G_*), \quad \hat{\rho}(G_*), \quad \hat{\Xi}_i(G_*)}, \quad (1136)$$

then into

$$\boxed{\hat{U}_{(II)}(G_*), \quad \hat{T}_{(II)}^1(G_*), \quad \hat{T}_{(II)}^2(G_*)}, \quad (1137)$$

and finally into

$$\boxed{H_{\text{eff}}(k) \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow (\lambda_c, \Lambda_E, \Lambda_O) \rightarrow (\lambda_{\parallel}, \alpha, \beta)}. \quad (1138)$$

CDLXXXIX. FINAL SUMMARY

For the selected BCC harmonic

$$G_* = \frac{q_*}{\sqrt{2}}(1, 1, 0),$$

the accepted Stage-IA branch extractor reduces to a finite list of longitudinal Fourier samples of the one-dimensional profile $u_{\text{IA}}(\xi)$.

The exact reductions are:

$$\hat{\Phi}(G_*) = \eta_+ \mathcal{U}_1(0) + \eta_- \mathcal{U}_1(2q_*),$$

$$\hat{\rho}(G_*) = B_{G_*} \mathcal{U}_2(0) + (B_0 + B_{2G_*}) \mathcal{U}_2(q_*) + B_{-G_*} \mathcal{U}_2(2q_*) + B_{-2G_*} \mathcal{U}_2(3q_*),$$

and

$$\hat{\Xi}_i(G_*) \approx \sum_{m=-3}^3 \left[n_i C_{mG_*}^{(0)} \mathcal{R}_1((1-m)q_*) + C_{mG_*,i}^{(1)} \mathcal{R}_2((1-m)q_*) \right].$$

Using parity, the full accepted-branch common extractor needs only the 15 independent one-dimensional samples

$$\mathcal{U}_1(0), \mathcal{U}_1(2q_*),$$

$$\mathcal{U}_2(0), \mathcal{U}_2(q_*), \mathcal{U}_2(2q_*), \mathcal{U}_2(3q_*),$$

$$\mathcal{R}_1(q_*), \mathcal{R}_1(2q_*), \mathcal{R}_1(3q_*), \mathcal{R}_1(4q_*),$$

$$\mathcal{R}_2(0), \mathcal{R}_2(q_*), \mathcal{R}_2(2q_*), \mathcal{R}_2(3q_*), \mathcal{R}_2(4q_*).$$

So the accepted-branch common extractor is no longer an open infinite-dimensional object. It is a finite one-dimensional profile-data computation on $u_{\text{IA}}(\xi)$.

CDXC. FINITE-MODE CLOSURE OF THE ACCEPTED-BRANCH COMMON EXTRACTOR

CDXCI. SHORT KOREAN EXPLANATION

The accepted Stage-IA branch has already reduced the branch-specific extractor to 15 one-dimensional longitudinal Fourier samples.

We now close the next step completely by writing the load-bearing Class II intervalley overlaps

$$\hat{U}_{(II)}(G_*), \quad \hat{T}_{(II)}^1(G_*), \quad \hat{T}_{(II)}^2(G_*)$$

as finite quadratic/bilinear forms in those samples.

This is the point at which the common extractor ceases to be an infinite-dimensional object and becomes a small finite-dimensional algebraic computation.

CDXCII. 1. THE FINITE ONE-DIMENSIONAL DATA VECTORS

For the selected intervalley harmonic

$$G_* = \frac{q_*}{\sqrt{2}}(1, 1, 0),$$

define the accepted-branch longitudinal sample vectors

$$\mathbf{U}_1 := \begin{pmatrix} \mathcal{U}_1(0) \\ \mathcal{U}_1(2q_*) \end{pmatrix}, \quad (1139)$$

$$\mathbf{U}_2 := \begin{pmatrix} \mathcal{U}_2(0) \\ \mathcal{U}_2(q_*) \\ \mathcal{U}_2(2q_*) \\ \mathcal{U}_2(3q_*) \end{pmatrix}, \quad (1140)$$

$$\mathbf{R}_1 := \begin{pmatrix} \mathcal{R}_1(q_*) \\ \mathcal{R}_1(2q_*) \\ \mathcal{R}_1(3q_*) \\ \mathcal{R}_1(4q_*) \end{pmatrix}, \quad (1141)$$

$$\mathbf{R}_2 := \begin{pmatrix} \mathcal{R}_2(0) \\ \mathcal{R}_2(q_*) \\ \mathcal{R}_2(2q_*) \\ \mathcal{R}_2(3q_*) \\ \mathcal{R}_2(4q_*) \end{pmatrix}. \quad (1142)$$

These are the only branch-dependent inputs that remain.

CDXCIII. 2. FINITE REDUCTION OF $\widehat{\Phi}(G_*)$

From the previous reduction,

$$\widehat{\Phi}(G_*) = \eta_+ \mathcal{U}_1(0) + \eta_- \mathcal{U}_1(2q_*). \quad (1143)$$

So define the geometry row vector

$$\mathbf{c}_\Phi^T := \begin{pmatrix} \eta_+ & \eta_- \end{pmatrix}, \quad (1144)$$

and write

$$\widehat{\Phi}(G_*) = \mathbf{c}_\Phi^T \mathbf{U}_1. \quad (1145)$$

CDXCIV. 3. FINITE REDUCTION OF $\widehat{\rho}(G_*)$

Previously we obtained

$$\widehat{\rho}(G_*) = B_{G_*} \mathcal{U}_2(0) + (B_0 + B_{2G_*}) \mathcal{U}_2(q_*) + B_{-G_*} \mathcal{U}_2(2q_*) + B_{-2G_*} \mathcal{U}_2(3q_*). \quad (1146)$$

Hence define

$$\mathbf{c}_\rho^T := \begin{pmatrix} B_{G_*} & B_0 + B_{2G_*} & B_{-G_*} & B_{-2G_*} \end{pmatrix}, \quad (1147)$$

so that

$$\boxed{\hat{\rho}(G_*) = \mathbf{c}_\rho^T \mathbf{U}_2.} \quad (1148)$$

More generally, for every reciprocal argument Q entering the finite intervalley sums, define a corresponding row vector

$$\boxed{\mathbf{c}_\rho(Q)^T} \quad (1149)$$

such that

$$\boxed{\hat{\rho}(Q) = \mathbf{c}_\rho(Q)^T \mathbf{U}_2.} \quad (1150)$$

Because only finitely many Q appear in the selected intervalley channel, this produces only finitely many such rows.

CDXCV. 4. FINITE REDUCTION OF $\widehat{\Xi}_i(G_*)$

The accepted-branch reduction gave

$$\boxed{\widehat{\Xi}_i(G_*) \approx \sum_{m=-3}^3 \left[n_i C_{mG_*}^{(0)} \mathcal{R}_1((1-m)q_*) + C_{mG_*,i}^{(1)} \mathcal{R}_2((1-m)q_*) \right].} \quad (1151)$$

Because $\mathcal{R}_1(0) = 0$, the $m = 1$ term contributes only through $\mathcal{R}_2(0)$.

Collect the coefficients multiplying $\mathbf{R}_1$ and $\mathbf{R}_2$ into row vectors

$$\boxed{\mathbf{c}_{\Xi_i}^{(1)} := \left(n_i C_{0G_*}^{(0)} \quad n_i C_{-G_*}^{(0)} \quad n_i C_{-2G_*}^{(0)} \quad n_i C_{-3G_*}^{(0)} \right),} \quad (1152)$$

$$\boxed{\mathbf{c}_{\Xi_i}^{(2)} := \left(C_{G_*,i}^{(1)} \quad C_{0G_*,i}^{(1)} \quad C_{-G_*,i}^{(1)} \quad C_{-2G_*,i}^{(1)} \quad C_{-3G_*,i}^{(1)} \right),} \quad (1153)$$

so that

$$\boxed{\widehat{\Xi}_i(G_*) \approx \mathbf{c}_{\Xi_i}^{(1)} \mathbf{R}_1 + \mathbf{c}_{\Xi_i}^{(2)} \mathbf{R}_2.} \quad (1154)$$

More generally, for every reciprocal argument Q' entering the transverse intervalley sum, define finite row vectors

$$\boxed{\mathbf{c}_{\Xi_i}^{(1)}(Q')^T, \quad \mathbf{c}_{\Xi_i}^{(2)}(Q')^T} \quad (1155)$$

such that

$$\boxed{\widehat{\Xi}_i(Q') \approx \mathbf{c}_{\Xi_i}^{(1)}(Q')^T \mathbf{R}_1 + \mathbf{c}_{\Xi_i}^{(2)}(Q')^T \mathbf{R}_2.} \quad (1156)$$

Again, only finitely many such Q' occur.

CDXCVI. 5. QUADRATIC FORM FOR $\widehat{U}_{(II)}(G_*)$

The scalar intervalley overlap is

$$\boxed{\widehat{U}_{(II)}(G_*) = \frac{\alpha_X^2 \nu_0}{M_X^2} \sum_{Q+Q'=G_*} (Q \cdot Q') \hat{\rho}(Q) \hat{\rho}(Q'), \quad \nu_0 := \sum_A (n_0^A)^2.} \quad (1157)$$

Substitute (1150):

$$\hat{U}_{(II)}(G_*) = \frac{\alpha_X^2 \nu_0}{M_X^2} \sum_{Q+Q'=G_*} (Q \cdot Q') (\mathbf{c}_\rho(Q)^T \mathbf{U}_2) (\mathbf{c}_\rho(Q')^T \mathbf{U}_2). \quad (1158)$$

Therefore define the symmetric 4×4 geometry matrix

$$\mathbb{M}_U := \sum_{Q+Q'=G_*} (Q \cdot Q') \mathbf{c}_\rho(Q) \mathbf{c}_\rho(Q')^T. \quad (1159)$$

Then the scalar intervalley overlap is exactly the quadratic form

$$\hat{U}_{(II)}(G_*) = \frac{\alpha_X^2 \nu_0}{M_X^2} \mathbf{U}_2^T \mathbb{M}_U \mathbf{U}_2. \quad (1160)$$

So $\hat{U}_{(II)}(G_*)$ depends only on the four samples in $\mathbf{U}_2$.

CDXCVII. 6. BILINEAR FORM FOR $\hat{T}_{(II)}^\alpha(G_*)$

The transverse intervalley overlap is

$$\hat{T}_{(II)}^\alpha(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \Gamma_\alpha \sum_{Q+Q'=G_*} iQ_i \hat{\rho}(Q) \hat{\Xi}_i(Q'), \quad (1161)$$

with

$$\Gamma_\alpha := \sum_A n_0^A \mathcal{Q}_\alpha^A. \quad (1162)$$

Substituting (1150) and (1156), we get

$$\hat{T}_{(II)}^\alpha(G_*) \approx -\frac{\alpha_X \beta_X}{M_X^2} \Gamma_\alpha \sum_{Q+Q'=G_*} iQ_i (\mathbf{c}_\rho(Q)^T \mathbf{U}_2) \left[\mathbf{c}_{\Xi_i}^{(1)}(Q')^T \mathbf{R}_1 + \mathbf{c}_{\Xi_i}^{(2)}(Q')^T \mathbf{R}_2 \right]. \quad (1163)$$

Define the geometry coupling matrices

$$\mathbb{M}_{T,1}^{(\alpha)} := \sum_{Q+Q'=G_*} iQ_i \mathbf{c}_\rho(Q) \mathbf{c}_{\Xi_i}^{(1)}(Q')^T, \quad (1164)$$

$$\mathbb{M}_{T,2}^{(\alpha)} := \sum_{Q+Q'=G_*} iQ_i \mathbf{c}_\rho(Q) \mathbf{c}_{\Xi_i}^{(2)}(Q')^T. \quad (1165)$$

Then the transverse intervalley overlap is the finite bilinear form

$$\hat{T}_{(II)}^\alpha(G_*) \approx -\frac{\alpha_X \beta_X}{M_X^2} \Gamma_\alpha \left[\mathbf{U}_2^T \mathbb{M}_{T,1}^{(\alpha)} \mathbf{R}_1 + \mathbf{U}_2^T \mathbb{M}_{T,2}^{(\alpha)} \mathbf{R}_2 \right]. \quad (1166)$$

Thus $\hat{T}_{(II)}^\alpha(G_*)$ depends only on the finite vectors $\mathbf{U}_2, \mathbf{R}_1, \mathbf{R}_2$.

CDXCVIII. 7. FINAL FINITE-MODE CLOSURE OF THE ACCEPTED-BRANCH EXTRACTOR

Collecting all pieces, the accepted Stage-IA branch now enters the common extractor only through the 15-component finite data vector

$$\boxed{\mathbf{d}_{\text{IA}} := \mathbf{U}_1 \oplus \mathbf{U}_2 \oplus \mathbf{R}_1 \oplus \mathbf{R}_2 \in \mathbb{C}^{15}.} \quad (1167)$$

The load-bearing overlaps are

$$\boxed{\widehat{\Phi}(G_*) = \mathbf{c}_\Phi^T \mathbf{U}_1,} \quad (1168)$$

$$\boxed{\widehat{U}_{(II)}(G_*) = \frac{\alpha_X^2 \nu_0}{M_X^2} \mathbf{U}_2^T \mathbb{M}_U \mathbf{U}_2,} \quad (1169)$$

$$\boxed{\widehat{T}_{(II)}^\alpha(G_*) \approx -\frac{\alpha_X \beta_X}{M_X^2} \Gamma_\alpha \left[\mathbf{U}_2^T \mathbb{M}_{T,1}^{(\alpha)} \mathbf{R}_1 + \mathbf{U}_2^T \mathbb{M}_{T,2}^{(\alpha)} \mathbf{R}_2 \right].} \quad (1170)$$

So the infinite-dimensional branch dependence has collapsed completely to a small finite-dimensional algebraic problem.

CDXCIX. 8. IMMEDIATE NEXT MOVE

Once the geometry matrices

$$\mathbb{M}_U, \quad \mathbb{M}_{T,1}^{(\alpha)}, \quad \mathbb{M}_{T,2}^{(\alpha)}$$

are evaluated for the selected G_* , the remaining chain is mechanical:

$$\widehat{U}_{(II)}(G_*), \widehat{T}_{(II)}^1(G_*), \widehat{T}_{(II)}^2(G_*) \rightarrow H_{\text{eff}}(k) \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow (\lambda_c, \Lambda_E, \Lambda_O) \rightarrow (\lambda_\parallel, \alpha, \beta).$$

Thus the actual next nonredundant task is:

$$\boxed{\text{evaluate the pure BCC geometry matrices } \mathbb{M}_U, \mathbb{M}_{T,1}^{(\alpha)}, \mathbb{M}_{T,2}^{(\alpha)} \text{ for the selected } G_*}. \quad (1171)$$

D. FINAL SUMMARY

The accepted Stage-IA common extractor is now fully closed at the finite-mode level.

Define the 15-component branch data vector

$$\mathbf{d}_{\text{IA}} = \mathbf{U}_1 \oplus \mathbf{U}_2 \oplus \mathbf{R}_1 \oplus \mathbf{R}_2,$$

with

$$\mathbf{U}_1 = \begin{pmatrix} \mathcal{U}_1(0) \\ \mathcal{U}_1(q_*) \\ \mathcal{U}_1(2q_*) \end{pmatrix}, \quad \mathbf{U}_2 = \begin{pmatrix} \mathcal{U}_2(0) \\ \mathcal{U}_2(q_*) \\ \mathcal{U}_2(2q_*) \\ \mathcal{U}_2(3q_*) \end{pmatrix},$$

$$\mathbf{R}_1 = \begin{pmatrix} \mathcal{R}_1(q_*) \\ \mathcal{R}_1(2q_*) \\ \mathcal{R}_1(3q_*) \\ \mathcal{R}_1(4q_*) \end{pmatrix}, \quad \mathbf{R}_2 = \begin{pmatrix} \mathcal{R}_2(0) \\ \mathcal{R}_2(q_*) \\ \mathcal{R}_2(2q_*) \\ \mathcal{R}_2(3q_*) \\ \mathcal{R}_2(4q_*) \end{pmatrix}.$$

Then the branch-dependent overlaps are

$$\hat{\Phi}(G_*) = \mathbf{c}_\Phi^T \mathbf{U}_1,$$

$$\hat{U}_{(II)}(G_*) = \frac{\alpha_X^2 \nu_0}{M_X^2} \mathbf{U}_2^T \mathbb{M}_U \mathbf{U}_2,$$

$$\hat{T}_{(II)}^\alpha(G_*) \approx -\frac{\alpha_X \beta_X}{M_X^2} \Gamma_\alpha \left[\mathbf{U}_2^T \mathbb{M}_{T,1}^{(\alpha)} \mathbf{R}_1 + \mathbf{U}_2^T \mathbb{M}_{T,2}^{(\alpha)} \mathbf{R}_2 \right].$$

Therefore the accepted-branch common extractor is no longer a functional-analytic problem. It is a finite-dimensional algebraic computation once the pure BCC geometry matrices

$$\mathbb{M}_U, \quad \mathbb{M}_{T,1}^{(\alpha)}, \quad \mathbb{M}_{T,2}^{(\alpha)}$$

are fixed.

DI. PURE BCC GEOMETRY MATRICES AT THE SELECTED INTERVALLEY HARMONIC

DII. SHORT KOREAN EXPLANATION

We now freeze the pure BCC geometry matrices for the selected shell vector

$$G_* = \frac{q_*}{\sqrt{2}}(1, 1, 0).$$

This is the first point where the accepted-branch common extractor becomes a completely finite algebraic object. The exact shell facts already established are:

1. the folded intervalley channel is controlled only by the single harmonic $Q = G_*$;
2. the ordered first-shell pair count is $N(G_*) = 4$;
3. every contributing ordered pair satisfies $Q_1 \cdot Q_2 = -q_*^2/2$.

So the remaining BCC part is just a 4×4 shell-pair combinatorics problem.

DIII. 1. THE CONTRIBUTING BCC SHELL QUARTET

For

$$G_* = \frac{q_*}{\sqrt{2}}(1, 1, 0),$$

the four ordered first-shell pairs satisfying

$$Q + Q' = G_*$$

are:

$$\boxed{q_1 := \frac{q_*}{\sqrt{2}}(1, 0, 1), \quad q_2 := \frac{q_*}{\sqrt{2}}(0, 1, -1),} \tag{1172}$$

$$\boxed{q_3 := \frac{q_*}{\sqrt{2}}(1, 0, -1), \quad q_4 := \frac{q_*}{\sqrt{2}}(0, 1, 1),} \tag{1173}$$

with ordered pair list

$$\boxed{(q_1, q_2), \quad (q_2, q_1), \quad (q_3, q_4), \quad (q_4, q_3)}. \quad (1174)$$

Equivalently,

$$\boxed{q_1 + q_2 = G_*, \quad q_3 + q_4 = G_*}. \quad (1175)$$

These are exactly the four ordered pairs counted in the established BCC shell combinatorics.

DIV. 2. PURE SHELL PAIRING MATRIX

Define the shell pairing permutation matrix

$$\boxed{P_* := \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}}. \quad (1176)$$

This exchanges each shell vector with its unique partner summing to G_* :

$$q_1 \leftrightarrow q_2, \quad q_3 \leftrightarrow q_4.$$

DV. 3. SCALAR GEOMETRY MATRIX $\mathbb{M}_U$

Let

$$\boxed{\boldsymbol{\rho}_* := \begin{pmatrix} \widehat{\rho}(q_1) \\ \widehat{\rho}(q_2) \\ \widehat{\rho}(q_3) \\ \widehat{\rho}(q_4) \end{pmatrix}}. \quad (1177)$$

The scalar intervalley overlap is

$$\widehat{U}_{(II)}(G_*) = \frac{\alpha_X^2 \nu_0}{M_X^2} \sum_{Q+Q'=G_*} (Q \cdot Q') \widehat{\rho}(Q) \widehat{\rho}(Q').$$

Since every contributing ordered pair has

$$Q \cdot Q' = -\frac{q_*^2}{2},$$

we obtain

$$\boxed{\widehat{U}_{(II)}(G_*) = \frac{\alpha_X^2 \nu_0}{M_X^2} \boldsymbol{\rho}_*^T \mathbb{M}_U \boldsymbol{\rho}_*}, \quad (1178)$$

with the exact pure BCC geometry matrix

$$\boxed{\mathbb{M}_U = -\frac{q_*^2}{2} P_*}. \quad (1179)$$

Explicitly,

$$\boxed{\widehat{U}_{(II)}(G_*) = -\frac{\alpha_X^2 \nu_0 q_*^2}{M_X^2} [\widehat{\rho}(q_1) \widehat{\rho}(q_2) + \widehat{\rho}(q_3) \widehat{\rho}(q_4)]}. \quad (1180)$$

DVI. 4. SHELL-ADAPTED COMPONENT MATRICES FOR THE TRANSVERSE OVERLAP

Choose the shell-adapted orthonormal frame

$$\hat{n} = \frac{1}{\sqrt{2}}(1, 1, 0), \quad e_1 = \frac{1}{\sqrt{2}}(1, -1, 0), \quad e_2 = (0, 0, 1). \quad (1181)$$

The shell-vector components are:

$$q_1 \cdot \hat{n} = q_2 \cdot \hat{n} = q_3 \cdot \hat{n} = q_4 \cdot \hat{n} = \frac{q_*}{2}, \quad (1182)$$

$$(q_1 \cdot e_1, q_2 \cdot e_1, q_3 \cdot e_1, q_4 \cdot e_1) = \frac{q_*}{2}(1, -1, 1, -1), \quad (1183)$$

$$(q_1 \cdot e_2, q_2 \cdot e_2, q_3 \cdot e_2, q_4 \cdot e_2) = \frac{q_*}{\sqrt{2}}(1, -1, -1, 1). \quad (1184)$$

Define the diagonal shell-component matrices

$$D_{\parallel} = \frac{q_*}{2} \mathbf{1}_4, \quad (1185)$$

$$D_1 = \frac{q_*}{2} \text{diag}(1, -1, 1, -1), \quad (1186)$$

$$D_2 = \frac{q_*}{\sqrt{2}} \text{diag}(1, -1, -1, 1). \quad (1187)$$

DVII. 5. TRANSVERSE GEOMETRY MATRICES $\mathbb{M}_{T,\lambda}$

Let

$$\Xi_{\lambda} := \begin{pmatrix} \widehat{\Xi}_{\lambda}(q_1) \\ \widehat{\Xi}_{\lambda}(q_2) \\ \widehat{\Xi}_{\lambda}(q_3) \\ \widehat{\Xi}_{\lambda}(q_4) \end{pmatrix}, \quad \lambda \in \{\parallel, 1, 2\}. \quad (1188)$$

Then the transverse intervalley overlap

$$\hat{T}_{(II)}^{\alpha}(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \Gamma_{\alpha} \sum_{Q+Q'=G_*} iQ_i \hat{\rho}(Q) \widehat{\Xi}_i(Q')$$

becomes

$$\hat{T}_{(II)}^{\alpha}(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \Gamma_{\alpha} \sum_{\lambda \in \{\parallel, 1, 2\}} \boldsymbol{\rho}_*^T \mathbb{M}_{T,\lambda} \Xi_{\lambda}, \quad (1189)$$

with the exact pure BCC geometry matrices

$$\mathbb{M}_{T,\parallel} = i D_{\parallel} P_*, \quad \mathbb{M}_{T,1} = i D_1 P_*, \quad \mathbb{M}_{T,2} = i D_2 P_*. \quad (1190)$$

Explicitly,

$$\mathbb{M}_{T,\parallel} = \frac{iq_*}{2} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad (1191)$$

$$\mathbb{M}_{T,1} = \frac{iq_*}{2} \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}, \quad (1192)$$

$$\mathbb{M}_{T,2} = \frac{iq_*}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \quad (1193)$$

DVIII. 6. EMBEDDING THE ONE-DIMENSIONAL STAGE-IA PROFILE DATA

Now define the finite density embedding matrix

$$C_\rho := \begin{pmatrix} \mathbf{c}_\rho(q_1)^T \\ \mathbf{c}_\rho(q_2)^T \\ \mathbf{c}_\rho(q_3)^T \\ \mathbf{c}_\rho(q_4)^T \end{pmatrix}, \quad (1194)$$

so that

$$\boldsymbol{\rho}_* = C_\rho \mathbf{U}_2. \quad (1195)$$

Likewise, for each shell-frame component $\lambda \in \{\parallel, 1, 2\}$, define

$$C_{\Xi,\lambda}^{(1)} := \begin{pmatrix} \mathbf{c}_{\Xi,\lambda}^{(1)}(q_1)^T \\ \mathbf{c}_{\Xi,\lambda}^{(1)}(q_2)^T \\ \mathbf{c}_{\Xi,\lambda}^{(1)}(q_3)^T \\ \mathbf{c}_{\Xi,\lambda}^{(1)}(q_4)^T \end{pmatrix}, \quad C_{\Xi,\lambda}^{(2)} := \begin{pmatrix} \mathbf{c}_{\Xi,\lambda}^{(2)}(q_1)^T \\ \mathbf{c}_{\Xi,\lambda}^{(2)}(q_2)^T \\ \mathbf{c}_{\Xi,\lambda}^{(2)}(q_3)^T \\ \mathbf{c}_{\Xi,\lambda}^{(2)}(q_4)^T \end{pmatrix}, \quad (1196)$$

so that

$$\boldsymbol{\Xi}_\lambda = C_{\Xi,\lambda}^{(1)} \mathbf{R}_1 + C_{\Xi,\lambda}^{(2)} \mathbf{R}_2. \quad (1197)$$

Therefore the fully reduced accepted-branch overlaps are

$$\widehat{U}_{(II)}(G_*) = \frac{\alpha_X^2 \nu_0}{M_X^2} \mathbf{U}_2^T \underbrace{(C_\rho^T \mathbb{M}_U C_\rho)}_{\mathbb{G}_U} \mathbf{U}_2, \quad (1198)$$

and

$$\widehat{T}_{(II)}^\alpha(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \Gamma_\alpha \sum_\lambda \mathbf{U}_2^T \underbrace{(C_\rho^T \mathbb{M}_{T,\lambda} C_{\Xi,\lambda}^{(1)})}_{\mathbb{G}_{T,\lambda}^{(1)}} \mathbf{R}_1 + \mathbf{U}_2^T \underbrace{(C_\rho^T \mathbb{M}_{T,\lambda} C_{\Xi,\lambda}^{(2)})}_{\mathbb{G}_{T,\lambda}^{(2)}} \mathbf{R}_2. \quad (1199)$$

So the remaining extractor is now a finite matrix multiplication problem.

DIX. 7. IMMEDIATE NEXT STEP

The actual next finite task is:

$$\boxed{\text{compute } C_\rho, C_{\Xi,\lambda}^{(1)}, C_{\Xi,\lambda}^{(2)} \text{ from the already-fixed BCC geometry, then evaluate } \mathbb{G}_U, \mathbb{G}_{T,\lambda}^{(1)}, \mathbb{G}_{T,\lambda}^{(2)}} \quad (1200)$$

Once these geometry matrices are fixed, the accepted Stage-IA branch enters the common extractor only through

$$\mathbf{U}_2, \quad \mathbf{R}_1, \quad \mathbf{R}_2,$$

and the rest is mechanical:

$$\hat{U}_{(II)}(G_*), \hat{T}_{(II)}^1(G_*), \hat{T}_{(II)}^2(G_*) \rightarrow H_{\text{eff}}(k) \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow (\lambda_c, \Lambda_E, \Lambda_O).$$

DX. FINAL SUMMARY

For the selected BCC shell vector

$$G_* = \frac{q_*}{\sqrt{2}}(1, 1, 0),$$

the only ordered first-shell pairs with $Q + Q' = G_*$ are

$$(q_1, q_2), (q_2, q_1), (q_3, q_4), (q_4, q_3),$$

with

$$q_1 = \frac{q_*}{\sqrt{2}}(1, 0, 1), \quad q_2 = \frac{q_*}{\sqrt{2}}(0, 1, -1), \quad q_3 = \frac{q_*}{\sqrt{2}}(1, 0, -1), \quad q_4 = \frac{q_*}{\sqrt{2}}(0, 1, 1),$$

and each contributing pair has

$$Q \cdot Q' = -\frac{q_*^2}{2}.$$

Hence the scalar pure-geometry matrix is

$$\boxed{\mathbb{M}_U = -\frac{q_*^2}{2} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}}.$$

In the shell-adapted frame

$$\hat{n} = \frac{1}{\sqrt{2}}(1, 1, 0), \quad e_1 = \frac{1}{\sqrt{2}}(1, -1, 0), \quad e_2 = (0, 0, 1),$$

the transverse pure-geometry matrices are

$$\boxed{\mathbb{M}_{T,\parallel} = \frac{iq_*}{2} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}},$$

$$\boxed{\mathbb{M}_{T,1} = \frac{iq_*}{2} \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}},$$

$$\mathbb{M}_{T,2} = \frac{iq_*}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Therefore the accepted-branch overlaps are

$$\hat{U}_{(II)}(G_*) = \frac{\alpha_X^2 \nu_0}{M_X^2} \mathbf{U}_2^T (C_\rho^T \mathbb{M}_U C_\rho) \mathbf{U}_2,$$

and

$$\hat{T}_{(II)}^\alpha(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \Gamma_\alpha \sum_\lambda \mathbf{U}_2^T (C_\rho^T \mathbb{M}_{T,\lambda} C_{\Xi,\lambda}^{(1)}) \mathbf{R}_1 + \mathbf{U}_2^T (C_\rho^T \mathbb{M}_{T,\lambda} C_{\Xi,\lambda}^{(2)}) \mathbf{R}_2.$$

So the branch dependence is fully compressed, and the remaining task is purely the finite BCC geometry embedding into these matrices.

DXI. EXPLICIT BCC GEOMETRY EMBEDDING AT THE SELECTED HARMONIC $G_* = (q_*/\sqrt{2})(1, 1, 0)$

DXII. 1. SELECTED SHELL QUARTET

Fix

$$G_* = \frac{q_*}{\sqrt{2}}(1, 1, 0). \quad (1201)$$

Inside the first BCC shell, the four ordered pair contributors satisfying

$$Q + Q' = G_*$$

are generated by the quartet

$$q_1 = \frac{q_*}{\sqrt{2}}(1, 0, 1), \quad q_2 = \frac{q_*}{\sqrt{2}}(0, 1, -1), \quad q_3 = \frac{q_*}{\sqrt{2}}(1, 0, -1), \quad q_4 = \frac{q_*}{\sqrt{2}}(0, 1, 1). \quad (1202)$$

Thus the only ordered pairs are

$$(q_1, q_2), \quad (q_2, q_1), \quad (q_3, q_4), \quad (q_4, q_3). \quad (1203)$$

DXIII. 2. EQUAL-AMPLITUDE LOCKED BCC DENSITY EMBEDDING

Assume the canonical locked shell weights

$$\eta_Q = 1 \quad (Q \in S_{\text{BCC}}). \quad (1204)$$

For each quartet momentum q_j , the exact admissible difference harmonics are only four:

$$\Delta = q_j, \quad \Delta = q_j - G_*, \quad \Delta = q_j + G_*, \quad \Delta = q_j - 2G_*.$$

Their corresponding multiplicities in

$$|S_{\text{BCC}}|^2 = \sum_{\Delta} B_{\Delta} e^{i\Delta \cdot x}$$

are

$$B_{q_j} = 4, \quad B_{q_j - G_*} = 4, \quad B_{q_j + G_*} = 2, \quad B_{q_j - 2G_*} = 2.$$

Therefore each quartet density harmonic is

$$\widehat{\rho}(q_j) = 4\mathcal{U}_2(0) + 6\mathcal{U}_2(q_*) + 2\mathcal{U}_2(2q_*), \quad j = 1, 2, 3, 4. \quad (1205)$$

Thus the quartet density vector

$$\boldsymbol{\rho}_* := \begin{pmatrix} \widehat{\rho}(q_1) \\ \widehat{\rho}(q_2) \\ \widehat{\rho}(q_3) \\ \widehat{\rho}(q_4) \end{pmatrix} \quad (1206)$$

has the explicit embedding

$$\boldsymbol{\rho}_* = C_\rho \mathbf{U}_2^{\text{red}}, \quad \mathbf{U}_2^{\text{red}} := \begin{pmatrix} \mathcal{U}_2(0) \\ \mathcal{U}_2(q_*) \\ \mathcal{U}_2(2q_*) \end{pmatrix}, \quad (1207)$$

with

$$C_\rho = \begin{pmatrix} 4 & 6 & 2 \\ 4 & 6 & 2 \\ 4 & 6 & 2 \\ 4 & 6 & 2 \end{pmatrix}. \quad (1208)$$

So the density embedding is already rank one:

$$\boldsymbol{\rho}_* = \rho_* \mathbf{1}_4, \quad \rho_* := 4\mathcal{U}_2(0) + 6\mathcal{U}_2(q_*) + 2\mathcal{U}_2(2q_*). \quad (1209)$$

DXIV. 3. SHELL-ADAPTED SPATIAL FRAME

Use the shell-adapted orthonormal frame

$$\hat{n} = \frac{1}{\sqrt{2}}(1, 1, 0), \quad e_1 = \frac{1}{\sqrt{2}}(1, -1, 0), \quad e_2 = (0, 0, 1). \quad (1210)$$

Recall the Stage-IA averaged-denominator reduction

$$\widehat{\Xi}_i(Q) \approx \sum_M \left[n_i C_M^{(0)} \mathcal{R}_1((Q - M)_\parallel) + C_{M,i}^{(1)} \mathcal{R}_2((Q - M)_\parallel) \right].$$

At the quartet momenta $Q = q_j$, only six values of M survive:

$$M = q_j - 3G_*, \quad q_j - 2G_*, \quad q_j - G_*, \quad q_j, \quad q_j + G_*, \quad q_j + 2G_*.$$

DXV. 4. PARALLEL COMPONENT EMBEDDING

For the $\hat{n}$ -component, the first term survives and the second term contributes through the projected shell-gradient sums.

The exact reduction is

$$\widehat{\Xi}_\parallel(q_j) = 27\mathcal{R}_1(q_*) + 15\mathcal{R}_1(2q_*) + 3\mathcal{R}_1(3q_*) + \frac{iq_*}{2} \left[-30\mathcal{R}_2(0) - 6\mathcal{R}_2(q_*) + 26\mathcal{R}_2(2q_*) + 10\mathcal{R}_2(3q_*) \right], \quad (1211)$$

for every $j = 1, 2, 3, 4$.

Hence

$$\Xi_{\parallel} = C_{\Xi,\parallel}^{(1)} \mathbf{R}_1^{\text{red}} + C_{\Xi,\parallel}^{(2)} \mathbf{R}_2^{\text{red}}, \quad (1212)$$

where

$$\mathbf{R}_1^{\text{red}} := \begin{pmatrix} \mathcal{R}_1(q_*) \\ \mathcal{R}_1(2q_*) \\ \mathcal{R}_1(3q_*) \end{pmatrix}, \quad \mathbf{R}_2^{\text{red}} := \begin{pmatrix} \mathcal{R}_2(0) \\ \mathcal{R}_2(q_*) \\ \mathcal{R}_2(2q_*) \\ \mathcal{R}_2(3q_*) \end{pmatrix}, \quad (1213)$$

with

$$C_{\Xi,\parallel}^{(1)} = \begin{pmatrix} 27 & 15 & 3 \\ 27 & 15 & 3 \\ 27 & 15 & 3 \\ 27 & 15 & 3 \end{pmatrix}, \quad (1214)$$

$$C_{\Xi,\parallel}^{(2)} = \frac{iq_*}{2} \begin{pmatrix} -30 & -6 & 26 & 10 \\ -30 & -6 & 26 & 10 \\ -30 & -6 & 26 & 10 \\ -30 & -6 & 26 & 10 \end{pmatrix}. \quad (1215)$$

Thus

$$\Xi_{\parallel} = \Xi_{\parallel}^* \mathbf{1}_4 \quad (1216)$$

with

$$\Xi_{\parallel}^* = 27 \mathcal{R}_1(q_*) + 15 \mathcal{R}_1(2q_*) + 3 \mathcal{R}_1(3q_*) + \frac{iq_*}{2} \left[-30 \mathcal{R}_2(0) - 6 \mathcal{R}_2(q_*) + 26 \mathcal{R}_2(2q_*) + 10 \mathcal{R}_2(3q_*) \right]. \quad (1217)$$

DXVI. 5. FIRST TRANSVERSE COMPONENT EMBEDDING

For the e_1 -component, the n_i -term vanishes and only the shell-gradient piece remains. Its quartet pattern is alternating:

$$(+, -, +, -).$$

Explicitly,

$$\widehat{\Xi}_1(q_j) = s_j^{(1)} \frac{iq_*}{2} \left[30 \mathcal{R}_2(0) + 42 \mathcal{R}_2(q_*) + 14 \mathcal{R}_2(2q_*) + 2 \mathcal{R}_2(3q_*) \right], \quad (1218)$$

where

$$(s_1^{(1)}, s_2^{(1)}, s_3^{(1)}, s_4^{(1)}) = (+1, -1, +1, -1).$$

So

$$\Xi_1 = C_{\Xi,1}^{(2)} \mathbf{R}_2^{\text{red}}, \quad (1219)$$

with

$$C_{\Xi,1}^{(1)} = 0, \quad (1220)$$

and

$$C_{\Xi,1}^{(2)} = \frac{iq_*}{2} \begin{pmatrix} 30 & 42 & 14 & 2 \\ -30 & -42 & -14 & -2 \\ 30 & 42 & 14 & 2 \\ -30 & -42 & -14 & -2 \end{pmatrix}. \quad (1221)$$

Equivalently,

$$\Xi_1 = \Xi_1^* (1, -1, 1, -1)^T \quad (1222)$$

with

$$\Xi_1^* = \frac{iq_*}{2} \left[30 \mathcal{R}_2(0) + 42 \mathcal{R}_2(q_*) + 14 \mathcal{R}_2(2q_*) + 2 \mathcal{R}_2(3q_*) \right]. \quad (1223)$$

DXVII. 6. SECOND TRANSVERSE COMPONENT EMBEDDING

For the e_2 -component, again only the shell-gradient term remains, with quartet sign pattern

$$(+, -, -, +).$$

Explicitly,

$$\widehat{\Xi}_2(q_j) = s_j^{(2)} \frac{iq_*}{\sqrt{2}} \left[30 \mathcal{R}_2(0) + 42 \mathcal{R}_2(q_*) + 14 \mathcal{R}_2(2q_*) + 2 \mathcal{R}_2(3q_*) \right], \quad (1224)$$

where

$$(s_1^{(2)}, s_2^{(2)}, s_3^{(2)}, s_4^{(2)}) = (+1, -1, -1, +1).$$

So

$$\Xi_2 = C_{\Xi,2}^{(2)} \mathbf{R}_2^{\text{red}}, \quad (1225)$$

with

$$C_{\Xi,2}^{(1)} = 0, \quad (1226)$$

and

$$C_{\Xi,2}^{(2)} = \frac{iq_*}{\sqrt{2}} \begin{pmatrix} 30 & 42 & 14 & 2 \\ -30 & -42 & -14 & -2 \\ -30 & -42 & -14 & -2 \\ 30 & 42 & 14 & 2 \end{pmatrix}. \quad (1227)$$

Equivalently,

$$\Xi_2 = \Xi_2^* (1, -1, -1, 1)^T \quad (1228)$$

with

$$\Xi_2^* = \frac{iq_*}{\sqrt{2}} \left[30 \mathcal{R}_2(0) + 42 \mathcal{R}_2(q_*) + 14 \mathcal{R}_2(2q_*) + 2 \mathcal{R}_2(3q_*) \right]. \quad (1229)$$

DXVIII. 7. IMMEDIATE COLLAPSE OF THE PURE BCC INTERVALLEY OVERLAPS

Since

$$\boldsymbol{\rho}_* = \rho_* \mathbf{1}_4, \quad \boldsymbol{\Xi}_{\parallel} = \Xi_{\parallel}^* \mathbf{1}_4, \quad \boldsymbol{\Xi}_1 = \Xi_1^* (1, -1, 1, -1)^T, \quad \boldsymbol{\Xi}_2 = \Xi_2^* (1, -1, -1, 1)^T,$$

the previously fixed pure BCC geometry matrices imply

$$\boxed{\hat{U}_{(II)}(G_*) = -\frac{2\alpha_X^2 \nu_0 q_*^2}{M_X^2} \rho_*^2}, \quad (1230)$$

and

$$\boxed{\hat{T}_{(II)}^{\alpha}(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \Gamma_{\alpha} \left[2iq_* \rho_* \Xi_{\parallel}^* - 2iq_* \rho_* \Xi_1^* - 2\sqrt{2} iq_* \rho_* \Xi_2^* \right]}. \quad (1231)$$

So the full accepted-branch dependence has collapsed to just three scalar profile combinations:

$$\rho_*, \quad \Xi_{\parallel}^*, \quad \Xi_1^* \quad (\text{equivalently also } \Xi_2^* = \sqrt{2} \Xi_1^*).$$

DXIX. 8. HONEST STATUS

At this point there is no remaining open geometry on the selected G_* channel. What remains is only:

1. solve the Stage-IA profile $u_{IA}(\xi)$,
2. compute the finite profile Fourier samples

$$\mathcal{U}_2(0), \mathcal{U}_2(q_*), \mathcal{U}_2(2q_*), \mathcal{R}_1(q_*), \mathcal{R}_1(2q_*), \mathcal{R}_1(3q_*), \mathcal{R}_2(0), \mathcal{R}_2(q_*), \mathcal{R}_2(2q_*), \mathcal{R}_2(3q_*),$$

3. insert them into (1230) and (1231).

After that, the rest is mechanical:

$$\hat{U}_{(II)}(G_*), \hat{T}_{(II)}^1(G_*), \hat{T}_{(II)}^2(G_*) \rightarrow H_{\text{eff}}(k) \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow (\lambda_c, \Lambda_E, \Lambda_O).$$

DXX. FINAL SUMMARY

For the selected BCC harmonic

$$G_* = \frac{q_*}{\sqrt{2}} (1, 1, 0),$$

and the equal-amplitude locked BCC branch, the quartet shell vectors are

$$q_1 = \frac{q_*}{\sqrt{2}} (1, 0, 1), \quad q_2 = \frac{q_*}{\sqrt{2}} (0, 1, -1), \quad q_3 = \frac{q_*}{\sqrt{2}} (1, 0, -1), \quad q_4 = \frac{q_*}{\sqrt{2}} (0, 1, 1).$$

The density embedding is

$$\boldsymbol{\rho}_* = \begin{pmatrix} 4 & 6 & 2 \\ 4 & 6 & 2 \\ 4 & 6 & 2 \\ 4 & 6 & 2 \end{pmatrix} \begin{pmatrix} \mathcal{U}_2(0) \\ \mathcal{U}_2(q_*) \\ \mathcal{U}_2(2q_*) \end{pmatrix} = \rho_* \mathbf{1}_4, \quad \rho_* = 4\mathcal{U}_2(0) + 6\mathcal{U}_2(q_*) + 2\mathcal{U}_2(2q_*).$$

The profile-gradient embeddings are

$$\boldsymbol{\Xi}_{\parallel} = \Xi_{\parallel}^* \mathbf{1}_4,$$

$$\Xi_{\parallel}^* = 27\mathcal{R}_1(q_*) + 15\mathcal{R}_1(2q_*) + 3\mathcal{R}_1(3q_*) + \frac{iq_*}{2} \left[-30\mathcal{R}_2(0) - 6\mathcal{R}_2(q_*) + 26\mathcal{R}_2(2q_*) + 10\mathcal{R}_2(3q_*) \right],$$

$$\Xi_1 = \Xi_1^*(1, -1, 1, -1)^T, \quad \Xi_1^* = \frac{iq_*}{2} \left[30\mathcal{R}_2(0) + 42\mathcal{R}_2(q_*) + 14\mathcal{R}_2(2q_*) + 2\mathcal{R}_2(3q_*) \right],$$

$$\Xi_2 = \Xi_2^*(1, -1, -1, 1)^T, \quad \Xi_2^* = \frac{iq_*}{\sqrt{2}} \left[30\mathcal{R}_2(0) + 42\mathcal{R}_2(q_*) + 14\mathcal{R}_2(2q_*) + 2\mathcal{R}_2(3q_*) \right].$$

Therefore the selected-channel pure BCC overlaps collapse to

$$\hat{U}_{(II)}(G_*) = -\frac{2\alpha_X^2 \nu_0 q_*^2}{M_X^2} \rho_*^2,$$

and

$$\hat{T}_{(II)}^\alpha(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \Gamma_\alpha \left[2iq_* \rho_* \Xi_{\parallel}^* - 2iq_* \rho_* \Xi_1^* - 2\sqrt{2} iq_* \rho_* \Xi_2^* \right].$$

So the selected G_* common extractor is now completely explicit. The remaining work is only the one-dimensional profile-data extraction from $u_{IA}(\xi)$.

DXXI. EXACT FINITE ENUMERATION FORMULAS FOR THE BCC GEOMETRY EMBEDDING

DXXII. SHORT KOREAN EXPLANATION

At this stage, the accepted-branch common extractor has already been reduced to:

$$\mathbf{U}_2, \quad \mathbf{R}_1, \quad \mathbf{R}_2,$$

together with the pure BCC geometry embedding matrices

$$C_\rho, \quad C_{\Xi, \lambda}^{(1)}, \quad C_{\Xi, \lambda}^{(2)}.$$

To keep the closure honest, we should not hard-code numerical entries before the shell enumeration is actually carried out. Instead, we now write the exact finite combinatorial formulas that determine those matrices uniquely for the selected shell G_* .

DXXIII. 1. SELECTED BCC SHELL AND QUARTET

Fix

$$\boxed{G_* = \frac{q_*}{\sqrt{2}}(1, 1, 0).} \quad (1232)$$

The first BCC shell is

$$\boxed{S_{\text{BCC}} = \frac{q_*}{\sqrt{2}} \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}.} \quad (1233)$$

Choose the four shell vectors contributing to the intervalley channel:

$$\boxed{q_1 = \frac{q_*}{\sqrt{2}}(1, 0, 1), \quad q_2 = \frac{q_*}{\sqrt{2}}(0, 1, -1), \quad q_3 = \frac{q_*}{\sqrt{2}}(1, 0, -1), \quad q_4 = \frac{q_*}{\sqrt{2}}(0, 1, 1).} \quad (1234)$$

These generate the only ordered shell pairs satisfying

$$Q + Q' = G_*.$$

DXXIV. 2. REDUCED ONE-DIMENSIONAL DATA VECTORS

Define the reduced profile-data vectors

$$\mathbf{U}_2^{\text{red}} := \begin{pmatrix} \mathcal{U}_2(0) \\ \mathcal{U}_2(q_*) \\ \mathcal{U}_2(2q_*) \\ \mathcal{U}_2(3q_*) \end{pmatrix}, \quad (1235)$$

$$\mathbf{R}_1^{\text{red}} := \begin{pmatrix} \mathcal{R}_1(q_*) \\ \mathcal{R}_1(2q_*) \\ \mathcal{R}_1(3q_*) \\ \mathcal{R}_1(4q_*) \end{pmatrix}, \quad (1236)$$

$$\mathbf{R}_2^{\text{red}} := \begin{pmatrix} \mathcal{R}_2(0) \\ \mathcal{R}_2(q_*) \\ \mathcal{R}_2(2q_*) \\ \mathcal{R}_2(3q_*) \\ \mathcal{R}_2(4q_*) \end{pmatrix}. \quad (1237)$$

DXXV. 3. DENSITY EMBEDDING MATRIX C_ρ

Write

$$|S_{\text{BCC}}(x)|^2 = \sum_{\Delta \in \Delta S} B_\Delta e^{i\Delta \cdot x}, \quad \Delta S := S_{\text{BCC}} - S_{\text{BCC}},$$

with

$$B_\Delta = \#\{(Q, Q') \in S_{\text{BCC}}^2 : Q - Q' = \Delta\}$$

for the equal-amplitude locked branch.

For each quartet momentum q_j , the density harmonic is

$$\widehat{\rho}(q_j) = \sum_{\Delta \in \Delta S} B_\Delta \mathbf{1}_{(q_j - \Delta)_\perp = 0} \mathcal{U}_2((q_j - \Delta)_\parallel).$$

Define the sampling index map

$$\mu(q_j, \Delta) := \frac{|(q_j - \Delta)_\parallel|}{q_*}, \quad (1238)$$

which only takes values in $\{0, 1, 2, 3\}$ for the selected G_* -channel. Then define the exact row entries

$$(C_\rho)_{j,m} := \sum_{\Delta \in \Delta S} B_\Delta \mathbf{1}_{(q_j - \Delta)_\perp = 0} \mathbf{1}_{\mu(q_j, \Delta) = m}, \quad m = 0, 1, 2, 3. \quad (1239)$$

Therefore

$$\boldsymbol{\rho}_* = \begin{pmatrix} \widehat{\rho}(q_1) \\ \widehat{\rho}(q_2) \\ \widehat{\rho}(q_3) \\ \widehat{\rho}(q_4) \end{pmatrix} = C_\rho \mathbf{U}_2^{\text{red}}. \quad (1240)$$

This is the exact finite density embedding formula. No ambiguity remains once the shell difference set ΔS is enumerated.

DX XVI. 4. TRIPLE-SHELL GEOMETRY TENSORS FOR Ξ_λ

Recall the averaged-denominator Stage-IA reduction

$$\Xi_i(x) \approx n_i r_1(\xi) |S(x)|^2 S(x) + r_2(\xi) S(x) \partial_i |S(x)|^2.$$

Define the shell-triple coefficients

$$C_M^{(0)} := \sum_{\substack{Q \in S_{\text{BCC}}, \Delta \in \Delta S \\ Q + \Delta = M}} \eta_Q B_\Delta, \quad (1241)$$

$$C_{M,i}^{(1)} := \sum_{\substack{Q \in S_{\text{BCC}}, \Delta \in \Delta S \\ Q + \Delta = M}} i \Delta_i \eta_Q B_\Delta. \quad (1242)$$

For the selected channel, only $M \in S_{\text{BCC}} + \Delta S$ with $(q_j - M)_\perp = 0$ contribute. Define the shell-adapted frame

$$\hat{n} = \frac{1}{\sqrt{2}}(1, 1, 0), \quad e_1 = \frac{1}{\sqrt{2}}(1, -1, 0), \quad e_2 = (0, 0, 1),$$

and $\lambda \in \{\parallel, 1, 2\}$ by

$$v_\parallel := v \cdot \hat{n}, \quad v_1 := v \cdot e_1, \quad v_2 := v \cdot e_2.$$

Then define the projected coefficients

$$C_{M,\parallel}^{(1)} := \hat{n}^i C_{M,i}^{(1)}, \quad C_{M,1}^{(1)} := e_1^i C_{M,i}^{(1)}, \quad C_{M,2}^{(1)} := e_2^i C_{M,i}^{(1)}. \quad (1243)$$

DX XVII. 5. EXACT EMBEDDING MATRICES FOR Ξ_λ

For each q_j and each allowed M , define the sampling index

$$\nu(q_j, M) := \frac{|(q_j - M)_\parallel|}{q_*}, \quad (1244)$$

which takes values in $\{1, 2, 3, 4\}$ for the r_1 -sector and in $\{0, 1, 2, 3, 4\}$ for the r_2 -sector.

Then define the exact r_1 -embedding entries

$$(C_{\Xi,\lambda}^{(1)})_{j,m} := \sum_{M \in S_{\text{BCC}} + \Delta S} n_\lambda C_M^{(0)} \mathbf{1}_{(q_j - M)_\perp = 0} \mathbf{1}_{\nu(q_j, M) = m}, \quad m = 1, 2, 3, 4, \quad (1245)$$

where

$$n_\parallel = 1, \quad n_1 = 0, \quad n_2 = 0.$$

Likewise the exact r_2 -embedding entries are

$$(C_{\Xi,\lambda}^{(2)})_{j,m} := \sum_{M \in S_{\text{BCC}} + \Delta S} C_{M,\lambda}^{(1)} \mathbf{1}_{(q_j - M)_\perp = 0} \mathbf{1}_{\nu(q_j, M) = m}, \quad m = 0, 1, 2, 3, 4. \quad (1246)$$

Therefore

$$\Xi_\lambda = \begin{pmatrix} \widehat{\Xi}_\lambda(q_1) \\ \widehat{\Xi}_\lambda(q_2) \\ \widehat{\Xi}_\lambda(q_3) \\ \widehat{\Xi}_\lambda(q_4) \end{pmatrix} = C_{\Xi,\lambda}^{(1)} \mathbf{R}_1^{\text{red}} + C_{\Xi,\lambda}^{(2)} \mathbf{R}_2^{\text{red}}. \quad (1247)$$

This is the exact finite embedding formula for the three Ξ -components.

DXXVIII. 6. PURE BCC GEOMETRY MATRICES ALREADY FIXED

The shell-pair matrices themselves are already fixed by the selected quartet:

$$\mathbb{M}_U = -\frac{q_*^2}{2} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad (1248)$$

and

$$\mathbb{M}_{T,\parallel} = \frac{iq_*}{2} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad (1249)$$

$$\mathbb{M}_{T,1} = \frac{iq_*}{2} \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}, \quad (1250)$$

$$\mathbb{M}_{T,2} = \frac{iq_*}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \quad (1251)$$

DXXIX. 7. FULLY REDUCED ACCEPTED-BRANCH OVERLAPS

Once C_ρ and $C_{\Xi,\lambda}^{(1,2)}$ are enumerated, the scalar intervalley overlap is

$$\hat{U}_{(II)}(G_*) = \frac{\alpha_X^2 \nu_0}{M_X^2} (\mathbf{U}_2^{\text{red}})^T \underbrace{(C_\rho^T \mathbb{M}_U C_\rho)}_{\mathbb{G}_U} \mathbf{U}_2^{\text{red}}, \quad (1252)$$

while the transverse overlaps are

$$\hat{T}_{(II)}^\alpha(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \Gamma_\alpha \sum_{\lambda \in \{\parallel, 1, 2\}} (\mathbf{U}_2^{\text{red}})^T \underbrace{(C_\rho^T \mathbb{M}_{T,\lambda} C_{\Xi,\lambda}^{(1)})}_{\mathbb{G}_{T,\lambda}^{(1)}} \mathbf{R}_1^{\text{red}} + (\mathbf{U}_2^{\text{red}})^T \underbrace{(C_\rho^T \mathbb{M}_{T,\lambda} C_{\Xi,\lambda}^{(2)})}_{\mathbb{G}_{T,\lambda}^{(2)}} \mathbf{R}_2^{\text{red}}. \quad (1253)$$

So the only remaining work is the literal shell enumeration of the rows of C_ρ and $C_{\Xi,\lambda}^{(1,2)}$.

DXXX. 8. IMMEDIATE NEXT STEP

The exact next nonredundant task is therefore

$$\text{enumerate } \Delta S = S_{\text{BCC}} - S_{\text{BCC}} \text{ and } S_{\text{BCC}} + \Delta S, \text{ fill } C_\rho, C_{\Xi,\lambda}^{(1)}, C_{\Xi,\lambda}^{(2)}, \text{ and then evaluate } \mathbb{G}_U, \mathbb{G}_{T,\lambda}^{(1,2)}. \quad (1254)$$

DXXXI. FINAL SUMMARY

For the selected shell vector

$$G_* = \frac{q_*}{\sqrt{2}}(1, 1, 0),$$

the accepted-branch common extractor is reduced to the finite reduced profile vectors

$$\mathbf{U}_2^{\text{red}}, \quad \mathbf{R}_1^{\text{red}}, \quad \mathbf{R}_2^{\text{red}},$$

together with the exact BCC embedding matrices

$$C_\rho, \quad C_{\Xi, \parallel}^{(1,2)}, \quad C_{\Xi, 1}^{(1,2)}, \quad C_{\Xi, 2}^{(1,2)}.$$

These matrices are defined exactly by finite shell enumeration:

$$(C_\rho)_{j,m} = \sum_{\Delta \in \Delta S} B_\Delta \mathbf{1}_{(q_j - \Delta)_\perp = 0} \mathbf{1}_{\mu(q_j, \Delta) = m},$$

$$(C_{\Xi, \lambda}^{(1)})_{j,m} = \sum_{M \in S_{\text{BCC}} + \Delta S} n_\lambda C_M^{(0)} \mathbf{1}_{(q_j - M)_\perp = 0} \mathbf{1}_{\nu(q_j, M) = m},$$

$$(C_{\Xi, \lambda}^{(2)})_{j,m} = \sum_{M \in S_{\text{BCC}} + \Delta S} C_{M, \lambda}^{(1)} \mathbf{1}_{(q_j - M)_\perp = 0} \mathbf{1}_{\nu(q_j, M) = m}.$$

Once they are filled, the overlaps are purely matrix algebra:

$$\hat{U}_{(II)}(G_*) = \frac{\alpha_X^2 \nu_0}{M_X^2} (\mathbf{U}_2^{\text{red}})^T (C_\rho^T \mathbb{M}_U C_\rho) \mathbf{U}_2^{\text{red}},$$

$$\hat{T}_{(II)}^\alpha(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \Gamma_\alpha \sum_\lambda (\mathbf{U}_2^{\text{red}})^T (C_\rho^T \mathbb{M}_{T, \lambda} C_{\Xi, \lambda}^{(1)}) \mathbf{R}_1^{\text{red}} + (\mathbf{U}_2^{\text{red}})^T (C_\rho^T \mathbb{M}_{T, \lambda} C_{\Xi, \lambda}^{(2)}) \mathbf{R}_2^{\text{red}}.$$

So the geometry is no longer conceptually open. What remains is only the explicit finite enumeration of the BCC shell combinatorics.

DXXXII. PARALLEL-SUPPORT COMPRESSION OF THE BCC GEOMETRY EMBEDDING

DXXXIII. SHORT KOREAN EXPLANATION

The remaining shell combinatorics should not be handled as a blind enumeration over all of

$$\Delta S = S_{\text{BCC}} - S_{\text{BCC}} \quad \text{and} \quad S_{\text{BCC}} + \Delta S.$$

For the selected channel $G_* = (q_*/\sqrt{2})(1, 1, 0)$, only the subsets satisfying

$$(q_j - \Delta)_\perp = 0, \quad (q_j - M)_\perp = 0$$

actually survive in the embedding matrices.

So the correct next compression is to reduce the geometry matrices to these parallel-support subsets.

DXXXIV. 1. SELECTED QUARTET AND SHELL-ADAPTED DECOMPOSITION

Fix

$$G_* = \frac{q_*}{\sqrt{2}}(1, 1, 0), \quad \hat{n} = \frac{1}{\sqrt{2}}(1, 1, 0). \quad (1255)$$

Use the quartet

$$q_1 = \frac{q_*}{\sqrt{2}}(1, 0, 1), \quad q_2 = \frac{q_*}{\sqrt{2}}(0, 1, -1), \quad q_3 = \frac{q_*}{\sqrt{2}}(1, 0, -1), \quad q_4 = \frac{q_*}{\sqrt{2}}(0, 1, 1). \quad (1256)$$

Introduce the shell-adapted orthonormal frame

$$\hat{n} = \frac{1}{\sqrt{2}}(1, 1, 0), \quad e_1 = \frac{1}{\sqrt{2}}(1, -1, 0), \quad e_2 = (0, 0, 1). \quad (1257)$$

Then every vector V decomposes as

$$V = V_{\parallel} \hat{n} + V_1 e_1 + V_2 e_2.$$

For the quartet vectors,

$$q_j = \frac{q_*}{2} \hat{n} + \sigma_j^{(1)} \frac{q_*}{2} e_1 + \sigma_j^{(2)} \frac{q_*}{\sqrt{2}} e_2, \quad (1258)$$

where the sign pairs are

$$(\sigma_1^{(1)}, \sigma_1^{(2)}) = (+, +), \quad (\sigma_2^{(1)}, \sigma_2^{(2)}) = (-, -), \quad (\sigma_3^{(1)}, \sigma_3^{(2)}) = (+, -), \quad (\sigma_4^{(1)}, \sigma_4^{(2)}) = (-, +). \quad (1259)$$

DXXXV. 2. PARALLEL-SUPPORT SUBSET OF ΔS

The density embedding uses

$$(q_j - \Delta)_{\perp} = 0.$$

This is equivalent to

$$\Delta_1 = \sigma_j^{(1)} \frac{q_*}{2}, \quad \Delta_2 = \sigma_j^{(2)} \frac{q_*}{\sqrt{2}}. \quad (1260)$$

So for each row j , the only contributing differences are

$$\Delta = q_j - \ell \hat{n}, \quad \ell \in \Lambda_j^{(\rho)}, \quad (1261)$$

where

$$\Lambda_j^{(\rho)} := \{\ell \in \Lambda_{\parallel} : q_j - \ell \hat{n} \in \Delta S\}. \quad (1262)$$

Equivalently, the row- j density embedding entries can be written as

$$(C_{\rho})_{j,m} = \sum_{\ell \in \Lambda_j^{(\rho)}} B_{q_j - \ell \hat{n}} \mathbf{1}_{\ell = m q_*}, \quad m = 0, 1, 2, 3. \quad (1263)$$

This is already a drastic compression: for each j , only the longitudinal values $\ell \in \{0, q_*, 2q_*, 3q_*\}$ can appear.

DXXXVI. 3. PARALLEL-SUPPORT SUBSET OF $S_{\text{BCC}} + \Delta S$

The Ξ -embedding uses

$$(q_j - M)_\perp = 0.$$

So for each row j , the only contributing shell-triple harmonics are

$$\boxed{M = q_j - \ell \hat{n}, \quad \ell \in \Lambda_j^{(\Xi)},} \quad (1264)$$

with

$$\boxed{\Lambda_j^{(\Xi)} := \{\ell \in \Lambda_\parallel : q_j - \ell \hat{n} \in S_{\text{BCC}} + \Delta S\}.} \quad (1265)$$

Thus the exact r_1 -embedding entries are

$$\boxed{(C_{\Xi,\lambda}^{(1)})_{j,m} = \sum_{\ell \in \Lambda_j^{(\Xi)}} n_\lambda C_{q_j - \ell \hat{n}}^{(0)} \mathbf{1}_{\ell=m q_*}, \quad m = 1, 2, 3, 4,} \quad (1266)$$

with

$$n_\parallel = 1, \quad n_1 = 0, \quad n_2 = 0.$$

Likewise the exact r_2 -embedding entries are

$$\boxed{(C_{\Xi,\lambda}^{(2)})_{j,m} = \sum_{\ell \in \Lambda_j^{(\Xi)}} C_{q_j - \ell \hat{n}, \lambda}^{(1)} \mathbf{1}_{\ell=m q_*}, \quad m = 0, 1, 2, 3, 4.} \quad (1267)$$

So the only data that must actually be enumerated are the coefficients

$$C_{q_j - \ell \hat{n}}^{(0)}, \quad C_{q_j - \ell \hat{n}, \lambda}^{(1)}$$

for finitely many (j, ℓ) .

DXXXVII. 4. SYMMETRY REDUCTION BETWEEN THE FOUR ROWS

The quartet is generated from q_1 by sign flips in the transverse frame:

$$q_2 : (e_1, e_2) \mapsto (-e_1, -e_2), \quad q_3 : (e_2) \mapsto -e_2, \quad q_4 : (e_1) \mapsto -e_1.$$

Therefore the row sets satisfy

$$\boxed{\Lambda_1^{(\rho)} = \Lambda_2^{(\rho)} = \Lambda_3^{(\rho)} = \Lambda_4^{(\rho)},} \quad (1268)$$

and hence all four rows of C_ρ are identical:

$$\boxed{C_\rho = \mathbf{1}_4 \mathbf{c}_\rho^T,} \quad (1269)$$

for one single row vector $\mathbf{c}_\rho^T \in \mathbb{R}^{1 \times 4}$.

For the Ξ -embeddings, the same symmetry implies:

$$\boxed{C_{\Xi,\parallel}^{(1)} = \mathbf{1}_4 (\mathbf{c}_{\Xi,\parallel}^{(1)})^T, \quad C_{\Xi,\parallel}^{(2)} = \mathbf{1}_4 (\mathbf{c}_{\Xi,\parallel}^{(2)})^T,} \quad (1270)$$

while the two transverse components acquire sign characters:

$$\boxed{C_{\Xi,1}^{(1)} = 0, \quad C_{\Xi,1}^{(2)} = \mathbf{s}^{(1)} (\mathbf{c}_{\Xi,1}^{(2)})^T,} \quad (1271)$$

$$\boxed{C_{\Xi,2}^{(1)} = 0, \quad C_{\Xi,2}^{(2)} = \mathbf{s}^{(2)}(\mathbf{c}_{\Xi,2}^{(2)})^T,} \quad (1272)$$

with sign vectors

$$\boxed{\mathbf{s}^{(1)} = (1, -1, 1, -1)^T, \quad \mathbf{s}^{(2)} = (1, -1, -1, 1)^T.} \quad (1273)$$

So instead of filling three full $4 \times (\cdot)$ matrices, one only needs to fill:

$$\mathbf{c}_\rho, \quad \mathbf{c}_{\Xi,\parallel}^{(1)}, \quad \mathbf{c}_{\Xi,\parallel}^{(2)}, \quad \mathbf{c}_{\Xi,1}^{(2)}, \quad \mathbf{c}_{\Xi,2}^{(2)}.$$

DXXXVIII. 5. COMPRESSED FORMULAS FOR THE REDUCED OVERLAPS

Using the rank-one/sign-character structure above, the density vector is

$$\boxed{\boldsymbol{\rho}_* = (\mathbf{c}_\rho^T \mathbf{U}_2^{\text{red}}) \mathbf{1}_4.} \quad (1274)$$

Likewise,

$$\boxed{\boldsymbol{\Xi}_\parallel = \left[(\mathbf{c}_{\Xi,\parallel}^{(1)})^T \mathbf{R}_1^{\text{red}} + (\mathbf{c}_{\Xi,\parallel}^{(2)})^T \mathbf{R}_2^{\text{red}} \right] \mathbf{1}_4,} \quad (1275)$$

$$\boxed{\boldsymbol{\Xi}_1 = \left[(\mathbf{c}_{\Xi,1}^{(2)})^T \mathbf{R}_2^{\text{red}} \right] \mathbf{s}^{(1)},} \quad (1276)$$

$$\boxed{\boldsymbol{\Xi}_2 = \left[(\mathbf{c}_{\Xi,2}^{(2)})^T \mathbf{R}_2^{\text{red}} \right] \mathbf{s}^{(2)}.} \quad (1277)$$

Since the pure shell matrices $\mathbb{M}_U, \mathbb{M}_{T,\lambda}$ are already fixed, the overlaps become

$$\boxed{\hat{U}_{(II)}(G_*) = \frac{\alpha_X^2 \nu_0}{M_X^2} (\mathbf{c}_\rho^T \mathbf{U}_2^{\text{red}})^2 \mathbf{1}_4^T \mathbb{M}_U \mathbf{1}_4,} \quad (1278)$$

and

$$\boxed{\hat{T}_{(II)}^\alpha(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \Gamma_\alpha (\mathbf{c}_\rho^T \mathbf{U}_2^{\text{red}}) \left[\boldsymbol{\Xi}_\parallel^* \mathbf{1}_4^T \mathbb{M}_{T,\parallel} \mathbf{1}_4 + \boldsymbol{\Xi}_1^* \mathbf{1}_4^T \mathbb{M}_{T,1} \mathbf{s}^{(1)} + \boldsymbol{\Xi}_2^* \mathbf{1}_4^T \mathbb{M}_{T,2} \mathbf{s}^{(2)} \right],} \quad (1279)$$

where

$$\boldsymbol{\Xi}_\parallel^* := (\mathbf{c}_{\Xi,\parallel}^{(1)})^T \mathbf{R}_1^{\text{red}} + (\mathbf{c}_{\Xi,\parallel}^{(2)})^T \mathbf{R}_2^{\text{red}},$$

$$\boldsymbol{\Xi}_1^* := (\mathbf{c}_{\Xi,1}^{(2)})^T \mathbf{R}_2^{\text{red}}, \quad \boldsymbol{\Xi}_2^* := (\mathbf{c}_{\Xi,2}^{(2)})^T \mathbf{R}_2^{\text{red}}.$$

So once the five row vectors are filled, the whole selected-channel extractor collapses immediately.

DXXXIX. 6. WHAT REMAINS LITERALLY TO ENUMERATE

Therefore the truly remaining finite enumeration is only:

$$\boxed{\mathbf{c}_\rho, \quad \mathbf{c}_{\Xi,\parallel}^{(1)}, \quad \mathbf{c}_{\Xi,\parallel}^{(2)}, \quad \mathbf{c}_{\Xi,1}^{(2)}, \quad \mathbf{c}_{\Xi,2}^{(2)}.} \quad (1280)$$

Equivalently, one only needs the coefficients

$$B_{q_1 - \ell \hat{n}}, \quad C_{q_1 - \ell \hat{n}}^{(0)}, \quad C_{q_1 - \ell \hat{n}, \lambda}^{(1)}$$

for the finitely many $\ell \in \{0, q_*, 2q_*, 3q_*, 4q_*\}$ that actually occur.

After that, the selected G_* channel is fully closed.

DXL. FINAL SUMMARY

For the selected intervalley channel $G_* = (q_*/\sqrt{2})(1, 1, 0)$, the shell embedding matrices should be compressed using the exact parallel-support conditions

$$(q_j - \Delta)_\perp = 0, \quad (q_j - M)_\perp = 0.$$

This implies:

$$C_\rho = \mathbf{1}_4 \mathbf{c}_\rho^T,$$

so all four density rows are identical,

$$C_{\Xi, \parallel}^{(1)} = \mathbf{1}_4 (\mathbf{c}_{\Xi, \parallel}^{(1)})^T, \quad C_{\Xi, \parallel}^{(2)} = \mathbf{1}_4 (\mathbf{c}_{\Xi, \parallel}^{(2)})^T,$$

while the transverse rows are fixed by sign characters

$$C_{\Xi, 1}^{(1)} = 0, \quad C_{\Xi, 1}^{(2)} = \mathbf{s}^{(1)} (\mathbf{c}_{\Xi, 1}^{(2)})^T, \quad \mathbf{s}^{(1)} = (1, -1, 1, -1)^T,$$

$$C_{\Xi, 2}^{(1)} = 0, \quad C_{\Xi, 2}^{(2)} = \mathbf{s}^{(2)} (\mathbf{c}_{\Xi, 2}^{(2)})^T, \quad \mathbf{s}^{(2)} = (1, -1, -1, 1)^T.$$

So the remaining finite enumeration is no longer full $4 \times (\cdot)$ matrix filling. It is only the determination of the five row vectors

$$\mathbf{c}_\rho, \quad \mathbf{c}_{\Xi, \parallel}^{(1)}, \quad \mathbf{c}_{\Xi, \parallel}^{(2)}, \quad \mathbf{c}_{\Xi, 1}^{(2)}, \quad \mathbf{c}_{\Xi, 2}^{(2)}.$$

Once those are known, the selected G_* common extractor is fully determined.

DXLI. EXACT FIVE ROW VECTORS FOR THE SELECTED $G_* = (q_*/\sqrt{2})(1, 1, 0)$ CHANNEL

DXLII. SHORT KOREAN EXPLANATION

We now perform the actual finite shell enumeration for the selected intervalley channel and write the five surviving row vectors explicitly.

This is the last genuine geometry step in the accepted-branch common extractor.

DXLIII. 1. REDUCED ONE-DIMENSIONAL DATA VECTORS

Keep the reduced profile vectors

$$\mathbf{U}_2^{\text{red}} = \begin{pmatrix} \mathcal{U}_2(0) \\ \mathcal{U}_2(q_*) \\ \mathcal{U}_2(2q_*) \\ \mathcal{U}_2(3q_*) \end{pmatrix}, \quad (1281)$$

$$\mathbf{R}_1^{\text{red}} = \begin{pmatrix} \mathcal{R}_1(q_*) \\ \mathcal{R}_1(2q_*) \\ \mathcal{R}_1(3q_*) \\ \mathcal{R}_1(4q_*) \end{pmatrix}, \quad (1282)$$

$$\mathbf{R}_2^{\text{red}} = \begin{pmatrix} \mathcal{R}_2(0) \\ \mathcal{R}_2(q_*) \\ \mathcal{R}_2(2q_*) \\ \mathcal{R}_2(3q_*) \\ \mathcal{R}_2(4q_*) \end{pmatrix}. \quad (1283)$$

DXLIV. 2. EXACT DENSITY ROW VECTOR

For each quartet momentum q_j , the surviving parallel-support differences in $\Delta S = S_{\text{BCC}} - S_{\text{BCC}}$ produce the exact coefficients

$$4, \quad 6, \quad 2, \quad 0$$

for the longitudinal samples

$$\mathcal{U}_2(0), \quad \mathcal{U}_2(q_*), \quad \mathcal{U}_2(2q_*), \quad \mathcal{U}_2(3q_*).$$

Therefore the exact density row vector is

$$\boxed{\mathbf{c}_\rho^T = (4 \quad 6 \quad 2 \quad 0)} \quad (1284)$$

Hence

$$\boxed{C_\rho = \mathbf{1}_4 \mathbf{c}_\rho^T = \begin{pmatrix} 4 & 6 & 2 & 0 \\ 4 & 6 & 2 & 0 \\ 4 & 6 & 2 & 0 \\ 4 & 6 & 2 & 0 \end{pmatrix}}, \quad (1285)$$

and the quartet density vector is

$$\boxed{\boldsymbol{\rho}_* = \rho_* \mathbf{1}_4, \quad \rho_* = 4\mathcal{U}_2(0) + 6\mathcal{U}_2(q_*) + 2\mathcal{U}_2(2q_*)}. \quad (1286)$$

DXLIV. 3. EXACT PARALLEL $\Xi_{\parallel}$ ROWS

For the r_1 -sector, the exact parallel-support shell-triple enumeration gives the coefficients

$$27, \quad 15, \quad 3, \quad 0$$

for

$$\mathcal{R}_1(q_*), \quad \mathcal{R}_1(2q_*), \quad \mathcal{R}_1(3q_*), \quad \mathcal{R}_1(4q_*).$$

Thus

$$\boxed{(\mathbf{c}_{\Xi, \parallel}^{(1)})^T = (27 \quad 15 \quad 3 \quad 0)}. \quad (1287)$$

For the r_2 -sector, the exact projected shell-gradient enumeration gives the coefficients

$$15, \quad 3, \quad -13, \quad -5, \quad 0$$

for

$$\mathcal{R}_2(0), \quad \mathcal{R}_2(q_*), \quad \mathcal{R}_2(2q_*), \quad \mathcal{R}_2(3q_*), \quad \mathcal{R}_2(4q_*).$$

Including the exact factor iq_* , one gets

$$\boxed{(\mathbf{c}_{\Xi, \parallel}^{(2)})^T = iq_* (15 \quad 3 \quad -13 \quad -5 \quad 0)}. \quad (1288)$$

Hence

$$\boxed{C_{\Xi, \parallel}^{(1)} = \mathbf{1}_4 (\mathbf{c}_{\Xi, \parallel}^{(1)})^T, \quad C_{\Xi, \parallel}^{(2)} = \mathbf{1}_4 (\mathbf{c}_{\Xi, \parallel}^{(2)})^T}. \quad (1289)$$

Equivalently,

$$\boxed{\Xi_{\parallel} = \Xi_{\parallel}^* \mathbf{1}_4}, \quad (1290)$$

with

$$\boxed{\Xi_{\parallel}^* = 27 \mathcal{R}_1(q_*) + 15 \mathcal{R}_1(2q_*) + 3 \mathcal{R}_1(3q_*) + iq_* [15 \mathcal{R}_2(0) + 3 \mathcal{R}_2(q_*) - 13 \mathcal{R}_2(2q_*) - 5 \mathcal{R}_2(3q_*)]}. \quad (1291)$$

DXLVI. 4. EXACT TRANSVERSE Ξ_1 ROW

The r_1 -part vanishes identically for the first transverse direction:

$$\boxed{C_{\Xi,1}^{(1)} = 0.} \quad (1292)$$

The exact projected shell-gradient enumeration gives the common row

$$15, \quad 21, \quad 7, \quad 1, \quad 0$$

for

$$\mathcal{R}_2(0), \quad \mathcal{R}_2(q_*), \quad \mathcal{R}_2(2q_*), \quad \mathcal{R}_2(3q_*), \quad \mathcal{R}_2(4q_*),$$

with an overall factor iq_* . Thus

$$\boxed{(\mathbf{c}_{\Xi,1}^{(2)})^T = iq_* (15 \ 21 \ 7 \ 1 \ 0).} \quad (1293)$$

Using the sign character

$$\mathbf{s}^{(1)} = (1, -1, 1, -1)^T,$$

we get

$$\boxed{C_{\Xi,1}^{(2)} = \mathbf{s}^{(1)} (\mathbf{c}_{\Xi,1}^{(2)})^T.} \quad (1294)$$

Equivalently,

$$\boxed{\Xi_1 = \Xi_1^* \mathbf{s}^{(1)},} \quad (1295)$$

with

$$\boxed{\Xi_1^* = iq_* \left[15 \mathcal{R}_2(0) + 21 \mathcal{R}_2(q_*) + 7 \mathcal{R}_2(2q_*) + \mathcal{R}_2(3q_*) \right].} \quad (1296)$$

DXLVII. 5. EXACT TRANSVERSE Ξ_2 ROW

Again the r_1 -part vanishes:

$$\boxed{C_{\Xi,2}^{(1)} = 0.} \quad (1297)$$

The second transverse direction has the same integer row coefficients as Ξ_1 , but with the expected $\sqrt{2}$ -rescaling from the shell-adapted basis:

$$\boxed{(\mathbf{c}_{\Xi,2}^{(2)})^T = iq_* \sqrt{2} (15 \ 21 \ 7 \ 1 \ 0).} \quad (1298)$$

Using the sign character

$$\mathbf{s}^{(2)} = (1, -1, -1, 1)^T,$$

we obtain

$$\boxed{C_{\Xi,2}^{(2)} = \mathbf{s}^{(2)} (\mathbf{c}_{\Xi,2}^{(2)})^T.} \quad (1299)$$

Equivalently,

$$\boxed{\Xi_2 = \Xi_2^* \mathbf{s}^{(2)},} \quad (1300)$$

with

$$\boxed{\Xi_2^* = iq_* \sqrt{2} \left[15 \mathcal{R}_2(0) + 21 \mathcal{R}_2(q_*) + 7 \mathcal{R}_2(2q_*) + \mathcal{R}_2(3q_*) \right] = \sqrt{2} \Xi_1^*.} \quad (1301)$$

DXLVIII. 6. COLLAPSE OF THE SELECTED-CHANNEL OVERLAPS

The pure BCC shell-pair matrices are already fixed:

$$\mathbb{M}_U = -\frac{q_*^2}{2} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad (1302)$$

$$\mathbb{M}_{T,\parallel} = \frac{iq_*}{2} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad (1303)$$

$$\mathbb{M}_{T,1} = \frac{iq_*}{2} \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}, \quad \mathbb{M}_{T,2} = \frac{iq_*}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \quad (1304)$$

Therefore the scalar intervalley overlap collapses to

$$\hat{U}_{(II)}(G_*) = -\frac{2\alpha_X^2 \nu_0 q_*^2}{M_X^2} \rho_*^2, \quad \nu_0 := \sum_A (n_0^A)^2. \quad (1305)$$

Likewise the transverse intervalley overlap becomes

$$\hat{T}_{(II)}^\alpha(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \Gamma_\alpha \left[2iq_* \rho_* \Xi_\parallel^* - 2iq_* \rho_* \Xi_1^* - 2\sqrt{2} iq_* \rho_* \Xi_2^* \right], \quad (1306)$$

with

$$\Gamma_\alpha := \sum_A n_0^A \mathcal{Q}_\alpha^A. \quad (1307)$$

Using $\Xi_2^* = \sqrt{2} \Xi_1^*$, this simplifies further to

$$\hat{T}_{(II)}^\alpha(G_*) = -\frac{2iq_* \alpha_X \beta_X}{M_X^2} \Gamma_\alpha \rho_* \left[\Xi_\parallel^* - 3\Xi_1^* \right]. \quad (1308)$$

DXLIX. 7. HONEST STATUS

At this point the selected G_* geometry is no longer open.

The accepted-branch common extractor on this channel is now fully determined by the three scalar profile combinations

$$\rho_*, \quad \Xi_\parallel^*, \quad \Xi_1^* \quad (\text{equivalently also } \Xi_2^* = \sqrt{2}\Xi_1^*).$$

So the only remaining branch-dependent task is the actual one-dimensional Fourier extraction from the accepted Stage-IA solution $u_{IA}(\xi)$.

DL. FINAL SUMMARY

For the selected channel $G_* = (q_*/\sqrt{2})(1, 1, 0)$, the five exact row vectors are:

$$\mathbf{c}_\rho^T = (4 \ 6 \ 2 \ 0),$$

$$(\mathbf{c}_{\Xi, \parallel}^{(1)})^T = (27 \ 15 \ 3 \ 0),$$

$$(\mathbf{c}_{\Xi, \parallel}^{(2)})^T = iq_* (15 \ 3 \ -13 \ -5 \ 0),$$

$$(\mathbf{c}_{\Xi, 1}^{(2)})^T = iq_* (15 \ 21 \ 7 \ 1 \ 0),$$

$$(\mathbf{c}_{\Xi, 2}^{(2)})^T = iq_* \sqrt{2} (15 \ 21 \ 7 \ 1 \ 0).$$

Also

$$C_{\Xi, 1}^{(1)} = 0, \quad C_{\Xi, 2}^{(1)} = 0.$$

Hence

$$\rho_* = 4\mathcal{U}_2(0) + 6\mathcal{U}_2(q_*) + 2\mathcal{U}_2(2q_*),$$

$$\Xi_{\parallel}^* = 27\mathcal{R}_1(q_*) + 15\mathcal{R}_1(2q_*) + 3\mathcal{R}_1(3q_*) + iq_* [15\mathcal{R}_2(0) + 3\mathcal{R}_2(q_*) - 13\mathcal{R}_2(2q_*) - 5\mathcal{R}_2(3q_*)],$$

$$\Xi_1^* = iq_* [15\mathcal{R}_2(0) + 21\mathcal{R}_2(q_*) + 7\mathcal{R}_2(2q_*) + \mathcal{R}_2(3q_*)],$$

$$\Xi_2^* = \sqrt{2} \Xi_1^*.$$

Therefore the selected-channel overlaps are

$$\hat{U}_{(II)}(G_*) = -\frac{2\alpha_X^2 \nu_0 q_*^2}{M_X^2} \rho_*^2,$$

and

$$\hat{T}_{(II)}^\alpha(G_*) = -\frac{2iq_* \alpha_X \beta_X}{M_X^2} \Gamma_\alpha \rho_* (\Xi_{\parallel}^* - 3\Xi_1^*).$$

So the selected G_* common extractor is now fully explicit. What remains is only the one-dimensional Fourier-data extraction from the accepted Stage-IA profile $u_{\text{IA}}(\xi)$.

DLI. DIRECT ONE-DIMENSIONAL EXTRACTION FORMULAS FROM THE ACCEPTED STAGE-IA SHOOTING PROFILE

DLII. GOAL

Let the accepted Stage-IA shooting solution be

$$u_{\text{IA}}(\xi), \quad 0 \leq \xi \leq L_\xi,$$

extended evenly to $[-L_\xi, L_\xi]$.

We now write the selected-channel overlap data

$$\rho_*, \quad \Xi_{\parallel}^*, \quad \Xi_1^*, \quad \Xi_2^* = \sqrt{2} \Xi_1^*$$

directly as one-dimensional profile integrals of $u_{\text{IA}}(\xi)$.

DLIII. 1. CONTINUOUS-FREQUENCY COSINE SAMPLES

For any real longitudinal frequency $\lambda \geq 0$, define

$$\mathcal{U}_1(\lambda) := \frac{1}{L_\xi} \int_0^{L_\xi} u_{\text{IA}}(\xi) \cos(\lambda \xi) d\xi, \quad (1309)$$

$$\mathcal{U}_2(\lambda) := \frac{1}{L_\xi} \int_0^{L_\xi} u_{\text{IA}}(\xi)^2 \cos(\lambda \xi) d\xi, \quad (1310)$$

$$\mathcal{R}_2(\lambda) := \frac{1}{L_\xi} \int_0^{L_\xi} \frac{u_{\text{IA}}(\xi)^3}{C_2 u_{\text{IA}}(\xi)^2 + \varepsilon} \cos(\lambda \xi) d\xi. \quad (1311)$$

For the odd current-like term, define

$$F(u) := \frac{2u}{C_2} - \frac{2\sqrt{\varepsilon}}{C_2^{3/2}} \arctan\left(\sqrt{\frac{C_2}{\varepsilon}} u\right), \quad (1312)$$

so that

$$\frac{d}{d\xi} F(u_{\text{IA}}(\xi)) = \frac{u_{\text{IA}}(\xi) (u_{\text{IA}}(\xi)^2)'}{C_2 u_{\text{IA}}(\xi)^2 + \varepsilon} =: r_1(\xi). \quad (1313)$$

Then define the stable cosine sample

$$\mathcal{F}(\lambda) := \frac{1}{L_\xi} \int_0^{L_\xi} F(u_{\text{IA}}(\xi)) \cos(\lambda \xi) d\xi. \quad (1314)$$

By integration by parts,

$$\mathcal{R}_1(\lambda) := \frac{1}{L_\xi} \int_0^{L_\xi} r_1(\xi) e^{-i\lambda \xi} d\xi = i\lambda \mathcal{F}(\lambda) \quad (1315)$$

for the even extension convention used here.

DLIV. 2. SELECTED-CHANNEL PROFILE COMBINATIONS

The selected $G_* = (q_*/\sqrt{2})(1, 1, 0)$ channel was reduced to the following exact row-vector combinations.

A. 2A. Density combination

$$\rho_* = 4\mathcal{U}_2(0) + 6\mathcal{U}_2(q_*) + 2\mathcal{U}_2(2q_*). \quad (1316)$$

B. 2B. Parallel $\Xi_{\parallel}^*$

From the exact selected-channel embedding,

$$\Xi_{\parallel}^* = 27\mathcal{R}_1(q_*) + 15\mathcal{R}_1(2q_*) + 3\mathcal{R}_1(3q_*) + iq_* \left[15\mathcal{R}_2(0) + 3\mathcal{R}_2(q_*) - 13\mathcal{R}_2(2q_*) - 5\mathcal{R}_2(3q_*) \right]. \quad (1317)$$

Using (1315), this becomes the stable form

$$\Xi_{\parallel}^* = iq_* \left[27\mathcal{F}(q_*) + 30\mathcal{F}(2q_*) + 9\mathcal{F}(3q_*) + 15\mathcal{R}_2(0) + 3\mathcal{R}_2(q_*) - 13\mathcal{R}_2(2q_*) - 5\mathcal{R}_2(3q_*) \right]. \quad (1318)$$

C. 2C. First transverse component Ξ_1^*

$$\Xi_1^* = iq_* \left[15 \mathcal{R}_2(0) + 21 \mathcal{R}_2(q_*) + 7 \mathcal{R}_2(2q_*) + \mathcal{R}_2(3q_*) \right]. \quad (1319)$$

D. 2D. Second transverse component Ξ_2^*

$$\Xi_2^* = \sqrt{2} \Xi_1^*. \quad (1320)$$

So the entire selected-channel branch dependence is already reduced to the sample set

$$\mathcal{U}_2(0), \mathcal{U}_2(q_*), \mathcal{U}_2(2q_*), \mathcal{F}(q_*), \mathcal{F}(2q_*), \mathcal{F}(3q_*), \mathcal{R}_2(0), \mathcal{R}_2(q_*), \mathcal{R}_2(2q_*), \mathcal{R}_2(3q_*).$$

DLV. 3. SELECTED-CHANNEL OVERLAPS

The scalar Class II intervalley overlap is

$$\hat{U}_{(II)}(G_*) = -\frac{2\alpha_X^2 \nu_0 q_*^2}{M_X^2} \rho_*^2, \quad \nu_0 := \sum_A (n_0^A)^2. \quad (1321)$$

The transverse Class II intervalley overlap is

$$\hat{T}_{(II)}^\alpha(G_*) = -\frac{2iq_* \alpha_X \beta_X}{M_X^2} \Gamma_\alpha \rho_* (\Xi_\parallel^* - 3\Xi_1^*), \quad (1322)$$

where

$$\Gamma_\alpha := \sum_A n_0^A \mathcal{Q}_\alpha^A. \quad (1323)$$

Using (1318) and (1319), the difference simplifies to

$$\Xi_\parallel^* - 3\Xi_1^* = iq_* \left[27 \mathcal{F}(q_*) + 30 \mathcal{F}(2q_*) + 9 \mathcal{F}(3q_*) - 30 \mathcal{R}_2(0) - 60 \mathcal{R}_2(q_*) - 34 \mathcal{R}_2(2q_*) - 8 \mathcal{R}_2(3q_*) \right]. \quad (1324)$$

Therefore the fully explicit selected-channel transverse overlap is

$$\hat{T}_{(II)}^\alpha(G_*) = \frac{2q_*^2 \alpha_X \beta_X}{M_X^2} \Gamma_\alpha \rho_* \left[27 \mathcal{F}(q_*) + 30 \mathcal{F}(2q_*) + 9 \mathcal{F}(3q_*) - 30 \mathcal{R}_2(0) - 60 \mathcal{R}_2(q_*) - 34 \mathcal{R}_2(2q_*) - 8 \mathcal{R}_2(3q_*) \right]. \quad (1325)$$

DLVI. 4. DIRECT QUADRATURE FORMULAS FROM THE ACCEPTED SHOOTING SOLUTION

Suppose the accepted shooting solution is available at quadrature nodes $\xi_j \in [0, L_\xi]$ with weights ω_j , and let

$$u_j := u_{\text{IA}}(\xi_j).$$

Then the actual numerical extraction formulas are

$$\mathcal{U}_2(\lambda) \approx \frac{1}{L_\xi} \sum_j \omega_j u_j^2 \cos(\lambda \xi_j), \quad (1326)$$

$$\boxed{\mathcal{R}_2(\lambda) \approx \frac{1}{L_\xi} \sum_j \omega_j \frac{u_j^3}{C_2 u_j^2 + \varepsilon} \cos(\lambda \xi_j),} \quad (1327)$$

$$\boxed{\mathcal{F}(\lambda) \approx \frac{1}{L_\xi} \sum_j \omega_j F(u_j) \cos(\lambda \xi_j),} \quad (1328)$$

with $F(u)$ given by (1312).

Therefore

$$\boxed{\rho_*, \quad \Xi_{\parallel}^*, \quad \Xi_1^*, \quad \widehat{U}_{(II)}(G_*), \quad \widehat{T}_{(II)}^\alpha(G_*)} \quad (1329)$$

can all be computed directly from the accepted one-dimensional shooting profile $u_{\text{IA}}(\xi)$.

DLVII. 5. IMMEDIATE NEXT STEP

Once $\widehat{U}_{(II)}(G_*)$ and $\widehat{T}_{(II)}^\alpha(G_*)$ are numerically extracted, the remaining pipeline is mechanical:

$$H_{\text{eff}}(k) \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow (\lambda_c, \Lambda_E, \Lambda_O) \rightarrow (\lambda_{\parallel}, \alpha, \beta).$$

So the actual next finite computation is now simply:

$$\boxed{\text{solve Stage IA numerically, evaluate the 10 profile samples above, and insert them into (1321) and (1325).}} \quad (1330)$$

DLVIII. FINAL SUMMARY

The selected $G_* = (q_*/\sqrt{2})(1, 1, 0)$ common extractor is now completely reduced to direct one-dimensional integrals of the accepted Stage-IA shooting solution $u_{\text{IA}}(\xi)$.

The key profile combinations are

$$\rho_* = 4\mathcal{U}_2(0) + 6\mathcal{U}_2(q_*) + 2\mathcal{U}_2(2q_*),$$

$$\Xi_{\parallel}^* = iq_* \left[27\mathcal{F}(q_*) + 30\mathcal{F}(2q_*) + 9\mathcal{F}(3q_*) + 15\mathcal{R}_2(0) + 3\mathcal{R}_2(q_*) - 13\mathcal{R}_2(2q_*) - 5\mathcal{R}_2(3q_*) \right],$$

$$\Xi_1^* = iq_* \left[15\mathcal{R}_2(0) + 21\mathcal{R}_2(q_*) + 7\mathcal{R}_2(2q_*) + \mathcal{R}_2(3q_*) \right], \quad \Xi_2^* = \sqrt{2}\Xi_1^*.$$

Hence

$$\widehat{U}_{(II)}(G_*) = -\frac{2\alpha_X^2 \nu_0 q_*^2}{M_X^2} \rho_*^2,$$

$$\widehat{T}_{(II)}^\alpha(G_*) = \frac{2q_*^2 \alpha_X \beta_X}{M_X^2} \Gamma_\alpha \rho_* \left[27\mathcal{F}(q_*) + 30\mathcal{F}(2q_*) + 9\mathcal{F}(3q_*) - 30\mathcal{R}_2(0) - 60\mathcal{R}_2(q_*) - 34\mathcal{R}_2(2q_*) - 8\mathcal{R}_2(3q_*) \right].$$

So the selected-channel extractor is no longer conceptually open. What remains is only the actual numerical Stage-IA solve and the evaluation of the 10 one-dimensional profile samples:

$$\mathcal{U}_2(0), \mathcal{U}_2(q_*), \mathcal{U}_2(2q_*), \mathcal{F}(q_*), \mathcal{F}(2q_*), \mathcal{F}(3q_*), \mathcal{R}_2(0), \mathcal{R}_2(q_*), \mathcal{R}_2(2q_*), \mathcal{R}_2(3q_*).$$

DLIX. STAGE IA ACTUAL NUMERICAL CONSTRUCTION: DIMENSIONLESS SHOOTING AND DIRECT PROFILE EXTRACTION

DLX. SHORT KOREAN EXPLANATION

The immediate bottleneck is the actual numerical construction of the accepted Stage-IA profile $u_{\text{IA}}(\xi)$.

So we now do the only nonredundant next step: we rewrite the Stage-IA scalar equation in dimensionless form, define the exact shooting variables and admissibility conditions, and connect the converged profile directly to the 10 required one-dimensional Fourier samples.

This is the actual numerical construction layer of the project.

DLXI. 1. STARTING STAGE-IA SCALAR EQUATION

The Stage-IA scalar amplitude equation on the accepted canonical branch was

$$\boxed{-YC_2 u'''' + A_2 u'' - M_2 u + 2\chi_{II} C_4 u^2 u'' + 2\chi_{II} C_4 u (u')^2 + U_3 u^3 - \gamma C_6 u^5 = 0,} \quad (1331)$$

with

$$\boxed{A_2 = C_2 \left(Z + \frac{10}{3} Y q_*^2 \right), \quad M_2 = C_2 (\mu^2 + Z q_*^2 + Y q_*^4), \quad U_3 = \chi_{II} C_\Delta - \lambda C_4.} \quad (1332)$$

The accepted Stage-IA branch is even:

$$\boxed{u'(0) = 0, \quad u'''(0) = 0,} \quad (1333)$$

and asymptotes to the nontrivial vacuum

$$\boxed{u(\xi) \rightarrow u_\infty, \quad u''(\xi) \rightarrow 0 \quad (\xi \rightarrow \infty),} \quad (1334)$$

where u_∞ is determined by

$$\boxed{-M_2 u_\infty + U_3 u_\infty^3 - \gamma C_6 u_\infty^5 = 0.} \quad (1335)$$

DLXII. 2. DIMENSIONLESS PROFILE VARIABLE

Introduce the vacuum-scaled amplitude

$$\boxed{u(\xi) = u_\infty f(s), \quad s := m_* \xi,} \quad (1336)$$

where $m_* > 0$ is the positive tail decay rate determined from the linearized far-field equation.

Then

$$u' = u_\infty m_* f_s, \quad u'' = u_\infty m_*^2 f_{ss}, \quad u''' = u_\infty m_*^3 f_{sss}, \quad u'''' = u_\infty m_*^4 f_{ssss}.$$

Substituting into (1331) and dividing by u_∞ , we obtain

$$\boxed{-\mathfrak{y} f_{ssss} + \mathfrak{a} f_{ss} - \mathfrak{m} f + \mathfrak{b} f^2 f_{ss} + \mathfrak{b} f (f_s)^2 + \mathfrak{u} f^3 - \mathfrak{g} f^5 = 0,} \quad (1337)$$

with the exact dimensionless couplings

$$\boxed{\mathfrak{y} := Y C_2 m_*^4, \quad \mathfrak{a} := A_2 m_*^2, \quad \mathfrak{m} := M_2,} \quad (1338)$$

$$\boxed{\mathfrak{b} := 2\chi_{II} C_4 u_\infty^2 m_*^2, \quad \mathfrak{u} := U_3 u_\infty^2, \quad \mathfrak{g} := \gamma C_6 u_\infty^4.} \quad (1339)$$

Using the vacuum equation

$$-\mathfrak{m} + \mathfrak{u} - \mathfrak{g} = 0,$$

the constant solution $f \equiv 1$ is automatic.

DLXIII. 3. EXACT DIMENSIONLESS SHOOTING EQUATION

The dimensionless Stage-IA equation is therefore

$$-\mathfrak{y} f_{ssss} + \mathfrak{a} f_{ss} + \mathfrak{b} f^2 f_{ss} + \mathfrak{b} f(f_s)^2 + \mathfrak{u}(f^3 - f) - \mathfrak{g}(f^5 - f) = 0. \quad (1340)$$

The even dark-type branch conditions become

$$f_s(0) = 0, \quad f_{ss}(0) = 0, \quad f(s) \rightarrow 1, \quad f_{ss}(s) \rightarrow 0 \quad (s \rightarrow \infty). \quad (1341)$$

Now define

$$a := f(0), \quad b := f_{ss}(0). \quad (1342)$$

Then the entire even branch is determined by the two dimensionless center data (a, b) .

DLXIV. 4. DIMENSIONLESS TWO-PARAMETER SHOOTING PROBLEM

Introduce

$$y_1 = f, \quad y_2 = f_s, \quad y_3 = f_{ss}, \quad y_4 = f_{sss}. \quad (1343)$$

Then

$$y'_1 = y_2, \quad y'_2 = y_3, \quad y'_3 = y_4, \quad (1344)$$

and

$$y'_4 = \frac{1}{\mathfrak{y}} \left[\mathfrak{a} y_3 + \mathfrak{b} y_1^2 y_3 + \mathfrak{b} y_1 y_2^2 + \mathfrak{u}(y_1^3 - y_1) - \mathfrak{g}(y_1^5 - y_1) \right]. \quad (1345)$$

The exact center data are

$$y_1(0) = a, \quad y_2(0) = 0, \quad y_3(0) = b, \quad y_4(0) = 0. \quad (1346)$$

Define the dimensionless shooting map

$$\mathcal{S}_{\text{dim}}(a, b) := \begin{pmatrix} f(S_{\text{max}}; a, b) - 1 \\ f_{ss}(S_{\text{max}}; a, b) \end{pmatrix}, \quad (1347)$$

where $S_{\text{max}} := m_* L_\xi$. The actual accepted branch solve is simply

$$\mathcal{S}_{\text{dim}}(a, b) = 0. \quad (1348)$$

DLXV. 5. LOCAL TAYLOR START AWAY FROM THE CENTER

To avoid the exact center in numerics, begin at a small $s = \delta_s$. Let

$$\Omega_s(a, b) := \frac{1}{\mathfrak{y}} \left[\mathfrak{a} b + \mathfrak{b} a^2 b + \mathfrak{u}(a^3 - a) - \mathfrak{g}(a^5 - a) \right]. \quad (1349)$$

Then

$$f(\delta_s) = a + \frac{b}{2} \delta_s^2 + \frac{\Omega_s}{24} \delta_s^4 + O(\delta_s^6), \quad (1350)$$

$$f_s(\delta_s) = b\delta_s + \frac{\Omega_s}{6}\delta_s^3 + O(\delta_s^5), \quad (1351)$$

$$f_{ss}(\delta_s) = b + \frac{\Omega_s}{2}\delta_s^2 + O(\delta_s^4), \quad (1352)$$

$$f_{sss}(\delta_s) = \Omega_s\delta_s + O(\delta_s^3). \quad (1353)$$

A practical choice is

$$\delta_s = 10^{-3}, \quad (1354)$$

or 5×10^{-3} if the Jacobian becomes noisy.

DLXVI. 6. DIMENSIONLESS DARK-SEED PREDICTOR

The simplest dark seed is

$$f^{(0)}(s) = \tanh^2 s. \quad (1355)$$

This gives the initial center data

$$a^{(0)} = 0, \quad b^{(0)} = 2. \quad (1356)$$

So the first actual shooting attempt is centered at

$$(a, b) \approx (0, 2).$$

DLXVII. 7. EXACT PROFILE SAMPLES REQUIRED BY THE SELECTED G_* EXTRACTOR

Once $f(s)$ is converged, all required one-dimensional data become direct integrals.

Let

$$\lambda_n := \frac{nq_*}{m_*}, \quad n = 0, 1, 2, 3. \quad (1357)$$

Then

$$\mathcal{U}_2(nq_*) = \frac{u_\infty^2}{S_{\max}} \int_0^{S_{\max}} f(s)^2 \cos(\lambda_n s) ds, \quad (1358)$$

and

$$\mathcal{R}_2(nq_*) = \frac{u_\infty^3}{S_{\max}} \int_0^{S_{\max}} \frac{f(s)^3}{C_2 u_\infty^2 f(s)^2 + \varepsilon} \cos(\lambda_n s) ds. \quad (1359)$$

For the stable odd-current term, define

$$\tilde{F}(f) := \frac{2u_\infty f}{C_2} - \frac{2\sqrt{\varepsilon}}{C_2^{3/2}} \arctan\left(\sqrt{\frac{C_2}{\varepsilon}} u_\infty f\right). \quad (1360)$$

Then

$$\mathcal{F}(nq_*) = \frac{1}{S_{\max}} \int_0^{S_{\max}} \tilde{F}(f(s)) \cos(\lambda_n s) ds, \quad n = 1, 2, 3. \quad (1361)$$

DLXVIII. 8. DIRECT SELECTED-CHANNEL COMBINATIONS

The density combination is

$$\boxed{\rho_* = 4\mathcal{U}_2(0) + 6\mathcal{U}_2(q_*) + 2\mathcal{U}_2(2q_*)}. \quad (1362)$$

The parallel combination is

$$\boxed{\Xi_{\parallel}^* = iq_* \left[27\mathcal{F}(q_*) + 30\mathcal{F}(2q_*) + 9\mathcal{F}(3q_*) + 15\mathcal{R}_2(0) + 3\mathcal{R}_2(q_*) - 13\mathcal{R}_2(2q_*) - 5\mathcal{R}_2(3q_*) \right]}. \quad (1363)$$

The first transverse combination is

$$\boxed{\Xi_1^* = iq_* \left[15\mathcal{R}_2(0) + 21\mathcal{R}_2(q_*) + 7\mathcal{R}_2(2q_*) + \mathcal{R}_2(3q_*) \right]}, \quad (1364)$$

and

$$\boxed{\Xi_2^* = \sqrt{2}\Xi_1^*}. \quad (1365)$$

DLXIX. 9. FINAL SELECTED-CHANNEL OVERLAPS

Therefore the selected-channel Class II overlaps are

$$\boxed{\hat{U}_{(II)}(G_*) = -\frac{2\alpha_X^2 \nu_0 q_*^2}{M_X^2} \rho_*^2, \quad \nu_0 := \sum_A (n_0^A)^2}, \quad (1366)$$

and

$$\boxed{\hat{T}_{(II)}^\alpha(G_*) = -\frac{2iq_* \alpha_X \beta_X}{M_X^2} \Gamma_\alpha \rho_* (\Xi_{\parallel}^* - 3\Xi_1^*)}, \quad (1367)$$

with

$$\boxed{\Gamma_\alpha := \sum_A n_0^A \mathcal{Q}_\alpha^A}. \quad (1368)$$

Substituting (1363) and (1364),

$$\boxed{\hat{T}_{(II)}^\alpha(G_*) = \frac{2q_*^2 \alpha_X \beta_X}{M_X^2} \Gamma_\alpha \rho_* \left[27\mathcal{F}(q_*) + 30\mathcal{F}(2q_*) + 9\mathcal{F}(3q_*) - 30\mathcal{R}_2(0) - 60\mathcal{R}_2(q_*) - 34\mathcal{R}_2(2q_*) - 8\mathcal{R}_2(3q_*) \right]}. \quad (1369)$$

DLXX. 10. IMMEDIATE EXECUTABLE ORDER

The actual next computation is now fully explicit:

$$\boxed{\text{(i) solve } \mathcal{S}_{\text{dim}}(a, b) = 0, \quad \text{(ii) recover } u_{\text{IA}}(\xi) = u_\infty f(m_* \xi)}, \quad (1370)$$

$$\boxed{\text{(iii) evaluate } \mathcal{U}_2(0), \mathcal{U}_2(q_*), \mathcal{U}_2(2q_*), \mathcal{F}(q_*), \mathcal{F}(2q_*), \mathcal{F}(3q_*), \mathcal{R}_2(0), \mathcal{R}_2(q_*), \mathcal{R}_2(2q_*), \mathcal{R}_2(3q_*)}, \quad (1371)$$

$$\boxed{\text{(iv) insert them into (1366) and (1369), then continue to } H_{\text{eff}}(k) \rightarrow K_i \rightarrow P_* K_i P_*}. \quad (1372)$$

DLXXI. FINAL SUMMARY

The selected-channel extractor has now been reduced to a direct numerical problem on the accepted Stage-IA shooting profile.

Using

$$u(\xi) = u_\infty f(s), \quad s = m_* \xi,$$

the dimensionless Stage-IA equation is

$$-\mathfrak{h} f_{ssss} + \mathfrak{a} f_{ss} + \mathfrak{b} f^2 f_{ss} + \mathfrak{b} f(f_s)^2 + \mathfrak{u}(f^3 - f) - \mathfrak{g}(f^5 - f) = 0,$$

with even boundary conditions

$$f_s(0) = 0, \quad f_{sss}(0) = 0, \quad f(s) \rightarrow 1, \quad f_{ss}(s) \rightarrow 0.$$

The actual shooting unknowns are

$$a = f(0), \quad b = f_{ss}(0),$$

and the first seed is

$$(a, b) \approx (0, 2).$$

Once $f(s)$ is obtained, the required selected-channel data are

$$\mathcal{U}_2(0), \mathcal{U}_2(q_*), \mathcal{U}_2(2q_*), \mathcal{F}(q_*), \mathcal{F}(2q_*), \mathcal{F}(3q_*), \mathcal{R}_2(0), \mathcal{R}_2(q_*), \mathcal{R}_2(2q_*), \mathcal{R}_2(3q_*),$$

from which

$$\rho_*, \quad \Xi_{\parallel}^*, \quad \Xi_1^*, \quad \Xi_2^* = \sqrt{2} \Xi_1^*$$

follow directly.

Therefore

$$\widehat{U}_{(II)}(G_*) = -\frac{2\alpha_X^2 \nu_0 q_*^2}{M_X^2} \rho_*^2,$$

and

$$\widehat{T}_{(II)}^\alpha(G_*) = \frac{2q_*^2 \alpha_X \beta_X}{M_X^2} \Gamma_\alpha \rho_* \left[27 \mathcal{F}(q_*) + 30 \mathcal{F}(2q_*) + 9 \mathcal{F}(3q_*) - 30 \mathcal{R}_2(0) - 60 \mathcal{R}_2(q_*) - 34 \mathcal{R}_2(2q_*) - 8 \mathcal{R}_2(3q_*) \right].$$

So the entire selected G_* common extractor is now a direct numerical post-processing layer on top of the Stage-IA shooting solve.

DLXXII. STAGE IA NUMERICAL WORKFLOW CLOSURE

DLXXIII. GOAL

The only remaining bottleneck is the actual numerical construction of the accepted Stage-IA profile $u_{\text{IA}}(\xi)$.

Once that profile is obtained, the selected-channel extractor is already reduced to direct one-dimensional integrals, and the branch-dependent inputs

$$\rho_*, \quad \Xi_{\parallel}^*, \quad \Xi_1^*, \quad \Xi_2^* = \sqrt{2} \Xi_1^*$$

follow immediately.

So the correct next step is not further symbolic reduction. It is the exact numerical workflow for the Stage-IA solve.

DLXXIV. 1. INPUTS

The numerical Stage-IA solver takes as inputs

$$\boxed{q_*, \quad C_2, \quad C_4, \quad C_6, \quad C_\Delta, \quad Y, \quad Z, \quad \mu^2, \quad \lambda, \quad \gamma, \quad \chi_{II}, \quad \varepsilon.} \quad (1373)$$

From these define

$$\boxed{A_2 = C_2 \left(Z + \frac{10}{3} Y q_*^2 \right), \quad M_2 = C_2 (\mu^2 + Z q_*^2 + Y q_*^4), \quad U_3 = \chi_{II} C_\Delta - \lambda C_4.} \quad (1374)$$

DLXXV. 2. VACUUM AND TAIL PREPROCESSING

First solve for the nontrivial vacuum $u_\infty > 0$ from

$$\boxed{-M_2 u_\infty + U_3 u_\infty^3 - \gamma C_6 u_\infty^5 = 0.} \quad (1375)$$

Equivalently, with $s = u_\infty^2$,

$$\boxed{\gamma C_6 s^2 - U_3 s + M_2 = 0.} \quad (1376)$$

Choose the positive physical root.

Next compute the positive tail rate $m_* > 0$ from the far-field linearization

$$\boxed{-Y C_2 m^4 + (A_2 + 2\chi_{II} C_4 u_\infty^2) m^2 + V_\infty = 0,} \quad (1377)$$

where

$$\boxed{V_\infty = -M_2 + 3U_3 u_\infty^2 - 5\gamma C_6 u_\infty^4.} \quad (1378)$$

Choose the positive root $m_* = +\sqrt{s_*}$.

DLXXVI. 3. DIMENSIONLESS REDUCTION

Introduce

$$\boxed{u(\xi) = u_\infty f(s), \quad s = m_* \xi.} \quad (1379)$$

Then the Stage-IA equation becomes

$$\boxed{-\mathfrak{y} f_{ssss} + \mathfrak{a} f_{ss} + \mathfrak{b} f^2 f_{ss} + \mathfrak{b} f (f_s)^2 + \mathfrak{u} (f^3 - f) - \mathfrak{g} (f^5 - f) = 0,} \quad (1380)$$

with

$$\boxed{\mathfrak{y} = Y C_2 m_*^4, \quad \mathfrak{a} = A_2 m_*^2, \quad \mathfrak{b} = 2\chi_{II} C_4 u_\infty^2 m_*^2,} \quad (1381)$$

$$\boxed{\mathfrak{u} = U_3 u_\infty^2, \quad \mathfrak{g} = \gamma C_6 u_\infty^4.} \quad (1382)$$

The boundary conditions are

$$\boxed{f_s(0) = 0, \quad f_{sss}(0) = 0, \quad f(s) \rightarrow 1, \quad f_{ss}(s) \rightarrow 0 \quad (s \rightarrow \infty).} \quad (1383)$$

DLXXVII. 4. SHOOTING UNKNOWNNS

Define the dimensionless center data

$$\boxed{a := f(0), \quad b := f_{ss}(0).} \quad (1384)$$

Introduce the first-order system

$$\boxed{y_1 = f, \quad y_2 = f_s, \quad y_3 = f_{ss}, \quad y_4 = f_{sss}.} \quad (1385)$$

Then

$$\boxed{y'_1 = y_2, \quad y'_2 = y_3, \quad y'_3 = y_4,} \quad (1386)$$

$$\boxed{y'_4 = \frac{1}{\mathfrak{y}} \left[\mathfrak{a} y_3 + \mathfrak{b} y_1^2 y_3 + \mathfrak{b} y_1 y_2^2 + \mathfrak{u}(y_1^3 - y_1) - \mathfrak{g}(y_1^5 - y_1) \right].} \quad (1387)$$

The exact center data are

$$\boxed{y_1(0) = a, \quad y_2(0) = 0, \quad y_3(0) = b, \quad y_4(0) = 0.} \quad (1388)$$

DLXXVIII. 5. LOCAL START AWAY FROM THE CENTER

To avoid the exact center in numerics, start at a small $\delta_s > 0$.

Define

$$\boxed{\Omega_s(a, b) = \frac{1}{\mathfrak{y}} \left[\mathfrak{a} b + \mathfrak{b} a^2 b + \mathfrak{u}(a^3 - a) - \mathfrak{g}(a^5 - a) \right].} \quad (1389)$$

Then initialize at $s = \delta_s$ with

$$\boxed{f(\delta_s) = a + \frac{b}{2} \delta_s^2 + \frac{\Omega_s}{24} \delta_s^4,} \quad (1390)$$

$$\boxed{f_s(\delta_s) = b \delta_s + \frac{\Omega_s}{6} \delta_s^3,} \quad (1391)$$

$$\boxed{f_{ss}(\delta_s) = b + \frac{\Omega_s}{2} \delta_s^2,} \quad (1392)$$

$$\boxed{f_{sss}(\delta_s) = \Omega_s \delta_s.} \quad (1393)$$

Use

$$\boxed{\delta_s = 10^{-3}} \quad (1394)$$

as the default start radius.

DLXXIX. 6. DIMENSIONLESS SHOOTING MAP

Choose a finite terminal point

$$\boxed{S_{\max} = 18} \quad (1395)$$

for the first pass, corresponding to the earlier $L_\xi = 18/m_*$ convention.

Define the shooting residual

$$\boxed{\mathcal{S}_{\dim}(a, b) = \begin{pmatrix} f(S_{\max}; a, b) - 1 \\ f_{ss}(S_{\max}; a, b) \end{pmatrix}.} \quad (1396)$$

The accepted branch solve is

$$\boxed{\mathcal{S}_{\dim}(a, b) = 0.} \quad (1397)$$

DLXXX. 7. NEWTON SHOOTING AT FIXED χ_{II}

Introduce the two variational fields

$$v := \partial_a y, \quad z := \partial_b y.$$

They satisfy

$$\boxed{v'_1 = v_2, \quad v'_2 = v_3, \quad v'_3 = v_4,} \quad (1398)$$

$$\boxed{v'_4 = \frac{1}{\mathfrak{y}} \left[(\mathfrak{a} + \mathfrak{b}y_1^2)v_3 + 2\mathfrak{b}y_1y_2v_2 + (2\mathfrak{b}y_1y_3 + \mathfrak{b}y_2^2 + 3uy_1^2 - 5\mathfrak{g}y_1^4 - \mathfrak{u} + \mathfrak{g})v_1 \right],} \quad (1399)$$

and the same system for z .

Their local initial data at $s = \delta_s$ are obtained by differentiating (1390)–(1393) with respect to a, b .

Then

$$\boxed{D\mathcal{S}_{\dim}(a, b) = \begin{pmatrix} v_1(S_{\max}) & z_1(S_{\max}) \\ v_3(S_{\max}) & z_3(S_{\max}) \end{pmatrix}.} \quad (1400)$$

The Newton step solves

$$\boxed{D\mathcal{S}_{\dim}(a^{(n)}, b^{(n)}) \begin{pmatrix} \delta a^{(n)} \\ \delta b^{(n)} \end{pmatrix} = -\mathcal{S}_{\dim}(a^{(n)}, b^{(n)})}.} \quad (1401)$$

Update with damping:

$$\boxed{a^{(n+1)} = a^{(n)} + \eta_n \delta a^{(n)}, \quad b^{(n+1)} = b^{(n)} + \eta_n \delta b^{(n)}, \quad 0 < \eta_n \leq 1,} \quad (1402)$$

choosing η_n by backtracking to reduce $\|\mathcal{S}_{\dim}\|_2$.

DLXXXI. 8. INITIAL SEED AND CONTINUATION

At $\chi_{II} = 0$, use the dark seed

$$\boxed{f^{(0)}(s) = \tanh^2 s} \quad (1403)$$

so the first center guess is

$$(a^{(0)}, b^{(0)}) = (0, 2). \quad (1404)$$

After the base solve converges, continue in χ_{II} with the predictor-corrector already derived:

$$\begin{pmatrix} \dot{a} \\ \dot{b} \end{pmatrix} = -[D\mathcal{S}_{\text{dim}}]^{-1} \partial_{\chi_{II}} \mathcal{S}_{\text{dim}}, \quad (1405)$$

and use

$$(a, b)_{\text{pred}} = (a, b) + \Delta\chi (\dot{a}, \dot{b}) \quad (1406)$$

as the next Newton seed.

DLXXXII. 9. ACCEPTANCE CRITERIA

At fixed χ_{II} , accept the Stage-IA profile if

$$\|\mathcal{S}_{\text{dim}}(a, b)\|_2 < 10^{-10}, \quad (1407)$$

$$|a^{(n+1)} - a^{(n)}| + |b^{(n+1)} - b^{(n)}| < 10^{-12}, \quad (1408)$$

and the first-integral defect satisfies

$$\|\mathcal{I}[f]\|_\infty < 10^{-8}, \quad (1409)$$

where $\mathcal{I}[f]$ is the dimensionless form of the exact first integral.

Then repeat the solve at

$$S_{\text{max}} = 18, 22$$

and accept only if the reconstructed profile changes by less than 10^{-4} in relative L^2 -norm.

DLXXXIII. 10. DIRECT PROFILE SAMPLE EXTRACTION

Once $f(s)$ is converged, reconstruct

$$u_{\text{IA}}(\xi) = u_\infty f(m_* \xi).$$

Now define

$$\lambda_n := \frac{nq_*}{m_*}, \quad n = 0, 1, 2, 3. \quad (1410)$$

Then compute directly:

$$\mathcal{U}_2(nq_*) = \frac{u_\infty^2}{S_{\text{max}}} \int_0^{S_{\text{max}}} f(s)^2 \cos(\lambda_n s) ds, \quad (1411)$$

$$\mathcal{R}_2(nq_*) = \frac{u_\infty^3}{S_{\text{max}}} \int_0^{S_{\text{max}}} \frac{f(s)^3}{C_2 u_\infty^2 f(s)^2 + \varepsilon} \cos(\lambda_n s) ds, \quad (1412)$$

and with

$$\tilde{F}(f) = \frac{2u_\infty f}{C_2} - \frac{2\sqrt{\varepsilon}}{C_2^{3/2}} \arctan\left(\sqrt{\frac{C_2}{\varepsilon}} u_\infty f\right), \quad (1413)$$

compute

$$\mathcal{F}(nq_*) = \frac{1}{S_{\max}} \int_0^{S_{\max}} \tilde{F}(f(s)) \cos(\lambda_n s) ds, \quad n = 1, 2, 3. \quad (1414)$$

DLXXXIV. 11. SELECTED-CHANNEL EXTRACTED COMBINATIONS

Then

$$\rho_* = 4\mathcal{U}_2(0) + 6\mathcal{U}_2(q_*) + 2\mathcal{U}_2(2q_*), \quad (1415)$$

$$\Xi_{\parallel}^* = iq_* \left[27\mathcal{F}(q_*) + 30\mathcal{F}(2q_*) + 9\mathcal{F}(3q_*) + 15\mathcal{R}_2(0) + 3\mathcal{R}_2(q_*) - 13\mathcal{R}_2(2q_*) - 5\mathcal{R}_2(3q_*) \right], \quad (1416)$$

$$\Xi_1^* = iq_* \left[15\mathcal{R}_2(0) + 21\mathcal{R}_2(q_*) + 7\mathcal{R}_2(2q_*) + \mathcal{R}_2(3q_*) \right], \quad (1417)$$

$$\Xi_2^* = \sqrt{2} \Xi_1^*. \quad (1418)$$

Therefore

$$\hat{U}_{(II)}(G_*) = -\frac{2\alpha_X^2 \nu_0 q_*^2}{M_X^2} \rho_*^2, \quad (1419)$$

and

$$\hat{T}_{(II)}^\alpha(G_*) = \frac{2q_*^2 \alpha_X \beta_X}{M_X^2} \Gamma_\alpha \rho_* \left[27\mathcal{F}(q_*) + 30\mathcal{F}(2q_*) + 9\mathcal{F}(3q_*) - 30\mathcal{R}_2(0) - 60\mathcal{R}_2(q_*) - 34\mathcal{R}_2(2q_*) - 8\mathcal{R}_2(3q_*) \right]. \quad (1420)$$

DLXXXV. 12. ACTUAL EXECUTABLE ORDER

The exact executable order is now:

$$(i) \text{ solve the dimensionless shooting problem (1397) at } \chi_{II} = 0, \quad (1421)$$

$$(ii) \text{ continue in } \chi_{II} > 0 \text{ to the target branch,} \quad (1422)$$

$$(iii) \text{ evaluate } \mathcal{U}_2(0), \mathcal{U}_2(q_*), \mathcal{U}_2(2q_*), \mathcal{F}(q_*), \mathcal{F}(2q_*), \mathcal{F}(3q_*), \mathcal{R}_2(0), \mathcal{R}_2(q_*), \mathcal{R}_2(2q_*), \mathcal{R}_2(3q_*), \quad (1423)$$

$$(iv) \text{ compute } \rho_*, \Xi_{\parallel}^*, \Xi_1^*, \Xi_2^*, \text{ then } \hat{U}_{(II)}(G_*), \hat{T}_{(II)}^\alpha(G_*), \quad (1424)$$

$$(v) \text{ pass those into } H_{\text{eff}}(k) \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow (\lambda_c, \Lambda_E, \Lambda_O). \quad (1425)$$

Nothing conceptual remains open at the selected-channel Stage-IA level.

DLXXXVI. FINAL SUMMARY

The Stage-IA construction is now a direct numerical workflow.

The actual shooting problem is the dimensionless even-branch equation

$$-\mathfrak{y} f_{ssss} + \mathfrak{a} f_{ss} + \mathfrak{b} f^2 f_{ss} + \mathfrak{b} f(f_s)^2 + \mathfrak{u}(f^3 - f) - \mathfrak{g}(f^5 - f) = 0,$$

with

$$f_s(0) = 0, \quad f_{sss}(0) = 0, \quad f(s) \rightarrow 1, \quad f_{ss}(s) \rightarrow 0.$$

The shooting unknowns are

$$a = f(0), \quad b = f_{ss}(0),$$

and the first seed is

$$(a, b) \approx (0, 2).$$

After convergence, reconstruct

$$u_{\text{IA}}(\xi) = u_\infty f(m_* \xi).$$

Then compute the ten one-dimensional samples

$$\mathcal{U}_2(0), \mathcal{U}_2(q_*), \mathcal{U}_2(2q_*),$$

$$\mathcal{F}(q_*), \mathcal{F}(2q_*), \mathcal{F}(3q_*),$$

$$\mathcal{R}_2(0), \mathcal{R}_2(q_*), \mathcal{R}_2(2q_*), \mathcal{R}_2(3q_*).$$

From them,

$$\rho_*, \quad \Xi_{\parallel}^*, \quad \Xi_1^*, \quad \Xi_2^* = \sqrt{2} \Xi_1^*$$

follow directly, and therefore

$$\widehat{U}_{(II)}(G_*) = -\frac{2\alpha_X^2 \nu_0 q_*^2}{M_X^2} \rho_*^2,$$

$$\widehat{T}_{(II)}^\alpha(G_*) = \frac{2q_*^2 \alpha_X \beta_X}{M_X^2} \Gamma_\alpha \rho_* \left[27\mathcal{F}(q_*) + 30\mathcal{F}(2q_*) + 9\mathcal{F}(3q_*) - 30\mathcal{R}_2(0) - 60\mathcal{R}_2(q_*) - 34\mathcal{R}_2(2q_*) - 8\mathcal{R}_2(3q_*) \right].$$

So the selected G_* extractor is now purely a numerical post-processing layer on top of the Stage-IA shooting solve.

DLXXXVII. ROUTE CORRECTION TO THE CURRENT MASTER AND IMMEDIATE EXECUTABLE CERTIFICATE WORKFLOW

DLXXXVIII. SHORT KOREAN EXPLANATION

According to the latest uploaded master notes, the current official forward-computable route is the explicit odd extra-carrier reference route, while the canonical mixed odd-active branch remains the structural target branch.

Therefore the correct immediate task is not to keep extending the direct Stage-IA symbolic branch construction indefinitely. The correct immediate task is to populate the already-closed carrier certificate by finite numerical extraction from the actual microscopic linearized operator.

In the current master, this is organized as:

$$\text{Gate 1} \rightarrow \text{Gate 2} \rightarrow \text{Gate 3} \rightarrow \text{Gate 4.}$$

DLXXXIX. 1. WHAT THE CURRENT MASTER SAYS IS THE REMAINING PROBLEM

The latest master fixes the current stage as:

$$\boxed{\text{Reference route: explicit odd extra-carrier route (forward-computable now),}} \quad (1426)$$

$$\boxed{\text{Target route: canonical mixed odd-active branch (structurally closed, two finite audits remain).}} \quad (1427)$$

It also states that “actual PDE completion” means:

$$\boxed{L(k) \rightarrow K_i := \partial_{k_i} L|_{G_*} \rightarrow P_* K_i P_*} \quad (1428)$$

and then finite coefficient extraction, rather than re-solving the full nonlinear PDE from scratch.

So the immediate computational goal is:

$$\boxed{\text{populate the already-derived certificate by actual finite branch/operator data.}} \quad (1429)$$

DXC. 2. CURRENT REFERENCE COEFFICIENT SHEET

The current reference low-order coefficient dictionary is

$$\boxed{\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = I_c \tilde{\Lambda}_E, \quad \beta = I_c \tilde{\Lambda}_O.} \quad (1430)$$

At one-mediator level,

$$\boxed{\tilde{\Lambda}_E^{(1)} = -\frac{\beta_X \gamma_E}{M_X^2}, \quad \tilde{\Lambda}_O^{(1)} = -\frac{\beta_X \gamma_O}{M_X^2}.} \quad (1431)$$

Under the fixed reference choice

$$\gamma_E = \gamma_O = \gamma_{\perp},$$

the current reference sheet is

$$\boxed{\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = \beta = -\frac{\beta_X \gamma_{\perp}}{M_X^2} I_c.} \quad (1432)$$

This is the actual current forward-computable reference PDE sheet.

DXCI. 3. THE FOUR CERTIFICATE MARGINS

The current master organizes the numerical completion problem through four margins.

A. Gate 1 margin

$$\boxed{\mathfrak{G}_1 := \Delta_{\text{bench},\rho}^{\text{fin}} - \eta_{R,\rho}.} \quad (1433)$$

Gate 1 passes iff

$$\boxed{\mathfrak{G}_1 > 0.} \quad (1434)$$

B. Gate 2 margin

$$\mathfrak{G}_2 := |\Delta_{00} \alpha_X L_* A_{G_*}| - |R_N|. \quad (1435)$$

Gate 2 passes iff

$$\mathfrak{G}_2 > 0 \implies \lambda_{\parallel} \neq 0. \quad (1436)$$

C. Gate 3 transport and calibration margins

$$\mathfrak{G}_{3,\text{tr}} := |c_E| + |c_O|, \quad \mathfrak{G}_{3,\text{cal}} := |C_E| + |C_O|. \quad (1437)$$

Gate 3 passes iff

$$\mathfrak{G}_{3,\text{tr}} > 0, \quad \mathfrak{G}_{3,\text{cal}} > 0. \quad (1438)$$

D. Gate 4 commutator audit

The current master requires

$$[Q_*, H_{\Psi\Psi}^{\text{const}}] = 0, \quad [Q_*, \delta H_{\text{Class II}}^{\text{const}}] = 0, \quad [Q_*, \Sigma(0, 0)] = 0. \quad (1439)$$

In the current working truncation this is already close to closed, so the numerical burden is mainly Gates 1–3.

DXCII. 4. MINIMAL FINITE DATA TABLE TO COMPUTE NEXT

So the actual next finite data table is:

$$\Delta_{\text{bench},\rho}^{\text{fin}}, \quad \eta_{R,\rho}, \quad \Delta_{00}, \quad L_*, \quad A_{G_*}, \quad (c_E, c_O), \quad (C_E, C_O). \quad (1440)$$

This is the smallest honest numerical payload that advances the current master route.

DXCIII. 5. IMMEDIATE EXECUTABLE ORDER

The correct immediate executable order is therefore:

$$\text{Step 1: evaluate } \Delta_{\text{bench},\rho}^{\text{fin}} \text{ and } \eta_{R,\rho}, \quad (1441)$$

$$\text{Step 2: evaluate } \Delta_{00}, L_*, A_{G_*} \text{ and compute } \mathfrak{G}_2, \quad (1442)$$

$$\text{Step 3: evaluate } (c_E, c_O) \text{ and } (C_E, C_O), \quad (1443)$$

$$\text{Step 4: confirm the constant-sector commutator audit.} \quad (1444)$$

Only after these are populated is the current reference-route certificate numerically complete.

DXCIV. 6. HONEST RELATION TO THE STAGE-IA ROUTE

The direct Stage-IA actual-branch construction we developed is still useful as a target-branch construction route. But according to the latest uploaded master, it is not the current shortest official forward route.

So the honest current alignment is:

$$\boxed{\text{keep the Stage-IA route as a target-branch construction path,}} \quad (1445)$$

$$\boxed{\text{but push the finite Gate-1/2/3/4 insertion as the present reference-route computation.}} \quad (1446)$$

DXCV. FINAL SUMMARY

According to the latest uploaded master, the present official forward route is

$$\boxed{\text{explicit odd extra-carrier reference route}}$$

rather than indefinite further symbolic extension of the direct Stage-IA branch construction.

The current actual numerical completion problem is the finite certificate insertion:

$$\boxed{\Delta_{\text{bench},\rho}^{\text{fin}}, \quad \eta_{R,\rho}, \quad \Delta_{00}, \quad L_*, \quad A_{G_*}, \quad (c_E, c_O), \quad (C_E, C_O).}$$

The four margins are

$$\mathfrak{G}_1 = \Delta_{\text{bench},\rho}^{\text{fin}} - \eta_{R,\rho},$$

$$\mathfrak{G}_2 = |\Delta_{00}\alpha_X L_* A_{G_*}| - |R_N|,$$

$$\mathfrak{G}_{3,\text{tr}} = |c_E| + |c_O|, \quad \mathfrak{G}_{3,\text{cal}} = |C_E| + |C_O|,$$

plus the Gate 4 commutator audit.

So the correct immediate computation order is

$$\boxed{\text{Gate 1} \rightarrow \text{Gate 2} \rightarrow \text{Gate 3} \rightarrow \text{Gate 4.}}$$

This keeps us aligned with the current master notes and moves directly toward the actual certificate completion.

DXCVI. IMMEDIATE ACTUAL PDE POPULATION ROUTE

The reduced PDE coefficients are

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = I_c \tilde{\Lambda}_E, \quad \beta = I_c \tilde{\Lambda}_O.$$

Under the symmetric reference choice $\gamma_E = \gamma_O = \gamma_{\perp}$,

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = \beta = -\frac{\beta_X \gamma_{\perp}}{M_X^2} I_c.$$

So the remaining finite microscopic tasks are:

$$\boxed{\text{Task A: compute } I_3, I_c}$$

from the defect zero-mode profiles $f_{\pm}(\xi)$ and $\theta(\xi)$,

Task B: compute λ_c

from the longitudinal projected low-block derivative,

Task C: compute $\tilde{\Lambda}_O$ (equivalently J_1 in canonical normalization)

from the explicit odd extra-carrier operator insertion,
and then assemble

$\lambda_{\parallel}, \alpha, \beta.$

The resulting reference reduced PDE is

$$\left[i\partial_t - \frac{\lambda_c}{2} I_3 \sigma_3 (-i\partial_{\parallel}) + \frac{\beta_X \gamma_{\perp}}{M_X^2} I_c (\sigma_1 (-i\partial_1) + \sigma_2 (-i\partial_2)) + \frac{\beta_X \gamma_{\perp}}{M_X^2} I_c (-\sigma_1 (-i\partial_2) + \sigma_2 (-i\partial_1)) \right] \chi = 0.$$

DXCVII. ACTUAL REDUCED PDE ON THE ACTIVE EXPLICIT ODD EXTRA-CARRIER ROUTE

The reduced coefficient dictionary is

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = I_c \tilde{\Lambda}_E, \quad \beta = I_c \tilde{\Lambda}_O,$$

with

$$I_3 = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3, \quad I_c = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) \cos \theta(\xi).$$

At one-mediator level,

$$\tilde{\Lambda}_E = -\frac{\beta_X \gamma_E}{M_X^2} C_E J_1, \quad \tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1.$$

Hence the general explicit lowest-order coefficient sheet is

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3,$$

$$\alpha = -\frac{\beta_X \gamma_E}{M_X^2} C_E J_1 I_c, \quad \beta = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1 I_c.$$

In canonical explicit normalization

$$C_{\parallel} = C_E = C_O = 1,$$

this becomes

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = -\frac{\beta_X \gamma_E}{M_X^2} J_1 I_c, \quad \beta = -\frac{\beta_X \gamma_O}{M_X^2} J_1 I_c.$$

Therefore the actual reduced PDE is

$$\left[i\partial_t - \frac{\lambda_c}{2} I_3 \sigma_3 (-i\partial_{\parallel}) + \frac{\beta_X \gamma_E}{M_X^2} J_1 I_c (\sigma_1 (-i\partial_1) + \sigma_2 (-i\partial_2)) + \frac{\beta_X \gamma_O}{M_X^2} J_1 I_c (-\sigma_1 (-i\partial_2) + \sigma_2 (-i\partial_1)) \right] \chi = 0.$$

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = \beta = -\frac{\beta_X \gamma_{\perp}}{M_X^2} J_1 I_c.$$

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = \beta = -\frac{\beta_X \gamma_{\perp}}{M_X^2} I_c.$$

$$I_3, \quad I_c, \quad \lambda_c, \quad J_1$$

DXCVIII. DIRECT MICROSCOPIC READING OF λ_c AND $\tilde{\Lambda}_O$

A. 1. Longitudinal coefficient λ_c

Write the background-dependent Hessian in block form and extract the first-order transport kernel

$$\Gamma^{(2)}[\bar{\Phi}_{\min}] = H_B(\xi) - iV^{\mu}(\xi)\partial_{y^{\mu}} + \dots$$

Restrict the OO -block to the tangent bundle Q_B :

$$V_{OO}^{\mu}(\xi) := \Pi_Q \mathcal{V}_{OO}^{\mu}(\xi) \Pi_Q.$$

The longitudinal projector functional is

$$\mathfrak{P}_{\parallel}[\mathcal{M}] := \frac{1}{2} \text{Tr}_{Q_B}(\Sigma_3 n_{\mu} \mathcal{M}^{\mu}).$$

Define

$$a_{3;OO}(\xi) = \mathfrak{P}_{\parallel}[V_{OO}] = \frac{1}{2} \text{Tr}_{Q_B}(\Sigma_3 n_{\mu} V_{OO}^{\mu}(\xi)).$$

Its principal odd expansion is

$$a_{3;OO}(\xi) = \ell_{0;OO}^{(1)} \theta'(\xi) + \ell_{0;OO}^{(3)} \theta'(\xi)^3 + O(\theta'(\xi)\theta(\xi)^2, \theta'(\xi)^5).$$

In the current concrete Class II-compatible branch one has

$$\Lambda_{\parallel}^{(0)} = 0,$$

so the longitudinal coefficient is Branch B and therefore

$$\lambda_c = 2 \ell_{0;OO}^{(3)}.$$

Equivalently, the executable limit formula is

$$\lambda_c = 2 \lim_{\theta' \rightarrow 0} \frac{\mathfrak{P}_{\parallel}[V_{OO}]}{\theta'(\xi)^3}.$$

Hence the reduced longitudinal coefficient is

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3 = I_3 \lim_{\theta' \rightarrow 0} \frac{\mathfrak{P}_{\parallel}[V_{OO}]}{\theta'(\xi)^3}.$$

B. 2. Odd transverse coefficient $\tilde{\Lambda}_O$

On the active explicit odd extra-carrier route, the decisive microscopic coupling is

$$\mathcal{I}_O = X_i^{(O)} \cos \theta(\xi) J_j D_j \psi.$$

Its induced tangent transport block is

$$V_{EO}^\mu = \theta'(\xi) \cos \theta(\xi) \left[\tilde{\Lambda}_E E^\mu{}_a \Sigma_a + \tilde{\Lambda}_O J^\mu{}_a \Sigma_a \right].$$

The odd projector functional is

$$\mathfrak{P}_O[\mathcal{M}] := \frac{1}{4} J^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_\mu).$$

Therefore the odd coefficient is read directly as

$$\tilde{\Lambda}_O = \frac{\mathfrak{P}_O[V_{EO}]}{\theta'(\xi) \cos \theta(\xi)}.$$

At one-mediator level this becomes

$$\tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1.$$

In canonical normalization $C_O = 1$,

$$\tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} J_1.$$

Hence the reduced odd transverse coefficient is

$$\beta = I_c \tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1 I_c,$$

and in canonical normalization

$$\beta = -\frac{\beta_X \gamma_O}{M_X^2} J_1 I_c.$$

C. 3. The actual reduced PDE with direct microscopic readout

Thus the active-route reduced PDE is

$$\left[i\partial_t - I_3 \left(\lim_{\theta' \rightarrow 0} \frac{\mathfrak{P}_\parallel[V_{OO}]}{\theta'^3} \right) \sigma_3(-i\partial_\parallel) + I_c \tilde{\Lambda}_E (\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) + I_c \tilde{\Lambda}_O (-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0.$$

If one also inserts the one-mediator odd coefficient,

$$\left[i\partial_t - I_3 \left(\lim_{\theta' \rightarrow 0} \frac{\mathfrak{P}_\parallel[V_{OO}]}{\theta'^3} \right) \sigma_3(-i\partial_\parallel) + I_c \tilde{\Lambda}_E (\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) - \frac{\beta_X \gamma_O}{M_X^2} C_O J_1 I_c (-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0.$$

D. 4. What remains literally to do

There is no new structural theorem left here.
The remaining microscopic tasks are exactly:

$$\boxed{\text{(A) compute } V_{OO}^\mu(\xi) \text{ from the full Hessian and evaluate } \mathfrak{P}_\parallel[V_{OO}],}$$

$$\boxed{\text{(B) compute } J_1 \text{ (equivalently } \tilde{\Lambda}_O) \text{ from the explicit odd extra-carrier insertion.}}$$

Once these are inserted, the reduced coefficient point

$$\boxed{(\lambda_\parallel, \alpha, \beta)}$$

is fully populated.

DXCIX. DIRECT SOURCE-BY-SOURCE READING OF THE REMAINING ACTUAL REDUCED PDE COEFFICIENTS

DC. 1. LONGITUDINAL COEFFICIENT λ_c FROM THE OO -BLOCK

Write the background-dependent Hessian block expansion

$$\mathcal{H}_{OO} = \mathcal{H}_{OO}^{(0)}(\xi) - i \mathcal{V}_{OO}^\mu(\xi) \partial_{y^\mu} + \mathcal{W}_{OO}^{\mu\nu}(\xi) \partial_{y^\mu} \partial_{y^\nu} + \cdots .$$

Restrict the first-order block to the tangent bundle:

$$\boxed{V_{OO}^\mu(\xi) := \Pi_Q \mathcal{V}_{OO}^\mu(\xi) \Pi_Q.}$$

The longitudinal projector functional is

$$\boxed{\mathfrak{P}_\parallel[\mathcal{M}] := \frac{1}{2} \text{Tr}_{Q_B}(\Sigma_3 n_\mu \mathcal{M}^\mu).}$$

Therefore define

$$\boxed{a_{3;OO}(\xi) = \mathfrak{P}_\parallel[V_{OO}] = \frac{1}{2} \text{Tr}_{Q_B}(\Sigma_3 n_\mu V_{OO}^\mu(\xi)).}$$

Its branch-sensitive odd expansion is

$$a_{3;OO}(\xi) = \ell_{0;OO}^{(1)} \theta'(\xi) + \ell_{0;OO}^{(3)} \theta'(\xi)^3 + O(\theta'(\xi) \theta(\xi)^2, \theta'(\xi)^5).$$

The exact identification is

$$\boxed{\Lambda_{\parallel;OO}^{(0)} = \ell_{0;OO}^{(1)}, \quad \frac{\lambda_c}{2} = \ell_{0;OO}^{(3)}.$$

In the current concrete Class II-compatible branch one has

$$\boxed{\Lambda_{\parallel}^{(0)} = 0,}$$

so the practical longitudinal coefficient is forced onto Branch B:

$$\boxed{\lambda_c = 2 \ell_{0;OO}^{(3)}.$$

Equivalently, the executable reading formula is

$$\lambda_c = 2 \lim_{\theta' \rightarrow 0} \frac{\mathfrak{P}_{\parallel}[V_{OO}]}{\theta'(\xi)^3}.$$

Hence the reduced longitudinal coefficient is

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3.$$

DCI. 2. WHY THE PRIMITIVE LONGITUDINAL SOURCE STARTS AT θ'^3

The microscopic primitive longitudinal channel is generated by

$$\mathcal{T}_{\xi} = i[P_B, \partial_{\xi} P_B].$$

Its direct tangent restriction vanishes,

$$\Pi_Q \mathcal{T}_{\xi} \Pi_Q = 0,$$

but its square is nontrivial:

$$\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q = \frac{\theta'(\xi)^2}{2} (\mathbf{1}_2 + \Sigma_3).$$

Taking the traceless part gives

$$S_{\xi} = \frac{\theta'(\xi)^2}{2} \Sigma_3.$$

Because the graded overlap kills the parity-even θ'^2 piece, one additional odd factor is required, so the surviving primitive transport seed is

$$V_{\parallel, \text{prim}}^{\mu}(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^{\mu} \Sigma_3.$$

This is why the active longitudinal reading is cubic in θ' , not linear.

DCII. 3. EVEN/ODD TRANSVERSE COEFFICIENTS FROM THE $TD + DT$ BLOCK

Define the mixed first-order transverse carrier

$$\mathcal{M}_{TD+DT}^{\mu} := \Pi_Q (\mathcal{V}_{TD}^{\mu} + \mathcal{V}_{DT}^{\mu}) \Pi_Q.$$

The canonical projector functionals are

$$\mathfrak{P}_E[\mathcal{M}] := \frac{1}{4} E^{\mu}{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_{\mu}), \quad \mathfrak{P}_O[\mathcal{M}] := \frac{1}{4} J^{\mu}{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_{\mu}).$$

So at the minimal mixed-block level,

$$\Lambda_E = \mathfrak{P}_E[\mathcal{M}_{TD+DT}^{\text{inv}}], \quad \Lambda_O = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}].$$

More generally, beyond the minimal truncation,

$$\Lambda_E = \sum_{AB \in \{TD, DT, OD, DO, DD\}} e_{0;AB}^{(1)}, \quad \Lambda_O = \sum_{AB \in \{TD, DT, OD, DO, DD\}} o_{0;AB}^{(1)},$$

but the sharp leading audit starts from $TD + DT$.

Hence the full remaining microscopic transport point is

$$\mathcal{H}_{OO} \rightarrow \lambda_c, \quad \mathcal{H}_{TD}, \mathcal{H}_{DT} \rightarrow \Lambda_E, \Lambda_O.$$

DCIII. 4. DIRECT ODD COEFFICIENT ON THE ACTIVE EXPLICIT ODD ROUTE

On the active explicit odd extra-carrier route,

$$\mathcal{I}_O = X_i^{(O)} \cos \theta(\xi) J_j D_j \psi.$$

Its induced tangent transport block is

$$V_{EO}^\mu = \theta'(\xi) \cos \theta(\xi) \left[\tilde{\Lambda}_E E^\mu{}_a \Sigma_a + \tilde{\Lambda}_O J^\mu{}_a \Sigma_a \right].$$

Therefore the local odd density is

$$b_O(\xi) = \tilde{\Lambda}_O \theta'(\xi) \cos \theta(\xi).$$

At operator level,

$$\tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1.$$

Under canonical shell normalization $C_O = 1$,

$$\tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} J_1.$$

Hence the reduced odd transverse coefficient is

$$\beta = I_c \tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1 I_c.$$

DCIV. 5. THE ACTUAL REDUCED PDE AFTER THESE TWO READINGS

Once λ_c and $\tilde{\Lambda}_O$ are populated, the active-route reduced PDE is

$$\left[i\partial_t - \frac{\lambda_c}{2} I_3 \sigma_3(-i\partial_{||}) + I_c \tilde{\Lambda}_E (\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) + I_c \tilde{\Lambda}_O (-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0.$$

If one also inserts the one-mediator odd coefficient explicitly,

$$\left[i\partial_t - \frac{\lambda_c}{2} I_3 \sigma_3(-i\partial_{||}) + I_c \tilde{\Lambda}_E (\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) - \frac{\beta_X \gamma_O}{M_X^2} C_O J_1 I_c (-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0.$$

DCV. 6. IMMEDIATE EXECUTABLE COMPUTATION ORDER

Nothing structural remains here. The actual microscopic tasks are exactly:

$$(A) \text{ compute } \mathcal{V}_{OO}^\mu(\xi) \text{ from the full Hessian, then evaluate } \mathfrak{P}_{||}[V_{OO}] \text{ and read } \lambda_c,$$

$$(B) \text{ compute } \mathcal{V}_{TD}^\mu + \mathcal{V}_{DT}^\mu \text{ (or use the explicit } EO\text{-route insertion), then read } \tilde{\Lambda}_O.$$

After that, the coefficient point

$$(\lambda_{||}, \alpha, \beta)$$

is fully populated.

DCVI. SOURCE-BY-SOURCE HESSIAN AUDIT FOR THE REMAINING ACTUAL REDUCED PDE COEFFICIENTS

DCVII. 1. EXACT SOURCE SPLIT TO AUDIT

The first-order tangent transport operator is obtained from the one-defect Hessian by keeping only the worldvolume first-derivative block:

$$\mathbb{H}_1^\mu(\xi) = \mathbb{H}_{1,\Psi}^\mu + \mathbb{H}_{1,P}^\mu + \mathbb{H}_{1,\sigma}^\mu + \mathbb{H}_{1,X}^\mu + \mathbb{H}_{1,J}^\mu + \mathbb{H}_{1,K}^\mu + \mathbb{H}_{1,EO}^\mu. \quad (1447)$$

Projecting to the tangent bundle gives

$$V^\mu(\xi) = \Pi_Q \mathbb{H}_1^\mu(\xi) \Pi_Q. \quad (1448)$$

The longitudinal and transverse projector functionals are

$$\mathfrak{P}_\parallel[\mathcal{M}] = \frac{1}{2} \text{Tr}_{Q_B}(\Sigma_3 n_\mu \mathcal{M}^\mu), \quad (1449)$$

$$\mathfrak{P}_E[\mathcal{M}] = \frac{1}{4} E^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_\mu), \quad \mathfrak{P}_O[\mathcal{M}] = \frac{1}{4} J^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_\mu). \quad (1450)$$

DCVIII. 2. LONGITUDINAL SOURCE TABLE INSIDE THE OO-BLOCK

The orientational diagonal block is

$$\mathcal{H}_{OO} = -Z_{O,k} \partial^2 + \mathcal{M}_{OO}^{\text{pot}}(\xi) + Y_k \mathcal{K}_{OO}^{\nabla \text{ani}}(\xi, \partial) + \mathcal{T}_{OO}(\xi, \partial_y). \quad (1451)$$

The exact solved lowest-order verdict is:

$$\mathfrak{P}_\parallel[V_{OO}^{(\text{kin})}] = 0, \quad \mathfrak{P}_\parallel[V_{OO}^{(\text{pot})}] = 0, \quad (1452)$$

because the kinetic Laplacian and local potential pieces do not generate a first-order worldvolume transport term.

Also,

$$\Lambda_\parallel^{(0)} = 0, \quad \ell_0^{(1)} = 0, \quad (1453)$$

so the derivative-anisotropy piece does not produce a Branch-A longitudinal coefficient in the current solved lowest-order class:

$$\mathfrak{P}_\parallel[V_{OO}^{(\nabla \text{ani})}] = 0. \quad (1454)$$

At principal shell order, the Schur-corrected source classification is

$$a_{3;OO}^{(JJ)} \neq 0 \text{ allowed}, \quad a_{3;OO}^{(JK)} = 0, \quad a_{3;OO}^{(KK)} = 0. \quad (1455)$$

Therefore the only surviving principal longitudinal source is

$$\mathcal{T}_{OO}^{(JJ)}. \quad (1456)$$

DCIX. 3. PRIMITIVE LONGITUDINAL SEED AND DIRECT READING OF λ_c

The primitive one-defect seed is

$$\boxed{\mathcal{T}_\xi = i[P_B, \partial_\xi P_B], \quad S_\xi = (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}}.} \quad (1457)$$

For the explicit defect representative,

$$\boxed{S_\xi = \frac{1}{2} \theta'(\xi)^2 \Sigma_3.} \quad (1458)$$

Hence the primitive longitudinal transport kernel is

$$\boxed{V_{OO,\text{prim}}^\mu(\xi) = \theta'(\xi) S_\xi n^\mu = \frac{1}{2} \theta'(\xi)^3 n^\mu \Sigma_3.} \quad (1459)$$

Thus the longitudinal coefficient profile is

$$\boxed{a_{3;OO}(\xi) = \mathfrak{P}_\parallel[V_{OO}] = \ell_{0;OO}^{(3)} \theta'(\xi)^3 + \dots, \quad \ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2}.} \quad (1460)$$

So the executable microscopic reading formula is

$$\boxed{\lambda_c = \lambda_{c;OO} = 2 \lim_{\theta' \rightarrow 0} \frac{\mathfrak{P}_\parallel[V_{OO}]}{\theta'(\xi)^3}.} \quad (1461)$$

After overlap insertion,

$$\boxed{\lambda_\parallel = \frac{\lambda_c}{2} I_{\theta'^3}.} \quad (1462)$$

DCX. 4. PRINCIPAL TRANSVERSE SOURCE FROM $TD + DT$

Define the mixed first-order transverse carrier

$$\boxed{\mathcal{M}_{TD+DT}^\mu = \Pi_Q (\mathcal{V}_{TD}^\mu + \mathcal{V}_{DT}^\mu) \Pi_Q.} \quad (1463)$$

The leading transverse coefficients are then read by

$$\boxed{\Lambda_E = \mathfrak{P}_E[\mathcal{M}_{TD+DT}^{\text{inv}}], \quad \Lambda_O = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}].} \quad (1464)$$

The sharp source classification is

$$\boxed{JK + KJ \rightarrow (\Lambda_E, \Lambda_O), \quad KK \rightarrow \text{same-shell residual / correction only}.} \quad (1465)$$

Therefore the actual leading transverse audit is not broad anymore: it is the explicit $TD + DT$ evaluation of the $JK + KJ$ -driven mixed transport block.

DCXI. 5. ODD COEFFICIENT ON THE ACTIVE EXPLICIT ODD ROUTE

On the active explicit odd extra-carrier route, the odd insertion is

$$\boxed{\mathcal{I}_O = X_i^{(O)} \cos \theta(\xi) J_j D_j \psi.} \quad (1466)$$

Its induced tangent transport block is

$$V_{EO}^\mu = \theta'(\xi) \cos \theta(\xi) \left[\tilde{\Lambda}_E E^\mu{}_a \Sigma_a + \tilde{\Lambda}_O J^\mu{}_a \Sigma_a \right]. \quad (1467)$$

Hence the odd projector reads directly

$$\tilde{\Lambda}_O = \frac{\Re_O[V_{EO}]}{\theta'(\xi) \cos \theta(\xi)}. \quad (1468)$$

At one-mediator level, in the current reference normalization,

$$\tilde{\Lambda}_O^{(1)} = -\frac{\beta_X \gamma_O}{M_X^2}. \quad (1469)$$

Thus

$$\beta = I_c \tilde{\Lambda}_O. \quad (1470)$$

DCXII. 6. ACTUAL REDUCED PDE ONCE THE TWO MICROSCOPIC READS ARE INSERTED

Once λ_c and $\tilde{\Lambda}_O$ are populated, the active-route reduced PDE is

$$\left[i\partial_t - \frac{\lambda_c}{2} I_3 \Sigma_3 (-i\partial_{||}) + I_c \tilde{\Lambda}_E (\Sigma_1 (-i\partial_1) + \Sigma_2 (-i\partial_2)) + I_c \tilde{\Lambda}_O (-\Sigma_1 (-i\partial_2) + \Sigma_2 (-i\partial_1)) \right] \chi = 0. \quad (1471)$$

DCXIII. 7. IMMEDIATE EXECUTABLE ORDER

So the literal next computation is:

$$\text{(A) differentiate the full one-defect Hessian block-by-block, keep only the first-order transport part,} \quad (1472)$$

$$\text{(B) in the } OO\text{-block, evaluate only the } JJ \text{ source and read } \lambda_c \text{ from (1461),} \quad (1473)$$

$$\text{(C) in the } TD + DT\text{-block, evaluate the } JK + KJ \text{ mixed transport and read } (\Lambda_E, \Lambda_O), \quad (1474)$$

$$\text{(D) on the active odd route, equivalently read } \tilde{\Lambda}_O \text{ from (1468).} \quad (1475)$$

DCXIV. ACTUAL LOCAL TENSOR FORM OF THE PURE JJ LONGITUDINAL SOURCE

DCXV. 1. STARTING POINT: SCHUR-REDUCED JJ SECTOR

After integrating out the heavy mediator X_i^a , the quadratic correction splits as

$$\Delta_X^{(2)} = -\frac{\alpha_X^2}{M_X^2} \delta J \delta J^\dagger - \frac{\alpha_X \beta_X}{M_X^2} (\delta J \delta K^\dagger + \delta K \delta J^\dagger) - \frac{\beta_X^2}{M_X^2} \delta K \delta K^\dagger. \quad (1476)$$

At principal longitudinal shell order, only the pure JJ -sector survives in the OO -longitudinal slot:

$$a_{3;OO}^{(JJ)} \neq 0, \quad a_{3;OO}^{(JK)} = 0, \quad a_{3;OO}^{(KK)} = 0. \quad (1477)$$

So the actual longitudinal microscopic source is

$$\mathcal{T}_{OO}^{(JJ)}. \quad (1478)$$

DCXVI. 2. LOCAL LOWEST-DERIVATIVE FORM OF THE PURE JJ SOURCE

At principal shell order, the background current takes the local form

$$\boxed{J_i^a[\phi_0, P_0] = m^a(P_0) \partial_i \phi_0.} \quad (1479)$$

The Bloch derivative insertion contributes only the spatial tensor δ_{ij} at leading shell order. Therefore the induced axial JJ -coefficient field must have the lowest-derivative local tensor form

$$\boxed{\mathcal{A}_j^{(JJ)}(x) = \sum_m C_{jm}^{(JJ)} \partial_m \phi_0(x).} \quad (1480)$$

Because no principal shell tensor survives except δ_{jm} , the coefficient tensor collapses to

$$\boxed{C_{jm}^{(JJ)} = \Xi_{JJ} \delta_{jm}.} \quad (1481)$$

Hence the exact local lowest-derivative form is

$$\boxed{\mathcal{A}_j^{(JJ)}(x) = \Xi_{JJ} \partial_j \phi_0(x).} \quad (1482)$$

This is the actual local tensor form of the pure longitudinal JJ -source.

DCXVII. 3. FIRST-SHELL FOURIER COEFFICIENT AT THE SELECTED HARMONIC

Let A_{G_*} be the first-shell Fourier amplitude of the locked background ϕ_0 at the selected shell vector G_* . Then

$$\boxed{\widehat{\mathcal{A}}_j^{(JJ)}(G_*) = i \Xi_{JJ} G_{*j} A_{G_*}.} \quad (1483)$$

Contracting with the shell axis $\hat{n} = G_*/|G_*|$, we obtain the scalar longitudinal first-shell amplitude

$$\boxed{\widehat{\mathcal{A}}_{JJ}(G_*) := \hat{n}^j \widehat{\mathcal{A}}_j^{(JJ)}(G_*) = i \Xi_{JJ} |G_*| A_{G_*}.} \quad (1484)$$

Thus the pure JJ longitudinal noncancellation problem is reduced to the finite scalar test

$$\boxed{\Xi_{JJ} \neq 0, \quad A_{G_*} \neq 0.} \quad (1485)$$

DCXVIII. 4. COLLAPSE TO THE INVARIANT L_*

The longitudinal scalar Ξ_{JJ} has already been reduced to the invariant

$$\boxed{\Xi_{JJ} = \mathcal{N}_{JJ} L_*, \quad L_* := m_{\parallel} u_{\parallel}.} \quad (1486)$$

Therefore

$$\boxed{\widehat{\mathcal{A}}_{JJ}(G_*) = i \mathcal{N}_{JJ} L_* |G_*| A_{G_*}.} \quad (1487)$$

So the pure- JJ longitudinal slot is completely reduced to the finite scalar package

$$\boxed{\mathcal{N}_{JJ}, \quad L_* = m_{\parallel} u_{\parallel}, \quad |G_*|, \quad A_{G_*}.} \quad (1488)$$

DCXIX. 5. READING λ_c

At the projected operator level, the longitudinal master slot is

$$\lambda_{\parallel}^{(\text{med})} := \frac{1}{2} \text{Tr} \left(\sigma_3 P_* (\hat{n}_i \partial_{k_i} H_{\text{eff}}^{(\text{med})}(k_*) P_*) \right). \quad (1489)$$

On the current working branch, the pure- JJ longitudinal representative satisfies

$$\lambda_c^{(JJ)} \sim \frac{\alpha_X^2}{M_X^2} \mathcal{N}_{JJ} m_{\parallel} u_{\parallel} |G_*| A_{G_*}. \quad (1490)$$

Equivalently,

$$\lambda_c^{(JJ)} \sim \frac{\alpha_X^2}{M_X^2} \Xi_{JJ} |G_*| A_{G_*}. \quad (1491)$$

This is the concrete actual-reading formula for the longitudinal coefficient on the current branch.

DCXX. 6. REDUCED ACTUAL PDE LONGITUDINAL COEFFICIENT

Once λ_c is read, the reduced longitudinal PDE coefficient is

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}. \quad (1492)$$

Using the explicit defect overlap,

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3, \quad (1493)$$

we get

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3. \quad (1494)$$

DCXXI. 7. HONEST REMAINING TASK

There is no structural ambiguity left in the longitudinal source classification.

What remains is only the finite scalar extraction of

$$\mathcal{N}_{JJ}, \quad m_{\parallel}, \quad u_{\parallel}, \quad A_{G_*} \quad (1495)$$

from the refined locked Bloch data, and then inserting them into (1490).